INTRODUCTION AND INSTRUCTIONS

📡 Overview Module 01: Introduction to GPS

GPS Module Page



🧭 Module Overview

Welcome to our exploration of satellite positioning, where we begin with the Global Positioning System, or **GPS**—a technology so deeply embedded in our daily lives that we often take it for granted. From navigating road trips to supporting precision agriculture and emergency response, GPS has transformed the way we interact with the world. But behind its apparent simplicity lies a sophisticated system of satellites, signals, and timing mechanisms that reflects decades of scientific innovation and collaboration. In this course, we'll uncover how GPS works, understand 2D and 3D coordinates it measures, error sources and methods to mitigate them, and examine the principles and practices that make accurate positioning possible anywhere on Earth.

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📍 What is GPS?

GPS (Global Positioning System) is a satellite-based navigation system that provides accurate location and time information anywhere on or near the Earth's surface. Originally developed by the U.S. Department of Defense for military purposes, GPS has become an essential civilian technology that enables precise positioning for countless applications.

📚 Module Learning Objectives

At the end of this module, learners will be able to:

- Describe the historical development of GPS and its predecessor systems.
- Explain the fundamental components and architecture of the GPS system, including the space, control, and user segments.
- Identify the key milestones in GPS policy decisions that led to its civilian accessibility.
- Recognize GPS objectives for global users.
- Interpret the significance of satellite geometry and DOP (Dilution of Precision) values in determining GPS accuracy.

📝 Module Instructions

- Review the pages in this module and watch the instructor videos embedded within the module pages.
- Review Chapter 3 of the textbook for relevant topics covered in this module.

- After you have review the content for this module, you will complete the assessments and activities to evaluate the learning objectives.

CONTENT

📚 GPS Overview and Governance



Section 01: GPS Overview and Governance

GISC 4315

📹 Instructor Video Lecture

GPS Overview and Governance

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📚 Video Topics Covered

- Introduction to NAVSTAR GPS
- GPS vs. GNSS relationship
- Signal evolution and modernization

- Key applications across industries
- U.S. governance structure
- International cooperation

> 💡 Study Tip
>
> Take notes on the different GPS signal types and their characteristics as you watch.

🛰️ What is GPS?

The NAVSTAR Global Positioning System

The **NAVSTAR Global Positioning System (GPS)** is a **satellite-based radio-navigation and time-transfer system**. It was designed, financed, deployed, and originally operated by the U.S. Department of Defense, and is now operated by the **United States Space Force**. Although developed for military purposes, GPS today supports a significantly larger base of **civilian users** than military users.

GPS is part of a broader category known as **Global Navigation Satellite Systems (GNSS)**. It enables **positioning, navigation, and timing (PNT)** anywhere on or near Earth, as long as the receiver has an unobstructed line of sight to at least four satellites.

> 💡 Key Concept: Passive Operation
>
> One of the distinguishing features of GPS is that it operates in a **passive mode**—the user does not need to transmit any signals, making it **independent of other technologies** and highly reliable.



A collage showing various GPS applications (navigation, surveying, agriculture, emergency response, etc.)

🌍 Key Applications of GPS

Diverse GPS Applications Across Industries

Navigation & Transportation

- Navigation and Route Guidance
- Transportation and Logistics

Professional Services

- Mapping and Surveying
- Scientific Research

Critical Infrastructure

- Timing and Synchronization
- Telecommunications
- Financial transactions
- Power grid operations

🚨 **Emergency**
Emergency Response

🌾 **Agriculture**
Agriculture (Precision Farming)

📍 **Tracking**
Tracking and Monitoring

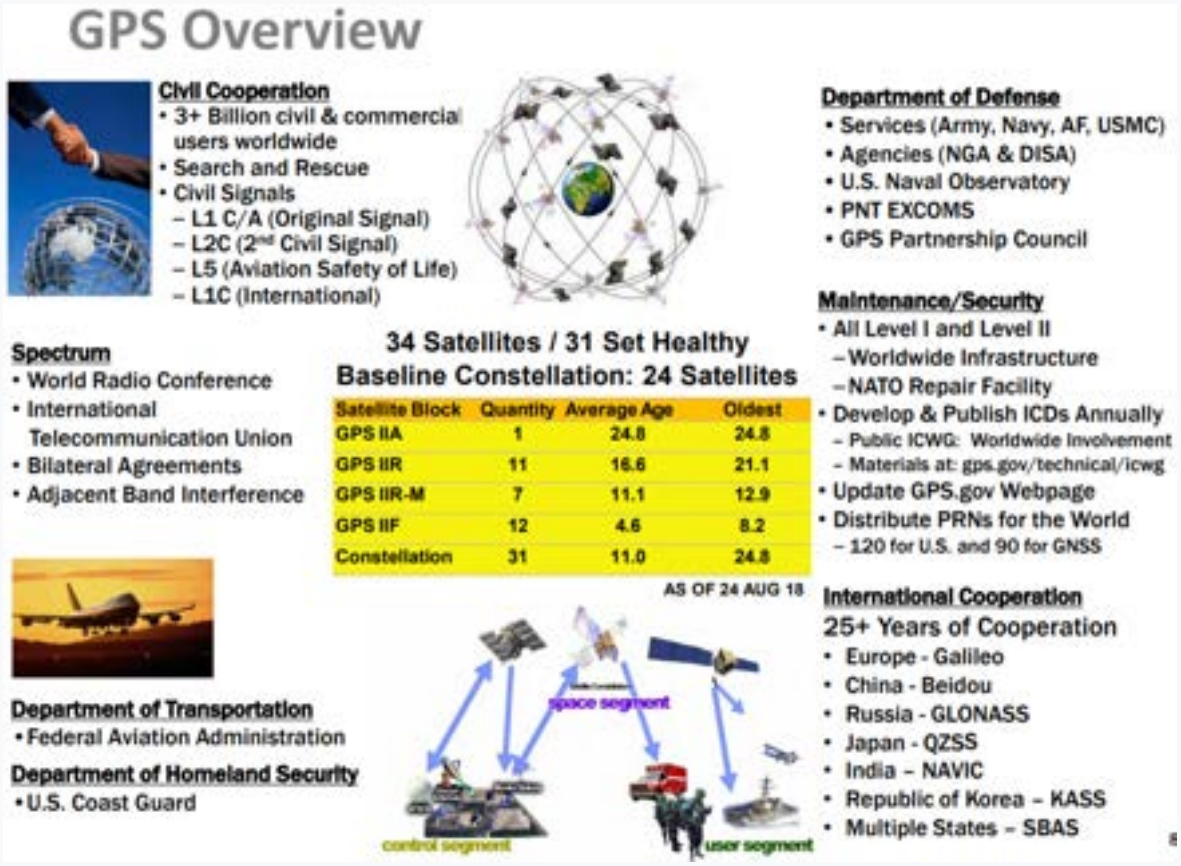
⚔️ **Military**
Military Operations

🔍 **Specialized Application**

Nuclear detonation detection - GPS satellites carry sensors as part of a global monitoring network.

📡 GPS Signals and Modernization

Evolution of GPS Signal Structure



GPS overview diagram showing satellite constellation, control segment, and user segment

L1 C/A Signal

- **Broadcast frequency:** 1575.42 MHz (L1 band)
- **Original GPS signal** for civilian users
- Uses **Coarse/Acquisition (C/A) code** for satellite ID and timing
- **Supported by all GPS satellites**
- Enables basic positioning with **single-frequency receivers**
- Widely used in **consumer devices** (phones, car GPS)
- Lower accuracy and more prone to interference than newer signals

L2C Signal

- **Increased sensitivity** for improved signal tracking
- Better performance in **unfavorable conditions** (obstructions, indoors)
- **Stronger tolerance to interference and multipath**

L5 Signal

- **Reserved exclusively for aviation safety services**
- Possibility to **track weaker signals**

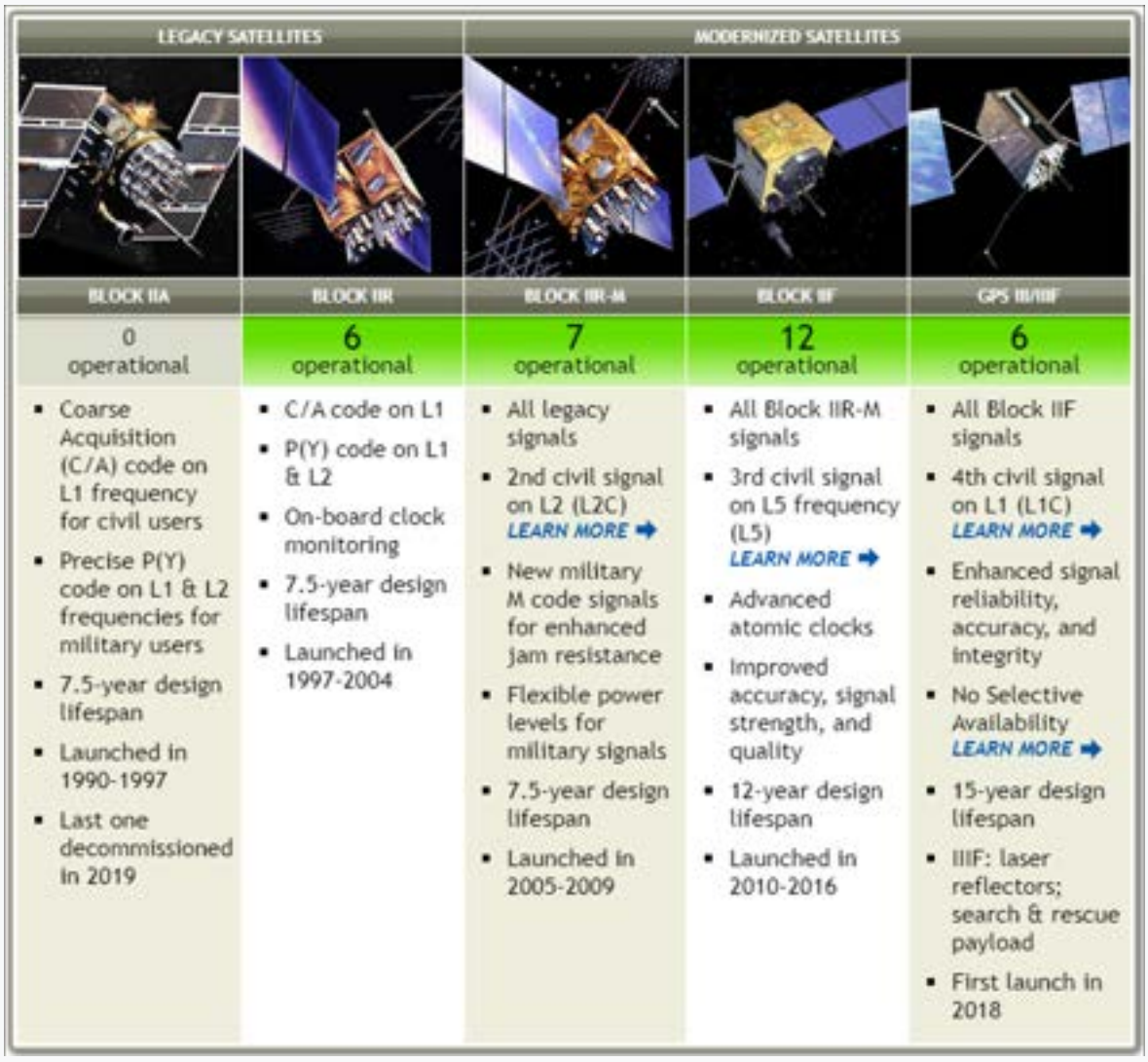
- **Better immunity to interference** with greater bandwidth
- **(L1/L2/L5) signal combination** for quick location fix

🚀 **L1C Signal Innovation**

L1C and L1 C/A signals are transmitted at the same frequency, but **L1C codes are 10 times longer** than L1 C/A. This makes the signal **more robust to interference** when multiple satellites are tracked on the same frequency band.

⚡ GPS Modernization Benefits

Enhanced Capabilities and Interoperability



LEGACY SATELLITES		MODERNIZED SATELLITES		
BLOCK IIA	BLOCK IIR	BLOCK IIR-M	BLOCK IIF	GPS III/IIIF
0 operational	6 operational	7 operational	12 operational	6 operational
• Coarse Acquisition (C/A) code on L1 frequency for civil users • Precise P(Y) code on L1 & L2 frequencies for military users • 7.5-year design lifespan • Launched in 1990-1997 • Last one decommissioned in 2019	• C/A code on L1 • P(Y) code on L1 & L2 • On-board clock monitoring • 7.5-year design lifespan • Launched in 1997-2004	• All legacy signals • 2nd civil signal on L2 (L2C) LEARN MORE ➡ • New military M code signals for enhanced jam resistance • Flexible power levels for military signals • 7.5-year design lifespan • Launched in 2005-2009	• All Block IIR-M signals • 3rd civil signal on L5 frequency (L5) LEARN MORE ➡ • Advanced atomic clocks • Improved accuracy, signal strength, and quality • 12-year design lifespan • Launched in 2010-2016	• All Block IIF signals • 4th civil signal on L1 (L1C) LEARN MORE ➡ • Enhanced signal reliability, accuracy, and integrity • No Selective Availability LEARN MORE ➡ • 15-year design lifespan • IIIF: laser reflectors; search & rescue payload • First launch in 2018

Legacy Satellites vs Modernized Satellites with Block designations and capabilities

🔧 Enhanced Performance

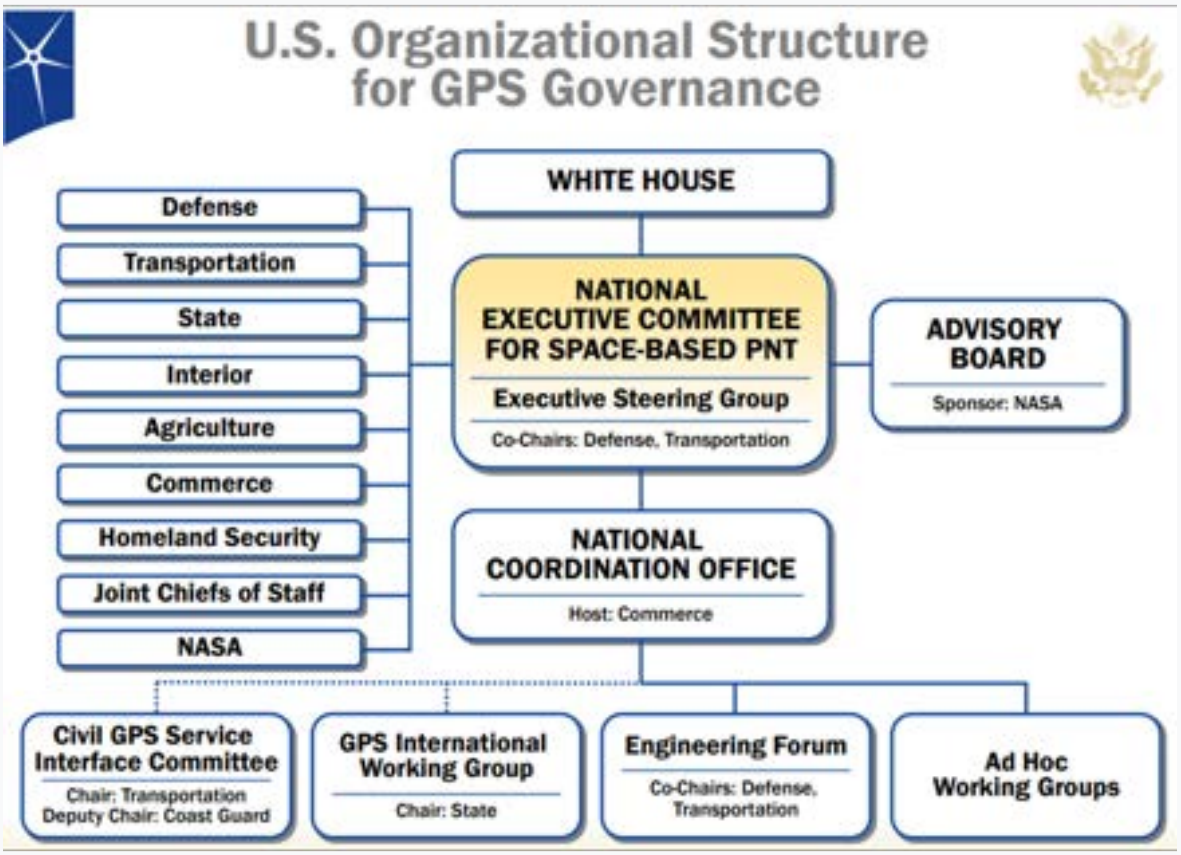
- **More robust and accurate positioning**
- **Resistant to interference and multipath effects**
- **Enhanced reliability** in challenging environments

🌐 Global Interoperability

- **Compatible with other GNSS systems**
- **L1C signal compatible with Galileo's E1 signal**
- **Multi-GNSS receivers** for improved accuracy and availability

🏛️ GPS Governance Structure

U.S. Organizational Structure for GPS Governance



Organizational chart showing White House, National Executive Committee, agencies, and supporting groups

Executive Leadership

- **White House** oversees GPS policy
- **National Executive Committee for Space-Based PNT** leads coordination, co-chaired by Defense and Transportation
- National Coordination Office (hosted by Commerce) supports implementation

Key Agencies Involved

- **Defense**
- **Transportation**
- **State**
- **Interior**
- **Agriculture**
- **Commerce**
- **Homeland Security**

Advisory Structure

- Advisory Board (sponsored by NASA) offers expert input
- **Joint Chiefs of Staff**
- **NASA**

🤝 **Supporting Groups**

- **Civil GPS Service Interface Committee**
- **GPS International Working Group**
- **Engineering Forum**
- **Ad Hoc Working Groups**

📝 Section Summary

Key Takeaways

In this page, we explored the fundamental aspects of GPS, from its definition as a satellite-based radio-navigation system to its complex governance structure. We examined the evolution of GPS signals from the original L1 C/A to modernized signals like L2C, L5, and L1C, understanding how these improvements enhance accuracy, reliability, and interoperability with other GNSS systems.

📚 Background & Origins

Module 01: Background & Origins

GISC 4315

📹 Instructor Video Lecture

GPS Historical Development and Predecessor Systems

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📚 Video Topics Covered

- GPS development challenges
- Doppler effect and Sputnik discovery
- TRANSIT system capabilities
- System 621B innovations
- Timation program contributions

- Integration into modern GPS

> 💡 Study Tip
>
> Pay attention to how each predecessor system contributed specific technologies to modern GPS.

⚠️ GPS Development Challenges

Critical Obstacles That Nearly Ended GPS

The development of the **Global Positioning System (GPS)** was not a straightforward process. It faced several critical obstacles that nearly caused the program to be **canceled on multiple occasions**:



Key Spacecraft from TRANSIT, Timation, and Pre-GPS Era

🔧 Technical Challenges

Early GPS development encountered major engineering hurdles:

- **Designing highly accurate atomic clocks** suitable for space
- **Creating reliable satellite communication links**
- **Developing robust signal structures** (pseudorandom noise codes)

💰 Budgetary Pressures

The GPS program suffered repeated funding cuts:

- **Funding cuts** in the 1970s and late 1970s
- **Economic and defense priorities shifted**
- Critics questioned **cost-effectiveness**
- Older systems like **TRANSIT still functioned adequately**

🏛️ Institutional Resistance

Military services were initially resistant:

- **Different branches** had their own navigation systems
- **Hesitant to adopt** a joint solution
- Concerns over **reliability and overdependence**
- Potential **vulnerabilities** of centralized system

- **Ensuring consistent global coverage**
- Integration into a **coherent, functional system**

✅ **GPS Survival**

Despite these challenges, GPS survived due to **strong support from a few visionary leaders** in government and defense, who recognized its potential to revolutionize navigation, positioning, and timing across both military and civilian sectors. Ultimately, their perseverance led to the successful deployment of one of the most widely used technologies in the world today.

📡 The Doppler Effect and Sputnik Discovery

Foundation of Satellite Navigation

🚂 Everyday Example

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An approaching train seems to get **higher** in pitch, and a receding train appears to get **lower** in pitch to a stationary observer.

The Doppler Effect

The **Doppler effect (or Doppler shift)** is the change in frequency of a wave in relation to an observer who is moving relative to the wave source. Named after Austrian physicist **Christian Doppler** (1842).

Sputnik Discovery (1957)

- Soviet Union launched **Sputnik**, first artificial satellite
- **Size of a basketball, ~180 pounds**
- **Transmitted radio signal** trackable from Earth
- **Johns Hopkins Applied Physics Laboratory** observed Doppler shift
- Realized they could **determine satellite position** by measuring frequency change

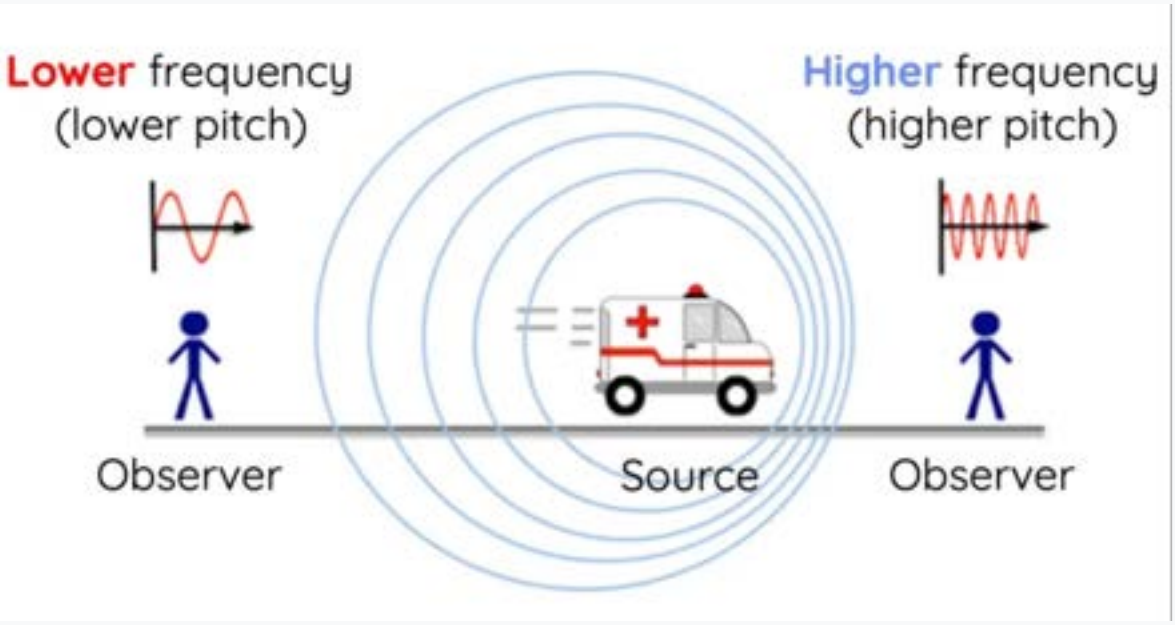


Diagram showing higher/lower frequency with moving source and stationary observers

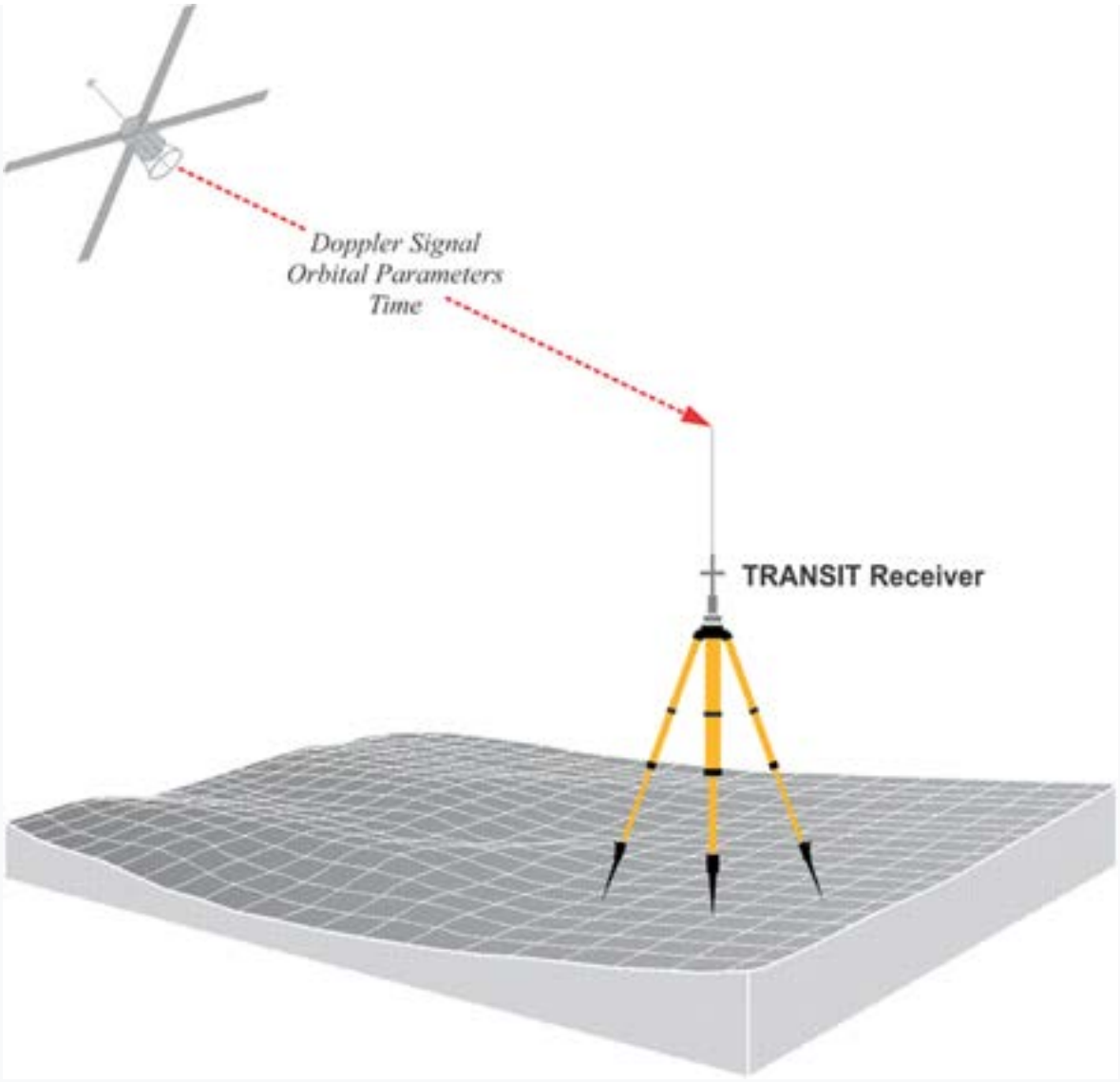
TRANSIT System - First Operational Satellite Navigation

U.S. Navy Navigational Satellite System

Building on the Sputnik discovery, the **U.S. Navy developed the TRANSIT system** (Navy Navigational Satellite System), which became the **first operational satellite navigation system**. Launched in the 1960s, it was designed to provide position fixes for military ships and submarines, especially those carrying **Polaris ballistic missiles**.







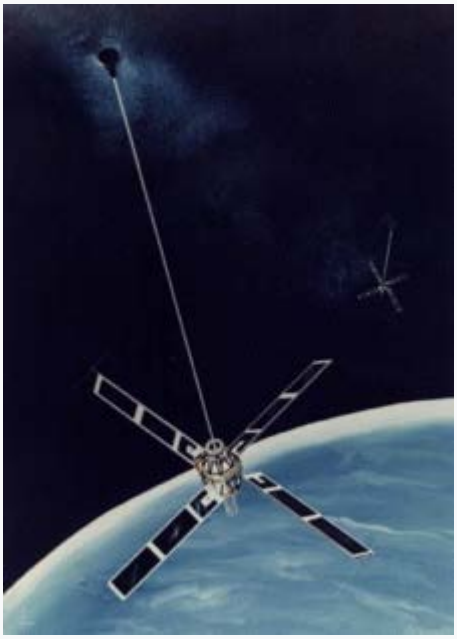
Sputnik satellite, Doppler effect visualization, and TRANSIT receiver setup

Key TRANSIT Features

- **Purpose:** Accurate positioning for naval operations
- **Satellite Visibility:** Only **one satellite** typically visible at a time
- **Observation Window:** 10-20 minutes per satellite
- Users often had to **wait several hours** between satellite passes
- **Frequencies:** 150 MHz and 400 MHz to correct ionospheric delays

TRANSIT Capabilities

- **2D positioning** (latitude and longitude) by tracking Doppler shifts
- **Required only one satellite** in view at a time
- Accuracy improved from **~120 meters (1960s)** to **~3 meters (1980s)**
- Used widely in **civilian applications** including hydrographic and geodetic surveying



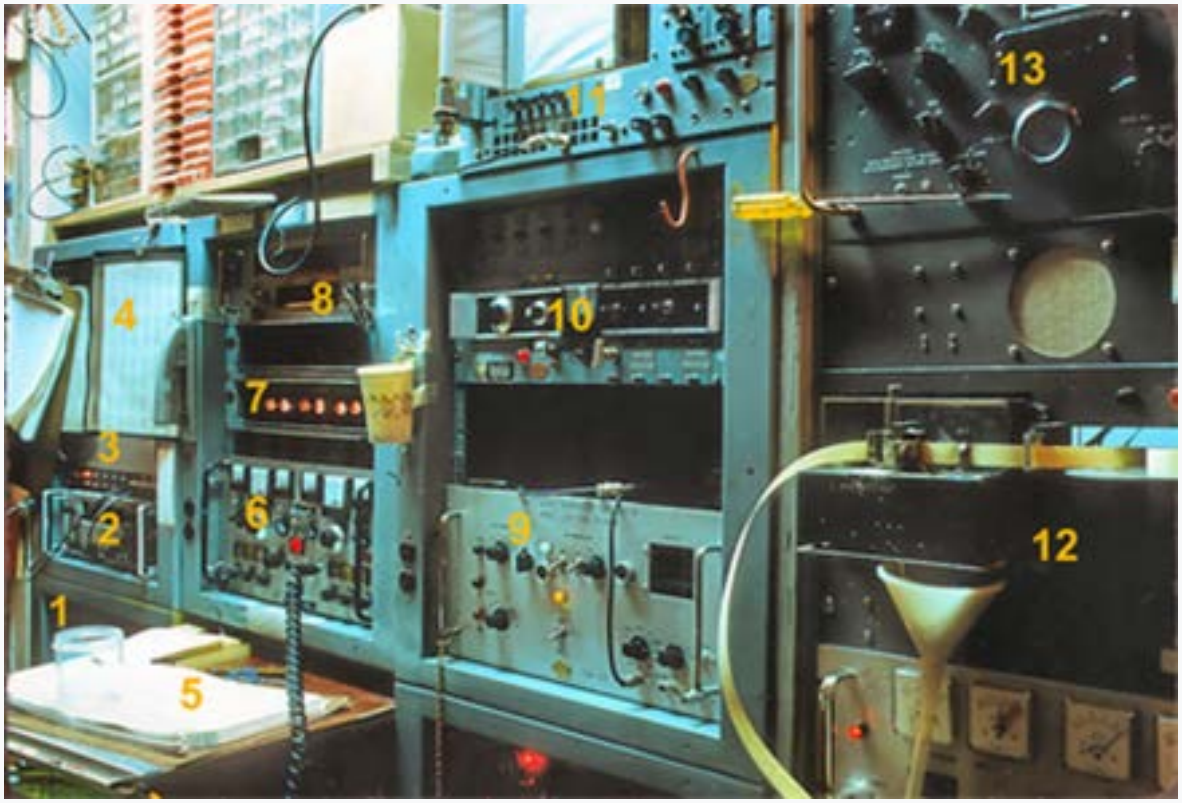
TRANSIT satellite in orbit: The first operational satellite navigation system

✅ **TRANSIT Positives**

- **Proved feasibility** of space-based navigation
- **Civilian accessibility** began in 1967
- **Operational longevity:** Over three decades of service
- Supported **hydrographic and geodetic surveying**
- Especially valuable in **remote oceanic regions**

❌ **TRANSIT Limitations**

- **Limited satellite visibility** - large coverage gaps
- Required **position interpolation** between passes
- **Velocity compensation** necessary for moving platforms
- **Only 2D positioning** - unsuitable for 3D applications
- **Not suitable for high-velocity platforms** (aircraft, missiles)



Equipment used for satellite signal tracking and Doppler shift analysis

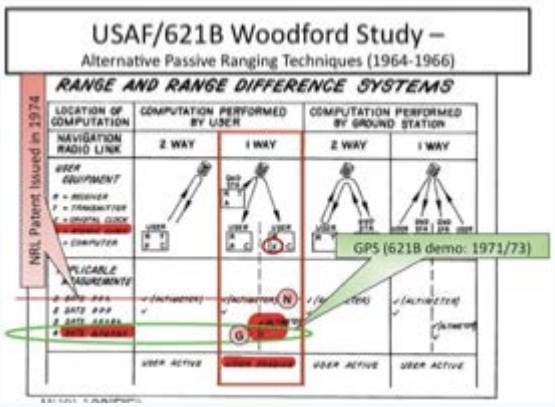
🏆 **TRANSIT Legacy**

The system remained in service for over three decades and was **officially decommissioned on December 31, 1996**, after GPS became fully operational. TRANSIT laid the **foundational concepts for GPS**, including satellite orbit prediction, Doppler-based ranging, and the importance of precise timing.

✈️ System 621B - Air Force Innovation

Pioneering 3D Navigation and PRN Codes

In the 1960s and early 1970s, while the U.S. Navy was advancing the TRANSIT system, the **U.S. Air Force developed its own satellite navigation concept** known as **System 621B**. This experimental program played a foundational role in shaping the architecture of modern GPS.



1960s USAF 621B Woodford Study comparing different ways to measure distance using radio signals



Engineers testing 621B receivers at White Sands Missile Range (1970)

🚀 Key Innovations

Three-Dimensional Navigation

System 621B was among the first satellite navigation concepts to offer **full 3D positioning**—providing latitude, longitude, and altitude—a major improvement over the 2D positioning of TRANSIT.

Pseudorandom Noise (PRN) Signals

By 1972, the system implemented **PRN-coded ranging signals**:

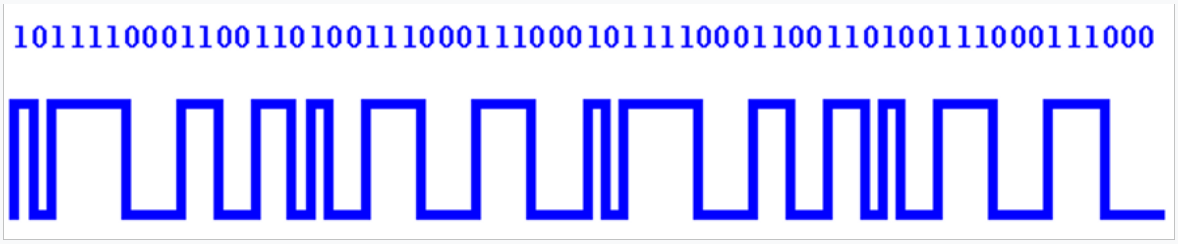
- Precise aircraft positioning to **within 16 meters**
- **Resistant to jamming and interference**
- **Multiple satellites** could broadcast on same frequency

⚠️ Primary Limitation

> **Dependence on Ground-Based Signals**
>
> Unlike fully autonomous satellite systems, System 621B **required uplink signals from Earth stations** to function, reducing its global utility and real-time independence.

🏆 Legacy

Although it never became operational, its technologies—especially **PRN codes and 3D positioning**—directly influenced the development of the **NAVSTAR GPS program initiated in 1973**.

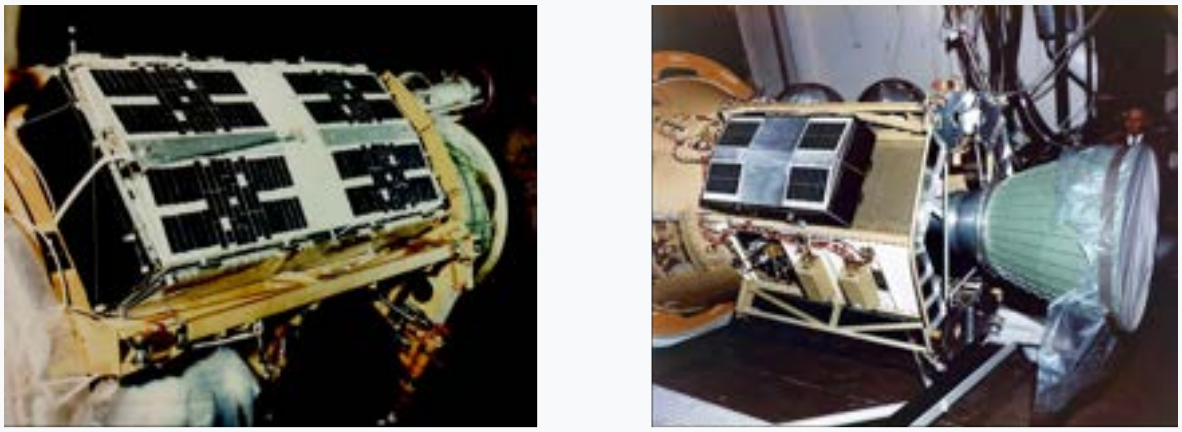


Pseudorandom Noise (PRN) Code Example
Binary sequence and corresponding signal waveform (e.g., 10111100011101101001110001111000111011010111000110 0)

⏰ Timation Program - Precision Timing Innovation

U.S. Navy Time Navigation Program

The Timation program (short for "Time Navigation") was initiated by the **U.S. Navy in 1964** as an experimental effort to explore the use of highly accurate space-based clocks for navigation purposes.



Early U.S. Navy prototype for space-based precision timing AND pre-launch configuration advancing space-based timekeeping

🔬 Key Contributions

First Precision Timing in Orbit

Timation focused on the development of **stable, space-qualified clocks**. Essential for enabling accurate **time-transfer**, a foundational requirement for satellite navigation.

⚗️ Clock Evolution

From Quartz to Atomic Precision

- First satellite: **high-stability quartz-crystal oscillator**
- Later models: **first space-qualified atomic clocks**
- Dramatically improved **frequency stability**

Launched First Timation Satellite (1967)

Successfully tested **two-dimensional navigation** using synchronized timing signals and validated early orbit-stabilized clock systems.

Paved the Way for GPS

Timation's innovations in **precise clock technology** directly influenced the timing architecture of the NAVSTAR GPS system.

Why Clock Stability Matters

- **1-microsecond timing error** = up to **300 meters** positional error
- Light takes **1.3 seconds** to travel Earth to Moon
- GPS requires **nanosecond-level precision**





Rubidium Frequency Standard (core timekeeping unit for early GPS) AND Cesium Frequency Standard (high-precision clock for GPS satellites)

⏱️ **Atomic Clock Precision**

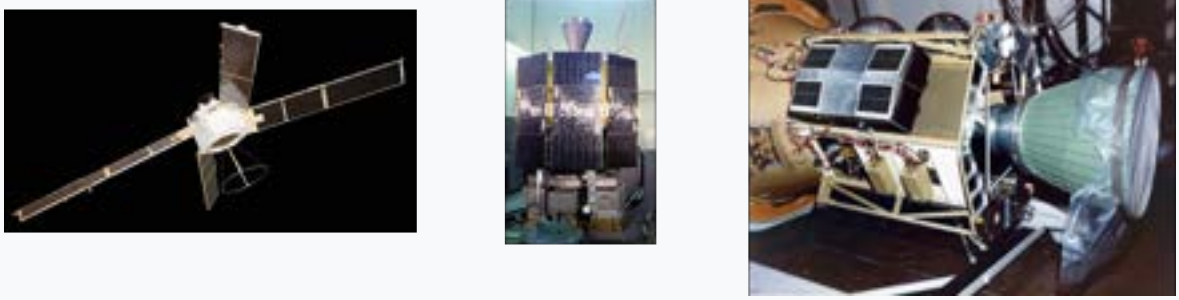
- **Rubidium atomic clocks:** Drift only **three seconds over a million years**
- **Passive hydrogen masers:** Even more precise—drifting just **one second in three million years**
- These technologies laid the **technical foundation** for highly accurate timing systems in modern GPS satellites

🎯 GPS Predecessors Summary

Integration of Technologies

By the **late 1960s**, the U.S. military recognized the strategic importance of reliable, all-weather, global navigation. Each branch of the armed forces independently pursued satellite-based navigation

systems to meet their specific operational needs. These efforts laid the groundwork for the eventual unification into the **NAVSTAR Global Positioning System (GPS) in the early 1970s**.



SECOR Satellite (Army's Sequential Correlation of Range), TRANSIT Satellite (Navy Navigation), Timation Satellite (Navy time-based navigation)

Key Pre-GPS Programs by Service Branch

🌊 U.S. Navy Programs

TRANSIT (with Johns Hopkins APL)

- First operational satellite navigation system
- Used Doppler shift to determine position
- 2D positioning, widely used by naval vessels

TIMATION (Naval Research Laboratory)

- Focused on space-based time-transfer
- Demonstrated precise timing for navigation
- Introduced early space-qualified atomic clocks

🏛️ Army & Air Force Programs

SECOR (U.S. Army)

- Sequential Correlation of Range system
- Aimed at surveying and geodetic applications
- Used ground-based transmitters for ranging

System 621B (U.S. Air Force)

- Developed 3D positioning concept
- Pioneered pseudorandom noise (PRN) codes
- Highly influential in GPS architecture

🎯 Combined Legacy

Though these systems were not integrated at the time, their **combined technologies**—accurate timing, satellite ranging, Doppler tracking, and PRN coding—formed the **technological and conceptual foundation** of the modern GPS system, officially launched as a joint program in **1973**.

📝 Module Summary

Key Historical Insights

This module revealed the complex origins of GPS, showing how multiple independent military programs overcame significant technical, budgetary, and institutional challenges. The convergence of TRANSIT's operational experience, System 621B's 3D positioning innovation, and Timation's precision timing breakthroughs created the foundation for modern satellite navigation.

🎯 Prepare for Next Topic

Next, we'll explore how these historical innovations evolved into the three-segment GPS architecture and examine the technical details of satellite constellation design, control systems, and user equipment that make modern GPS possible.

CONTENT

📚 Development of GPS



Module 01: Development of GPS

GISC 4315

📹 Instructor Video Lecture

GPS Development and Modern System Evolution

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📚 Video Topics Covered

- Integration of technologies from predecessor systems
- Early GPS satellite development and testing
- Program challenges and near-cancellation
- Presidential policy decisions (1983 & 2000)
- Selective Availability implementation and removal

• GPS constellation evolution and modernization

> 💡 Study Tip
>
> Focus on understanding how policy decisions shaped GPS accessibility and accuracy for civilian users.

🚀 The Birth of Modern GPS (1973)

Integration of Key Technologies

The **Global Positioning System (GPS)**, **formally initiated in 1973**, was the result of integrating key technologies from earlier military satellite navigation efforts. Rather than starting from scratch, GPS leveraged the most successful innovations from predecessor programs:

⏰ Timation Contribution

Atomic clock technology from the Timation program enabled precise timekeeping in orbit—the foundation of all GPS operations.

🌊 TRANSIT Contribution

The **ephemeris and orbital prediction system** was derived from the TRANSIT satellite network's operational experience.

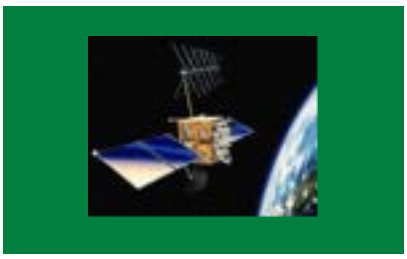
✈️ System 621B Contribution

The **pseudorandom noise (PRN) signal design** was pioneered by the Air Force's System 621B, allowing accurate ranging and multiple satellite signals on the same frequency.

🛰️ Early GPS Satellites and System Design

Prototype Satellites and Initial Implementation

Early GPS Satellite Evolution





GPS Block I
1978-1985

GPS Block II
1989-1997



GPS Block IIR
1997-2016

Early GPS Satellite Evolution
From Test Satellites to Modernized Design with Improved Accuracy and Reliability

🔧 Initial GPS System Components

The last two Timation satellites were modified and launched as **early GPS prototypes**, validating the use of atomic clocks and signal transmission in medium Earth orbit (MEO).

Initial Funding: $150 Million (1973)

- **4 initial satellites** (later expanded to 6)
- **4 launch vehicles** and related operations
- **Ground control segment** with master control station and 6 global monitoring stations
- **7 types of user receiver equipment**
- **Development timeline:** ~4 years, plus 1 year of system testing

⚠️ Program Challenges and Near-Cancellation

The Program That Almost Didn't Survive

🚫 1979 Cancellation Attempt

In **1979**, despite successful early testing, the **U.S. Air Force attempted to cancel GPS** due to:

- Budget pressures and funding constraints
- Internal resistance within military branches
- Questions about long-term viability

✅ Civilian Leadership Intervention

The program was preserved thanks to **intervention by civilian leadership**, who recognized:

- GPS's long-term strategic value
- Commercial potential and applications
- Global technological leadership opportunity

🌍 GPS Orbital Characteristics

Medium Earth Orbit Configuration

GPS satellites are deployed in **medium Earth orbit (MEO)** at an altitude of approximately **20,200 km**. This orbital design ensures global coverage and optimal signal geometry.

Orbital Planes	Satellites per Plane	Inclination Angle	Orbital Period
3	8	63°	12h
Evenly spaced planes	Total: 24 satellites	Optimal coverage	Repeated passes

🎯 Global Coverage Design

The 12-hour orbital period ensures that satellites appear in nearly the same positions relative to Earth every day, enabling **global coverage with repeated satellite passes** for consistent positioning worldwide.

📅 Two Defining Policy Events

Presidential Decisions That Changed GPS Forever

🇺🇸 Civilian Access Enabled – 1983



President Ronald Reagan signing

🔓 Selective Availability Disabled – 2000



President Bill Clinton directing the U.S.

executive order authorizing civilian access to GPS

In **1983**, following the tragic shootdown of **Korean Air Lines Flight 007**, President Ronald Reagan signed an executive order authorizing civilian access to GPS.

- Intended primarily for **commercial aviation**
- Enhanced navigation safety to prevent future accidents
- Shifted GPS from strictly military to **dual-use global utility**

Department of Defense to disable Selective Availability (SA)

On **May 1, 2000 (UTC)**, President Bill Clinton directed the U.S. Department of Defense to disable Selective Availability (SA).

- Improved civilian accuracy from **~100m to ~20m**
- Opened door for high-precision commercial applications
- Marked beginning of GPS as **globally trusted public service**

📈 **Early Commercial Impact**

By **1989**, the first commercially available handheld GPS receivers appeared: weighing **1.5 pounds**, providing only **a few hours of battery life**, and priced at around **$3,000 USD**.

🔒 Understanding Selective Availability (SA)

Intentional Degradation for National Security

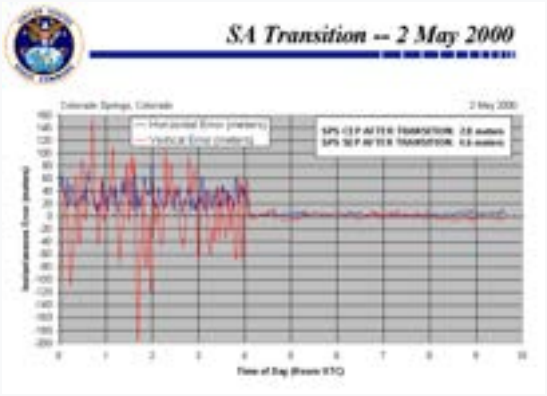
What Was Selective Availability?

Selective Availability (SA) was a U.S. Department of Defense feature that intentionally degraded the accuracy of GPS signals available to civilian users. It introduced random timing and orbital errors into:

- The satellite's **atomic clock data**
- The broadcast **orbital positions (ephemerides)**

The goal was to prevent adversaries from using high-precision GPS while maintaining full accuracy for military users with encrypted access.

At **midnight UTC on May 2, 2000**, the change was immediate and dramatic—civilian GPS accuracy improved significantly to **Horizontal accuracy (CEP): ~2.8 meters** and **Vertical accuracy (SEP): ~4.6 meters**.



SA Transition Graph – May 2, 2000
Showing dramatic improvement in GPS accuracy when Selective Availability was turned off

🌟 Impact on Civilian Applications

Democratization of High-Precision GPS

✅ Revolutionary Policy Shift

This policy shift **democratized high-precision GPS**, allowing commercial, academic, and public users to benefit from near-military-grade accuracy.

It accelerated the growth of GPS-based industries such as **agriculture, mapping, surveying, aviation, and mobile navigation**.

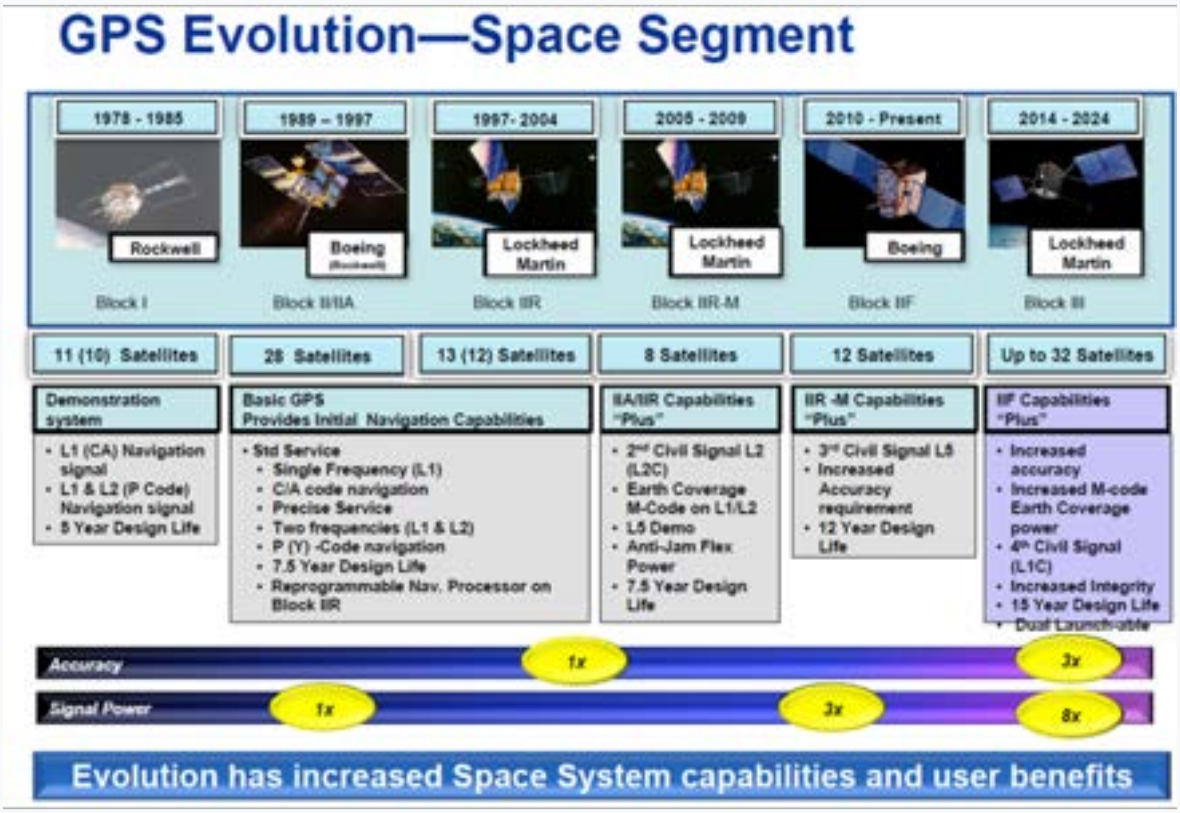
🔮 Future of SA

Though SA could have been reactivated at any time, it was **permanently disabled in GPS Block III satellites**, ensuring open high-accuracy access worldwide.

This commitment solidified GPS as a globally trusted, high-accuracy public service.

📈 GPS Space Segment Evolution

From Demonstration to Multi-Signal, High-Power Constellation



GPS Evolution—Space Segment

Timeline showing progression from Block I (1978-1985) through Block III (2014-2024) Including satellite images, capabilities, and evolution of accuracy, signal power, and features

🚀 `Continuous Improvement`

GPS Space Segment Evolution demonstrates how the system has continuously evolved from a demonstration project to a multi-signal, high-power constellation with enhanced capabilities, increased accuracy, improved coverage, and stronger signals—showing **evolution has increased Space System capabilities and user benefits**.

📝 Module Summary

From Military Project to Global Infrastructure

This module traced GPS development from its 1973 inception through key policy decisions that transformed it into today's global utility. The integration of technologies from Timation, TRANSIT, and System 621B created a revolutionary positioning system, while presidential decisions in 1983 and 2000 democratized access and established GPS as essential global infrastructure.

🎯 Prepare for Next Topic

Next, we'll explore the three-segment GPS architecture in detail, examining how the space, control, and user segments work together to provide positioning, navigation, and timing services worldwide.

📚 GPS Objectives

Module 01: GPS Objectives and Performance

GISC 4315

📹 Instructor Video Lecture

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🎯 Primary GPS System Objectives

Foundation Design Requirements (1973)

When GPS was formally initiated in 1973, the system was designed with **comprehensive objectives** that would make it suitable for diverse users while maintaining strategic military advantages. These foundational objectives continue to guide GPS evolution today.

🌐 Universal Platform Support

- **Aircraft, ship, land-based and space platforms** (missiles and satellites)
- Able to handle a **wide variety of dynamics** from stationary to high-speed applications

🌍 Global Coverage & Standards

- Positioning results available on a **single global geodetic datum**
- **World-wide coverage** for all user types
- Highest accuracy **restricted to certain classes of users** (military applications)

- **Real-time positioning, velocity and time determination** capability to appropriate accuracy levels

🛡️ Security & Resilience

- **Resistant to jamming/interference** (intentional and unintentional)
- **Redundancy provisions** to ensure survivability of the system
- Secure military access while providing civilian services

⚡ Efficiency & Scalability

- **Passive positioning system** - no transmission from user to satellite required
- Able to provide service to an **unlimited number of users**
- **Low cost, low power** user equipment design priority

🚀 Advanced System Objectives

Strategic Implementation Goals

🔧 System Complexity Management

Minimize complexity in the satellite segment - as much complexity as possible should be built into ground-based systems and user equipment rather than the satellites themselves.

🔄 System Replacement Strategy

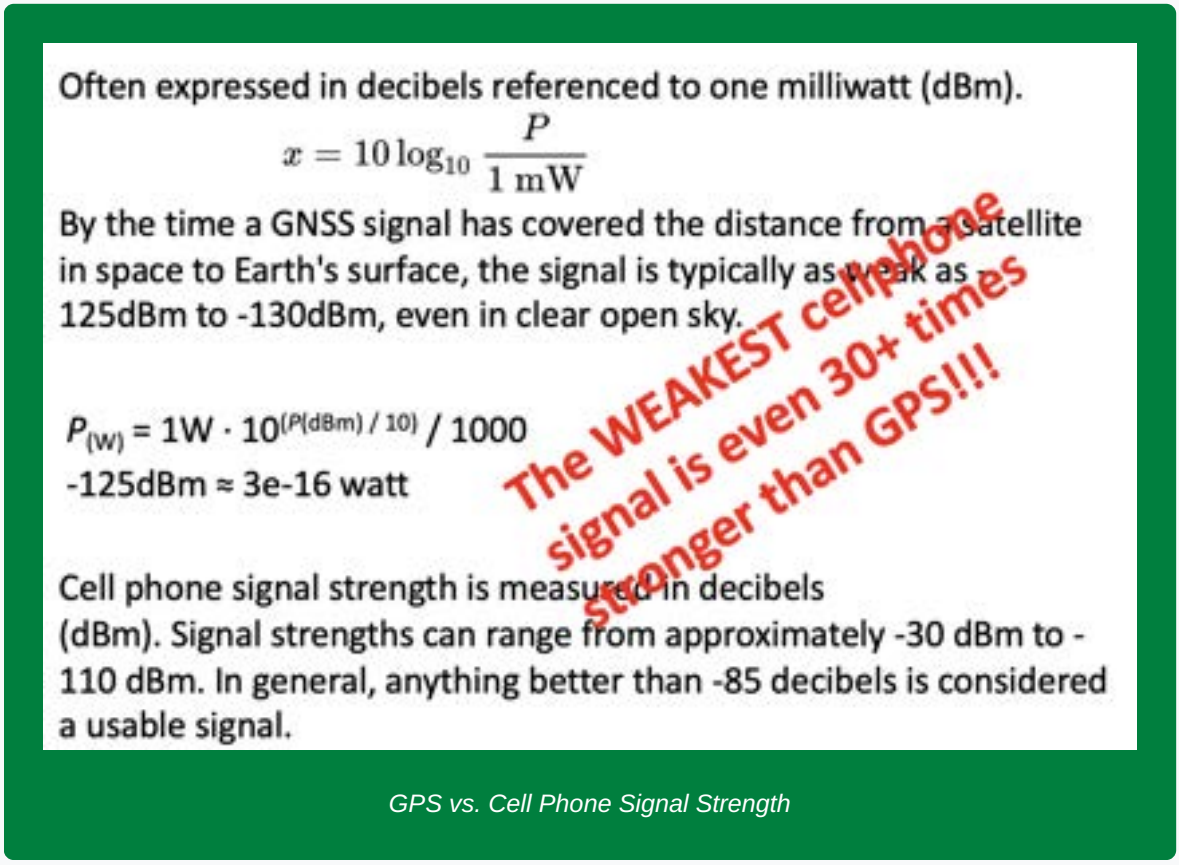
Total replacement of Transit satellite and other terrestrial systems - GPS was designed to supersede existing navigation infrastructure completely.

💡 Design Philosophy Impact

These objectives shaped GPS into a **revolutionary dual-use system** that balances military security requirements with civilian accessibility, creating the foundation for today's global positioning infrastructure.

📡 GPS Signal Strength Characteristics

Understanding GPS Signal Power



GPS vs. Cell Phone Signal Strength

GPS signals (-125dBm to -130dBm) vs. cell phone signals (-30dBm to -110dBm)

📐 Signal Strength Mathematics

GPS signal strength is often expressed in **decibels referenced to one milliwatt (dBm)**:

$$x = 10 \log_{10}(P / 1\ mW)$$

> ⚠️ **Extremely Weak Signals**
>
> **P[W] = 1W × 10^(P[dBm]/10) / 1000**
> At -125dBm ≈ 3e-16 watt
> *The WEAKEST cellular signal is even 30+ times stronger than GPS!*

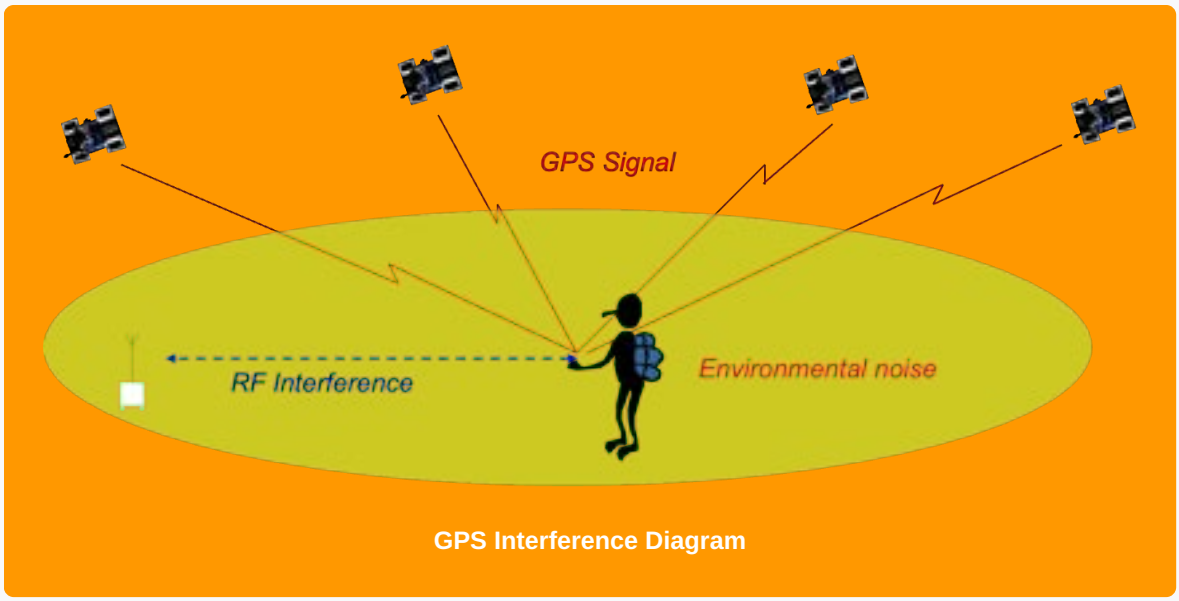
> 📱 **Signal Strength Context**
>
> By the time a GNSS signal travels from satellite to Earth's surface, it is typically as weak as **-125dBm to -130dBm**, even in clear open sky. For comparison, cell phone signal strength ranges from approximately **-30 dBm to -110 dBm**, with anything better than **-85 decibels considered a usable signal**.

⚠️ GPS Interference Challenges

Understanding GPS Vulnerability

GPS interference refers to any disruption or degradation of the signals received by GPS receivers, which can impair the accuracy, reliability, and availability of GPS-based services. GPS signals are extremely weak by the time they reach Earth's surface, making them vulnerable to various interference sources.

🔧 Types of GPS Interference



GPS Interference Diagram

📡 GPS Signal

L-band frequencies (L1 at 1575.42 MHz) containing ephemerides, timing, and navigation data

🔊 Noise

Random electrical fluctuations limiting signal-to-noise ratio, especially problematic in urban settings

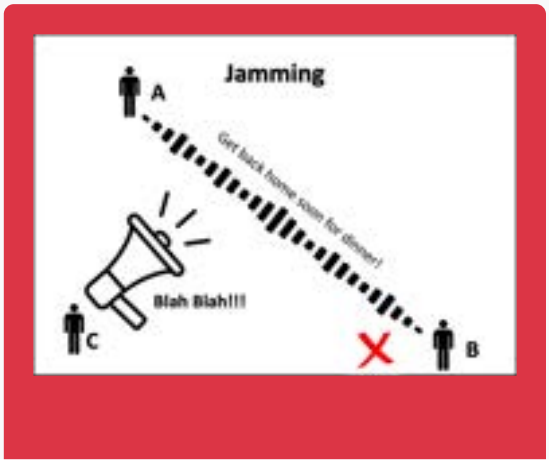
🏢 Multipath

Signal reflections off buildings/terrain causing delayed, distorted signals and positioning errors

📻 RF Interference

Jamming (noise overwhelming) and spoofing (false signals) that disrupt or mislead receivers

🚫 GPS Jamming



🎭 GPS Spoofing



Jamming Analogy Diagram
Person A → Person B communication blocked by Person C with megaphone

Overwhelms GPS signals with stronger noise, **denying service** by preventing receivers from acquiring or tracking satellite signals.

Spoofing Analogy Diagram
Person C intercepts and provides false message to Person B

Transmits false GPS-like signals to **mislead receivers** into computing incorrect positions or times.

🌐 RF Interference Impact

Why RF Interference is a Critical Threat



CORS Station & RFI Impact

⚡ Primary RFI Effects

- **Reduces Signal-to-Noise Ratio (SNR)** - weakens satellite signals relative to background interference
- **Degrades Measurement Quality** - diminishes pseudorange and carrier phase accuracy
- **Impairs Positioning Accuracy** - can cause loss of satellite lock or complete receiver outage
- **Widespread GNSS Infrastructure Impact** - affects CORS stations and differential correction services

📰 Real-World Example: London Stock Exchange (2013)

For nearly ten minutes each day near the London Stock Exchange, someone was blocking GPS signals. Navigation systems in cars stopped working and time stamps on financial trades were affected. **The offender was a delivery driver using a GPS jammer to hide from his**

management - demonstrating how even unintentional interference can have widespread consequences.

📝 Module Summary

GPS Objectives and Performance Fundamentals

This module explored the foundational objectives that shaped GPS design, from universal platform support to global coverage and interference resistance. Understanding these objectives provides context for GPS capabilities, limitations, and the engineering challenges overcome to create a globally trusted positioning system.

🎯 Prepare for Next Module

Next, we'll examine GPS signal structure and modulation techniques, exploring how satellites encode navigation data and timing information within the carrier waves that enable precise positioning calculations.

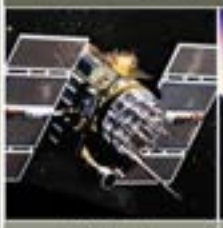
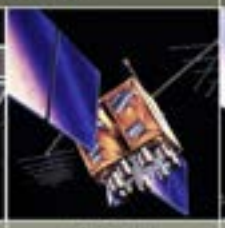
CONTENT

📚 GPS Signals and Modernization



Current GPS Constellation Status

Operational Satellites and Capabilities

LEGACY SATELLITES		MODERNIZED SATELLITES		
				
BLOCK IIA	BLOCK IIR	BLOCK IIR-M	BLOCK IIF	GPS III/IIIF
0 operational	6 operational	7 operational	12 operational	6 operational
• Coarse Acquisition (C/A) code on L1 frequency for civil users • Precise P(Y) code on L1 & L2 frequencies for military users • 7.5-year design lifespan • Launched in 1990-1997 • Last one decommissioned in 2019	• C/A code on L1 • P(Y) code on L1 & L2 • On-board clock monitoring • 7.5-year design lifespan • Launched in 1997-2004	• All legacy signals • 2nd civil signal on L2 (L2C) *LEARN MORE* • New military M code signals for enhanced jam resistance • Flexible power levels for military signals • 7.5-year design lifespan • Launched in 2005-2009	• All Block IIR-M signals • 3rd civil signal on L5 frequency (L5) *LEARN MORE* • Advanced atomic clocks • Improved accuracy, signal strength, and quality • 12-year design lifespan • Launched in 2010-2016	• All Block IIF signals • 4th civil signal on L1 (L1C) *LEARN MORE* • Enhanced signal reliability, accuracy, and integrity • No Selective Availability *LEARN MORE* • 15-year design lifespan • IIIF: laser reflectors; search & rescue payload • First launch in 2018

GPS Constellation Chart

Legacy Satellites vs. New Satellites by Block Type

Block IIA | Block IIR | Block IIR-M | Block IIF | GPS IIIF

📊 Constellation Facts (Sep 2024)

- **29 operational satellites** in GPS constellation
- **Not including decommissioned** or on-orbit spares
- **Multiple satellite blocks** with varying capabilities
- **Legacy and modernized signals** broadcast simultaneously

🔗 Live Status

GPS constellation status:
https://www.navcen.uscg.gov/gps-constellation

💡 *Check it out:* The diagram to the left shows the GPS constellation status, showing 29 operational satellites as of September 3, 2024, with a breakdown by satellite block and capabilities.

🚀 GPS Modernization: Underway, Not Future

Current Reality of GPS Enhancement

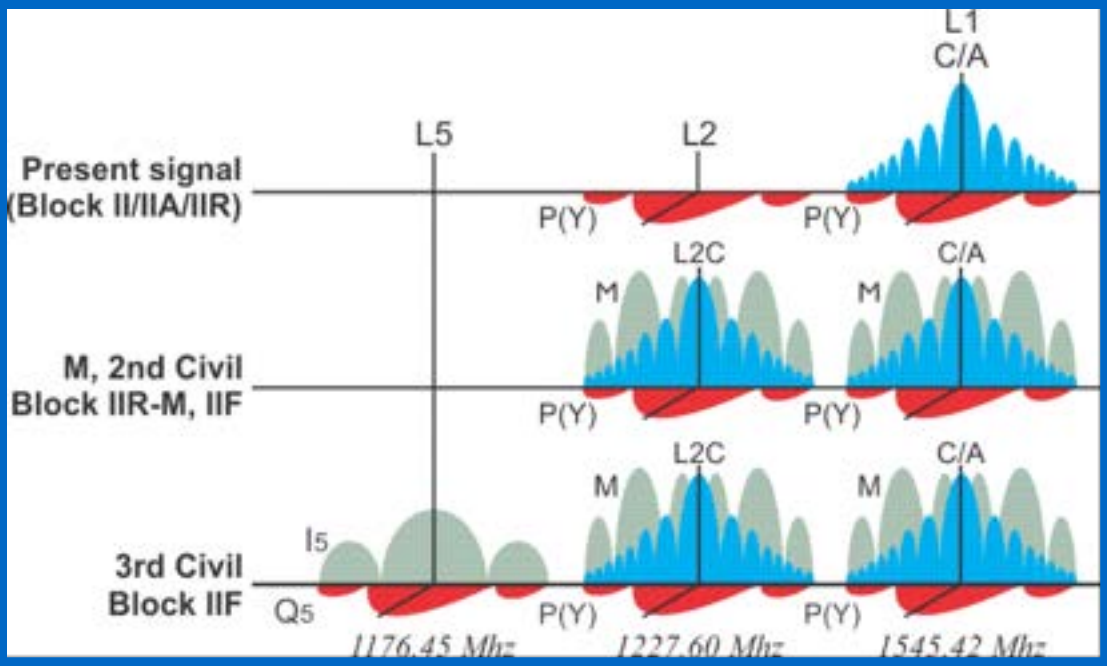
GPS modernization is no longer a future development, it is underway. New spacecraft with better electronics, better navigation messages, newer and better clocks are just part of the story. Beginning with the launch of the first IIR-M satellite, new civil signals began to appear, starting with L2C which was followed by L5 on the Block IIF satellites.

🔧 Modernization Components

- **Better electronics** - Enhanced satellite systems and components
- **Better navigation messages** - Improved data accuracy and reliability
- **Newer and better clocks** - Enhanced timing precision
- **New civil signals** - L2C, L5, and L1C for civilian users
- **Enhanced military capabilities** - M-code for secure military access

📶 Evolution of GPS Signal Structure

Frequency Bands and Satellite Block Evolution



Evolution of GPS Signal Structure
Across Frequency Bands and Satellite Blocks

L5 (1176.45 MHz) | L2 (1227.60 MHz) | L1 (1545.42 MHz)

📡 Signal Frequency Summary

L5 Band	L2 Band	L1 Band
1176.45 MHz	**1227.60 MHz**	**1575.42 MHz**
Aviation safety services	L2C civilian signal	C/A code and L1C

📶 L2C: Second Civilian GPS Signal

Commercial-Grade Signal for Professional Users

L2C is the second civilian GPS signal, designed specifically to meet commercial needs. When combined with L1 C/A in a dual-frequency receiver, L2C enables ionospheric correction, a technique that boosts accuracy.

✅ L2C Advantages

- **Dual-frequency capability** - enables ionospheric correction with L1 C/A
- **Faster signal acquisition** - enhanced reliability for professional users
- **Higher effective power** - easier to receive under trees and indoors
- **Improved interference resistance** - more sophisticated than C/A code

🔧 Technical Improvements

- **Not a copy of C/A code** - despite original design intention
- **Less susceptible to waveform distortion** than legacy C/A code
- **Better narrow-band interference resistance** than C/A
- **More sophisticated signal structure** than legacy signals

📊 L2C Power Spectrum Characteristics

L2C looks very similar to power spectrum to the C/A code, but it is important to note that **L2C is not a copy of the C/A-code** even though that was the original idea. The C/A code is somewhat susceptible to both waveform distortion and narrow-band interference. L2C is more sophisticated than C/A, so **the L2C is actually a different code than the C/A**.

🔒 M-Code: Encrypted Military Signal

Advanced Military Signal with Enhanced Security

M-code is an encrypted signal. It uses a different phase modulation which makes it appear in two lobes of the spread spectrum around the center carrier frequency.

🔐 M-Code Characteristics

- **Split-spectrum signal** around the carrier frequency
- **Two distinct peaks** in the signal spectrum
- **Encrypted for military use** - secure access only

💪 Enhanced Capabilities

- **Stronger signal power** than legacy signals
- **Superior interference resistance** capabilities
- **Enhanced security features** for military operations

- **Different phase modulation** from civilian signals

- **Improved jamming resistance** over legacy military signals

📡 **Signal Structure**

Those two peaks in the M-code represent a **split-spectrum signal about the carrier**. In general, it has stronger signal power than legacy signals and has ability to resist interference.

✈️ L5: Aviation Safety Signal

Third Civilian Signal for Safety-of-Life Transportation

L5 is the third civilian GPS signal, designed to meet demanding requirements for safety-of-life transportation. It is broadcast in a radio band reserved exclusively for aviation safety services. Future aircraft will use L5 in combination with L1 C/A to improve accuracy (via ionospheric correction) and robustness (via signal redundancy).

🛩️ Aviation Applications

- **Safety-of-life transportation** requirements
- **Exclusive aviation safety radio band** allocation
- **Future aircraft integration** with L1 C/A
- **Enhanced accuracy through ionospheric correction**
- **Signal redundancy** for improved robustness

🔧 Technical Enhancements

- **Lower interference risk** than other frequency bands
- **Improved multipath protection** for urban environments
- **Easier signal acquisition** in obstructed conditions
- **Improved resolution** for positioning calculations

📈 **Performance Benefits**

L5 lowers the risk of interference and improves multipath protection. It also makes signal easier to acquire in unfavorable and obstructed conditions. Resolution is improved.

🌐 L1C: International Interoperability Signal

Fourth Civil Signal for Global Navigation Cooperation

L1C designed to enable interoperability between GPS and international satellite navigation systems. L1C isn't drawn here, it's the 4th and newest civil signal. It enables international cooperation while protecting U.S. national security interests.

🌍 International Cooperation

- **GNSS interoperability** with international systems
- **Galileo, GLONASS, BeiDou compatibility**
- **Global navigation standards** harmonization
- **Protects U.S. national security interests**

📱 Mobile & Urban Performance

- **Improved mobile GPS reception** in cities
- **Enhanced performance** in challenging environments
- **Better urban canyon performance**
- **Advanced signal design** for modern applications

🏙️ Urban Environment Enhancement

The design will **improve mobile GPS reception in cities and other challenging environments**, addressing the growing need for reliable positioning in urban areas with tall buildings and signal obstructions.

📝 Module Summary

GPS Signal Evolution and Modernization Progress

This module explored the current state of GPS modernization, demonstrating that it is no longer a future development but an ongoing reality. The evolution from legacy C/A code to multiple civilian signals (L2C, L5, L1C) and enhanced military capabilities (M-code) represents a fundamental transformation of GPS from a single-signal system to a multi-frequency, multi-purpose global infrastructure supporting both civilian commerce and military operations.

🎯 **Prepare for Next Module**

Next, we'll examine GPS receiver architecture and processing techniques, exploring how receivers acquire, track, and decode the multiple signal types we've discussed to compute precise positioning solutions.

CONTENT

📚 GPS Segments

Module 01: GPS Segments

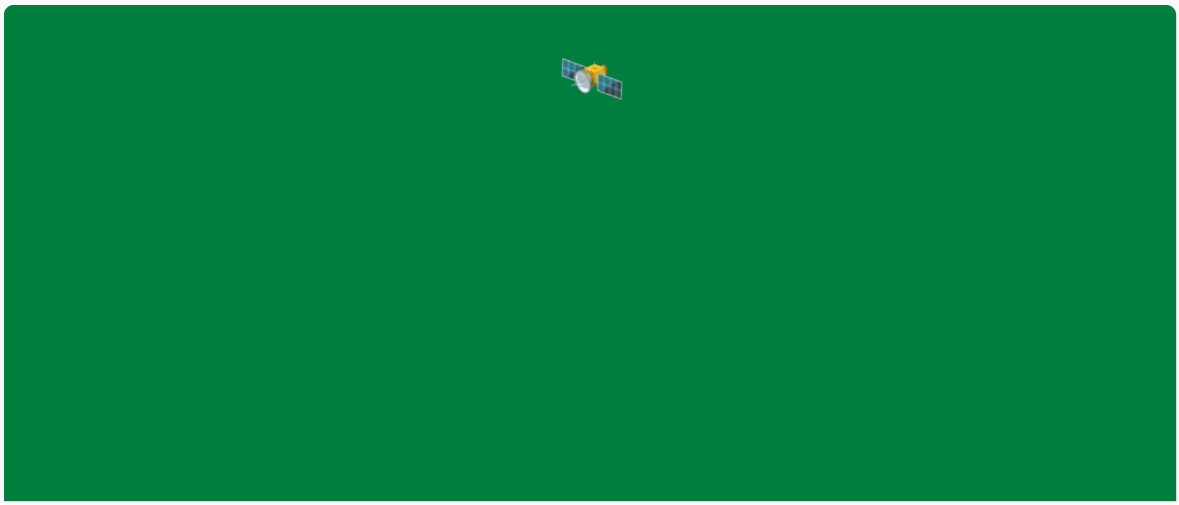
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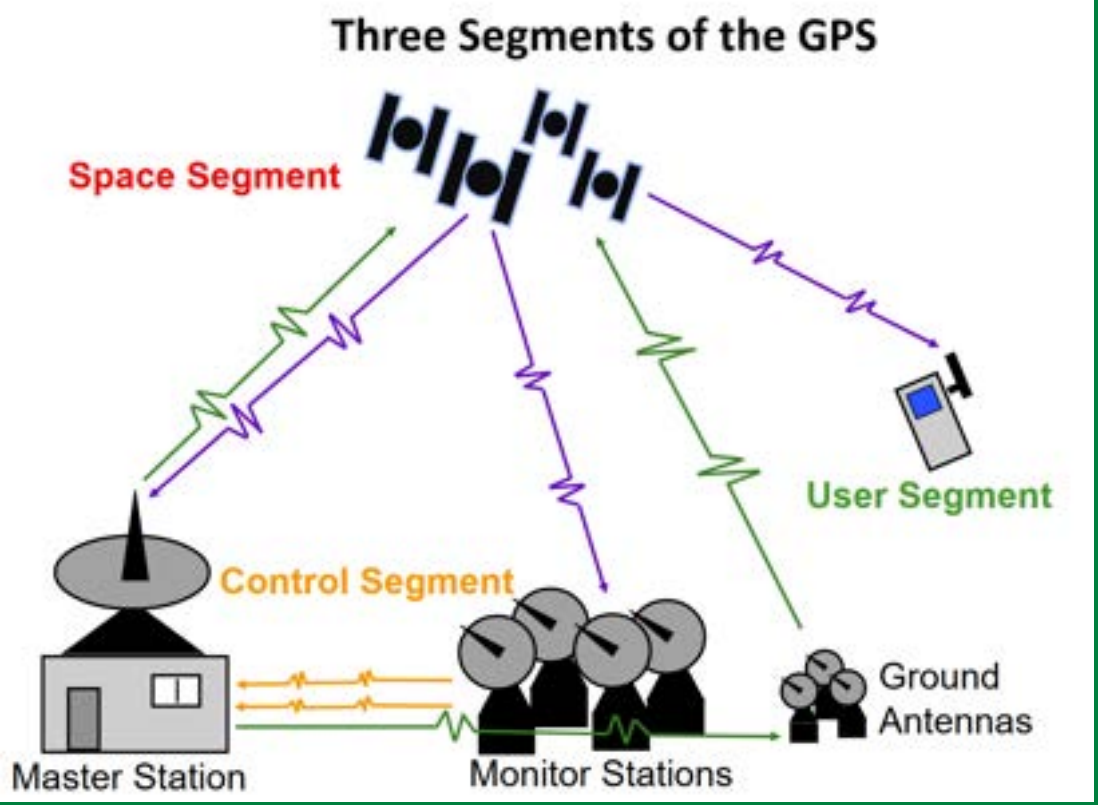
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🏗️ Three Segments of the GPS

Integrated System Architecture



Three Segments of the GPS



Satellites → Master Station → Monitor Stations → Ground Antennas → User Receivers

🛰️ Space Segment

- Comprises the **constellation of GPS satellites** orbiting Earth
- **Continuously transmits radio signals** containing positioning and timing data
- Enables **global, all-weather coverage**

🎛️ Control Segment

- Includes the **Master Control Station, Monitor Stations, and Ground Antennas**
- **Tracks satellite positions and health**
- Updates satellite navigation messages and **synchronizes satellite clocks**
- **Ensures satellites broadcast accurate data** to users

📱 User Segment

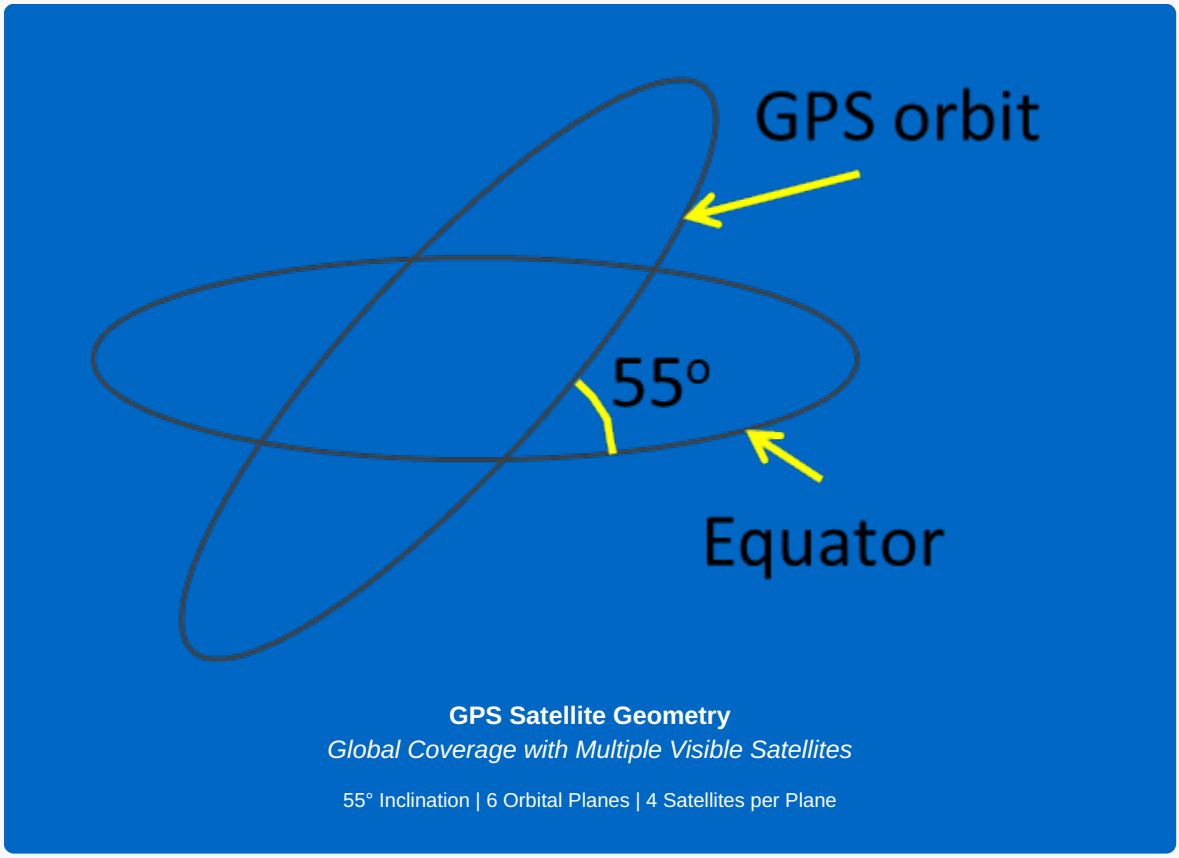
- Made up of **GPS receivers** (e.g., phones, vehicles, aircraft, survey equipment)
- Receives satellite signals to compute **position, velocity, and time (PVT)**
- Used by **civilians and military worldwide** for navigation, mapping, surveying, and timing

🔄 **System Integration**

In summary, the space segment broadcasts signals, the control segment maintains and corrects them, and the user segment processes the signals for positioning. All three components must work together to deliver accurate and reliable GPS services.

🛰️ Space Segment: GPS Constellation

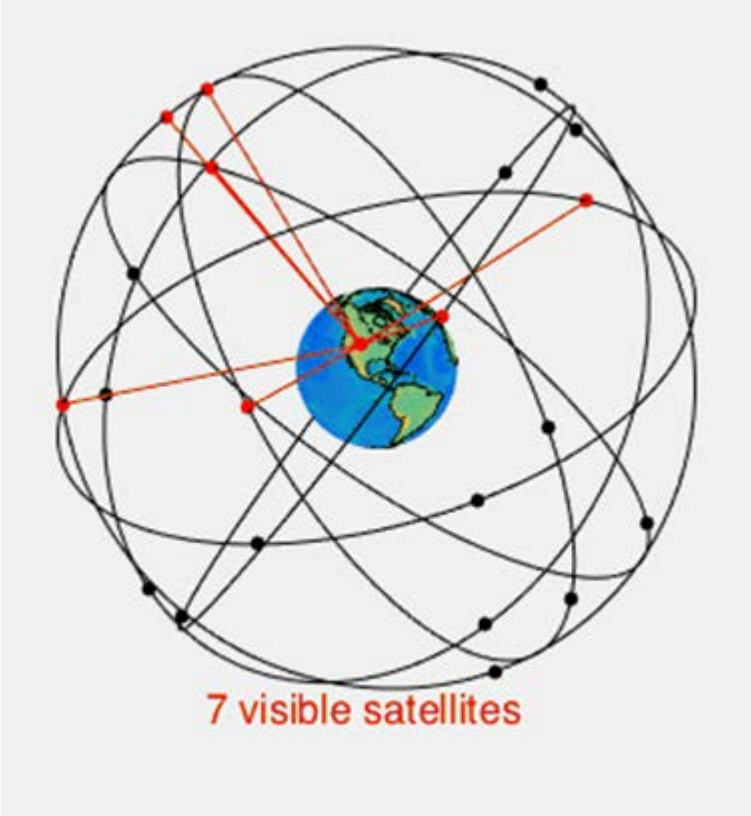
Orbital Configuration and Satellite Characteristics



GPS Satellite Geometry

Global Coverage with Multiple Visible Satellites

55° Inclination | 6 Orbital Planes | 4 Satellites per Plane



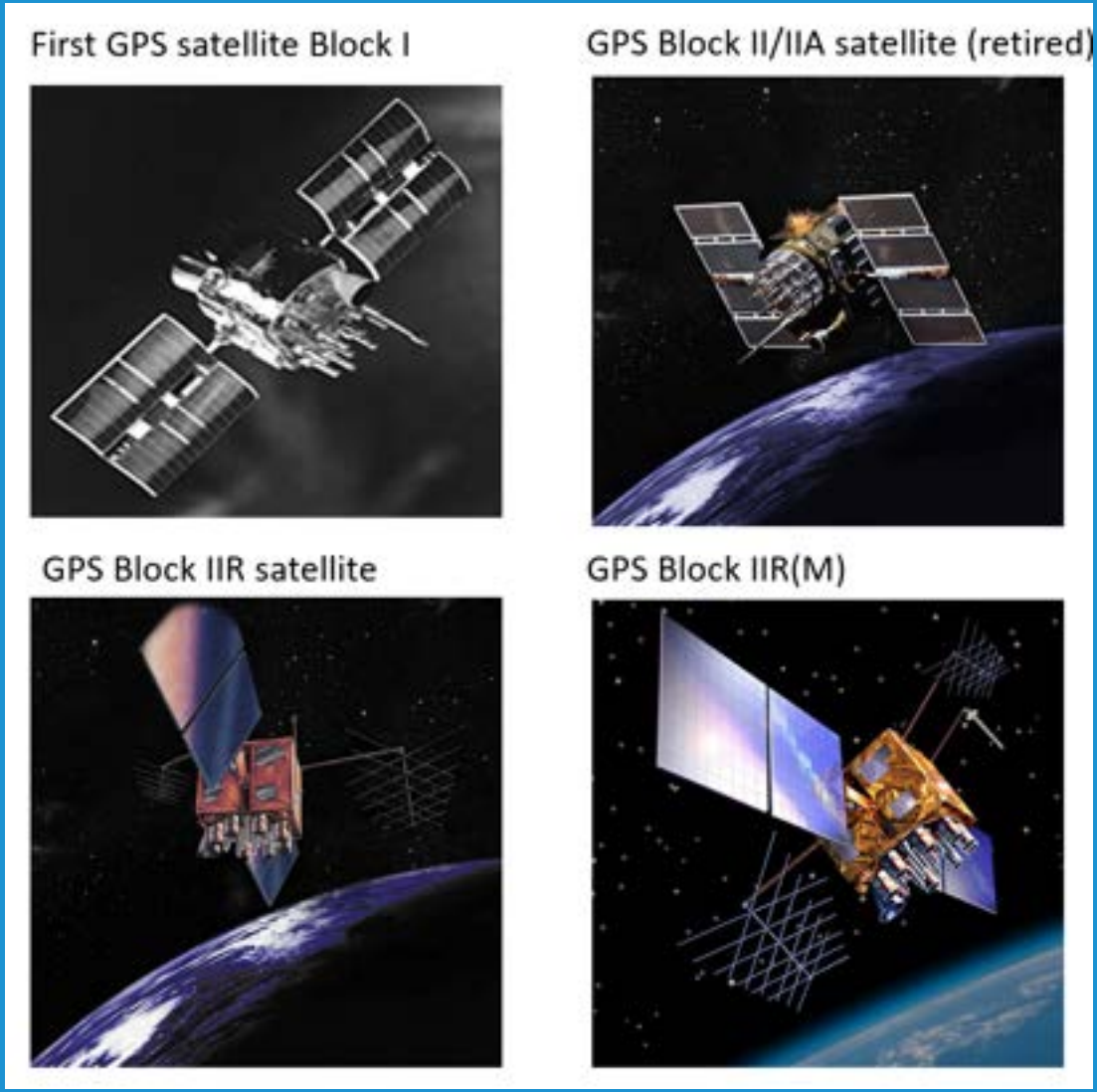
📊 Constellation Specifications

- **GPS satellites are controlled and operated by the DoD**
- **Satellites constantly transmit signals**
- **24 satellites in the sky** (minimum requirement for global coverage)
- **6 orbits, each orbit runs 4 satellites**
- The orbit inclination angle between the equator plane and the orbit plane is **55°**
- **A GPS satellite travels around the Earth in 11h 58m**
- **The GPS satellites are about 20,000 km above from the Earth's surface**
- **Orbit twice daily**

🚀 GPS Satellite Evolution

From Block I to GPS III: Four Decades of Innovation

📡 Early Generation Satellites (1978-2009)



Historical GPS Satellite Evolution
Block I (1978) → Block II/IIA (retired) → Block IIR → Block IIR(M)

Block I
1978
First GPS satellite launched

Block II/IIA
Retired
Legacy operational satellites

Block IIR
Enhanced
Improved reliability

Block IIR(M)
Modernized
L2C signal capability

🌟 Modern Generation Satellites (2009-Present)



Advanced GPS Satellites & Launch Operations
Block IIF & GPS III with Launch Vehicle Images

GPS Block IIF

- **Enhanced navigation accuracy**
- **L5 signal capability** for aviation safety
- **Improved atomic clocks**
- **Final launch via Delta II rocket** (August 2009)

GPS III

- **Next-generation satellites**
- **Enhanced signal power and accuracy**
- **Improved jam resistance**
- **First launch via Falcon 9** (December 2018)

📱 User Segment: Global Applications

Worldwide GPS Receiver Applications

The **User Segment** **includes all GPS receivers used by individuals, organizations, and military forces worldwide to determine position, velocity, and time (PVT)**.



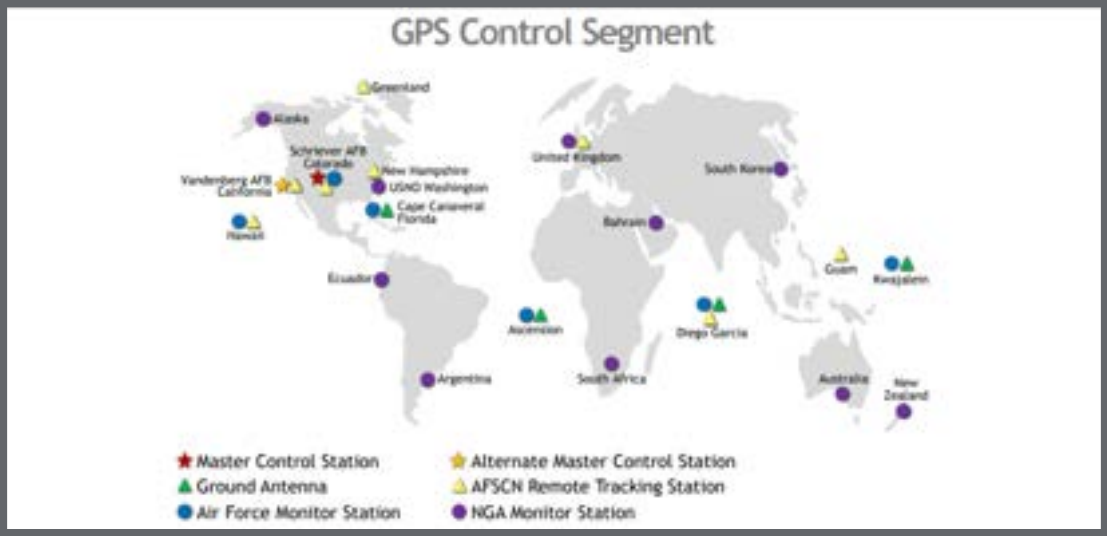
GPS Receiver Applications
Various GPS receiver types and applications

🌍 Application Categories

- **Military**
- **Search and rescue**
- **Disaster relief**
- **Surveying**
- **Marine, aeronautical and terrestrial navigation**
- **Remote controlled vehicle and robot guidance**
- **Satellite positioning and tracking**
- **Shipping**
- **Geographic Information Systems (GIS)**
- **Recreation**

🎛️ Control Segment: System Operations

Global Control Network Architecture



GPS Control Segment Network

Global distribution of Master Control Station, Monitor Stations, and Ground Antennas

Worldwide coverage with redundancy and backup systems

🏢 Master Control Station (MCS)

The **Master Control Station (MCS)** **is responsible for satellite control and overall system operations.**

- Primary command and control facility
- Located in Colorado Springs
- Backed up by alternate master control station

📡 Monitor Stations

Monitor Stations continuously receive GPS satellite transmissions, and relay this information in real time to the Master Control Station in Colorado.

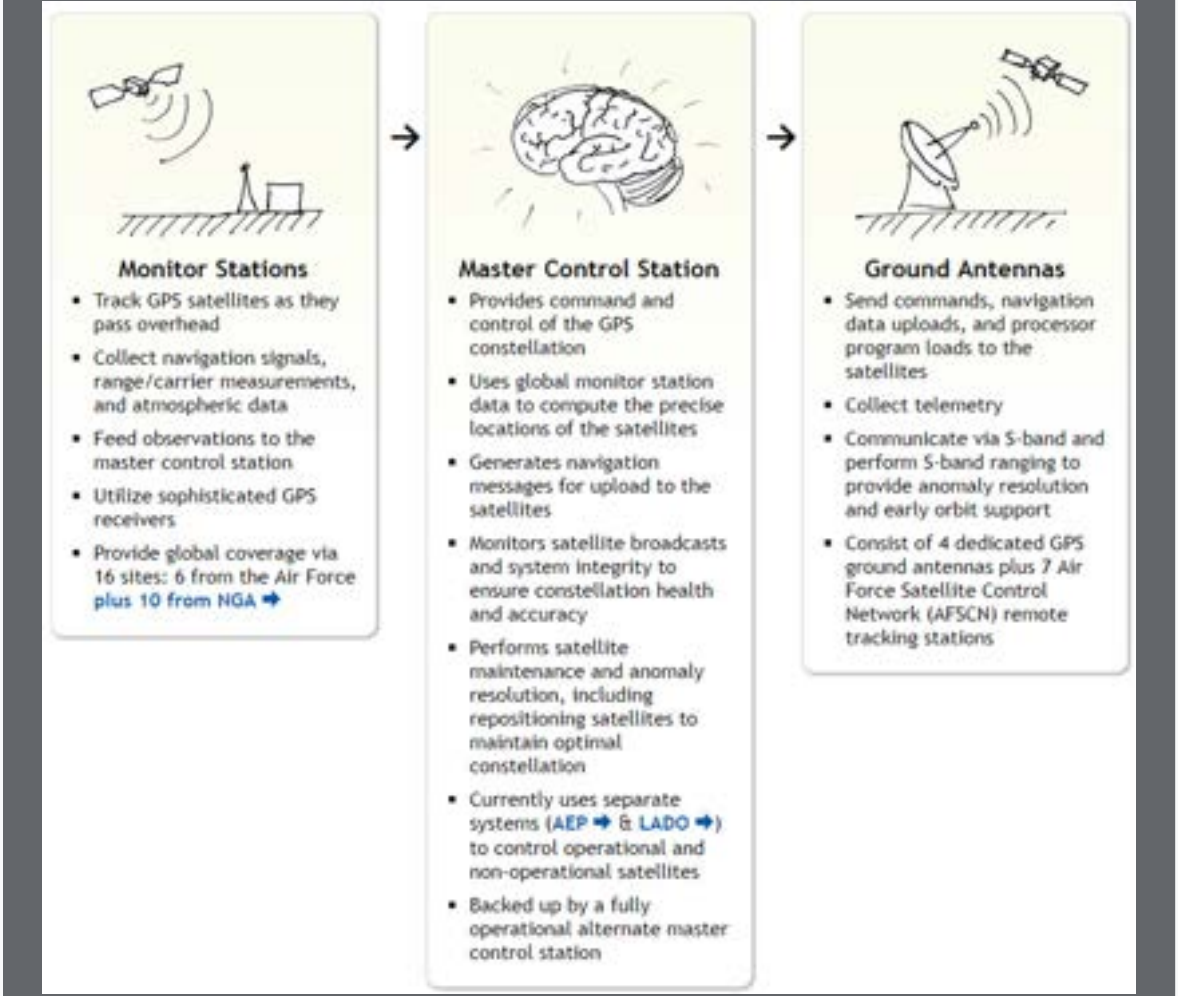
- Unmanned remote sensors
- Passively collect raw satellite signal data
- Very precise radio receivers

📶 Ground Antennas

Ground antennas are remotely controlled by the MCS. Ground antennas transmit data commands received from the MCS to GPS satellites within their sky view.

- Uplink capabilities to satellites
- Remote operation from MCS
- Strategic global positioning

🔧 Control Segment Components Workflow



GPS Control Segment Components
Workflow diagram showing Monitor Stations → Master Control Station → Ground Antennas

📊 Operational Functions

These stations are unmanned remote sensors that **passively collect raw satellite signal data and re-transmit it in real time to the MCS for evaluation**. Monitor stations basically function as very precise radio receivers, tracking each satellite as it comes into view.

📝 Module Summary

Integrated GPS System Architecture

This module examined the three-segment GPS architecture that enables global positioning services. The **Space Segment** provides continuous satellite coverage through 24+ satellites in medium Earth orbit, the **Control Segment** maintains system accuracy through global monitoring and satellite command networks, and the **User Segment** encompasses the diverse applications from military operations to civilian navigation that depend on GPS positioning, velocity, and timing services.

🎯 Prepare for the Next Topic

Next, we'll dive deeper into GPS positioning principles, exploring how receivers use signals from multiple satellites to calculate precise three-dimensional positions through trilateration and timing calculations.

📚 Application and Accuracy



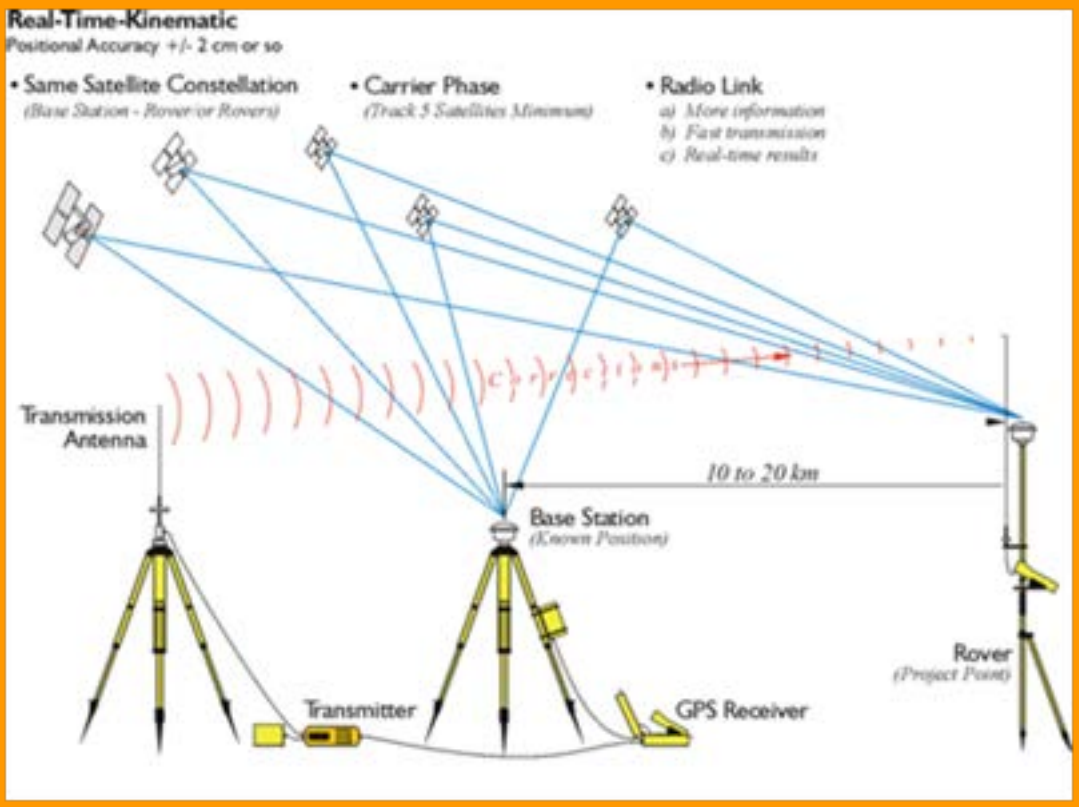
📹 Instructor Video Lecture

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🎯 How Accurate is GPS?

Accuracy Range: From Meters to Centimeters

Depending on the design of the GPS receiver and the measurement techniques employed, the accuracy is from 100 meters under Selective Availability (SA) policy (below 10 m with Selective Availability turned to zero) to better than 1 centimeter.



Real-Time Kinematic (RTK) GPS Setup
for Centimeter-Level Accuracy

Satellite Constellation • Carrier Phase • Radio Link

📊 GPS Accuracy Spectrum

Basic GPS (SA Enabled)

~100 meters - Historical accuracy with Selective Availability

Standard GPS (SA Disabled)

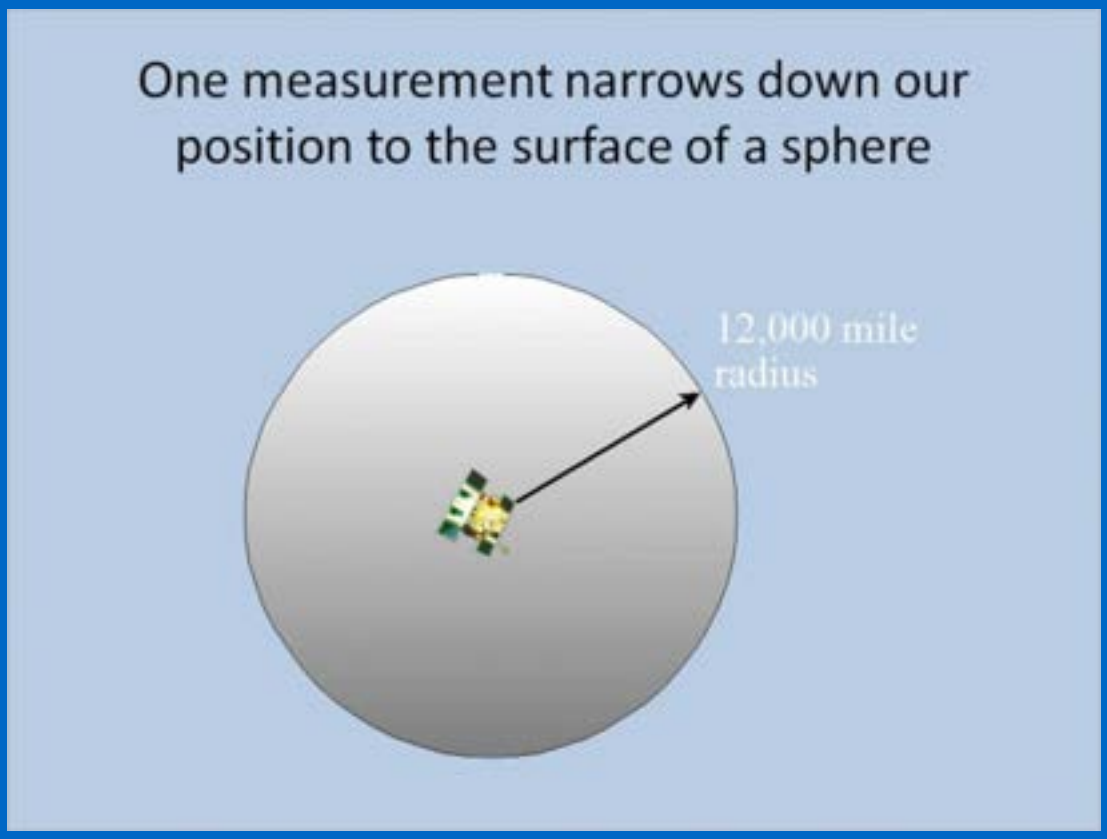
Below 10 meters - Modern civilian GPS accuracy

High-Precision GPS (RTK)

Better than 1 centimeter - Professional surveying applications

🛰️ Signal From One Satellite

Spherical Position Estimation



One measurement narrows down our position to the surface of a sphere

12,000-mile radius sphere centered at satellite

⚡ Single Satellite Limitations

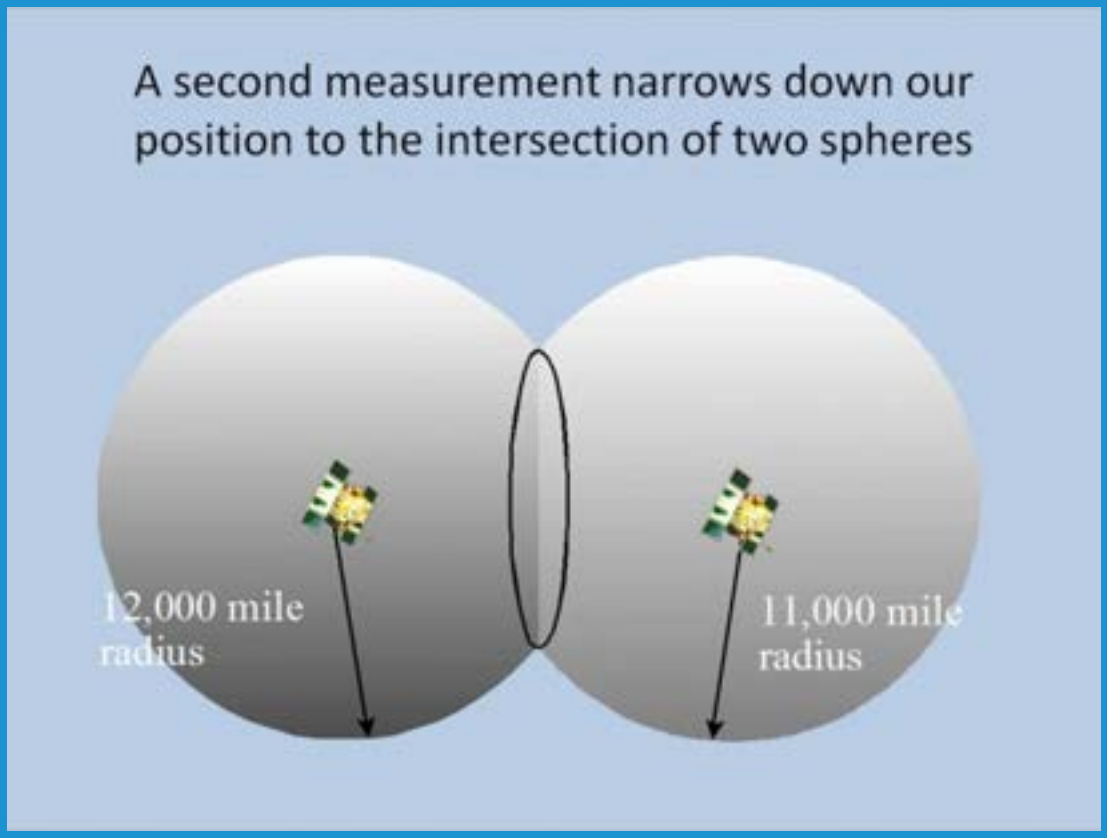
- Receiver lies **somewhere on a sphere centered at that satellite**
- Radius equals **distance signal traveled—typically around 12,000 miles**
- Distance calculated by **multiplying signal's travel time by speed of light**
- Need signals from **at least four satellites** for exact location
- Requires **strong signal lock, accurate timing, and good satellite geometry**

⏱️ **Cold Start Requirements**

If the receiver has been turned off for a while, it must **download a new almanac/ephemeris, which takes about 12–13 minutes after acquiring a signal**. Until multiple signals are received, the receiver can only estimate a rough position on a spherical shell.

Signals From Two Satellites

Circular Intersection: Improved Position Estimation



A second measurement narrows down our position to the intersection of two spheres

12,000-mile radius | 11,000-mile radius

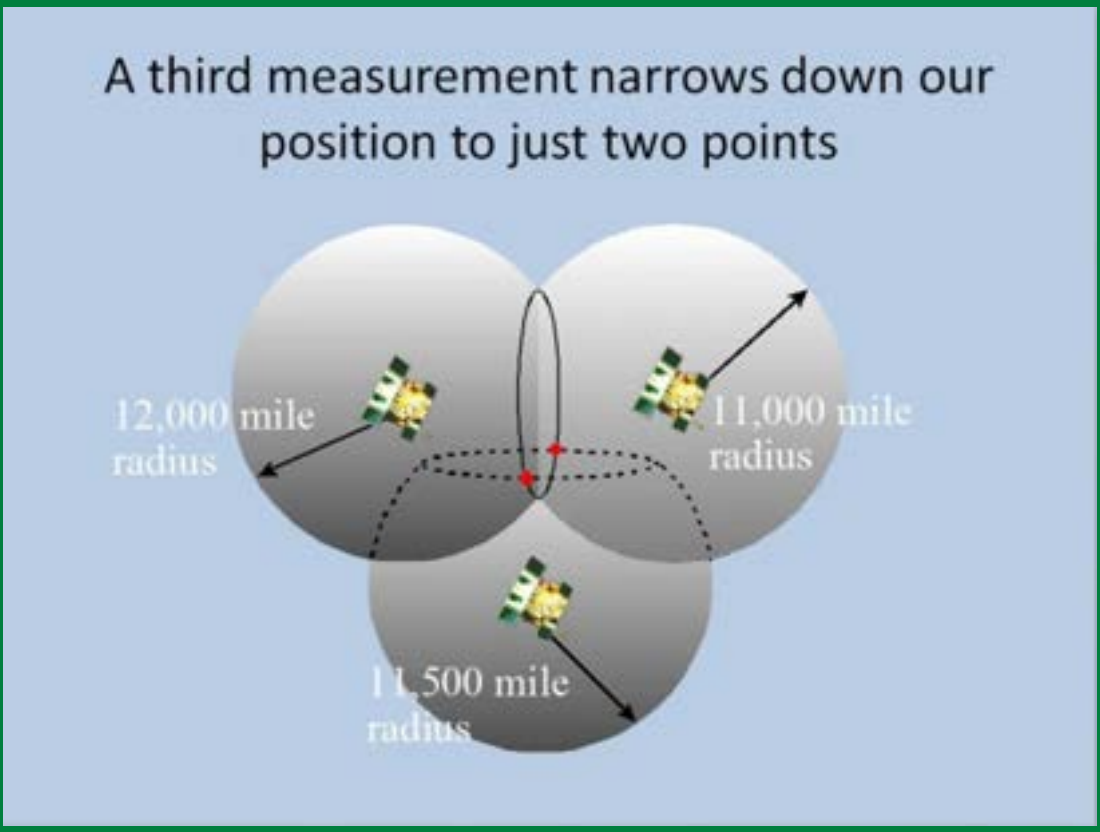
🔄 Refinement Process

With **signals from two GPS satellites, the receiver can refine its position from being somewhere on a single sphere to being located along the circle where two spheres intersect.**

- **Each sphere represents possible distance** from one satellite based on signal travel time
- **Overlap narrows possible receiver location** to the edge they share
- **Significant improvement** in position estimation
- **Still not enough** to pinpoint exact location
- **More satellite signals needed** for precise location

Signals From Three Satellites

Trilateration: Precise Position Determination



A third measurement narrows down our position to just two points

12,000-mile | 11,000-mile | 11,500-mile radius

🎯 Trilateration Process

In the absence of clock error, a GPS receiver uses trilateration to determine its position.

- **Three spheres defined**—each centered at a satellite with radius equal to distance to receiver
- **Intersection occurs at two points** in space
- **Only one point is physically plausible**—typically near Earth's surface
- **Other point often in space** or unrealistic location
- **Receiver selects correct location** based on logic, context, or signal strength consistency

🧠 **Location Selection Logic**

The receiver selects the correct location from the two intersections based on **logic, context (e.g., known altitude), or signal strength consistency**. The unrealistic point is often far above Earth or underground and can be discarded.

📐 Dilution of Precision (DOP)

Geometric Strength of Satellite Distribution

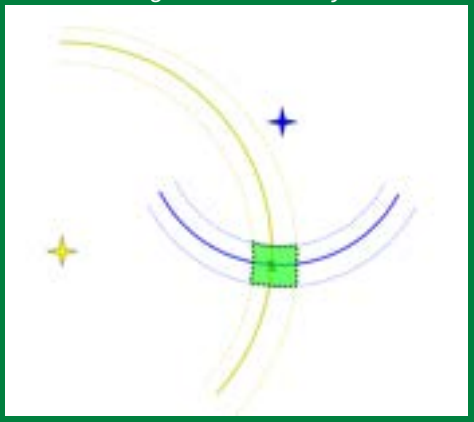
Dilution of Precision (DOP) concerns the geometric strength of the satellite distribution in space with respect to one another and the GPS receiver.

🎯 Satellite Geometry Impact on Accuracy

✅

Good Geometry (Low DOP)

Small green uncertainty box

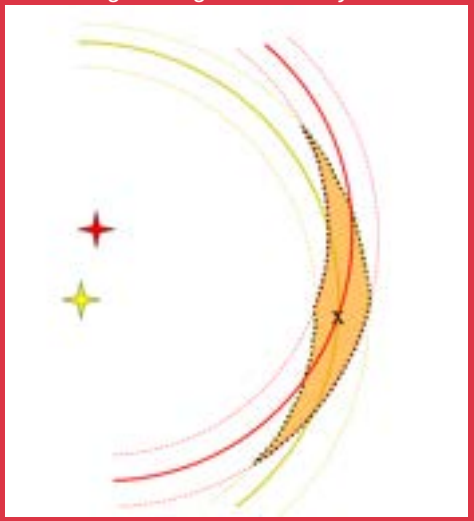


Well-distributed satellites across the sky result in tighter intersection areas and **better positional accuracy**.

❌

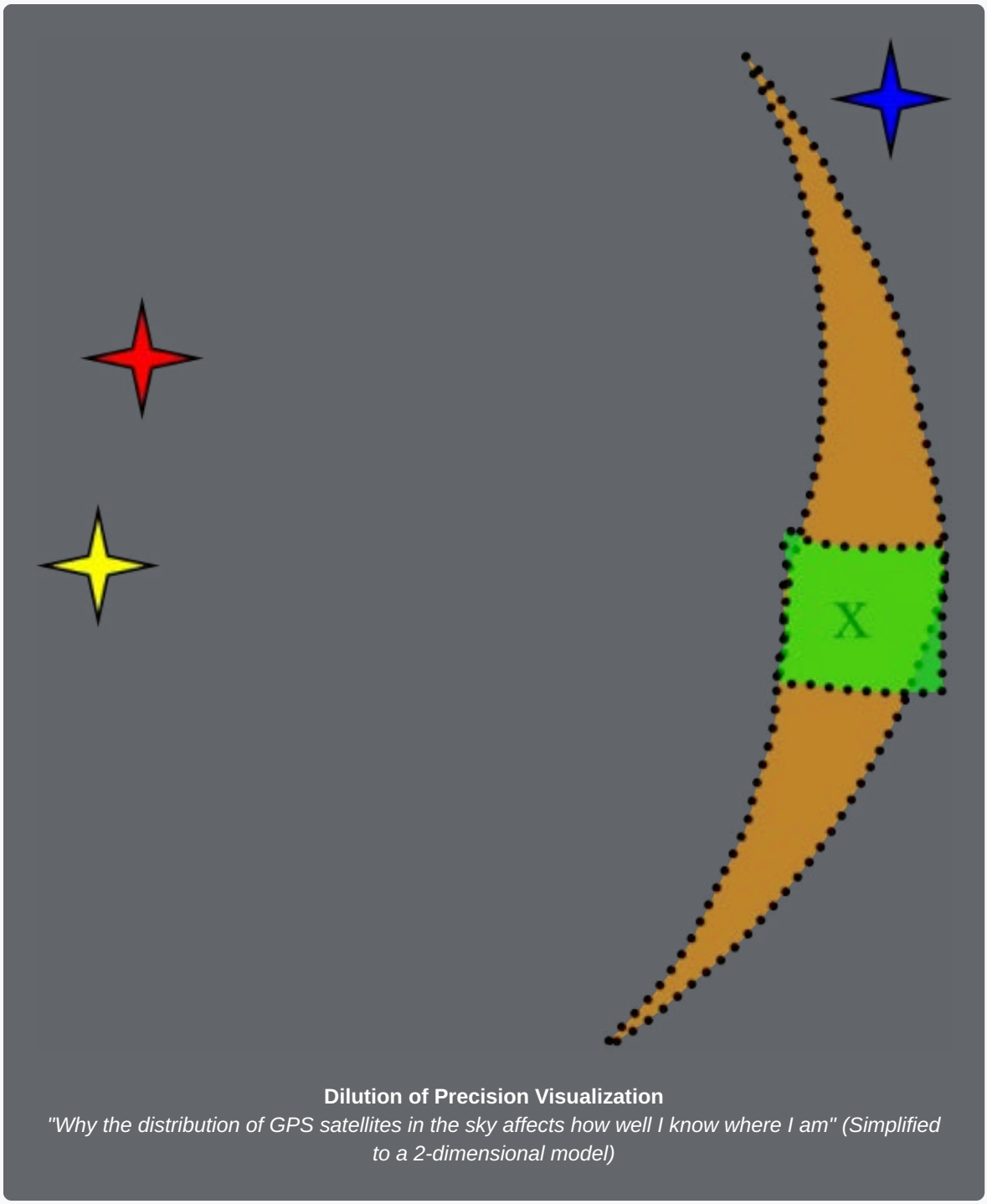
Poor Geometry (High DOP)

Large orange uncertainty box



Clustered satellites create larger or elongated overlapping regions, **increasing uncertainty**.

🔧 DOP Diagram Elements



Dilution of Precision Visualization
"Why the distribution of GPS satellites in the sky affects how well I know where I am" (Simplified to a 2-dimensional model)

⭐ **Yellow/Blue Stars**

GPS satellites - source of range measurements for positioning

❌ **X Symbol**

Actual location of GPS receiver (user's position on ground)

🟢 **Green Region**

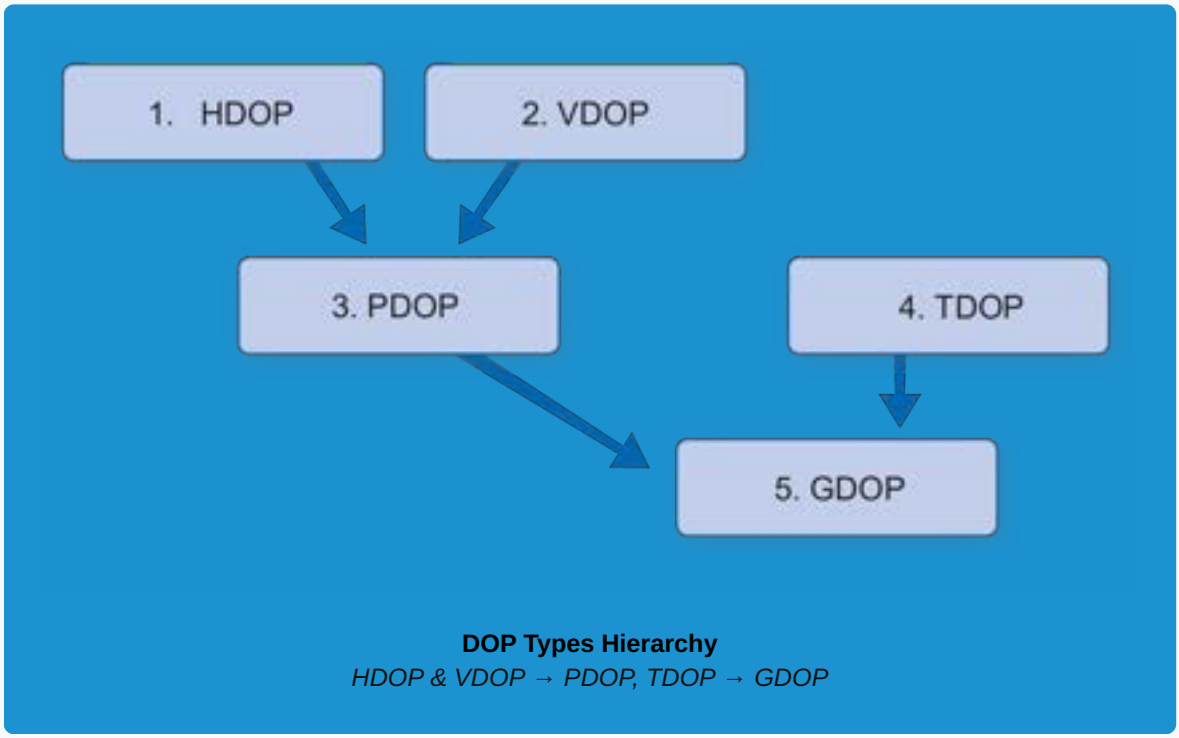
Intersection of range arcs - smaller and more orthogonal = more precise

📊 DOP Relationship Summary

When satellites are well distributed across the sky, their range circles intersect in a tighter area, resulting in **better positional accuracy**. This concept shows that **a wider satellite spread leads to a lower DOP and higher accuracy, while poor distribution leads to higher DOP and lower accuracy**.

📊 Types of Dilution of Precision

Comprehensive DOP Measurement Framework



DOP Types Hierarchy
HDOP & VDOP → PDOP, TDOP → GDOP

🏔️ Horizontal & Vertical DOP

HDOP (Horizontal DOP)

Quantifies potential error in the horizontal plane—latitude and longitude—based on satellite geometry. A lower HDOP means

📍 Position & Time DOP

PDOP (Position DOP)

Combines both HDOP and VDOP using the formula:

$$PDOP = \sqrt{(HDOP^2 + VDOP^2)}$$

satellites are well spread out in the sky, leading to more accurate horizontal positioning.

VDOP (Vertical DOP)

Reflects potential error in vertical positioning (elevation or altitude). Poor vertical satellite geometry (e.g., satellites clustered at similar elevation angles) leads to higher VDOP and less reliable height estimation.

PDOP is perhaps the most commonly reported DOP value. It represents overall 3D position error.

TDOP (Time DOP)

Indicates possible timing error due to satellite-receiver clock discrepancies and how satellite placement affects time accuracy.

🌐 GDOP (Geometric DOP)

GDOP (Geometric DOP) is the most comprehensive measure, combining PDOP and TDOP:

$$GDOP = \sqrt{(PDOP^2 + TDOP^2)}$$

It reflects the **total impact of satellite geometry on the accuracy of both position (x, y, z) and time**.

🛠️ Practical Applications

For practical use, **most GPS units provide PDOP and GDOP values**, and surveyors or UAV operators often **avoid measurements when DOP values exceed acceptable thresholds (e.g., PDOP > 6)**.

📝 Module Summary

From Basic Positioning to Precision Applications

This module explored GPS accuracy fundamentals, from the basic trilateration process using signals from multiple satellites to the sophisticated concepts of Dilution of Precision that affect positioning quality. Understanding how satellite geometry impacts accuracy—from single satellite spherical estimates to precise trilateration with optimal satellite distribution—provides the foundation for professional GPS applications requiring centimeter-level precision.

Texas A&M University - Corpus Christi
College of Engineering & Computer Science
GISC 4315 - Satellite Positioning
Fall 2026

ACTIVITIES AND ASSESSMENTS

📰 Part 1: The Origins of GPS, and the Pioneers Who Launched the System



🧭 About This Activity

After you read the article "Part 1: The Origins of GPS, and the Pioneers Who Launched the System", click **Start Course** in the embedded activity below to explore the evolution of GPS through Cold War breakthroughs, military collaboration, and satellite innovations.

Embedded content: Embedded content link

Embedded content: Embedded content link

📰 Part 2: The Origins of GPS, Fighting to Survive



🧭 About This Activity

"Part 2: The Origins of GPS, Fighting to Survive" builds on the GPS origin story by focusing on the challenges and turning points of the 1970s. Learn how visionaries, political obstacles, and Cold War dynamics shaped the fight to keep GPS alive and funded — and how that perseverance paid off. After you have read the article, click **Start Course** to begin.

Embedded content: Embedded content link

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ACTIVITIES AND ASSESSMENTS

📰 Part 3: The Legal, Ethical, and Regulatory Aspects of GPS Jamming



🧭 About This Activity

After you read the article "The Legal, Ethical, and Regulatory Aspects of GPS Jamming," click **Start Course** in the embedded activity below to explore how misuse of GPS technology presents modern-day risks—and how global authorities are responding with evolving laws and safeguards.

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📡 Module 01 Summary



📋 Module 01 Summary: Introduction to GPS

Key Concepts and Learning Review

🎯 Learning Objectives - What You've Learned

✅ Module Objectives Achieved

You should now be able to:

- **Describe the historical development of GPS** and its predecessor systems (TRANSIT, System 621B, Timation)
- **Explain the fundamental components and architecture** of the GPS system, including the space, control, and user segments
- **Identify key milestones in GPS policy decisions** that led to civilian accessibility (Reagan's authorization, Clinton's SA deactivation)
- **Recognize GPS objectives** for global users and understand system design goals
- **Interpret the significance of satellite geometry** and DOP (Dilution of Precision) values in determining GPS accuracy

🔑 Key Concepts Covered

🕰️ Historical Development

- **TRANSIT System:** First operational satellite navigation using Doppler shift
- **System 621B:** Introduced "passive user" concept and PRN codes
- **Timation:** Developed precise space-based timing with atomic clocks

🏗️ System Architecture

- **Space Segment:** 24+ satellites, 6 orbits, 20,000 km altitude
- **Control Segment:** Master control, monitor stations, ground antennas
- **User Segment:** GPS receivers for positioning, velocity, and time

- **GPS Launch:** NAVSTAR program initiated in 1973
- **Signal Evolution:** L1 C/A to modern multi-frequency signals

📜 Critical Policy Milestones

1983 - Reagan Administration

Authorized civilian access to GPS following Korean Air Lines Flight 007 incident

2000 - Clinton Administration

Disabled Selective Availability, improving civilian accuracy from 100m to <10m

📊 GPS Performance and Applications

🎯 System Objectives

- Global coverage
- All-weather operation
- Passive positioning
- Real-time capability
- Multiple platform support

📏 Accuracy Ranges

- Standard GPS: <10m
- DGPS: 1-3m
- RTK: <1cm
- Survey grade: mm-level

🌍 Applications

- Navigation
- Surveying
- Precision agriculture
- Emergency response
- Scientific research

📐 Understanding Dilution of Precision (DOP)

DOP measures how satellite geometry affects positioning accuracy:

- **HDOP:** Horizontal Dilution of Precision
- **VDOP:** Vertical Dilution of Precision
- **PDOP:** Position (3D) Dilution of Precision
- **TDOP:** Time Dilution of Precision
- **GDOP:** Geometric Dilution of Precision
- **Lower values = Better accuracy**

💡 Key Takeaways

🎯 Bottom Line Up Front

- **GPS evolved from multiple predecessor systems** (TRANSIT, 621B, Timation) that each contributed essential technologies

- **The system architecture consists of three segments** working together to provide global positioning capabilities
- **Political decisions significantly impacted civilian access** and accuracy improvements over time
- **Signal strength is extremely weak** (-125 to -130 dBm) making GPS vulnerable to interference
- **Satellite geometry directly affects positioning accuracy** through DOP values

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INTRODUCTION AND INSTRUCTIONS

📡 Overview Module 02: GPS Signals



🧭 Module Overview

This section of the course introduces **the foundational concepts of GPS signal structureand behavior**. It explains that GPS is a passive system in which satellites continuously broadcast signals that receivers use to determine position through trilateration. These signals include a carrier wave, pseudo-random codes (like the C/A code), and a navigation message containing satellite clock and orbit information. Module 02 details how each satellite transmits a unique code using spread-spectrum technology, allowing receivers to distinguish signals even though all satellites share the same frequencies, such as L1 and L2. The process of signal acquisition and tracking is described, emphasizing how receivers search for the correct satellite code, frequency, and timing phase before locking on and extracting navigation data. Overall, this section of our course lays the groundwork for understanding how GPS signals enable precise positioning in surveying applications.

📍 What are GPS Signals?

GPS Signals are radio frequency transmissions broadcast continuously by Global Positioning System satellites that enable receivers on Earth to determine their precise location. These signals are the fundamental communication mechanism that makes satellite positioning possible.

📚 Module Learning Objectives

At the end of this module, learners will be able to:

- Demonstrate understanding of the basic GPS signal structure
- Discuss the similarities between GPS and trilateration
- Describe the pertinence of the navigation code
- Explain the structure of the P and C/A codes
- Define the creation of the GPS modulated carrier wave
- Identify the two GPS Observables
- Describe the role of autocorrelation and the lock and time shift associated with GPS pseudoranging
- Recognize the pseudorange equation (This is the code phase)
- Discuss the role of carrier phase ranging in high-accuracy GPS.

📝 Module Instructions

- **Read Chapter 1 in *GPS and GNSS for Land Surveyors*.**
- Review the pages in this module and watch the instructor videos embedded within the module pages.
- After you have review the content for this module, you will complete the assessments and activities to evaluate the learning objectives.

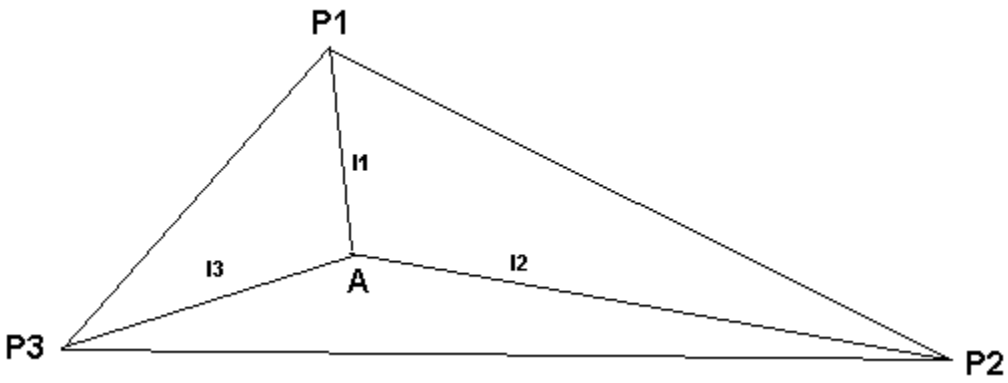
📚 Trilateration, Unknowns, and One-way Ranging

Module 02: GPS Trilateration, Unknowns, and One-Way Ranging

GISC 4315

📐 Trilateration: The Foundation of GPS

Understanding Distance-Based Positioning



Trilateration diagram showing control points P1, P2, P3 and unknown point A with distance measurements L1, L2, L3

Trilateration: Using distance measurements to three control points to determine position

🔍 Trilateration

- **Based on distance measurements**
- Uses **electronic distance measurement (EDM)**
- Relies on **electromagnetic signal travel time**

📐 Triangulation

- **Based on angle measurements**
- Uses **angular observations**
- Requires **theodolites or total stations**
- Control points are **stationary on Earth's surface**

- GPS satellites serve as **moving control points**

💡 Key Concept

GPS uses **trilateration, not triangulation**. Both techniques rely exclusively on the measurement of distances to fix positions, but GPS measures distances to satellites orbiting approximately 20,000 km above Earth.

🔢 The Four Unknowns in GPS

Why GPS Needs Four Satellites

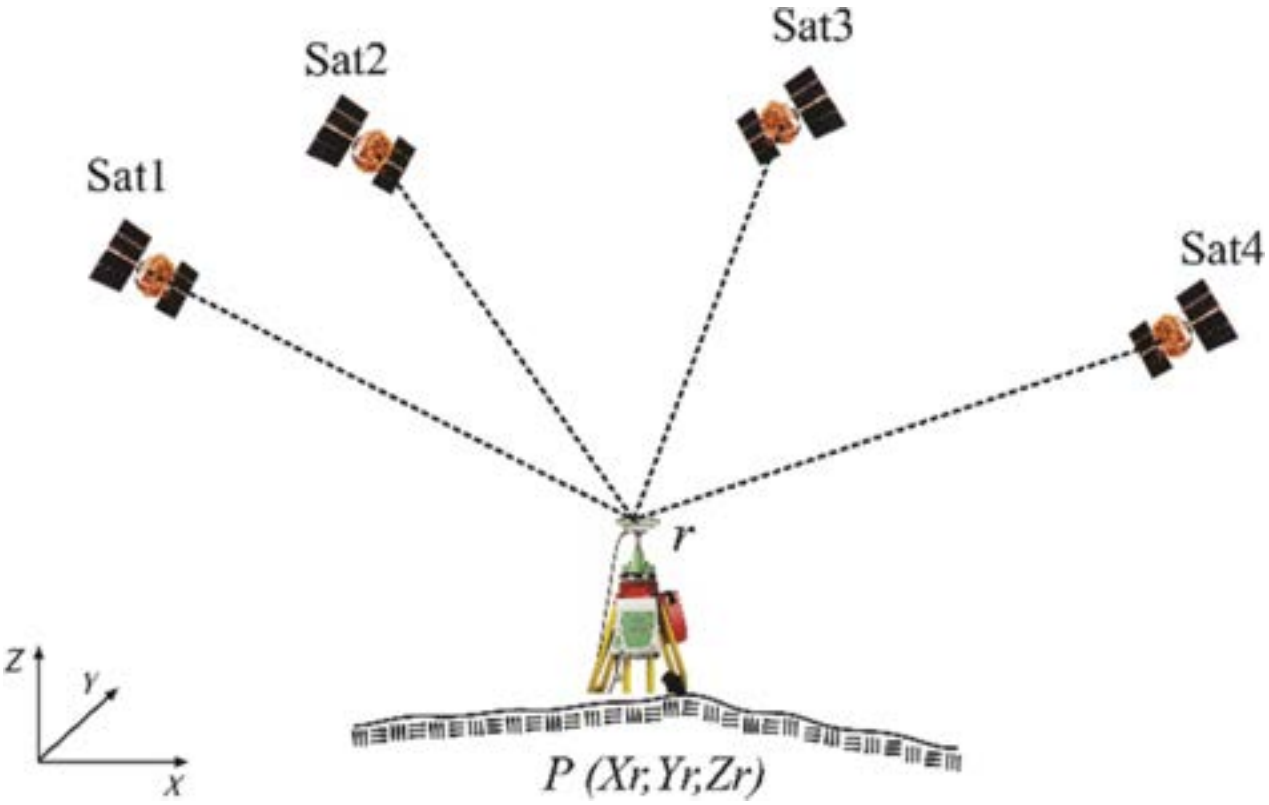


Diagram showing 4 satellites observed from a single point simultaneously

Four satellites observed simultaneously for 3D positioning and time resolution

📍 X Coordinate

East-West Position

📍 Y Coordinate

North-South Position

📍 Z Coordinate

Vertical Position (Height)

⏰ Time

Clock Synchronization

🔄 Why Four Instead of Three?

Unlike terrestrial trilateration which uses three control points for 2D positioning, GPS requires **four satellites** because it provides **3D positioning** and must resolve **time synchronization** between satellite atomic clocks and receiver clocks.

📡 GPS as a Passive System

One-Way Signal Flow

🔄 Passive Operation

- **Satellites transmit** signals continuously
- **Receivers only listen** - no transmission back
- Similar to **car radio receiving broadcasts**
- **Unlimited users** can access simultaneously

⚡ Time Critical Elements

- **Range calculation:** Distance = Speed × Time
- **Satellite position** at signal transmission
- **Clock synchronization** between systems
- **Signal travel time** measurement

🚀 **Advantages of Passive Operation**

This passive nature allows GPS to serve **millions of users simultaneously** without system overload or revealing user locations, making it ideal for both civilian and military applications.

📶 One-Way Ranging Requirements

Essential Signal Information

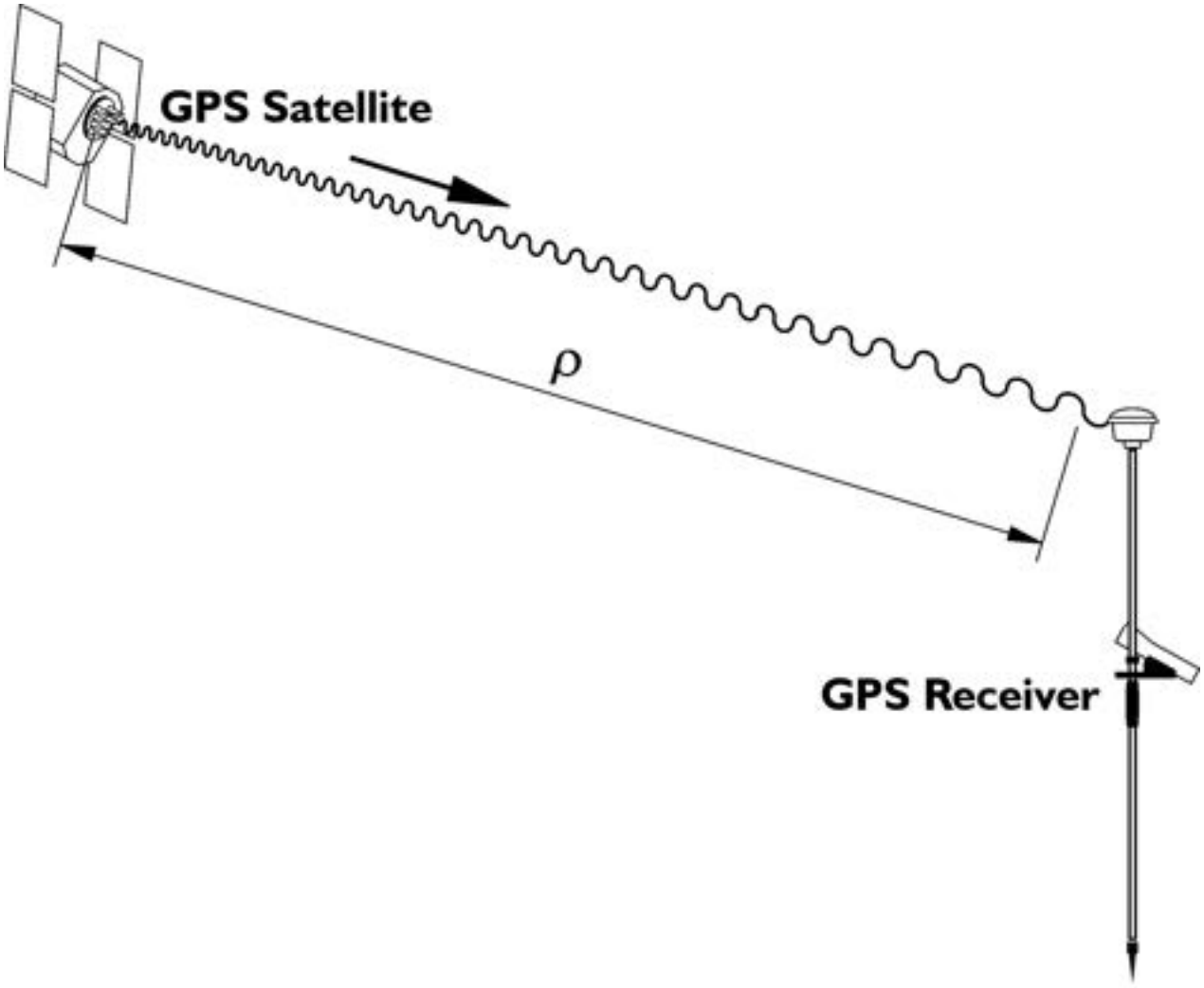


Diagram showing GPS signal path from satellite to receiver with range measurement

One-way ranging: GPS signal travels from satellite to receiver

For one-way ranging to work, the GPS signal must communicate **four essential types of information**:

⏰ `Satellite Time`

Precise time when signal was transmitted from the satellite

📍 `Satellite Position`

Instantaneous coordinates of the moving satellite (ephemeris)

🌍 Atmospheric Data

Information about ionosphere and troposphere corrections

🆔 Satellite ID

Unique identifier and information about other satellites

🚀 Moving Control Points

Unlike stationary terrestrial control points, GPS satellites move at tremendous speeds. This requires **real-time ephemeris data** to know exactly where each satellite is at the moment of signal transmission.

📝 Section Summary

Key Takeaways

In this section, we explored the fundamental positioning principles behind GPS. We learned that GPS uses **trilateration** (distance-based measurements) rather than triangulation (angle-based measurements), requires **four satellites** to resolve four unknowns (X, Y, Z, and time), and operates as a **passive system** with one-way signal flow from satellites to receivers.

🔍 Trilateration vs. Triangulation

GPS uses distance measurements to moving satellites, not angle measurements to fixed points.

📡 Passive Operation

One-way signal flow enables unlimited users without system overload.

CONTENT

📚 The Navigation Message

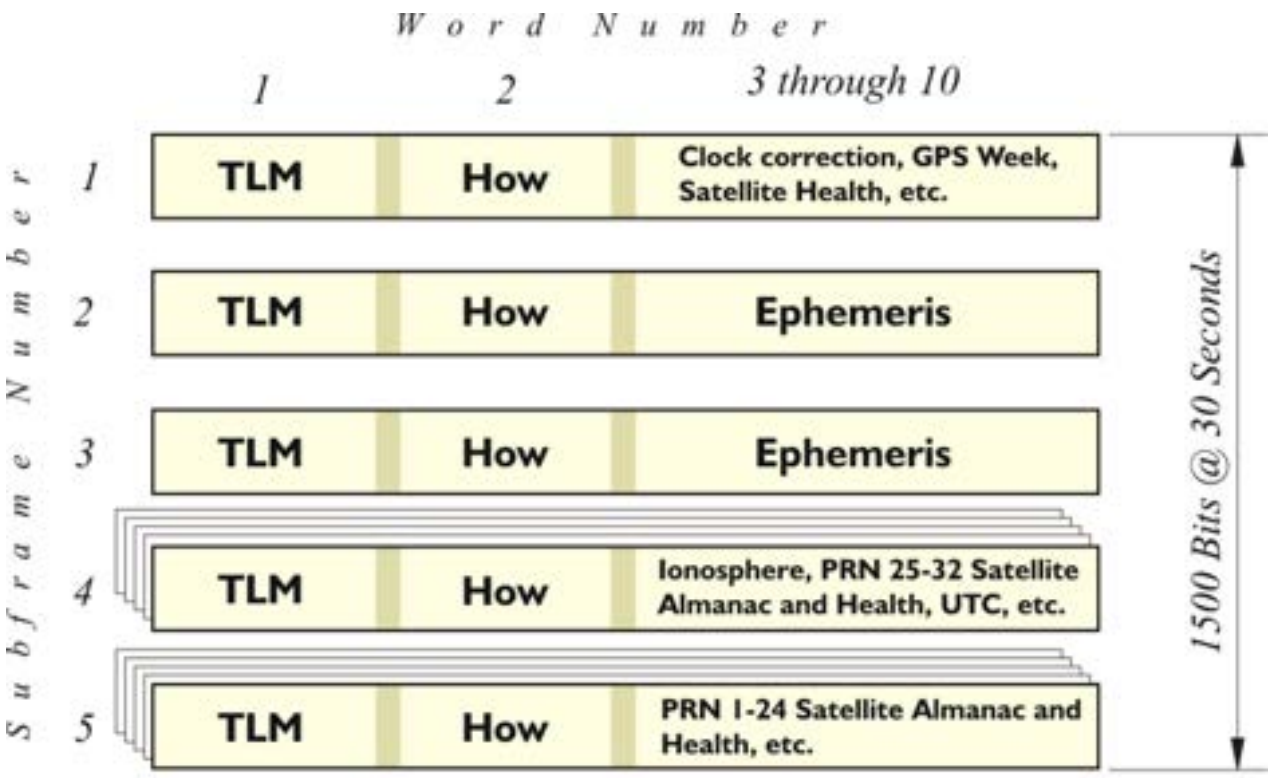


GPS Navigation Message Structure

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Understanding the NAV Message



Each subframe has 10 30-Bit Words, 6 Seconds. The entire Navigation message. The Master Frame, contains 37,500 bits - 12 ½ min to broadcast and receive at 50 bps.

Navigation Message

An image of the following table, representing a navigation message: 1500 bits @ 30 seconds

The Navigation Message

Click here to see a text description.

An image of the following table, representing a navigation message: 1500 bits @ 30 seconds

Subframe	1 word	2 words	3–10 words
1	TLM	How	Clock correction, GPS week, Satellite Health, etc.
2	TLM	How	Ephemeris
3	TLM	How	Ephemeris
4	TLM	How	Ionosphere, PRN 25–32 Satellite Almanac and Health, UTC, etc. (Subframe 4 contains 25 subcommutated pages)

5	TLM	How	PRN 1–24 Satellite Almanac and Health, etc. (Subframe 5 contains 25 subcommutated pages)

At the bottom of the table a description says: Each word = 30 bits, Each subframe = 10 words = 300 bits, Each frame = 5 subframes = 1500 bits, Navigation Message = 25 Frames = 37,500 bits

📡 Legacy NAV Message Overview

This is the primary vehicle for communicating the NAV message to GPS receivers. The NAV message is also known as the GPS message. It includes some of the information the receivers need to determine positions. Today, there are several NAV messages being broadcast by GPS satellites, but we will look at the oldest of them first.

The legacy NAV (NAV) message continues to be one of the mainstays on which GPS relies. The NAV code is broadcast at a low frequency of 50 Hz on both the L1 and the L2 GPS carriers. It carries information about the location of the GPS satellites called the ephemeris and data used in both time conversions and offsets called clock corrections.

⏰ Clock, Health & Ionosphere Data

Both GPS satellites and receivers have clocks on board. It also communicates the health of the satellites on orbit and information about the ionosphere. The ionosphere, along with the troposphere, is a layer of atmosphere through which the GPS signals must travel to get to the user.

It includes data called almanacs that provide a GPS receiver with enough little snippets of ephemeris information to calculate the coordinates of all the satellites in the constellation with an approximate accuracy of a couple of kilometers.

📦 Message Size and Timing

The Navigation code, or message, is the vehicle for telling the GPS receivers some of the most important things they need to know. Here are some of the parameters of its design.

The entire Navigation message, the *Master Frame*, contains 25 frames. Each frame is 1500 bits long and is divided into five subframes. Each subframe contains 10 words, and each word is comprised of 30 bits.

Therefore, the entire Navigation message contains 37,500 bits and at a rate of 50 bits-per-second takes 12½ minutes to broadcast and to receive on a completely cold start. In other words, getting the

whole thing is not instantaneous. It does take a bit of time for the receiver to update its Navigation Message.

🔍 Subframe Content Details

In the five sub-frames of the legacy Navigation Message. TLM stands for telemetry. HOW stands for handover word. Over on the right-hand side in the illustration, you see the clock correction, GPS satellite health, et cetera, in sub frame one. Two and three are devoted to the ephemeris.

In four and five, you see ionosphere, and then PRN (Pseudo Random Noise) satellite numbers and almanac are mentioned. The PRN term is used because the GPS signals that the receiver uses for positioning appear to be random noise, but in fact, the signal is pseudo (false) random noise because in truth, the signals are very carefully designed and consistent.

📍 Almanac and Satellite PRNs

They are not noise at all. They just seem to be irregular. The PRN numbers 25 to 32 in sub-frame number four mean that satellite's almanacs from number 25 to number 32 are to be found there.

In subframe five the PRNs from 1 to 24, those satellites have their almanacs, in other words, a little bit of their ephemerides. You might wonder why they are there. This is that identification system. In other words, when a receiver acquires the Navigation Message from one satellite - embedded in that message - there's a bit of information, just a bit, that will tell the receiver where it can find the rest of the entire constellation in the sky. This helps it acquire the additional satellites after it's got the first one. That's what the satellite almanac does.

🛰️ Final Summary

The essential point here is that this message is the fundamental vehicle for the satellite to communicate important information to the receiver. After the receiver has acquired the signal from that satellite, the NAV message tells the receiver where the satellite is.

The ephemeris is the satellite coordinate system. It tells the receiver where the satellite is at an instant of time. The clock correction is one of the ways that the satellite can tell the receiver what time it is on-board the satellite. The ionosphere is that information that will allow the receiver to make some atmospheric corrections on the signal it receives from a particular satellite.

📝 Section Summary

Key Takeaways

In this section, we explored the structure and contents of the GPS Navigation Message. We learned how it is divided into subframes that communicate critical information such as satellite position (ephemeris), system time (clock correction), satellite health, and almanac data. These messages are crucial for accurate positioning and efficient satellite acquisition.

📶 Ephemeris & Clock Correction

Provide satellite coordinates and time synchronization data.

🛰️ Almanac & Satellite Health

Used to track and acquire other satellites quickly.

📚 The Control Segment



The Control Segment

GISC 4315 – Satellite Positioning

⏱️ Why the Control Segment is Needed

Unfortunately, the accuracy of some aspects of the information included in the NAV message deteriorates with time. Translated into the rate of change in the three-dimensional position of a GPS receiver, it is about 4 cm per minute. Therefore, mechanisms are in place to prevent the message from getting too old. For example, every two hours, the data in subframes 1, 2 and 3, the ephemeris and clock's parameters, are updated. The data in subframes 4 and 5, the almanacs, are renewed every six days. These updates are provided by the government uploading facilities around the world which are known, along with their tracking and computing counterparts, as the Control Segment. The information sent to each satellite from the Control Segment makes its way through the satellites and back to the users in the NAV message.



Map showing the control stations and types around the world.

📡 Origins of the Navigation Message

The question might arise in your mind, where does this Navigation Message and the information embedded there come from? It comes from the Control Segment. There are three segments in the GPS system: the User Segment, the Control Segment, and the Space Segment. The Space Segment is obviously the satellites themselves, the User Segment are all of us with our GPS receivers, and the Control Segment, well, here's an illustration that shows you the expanded Control Segment that is currently operational.

🌍 How Ground Stations Work

It's a good thing that we have the Control Segment a little bit modernized from what it once was. We'll talk more about that in the future, but here we have the stations, some of them monitor stations and then some of them actually have upload capability as well. In general, what happens is this. The GPS satellites are tracked by at least, now, three of these stations simultaneously all the time. Now, by tracking them, they are able to do several things. For example, since the stations already know their positions on the surface of the Earth, they are able to collect the signals from the satellites and know how much they have been delayed or attenuated by the atmosphere. They're also able to track the satellites and determine their path, their orbit. From this, their ephemerides

can be calculated. The control station is also able to compare the signal coming from the satellite with the station's atomic clock, see the difference between the two, and then upload a clock correction to the satellite.

🏢 Master Control Station

This is basically how the information is derived. It then goes from the tracking stations to the Master Control Station in Colorado Springs at Schriever Air Force Base, where calculations are done and the components that we've just seen in the Navigation Message are generated.

📶 Single vs. Multi-Frequency Receivers

🔁 Uploading the Corrections

Those corrections go back to the upload stations with the ground antennas and are uploaded back to the satellites. That's where the information that is in the Navigation Message comes from. It's the control stations in the Control Segment on the surface of the Earth comparing the received signal from the satellite with their satellite clock, with their atomic clock and with their actual known position. This way, they can derive the corrections that go into the satellite's Navigation Message.

📝 Section Summary

Key Takeaways

This section explained the GPS Control Segment and how it supports the accuracy and reliability of the Navigation Message. We learned how monitor and upload stations continuously track satellites, calculate corrections, and update satellite messages with essential information such as clock offsets, ephemeris, and almanacs.

📍 Ground Station Network

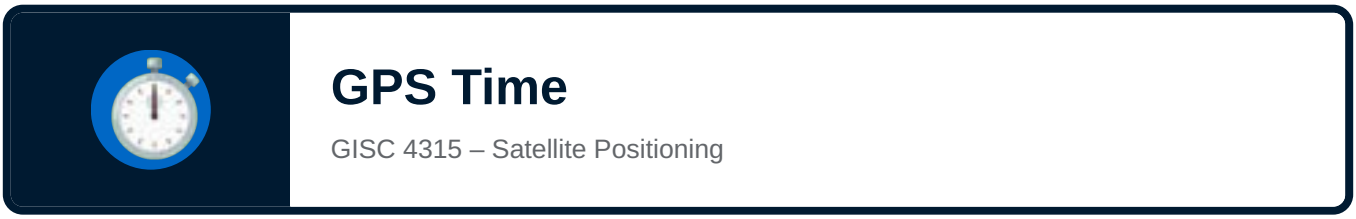
Control stations track satellite positions and measure signal delay through the atmosphere.

🧭 Role of Master Control

Coordinates data processing and uploads corrections to satellites for reliable NAV messages.

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📚 GPS Time



GPS Time

System	Time	Day	Additional Info	Remarks
Local	2020-09-09 14:37:04	Wednesday	day 253	timezone UTC-6
UTC	2020-09-09 20:37:04	Wednesday	day 253	MJD 59101.85907
GPS	2020-09-09 20:37:22	week 2122	333442 s	cycle 2 week 0074 day 3
Loran	2020-09-09 20:37:31	GRI 9940	47 s until	next TOC 20:37:51 UTC
TAI	2020-09-09 20:37:41	Wednesday	day 253	37 leap seconds

⏰ Time Standards in the NAV Message

There is time-sensitive information in the NAV message in both subframe 1 and subframe 4. The information in subframe 4 helps a receiver relate two different time standards to one another. One of them is GPS Time and the other is Coordinated Universal Time (UTC).

🛰️ GPS Time vs UTC

GPS Time is the time standard of the GPS system. It is also known as GPS System Time (GPST). Coordinated Universal Time is the time standard for the world. The rates of these two standards are virtually the same. Specifically, the **rate** of GPS Time is kept within 1 microsecond, and usually less than 25 nanoseconds, of the **rate** of Coordinated Universal Time (UTC). The exact difference is in two constants, A0 and A1 in the NAV message, which give the time difference and rate of system time against UTC.

🌐 The Role of Leap Seconds

The rate of UTC itself is carefully determined. It is steered by about 65 timing laboratories and hundreds of atomic clocks around the world and is remarkable in its stability. In fact, it is more stable than the rotation of the earth itself, such that UTC and the rotation gradually get out of sync with one another. Therefore, in order to keep the discrepancy between UTC and the earth's actual motion under 0.9 seconds, corrections of 1 second, called leap seconds, are periodically introduced into UTC.

However, leap seconds are not used in GPS Time. It is a continuous timescale. Nevertheless, there was a moment when GPS Time was identical to UTC. It was midnight, January 6, 1980. Since then, many leap seconds have been added to UTC, but none have been added to GPS Time. So, even though their rates are virtually identical, the numbers expressing a particular instant in GPS Time are different by some seconds from the numbers expressing the same instant in UTC. For example, GPS time was 16 seconds ahead of UTC on July 1, 2012 and 18 seconds ahead of UTC on September 11, 2020.

📡 NAV Message & Future Corrections

Information in subframe 4 of the NAV message includes the relationship between GPS time and UTC, and it also notes future scheduled leap seconds. In this area, subframe 4 can accommodate 8 bits, 255 leap seconds, which should suffice until about 2330. The NAV message also contains information the receiver needs to come close to correlating its clock with that of the clock on the satellite. But because the time relationships in GPS are changing constantly, they can only be partially defined in these subframes. It takes more than a portion of the NAV message to define those relationships to the necessary degree of accuracy.

🛠️ Satellite Clock Corrections

The Control Segment keeps track of time and uploads clock corrections to the satellites. However, the oscillators (clocks) in satellites have a tendency to drift. They're not perfect timekeepers, and they're affected by several things, moving in and out of the shadow of the Earth, gravitational changes, and so on. These effects cause the satellite's clocks to not oscillate perfectly. Their rate can vary. Nevertheless, the Control Segment does **not** constantly tweak the clocks in the satellites to keep them perfectly on GPS time. They let them wander a little bit but keep them within the limit of one millisecond of GPS Time. They find that by doing this, it keeps the clock in the satellites healthy longer. It extends their life. You will note that the first things to go on a satellite are its clocks. This is one reason that many of them have multiple oscillators.

So, the message is that the Control Segment introduces a clock correction into the Navigation Message that gives your receiver a way to correlate the drift of the oscillator(s) in the satellite with GPS time. This is one way to keep the time standard within limits.

📝 Section Summary

Key Takeaways

This section explored GPS Time, how it differs from Coordinated Universal Time (UTC), and how the NAV message communicates time data and clock corrections. We learned about the role of the Control Segment in tracking clock drift, the use of constants A0 and A1 to relate GPS time to UTC, and the impact of leap seconds.

⏱️ GPS vs UTC

GPS Time is continuous and does not account for leap seconds, unlike UTC.

🛰️ Clock Corrections

The Control Segment uploads time corrections to keep satellite clocks within milliseconds of GPS Time.

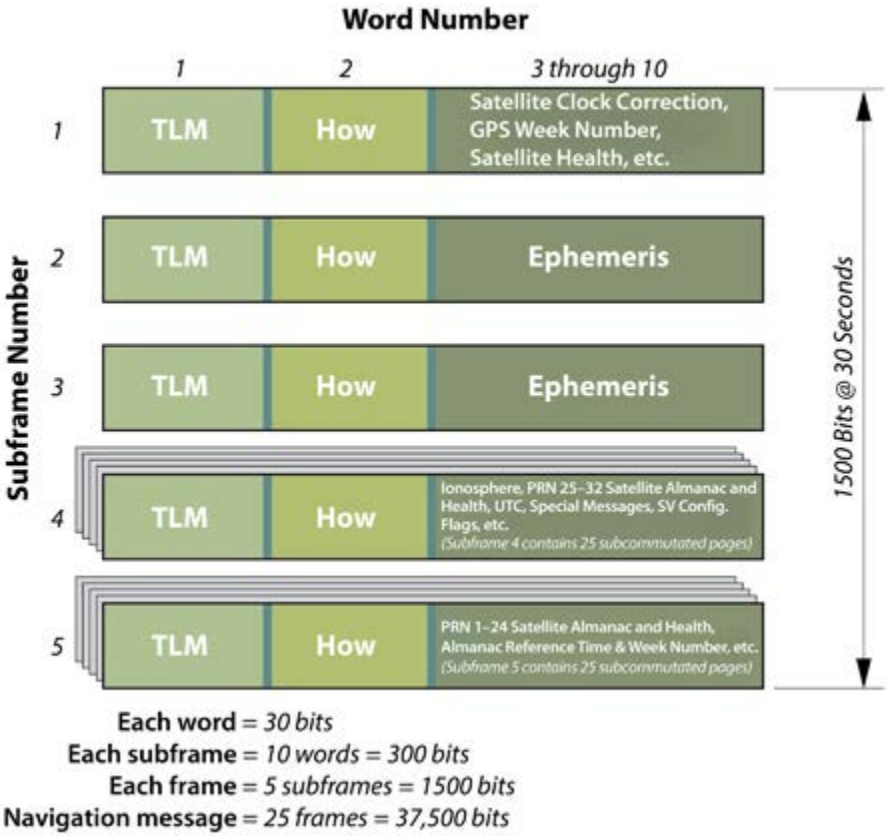
CONTENT

📚 Ephemerides



The Broadcast Ephemeris

GISC 4315 – Satellite Positioning



Relationships of the elements defining the ephemerides

📘 Understanding Broadcast Ephemeris

Another example of time-sensitive information is found in subframes 2 and 3 of the NAV Message. They contain information about the position of the satellite, with respect to time. This is called the satellite's ephemeris. The ephemeris that each satellite broadcasts to the receivers provides information about its position relative to the earth. In other words, these are the coordinates of the satellite in space at the instant the Control Segment uploads the ephemerides to the Navigation

Message for each individual satellite. Most particularly, it provides information about the position of the satellite antenna's phase center.

🛰️ Orbital Elements

The ephemeris is given in a right ascension (RA) system of coordinates. There are six orbital elements; among them are the size of the orbit, that is its semimajor axis, a, and its shape, that is the eccentricity, e. However, the orientation of the orbital plane in space is defined by other things, specifically the right ascension of its ascending node, Ω, and the inclination of its plane, i. These parameters along with the argument of the perigee, ω, and the description of the position of the satellite on the orbit, known as the true anomaly, provides all the information the user's computer needs to calculate earth-centered, earth-fixed, World Geodetic System 1984, GPS Week 1762 (WGS84 [G1762]) coordinates of the satellite at any moment.

📉 Accuracy and Perturbations

The broadcast ephemeris, however, is far from perfect. It is expressed in parameters named for the seventeenth century German astronomer Johann Kepler. The ephemerides may appear Keplerian, but in this case, the orbits of the GPS satellites deviate from nice smooth elliptical paths because they are unavoidably perturbed by gravitational and other forces. They are affected by lots of biases. Therefore, the orbits change with time so their actual paths through space are found in the result of least-squares, curve-fitting analysis. The accuracies of both the broadcast clock correction and the broadcast ephemeris deteriorate with time, so the ephemeris needs to be updated periodically to keep it within the accuracy required to get good positions on the Earth's surface.

🕒 Issue of Data Ephemeris (IODE)

As a result, one of the most important parts of this portion of the NAV message is called IODE. IODE is an acronym that stands for Issue of Data Ephemeris, sort of a time stamp on the ephemeris that the receiver gets from the navigation message, and it appears in both subframes 2 and 3. Remember, the satellites are the control points from which the distances must be derived.

📝 Section Summary

Key Takeaways

This section introduced the Broadcast Ephemeris and its role in satellite positioning. We learned that it contains essential orbital parameters required by GPS receivers to compute the satellite's location in space. These parameters are transmitted through the NAV message and are periodically updated by the Control Segment to ensure reliability.

🛰️ Orbital Parameters

Provide satellite trajectory data used by GPS receivers to calculate positions accurately.

🔄 Updated Regularly

Refreshed approximately every two hours to reflect any changes in the satellite's orbit.

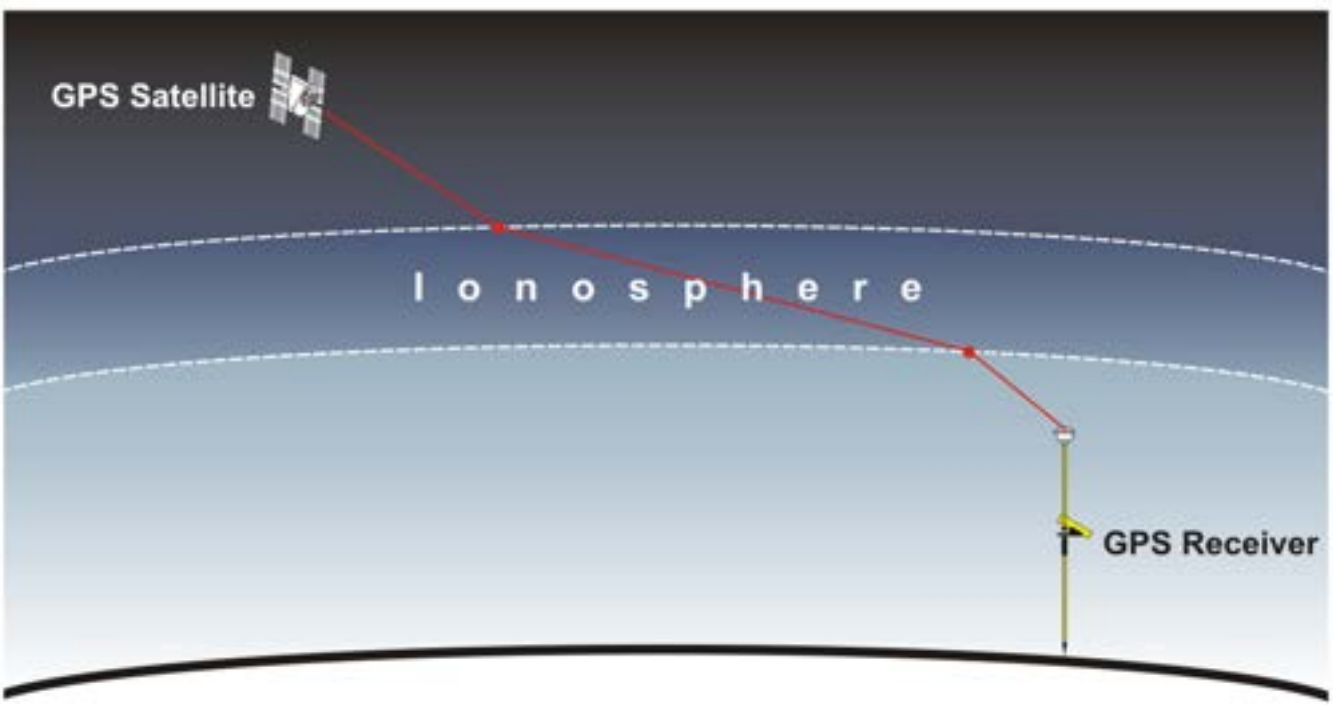
📚 Atmospheric Correction

Atmospheric Correction

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🛰️ Ionospheric Delay and Correction

Subframe 4 addresses atmospheric correction. As with subframe 1, the data there offer only a partial solution to a problem. The Control Segment's monitoring stations find the apparent delay of a GPS signal caused by its trip through the ionosphere through an analysis of the different propagation rates of the carrier frequencies broadcast by GPS satellites, L1, L2 and L5. These frequencies and the effects of the atmosphere on the GPS signal will be discussed later. For now, it is sufficient to say that a single-frequency receiver depends on the ionospheric correction in subframe 4 of the NAV message to help remove part of the error introduced by the atmosphere, whereas a receiver that can track more than one carrier has a bit of an advantage by comparing the differences in the frequency dependent propagation rates.



Distortion in the GPS signal caused by the ionosphere

📡 Atmospheric Effects on GPS Signals

As the illustration indicates, the signal from the GPS satellite appears to slow down as it goes through the ionosphere. The atmospheric correction has been uploaded to the satellite from the Control Segment. The Control Segment can quantify the slowing because the tracking station is at a known point.

The ionosphere is dispersive. Since the GPS satellite broadcasts signals at different frequencies, it is important to understand that these frequencies are affected differently as they pass through the ionosphere. Their propagation rates are slowed differently in the ionosphere. They're not slowed at exactly the same rate. We'll talk more about that in a little bit.

📶 Single vs. Multi-Frequency Receivers

For the moment, it's sufficient to say that a single frequency receiver, and there are such things, of course, a single frequency GPS receiver is somewhat handicapped by the fact that it cannot model the ionosphere. If the receiver has only one frequency to work with, it can't take advantage of the fact that it could quantify how much the signals have been affected by the ionosphere for itself, rather than relying exclusively on the Atmospheric Correction in the Navigation Message. A single frequency GPS receiver must rely almost completely on the Navigation Message Atmospheric Correction. This is a bit of a handicap. The atmosphere over Kwajalien in the Pacific might be quite different than the atmosphere above a single frequency GPS receiver in the continental US. This is one reason that the Navigation Message is not as perfect as we would like it to be.

📝 Section Summary

Key Takeaways

This section covered the importance of atmospheric correction in GPS. It explained how ionospheric delay affects signal accuracy, how control segments calculate corrections, and the limitations of single-frequency receivers versus multi-frequency ones.

📍 Ionospheric Delay

🔄 Role of the Control Segment

Slows GPS signal; varies with frequency; calculated by comparing signal propagation at L1, L2, L5.	Uploads atmospheric correction data so single-frequency receivers can partially correct for ionospheric effects.

CONTENT

📚 The Almanac, Time to First Fix and Satellite Health



📘 The Almanac

As mentioned briefly earlier, the information in the almanac in subframes 4 and 5 tells the receiver where to find all the GPS satellites. Subframe 4 contains the almanac data for satellites with pseudorandom noise (PRN) numbers from 25 through 32, and subframe 5 contains almanac data for satellites with PRN numbers from 1 through 24. The Control Segment generates and uploads a new almanac every day to each satellite, at least every six days, but typically on a daily basis.

While a GPS receiver must collect a complete ephemeris from each individual GPS satellite to know its correct orbital position, it is convenient for a receiver to be able to have some information about where all the satellites in the constellation are by reading the almanac from just one of them. The almanacs are much smaller than the ephemerides because they contain coarse orbital parameters and incomplete ephemerides, but they are still accurate enough for a receiver to generate a list of visible satellites at power-up. They, along with a stored position and time, allow a receiver to find its first satellite.

⏱️ Time to First Fix (TTFF)

If a receiver has been in operation recently and has some leftover almanac and position data in its non-volatile memory from its last observations, it can begin its search with what is known as a warm start. A warm start is also known as a normal start. In this condition, the receiver might begin by knowing the time within about 20 seconds and its position within 100 km or so, and this approximate information helps the receiver estimate the range to satellites. For example, it will be able to restrict its search for satellites to those likely above its horizon, rather than wasting time on those below it. Limiting the range of the search decreases the time to first fix (TTFF). It can be as short as 30 seconds with a warm start.

On the other hand, if a receiver has no previous almanac or ephemeris data in its memory, it will have to perform a cold start, also known as a factory start. Without previous data to guide it, the receiver in a cold start must search for all the satellites without knowledge of its own position, velocity, or the time. When it does finally manage to acquire the signal from one, it gets some help

and can begin to download an almanac. That almanac data will contain information about the approximate location of all the other satellites. The period needed to receive the full information is 12.5 minutes.

The time to first fix (TTFF) is longest at a cold start, less at warm, and least at hot. A receiver that has a current almanac, a current ephemeris, time and position can have a hot start. A hot start can take from 1/2 to 20 seconds.

Estimating how long each type of start will actually take is difficult; overhead obstructions interrupting the signal from the satellites, the GPS signals reflecting from nearby structures, etc., can delay the loading of the ephemeris necessary to lock onto the satellite's signals.

🛰️ Satellite Health

Subframe 1 contains information about the health of the satellite the receiver is tracking when it receives the NAV message and allows it to determine if the satellite is operating within normal parameters. Subframes 4 and 5 include health data all of the satellites, data that is periodically uploaded by the Control Segment. These subframes inform users of any satellite malfunctions before they try to use a particular signal. The codes in these bits may convey a variety of conditions. They may tell the receiver that all signals from the satellite are good and reliable, or that the receiver should not currently use the satellite because there may be tracking problems or other difficulties. They may even tell the receiver that the satellite will be out of commission in the future; perhaps it will be undergoing a scheduled orbit correction. GPS satellites' health is affected by a wide variety of breakdowns, particularly clock trouble. That is one reason they carry multiple clocks.

The almanac in subframes four and five tell the receiver where to find the satellites. The little truncated ephemerides in the almanac help it do that. There is also information about the satellite health in the Navigation Message. GPS satellites operate in a rather hostile environment, and if they're having trouble, i.e., if some of the clocks are not operating within acceptable parameters, the health data allows the receiver to have that information. For example, a receiver might receive information that a satellite is unhealthy and shouldn't be used in the positioning.

📝 Section Summary

Key Takeaways

This section explained how the almanac data and health flags are used to help the receiver determine the location and status of satellites. It also discussed how Time to First Fix (TTFF) varies by start type and what factors affect that timing.

📍 Almanac Usage

Provides coarse orbital parameters allowing quicker satellite acquisition.

⏱️ TTFF Types

Cold, warm, and hot starts define how quickly the receiver locks onto satellites.

🛰️ Satellite Health Info

Health flags alert receivers about issues or maintenance in satellites.

CONTENT

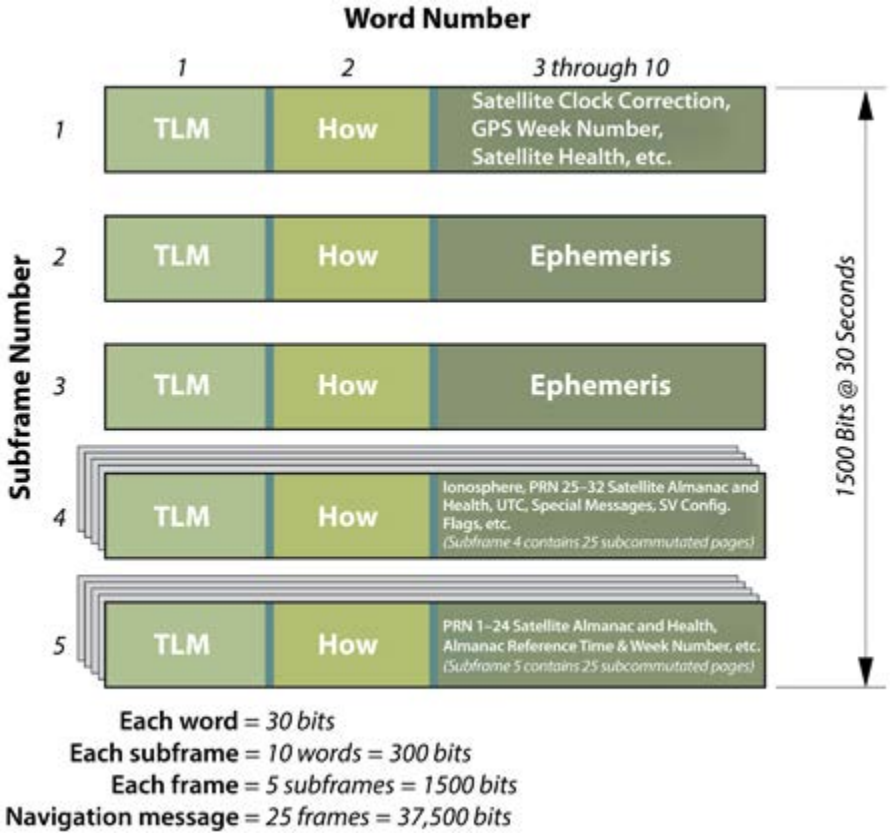
📚 GPS Navigation Message Structure

GPS Navigation Message Structure

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📊 The Navigation Message Structure

The GPS Navigation Message is a carefully structured data stream that provides all the information needed for positioning. It consists of **5 subframes**, each containing **10 words of 30 bits each**, for a total of **1,500 bits transmitted every 30 seconds**.



Navigation Message structure showing 5 subframes with TLM, HOW, and data content

Navigation Message: 1500 bits transmitted every 30 seconds

📐 Message Structure

- **Each word:** 30 bits
- **Each subframe:** 10 words = 300 bits
- **Each frame:** 5 subframes = 1,500 bits
- **Complete navigation message:** 25 frames = 37,500 bits

🗂️ Subframe Content Breakdown

Subframe	Word 1	Word 2	Words 3-10
1	TLM	HOW	Clock correction, GPS week, Satellite Health, etc.
2	TLM	HOW	Ephemeris data
3	TLM	HOW	Ephemeris data
4	TLM	HOW	Ionosphere, PRN 25-32 Satellite Almanac and Health, UTC, etc. (25 subcommutated pages)
5	TLM	HOW	PRN 1-24 Satellite Almanac and Health, etc. (25 subcommutated pages)

⏰ GPS Time System

GPS time restarts each **Sunday at midnight (0:00 o'clock)**. The navigation message contains the time since the last restart of GPS time on the previous Sunday, providing precise timing for all positioning calculations.

🔧 Telemetry (TLM) and Handover (HOW) Words

📡 Telemetry Word (TLM)

The **first word in each subframe**, generated by the satellite itself:

- **Upload status** from Control Segment
- **Ephemeris data age** information
- **8-bit preamble** (10001011) for synchronization
- **Subframe identification** for receivers

🔄 Handover Word (HOW)

The **second word in each subframe**, providing timing information:

- **Time of Week (TOW)** data
- **Subframe number** identification
- **Z count** (1.5 second epoch timing)
- **Code generation status** information

🎯 Key Function

The TLM and HOW words are **generated by the satellite itself**, unlike most other navigation message content. The HOW's Z count tells the receiver exactly where the satellite stands in the generation of positioning codes, which is crucial for proper signal synchronization.

📋 Data Organization and Transmission

📊 `Subframes 1-3`

Primary data: Clock corrections, satellite health, and detailed ephemeris for precise orbital positioning.

🌐 `Subframes 4-5`

Constellation data: Almanac information for all satellites, ionospheric corrections, and UTC parameters.

Each of the five subframes begins with the same two words: the **telemetry word (TLM)** and the **handover word (HOW)**. These words help the receiver reliably find the beginning of each subframe and understand the satellite's current status in code generation.

📡 Synchronization Process

> The TLM's constant 8-bit preamble (10001011) provides a reliable marker for receivers to identify the start of each subframe, while the HOW provides precise timing information needed for code synchronization and positioning calculations.

📝 Section Summary

Key Takeaways

This section explained the structure and function of the GPS Navigation Message, including how the 5 subframes organize different types of positioning data and the critical role of TLM and HOW words in signal synchronization and timing.

📊 Message Structure

5 subframes, each with 10 words of 30 bits, transmitted every 30 seconds.

🔧 TLM & HOW

Critical words for synchronization and timing information.

📡 Data Organization

Ephemeris in subframes 2-3, almanac and corrections in 4-5.

📚 The P and C/A Codes



📹 Instructor Video Lecture

Understanding PRN, P Code, and C/A Code

Embedded content: Embedded content link

📋 Video Topics Covered

- What are PRN codes?
- Generation and modulation of P code
- Structure and role of the C/A code
- Anti-spoofing measures and military access
- SPS and PPS services

> **💡 Study Tip**
>
> Focus on how phase modulation distinguishes the code signals and how P(Y) and C/A codes affect GPS accuracy for different users.

🔍 Introduction to PRN Codes

The P and C/A codes are complicated, so complicated that they appear to be noise at first. In fact, they are known as pseudorandom noise, or PRN, codes. Actually, they are carefully designed. They have to be. They must be capable of repetition and replication. However, unlike the Navigation Message, the P and C/A codes

are not vehicles for broadcasting information that has been uploaded by the Control Segment. They carry the raw data from which GPS receivers derive their time and distance measurements.

📡 The P Code (Precise Code)

P Code @ 10.23 MBPS is Fsub0=10.23 MHz, see text surrounding image

The P code is called the Precise code. It is a particular series of ones and zeroes generated at a rate of 10.23 million bits per second. It is carried on both L1 and L2, and it is very long, 37 weeks ($2x10^{14}$ bits in code). Each GPS satellite is assigned a part of the P code all its own, and then repeats its portion every 7 days. This assignment of one particular week of the 37-week-long P code to each satellite helps a GPS receiver distinguish one satellite's transmission from another. For example, if a satellite is broadcasting the fourteenth week of the P code, it must be Space Vehicle 14 (SV 14). The encrypted P code is called the P(Y) code.

> 🔐 **Note:** A flag in subframe 4 of the NAV message tells a receiver when the P code is encrypted into the P(Y) code. This Antispoofing (AS) has been active since 1994, using W-Code to generate the more secure Y-Code. Civilian receivers cannot use the P(Y) code directly.

Commercial GPS receiver manufacturers are not authorized to use the P(Y) code directly. Therefore, most have developed proprietary techniques both for carrier wave and pseudorange measurements on L2 indirectly. There is now a civilian code on L2 (L2C). There is also now a military code known as the M code. We will discuss both of these later.

The Navigation Message can be thought of as the NAV Code, but there are others. Positioning, one of the primary objectives of GPS, is really the office of the P-Code, the C/A Code and some others that are newer than these legacy codes. The P code is the Precise code, The C/A code is the Civilian Access code. They're modulated onto carrier waves.

> 📻 **Analogy:** Like how FM/AM radios carry voices on a frequency, GPS signals use phase modulation to carry positioning data. You don't hear the 760 MHz; you hear what's modulated onto it.

In GPS, the modulation is done differently. Phase modulation is used. The image here is intending to show that. The P code in the image is a sine wave. It has particularly sharp peaks, but it is still a sine wave. Please note that up at the top of the blue line, there's a sequence of 1's and 0's. These indicate code chips of a binary code.

Please notice the way these code chips transition from 1 to 0 and back to 1. At those instances the sine wave, the blue line, does not go all the way to the top or bottom. When there is a transition from a 1 to a 0 or from a 0 to a 1, the blue line stops in the middle and reverses direction. However, when the code chip

just goes from a 1 to a 1, or from a 0 to a 0, there is no interruption of the normal sine wave path. In those cases, the blue line does go all the way up to the top, and it goes all the way down to the bottom.

> 🌀 **Technique Highlight:** This approach is called **phase modulation**. It allows information to be encoded in changes to the phase of the carrier wave, enabling robust and accurate signal interpretation.

🌐 The C/A Code and Positioning Services

The C/A code is also a particular series of ones and zeroes, but the rate at which it is generated is 10 times slower than the P(Y) code. The C/A code rate is 1.023 million bits per second. Here, satellite identification is quite straightforward. Not only does each GPS satellite broadcast its own completely unique 1023 bit C/A code, it repeats its C/A code every millisecond. The legacy C/A code is broadcast on L1 only. It used to be the only civilian GPS code, but no longer; it has been joined by a new civilian signal known as L2C that is carried on L2.

diagram showing C/A Code @ 1.023 MBPS is F0/10 = 10.23 MHz/10, see surrounding text

📶 The C/A Code

The C/A Code or Civilian Acquisition or Access Code is generated 10 times slower than the P-Code. The GPS fundamental clock rate is 10.23 megahertz, but C/A Code is generated at 1.023 megabits per second.

The C/A Code is modulated onto the carrier by phase modulation, too.

> 🌀 **Modulation Insight:** The image shows a green line, a sine wave that only transitions when it reaches the center line and reverses direction. That is a phase shift, and then the code goes from the 1 to the 0 or from the 0 to the 1.

📡 SPS and PPS

Still, the C/A code is the vehicle for the Standard Positioning Service, SPS, which is used for most civilian surveying applications. The P(Y) code on the other hand provides the same service for the precise positioning servicer, PPS. The idea of SPS and PPS was developed by the Department of Defense many years ago. SPS was designed to provide a minimum level of positioning capability considered consistent with national security, ±100m, 95% of the time, when intentionally degraded through Selective Availability (SA).

> ⚠️ **Selective Availability (SA):** Instituted in 1989, SA intentionally degraded clock accuracy because C/A point positioning was more accurate than intended — often ±20 to ±40 meters. SA was turned off on May 2, 2000 by presidential order.

Actually, Selective Availability never did hinder the surveying applications of GPS. We will delve into the reason a bit later. In any case, satellite clock errors, the unintentional kind, still contribute error to GPS positioning.

PPS is designed for higher positioning accuracy and was originally available only to users authorized by the Department of Defense; that has changed somewhat, more about that later in this chapter. It used to be that the P(Y) code was the only military code. That is no longer the case. It has been joined by a new military signal called the M-code.

📝 Section Summary

Key Takeaways

This section explained the **C/A Code** and its role in the **Standard Positioning Service (SPS)** for civilian GPS applications. It also contrasted this with the **P(Y) Code** used in the **Precise Positioning Service (PPS)** for military-grade accuracy. Key concepts like Selective Availability (SA), phase modulation, and the unique identification of satellites were discussed.

📶 C/A Code Rate

Runs at 1.023 Mbps — 10 times slower than the P Code. Broadcasts on L1 and repeats every millisecond.

🛰️ SPS vs PPS

SPS uses the C/A code; PPS uses the encrypted P(Y) code. PPS is more precise and was originally for DoD use only.

🛡️ Selective Availability

Deliberately degraded GPS accuracy from 1989–2000 to limit civilian precision. Removed by presidential order.

🔄 Modulation Type

C/A and P Codes are modulated using phase modulation onto carrier signals like L1 and L2.

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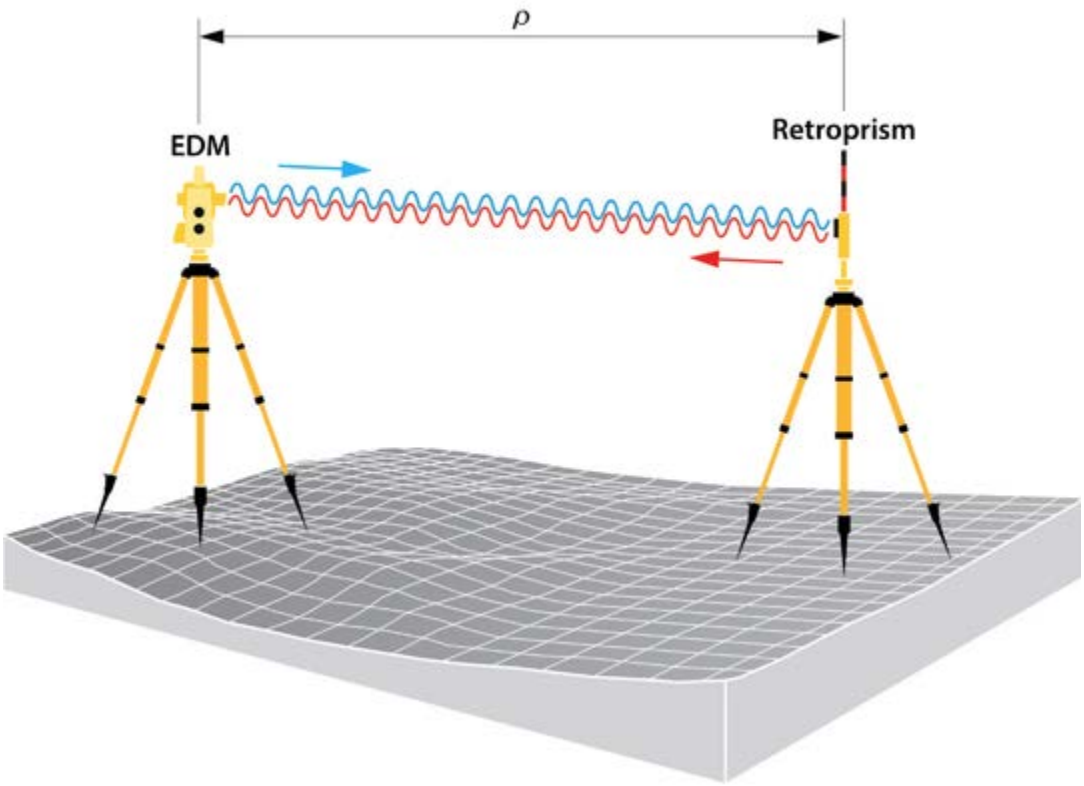
CONTENT

📚 GPS Ranging

EDM vs GPS Ranging Methods

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📏 EDM Ranging: Two-Way System



EDM sending signal to reflect off retroprism and return to EDM, distance LaTeX: \rho

EDM Ranging: Signal reflects off retroprism and returns to instrument

Distance measurement in modern surveying is done **electronically**. Distance is measured as a function of the **speed of light**, an electromagnetic signal of stable frequency and elapsed time.

🔄 Two-Way Operation

- **Signal bounces off retroprism** and returns
- **EDM is both transmitter and receiver**
- **Only one oscillator needed** at point of origin
- **Round-trip timing** simplifies calculations

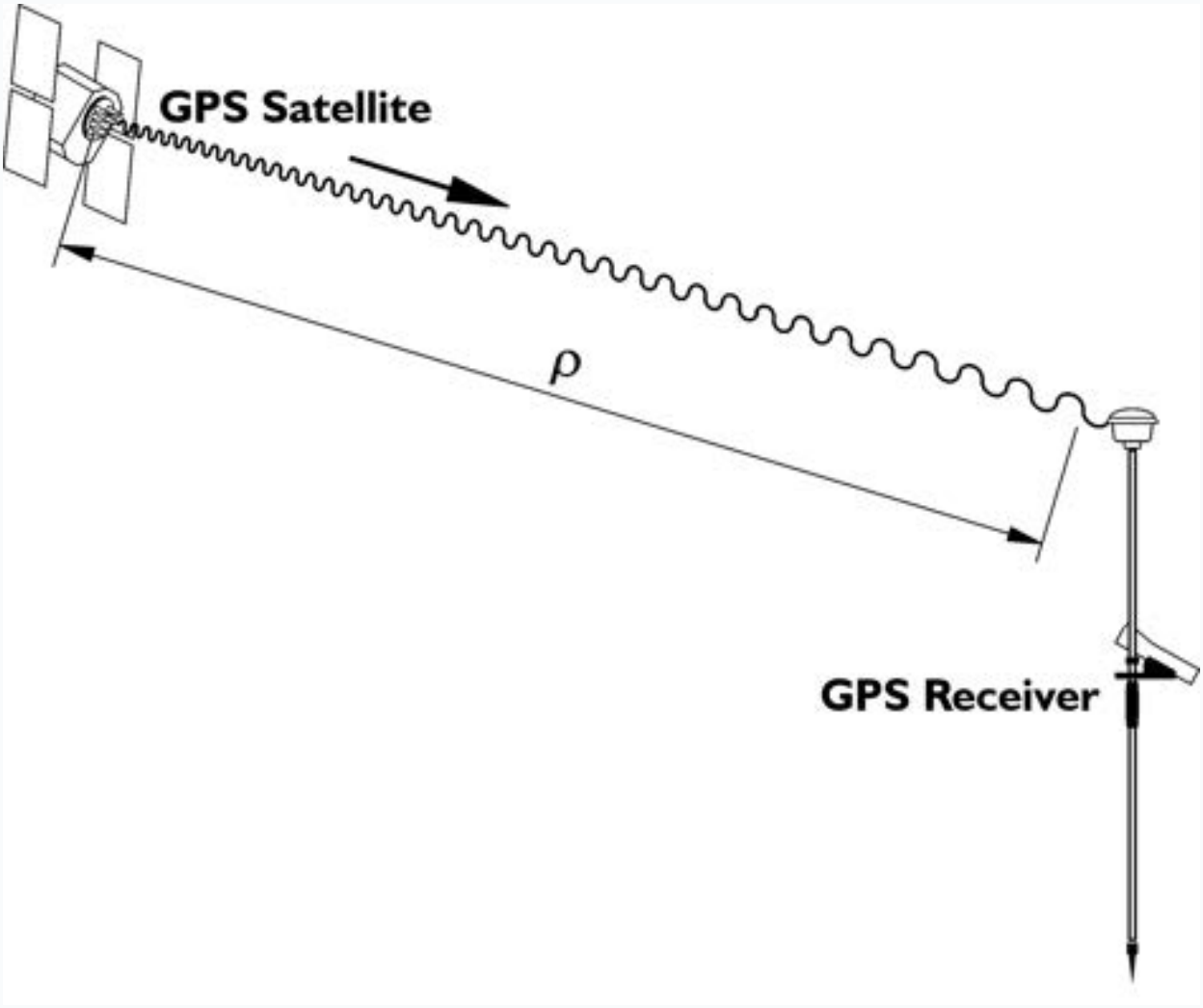
⚡ Calculation Method

- **Measure elapsed time** between transmission and reception
- **Take half the travel time** (round trip)
- **Multiply by speed of light**
- **Formula:** Distance = (Elapsed Time ÷ 2) × Speed of Light

🎯 Key Advantage

EDM uses frequencies generated within the device to determine elapsed travel time because the signal returns to its starting point, allowing for precise distance calculations with a single, synchronized timing system.

📡 GPS Ranging: One-Way System



One-Way Ranging: GPS Satellite transmitting a message to a GPS receiver over distance LaTeX: \rho

GPS Ranging: One-way signal from satellite to receiver

The **one-way ranging** used in GPS is more complicated. The broadcast signals from the satellites are **collected by the receiver, not reflected**. Nevertheless, the same measurement concept is used.

⚡ Basic Principle

The full time elapsed between the instant a GPS signal leaves a satellite and arrives at a receiver, **multiplied by the speed of light**, is the distance between them.

⚠️ Major Challenge

Unlike EDM, a GPS signal **cannot be analyzed at its point of origin**. This requires:

- **Two separate clocks** (satellite and receiver)
- **Perfect synchronization** (impossible)
- **Mathematical solutions** for timing errors

$$\rho = \Delta t \times c$$

📐 Mathematical Formula

In the calculation of range (ρ) from GPS receiver to satellite, the basis is the multiplication of time elapsed between signal transmission and reception (Δt) by the speed of light (c).

⏰ Clock Synchronization Challenge

⚠️ Critical Timing Accuracy

1 microsecond error = 299.79 meters range error

(1 millionth of a second timing discrepancy)

🎯 The Problem

- **Perfect synchronization is impossible**
- **Satellite atomic clocks** vs. receiver quartz clocks
- **Microsecond discrepancies** create meter-level errors
- **Unacceptable for surveying work**

🧮 The Solution

- **Mathematical approach** to timing errors
- **Time becomes a fourth unknown**
- **Four satellites minimum** required
- **Simultaneous equation solving**

🔬 Engineering Solution

Since perfect synchronization is not achievable, GPS addresses this mathematically by **solving for time as an unknown variable**, which is why GPS requires at least four satellite measurements instead of the three used in traditional trilateration.

📝 Section Summary

Key Takeaways

This section compared EDM two-way ranging with GPS one-way ranging, highlighting the fundamental challenge of clock synchronization in GPS and how it requires mathematical solutions to achieve accurate positioning.

📏 EDM Two-Way

Simple, reliable with one clock and retroprism reflection.

📡 GPS One-Way

Complex, requires two clocks and mathematical time solutions.

⏰ Timing Precision

Microsecond accuracy critical - 1μs error = 300m range error.

CONTENT

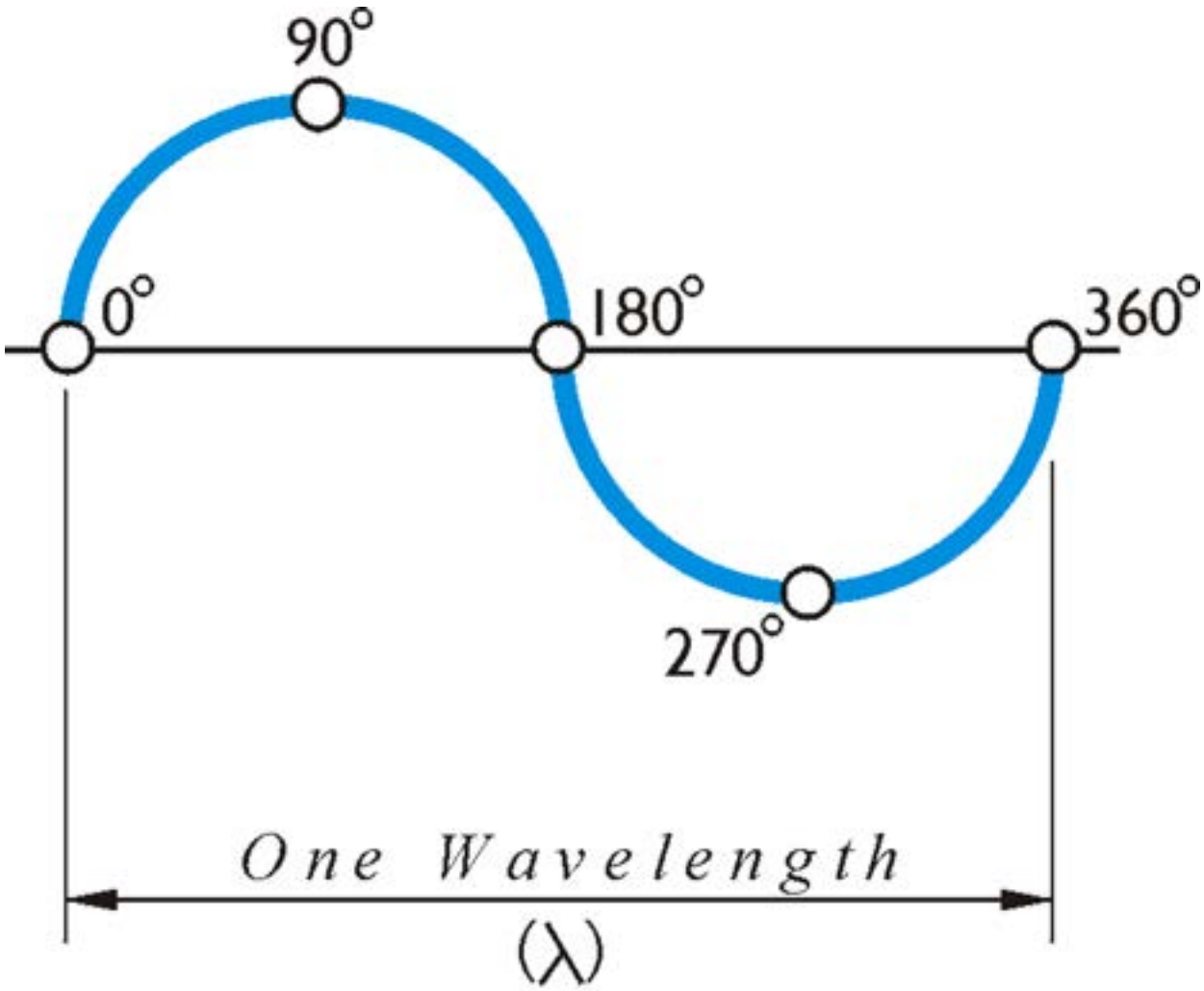
📚 Phase Angles and Shifts



Wavelength and Phase Shift in GPS

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📊 Wavelength and Phase Angles



0° to 360° = One Wavelength

One Wavelength: crosses x axis at 0, 180, 360 degrees, peak at 90, trough at 270

Single wavelength of 1 Hertz showing phase angles

Here's a sine wave illustrating **a single wavelength of 1 Hertz**, that is a wavelength that takes 1 second to cycle through 360 degrees. The **0°, 90°, 180°, 270°, and 360°** are known as **phase angles**.

📐 Phase Angles

- **0° and 360°:** Starting/ending points

🎯 Importance

- **Critical for GPS modulation**

- **90°:** Peak of the wave
- **180°:** Zero crossing (descending)
- **270°:** Trough of the wave

- **Method of attaching codes** to carriers
- **Phase comparison** enables distance measurement
- **Foundation of carrier phase ranging**

> ⚡ **Key Concept**
>
> Phase angles are important to the **modulation of the carrier by phase** and that is the method of attaching the codes to the GPS carriers.

⚙️ Oscillators and Frequency Standards

The time measurement devices used in both EDM and GPS measurements are **clocks only in the most general sense**. They are more correctly called **oscillators**, or **frequency standards**.

🔄 How Oscillators Work

Rather than producing a steady series of ticks, they keep time by:

- **Chopping continuous electromagnetic energy**
- **Creating extremely regular intervals**
- **Producing steady series of wavelengths**
- **Forming the foundation** of modulated carriers

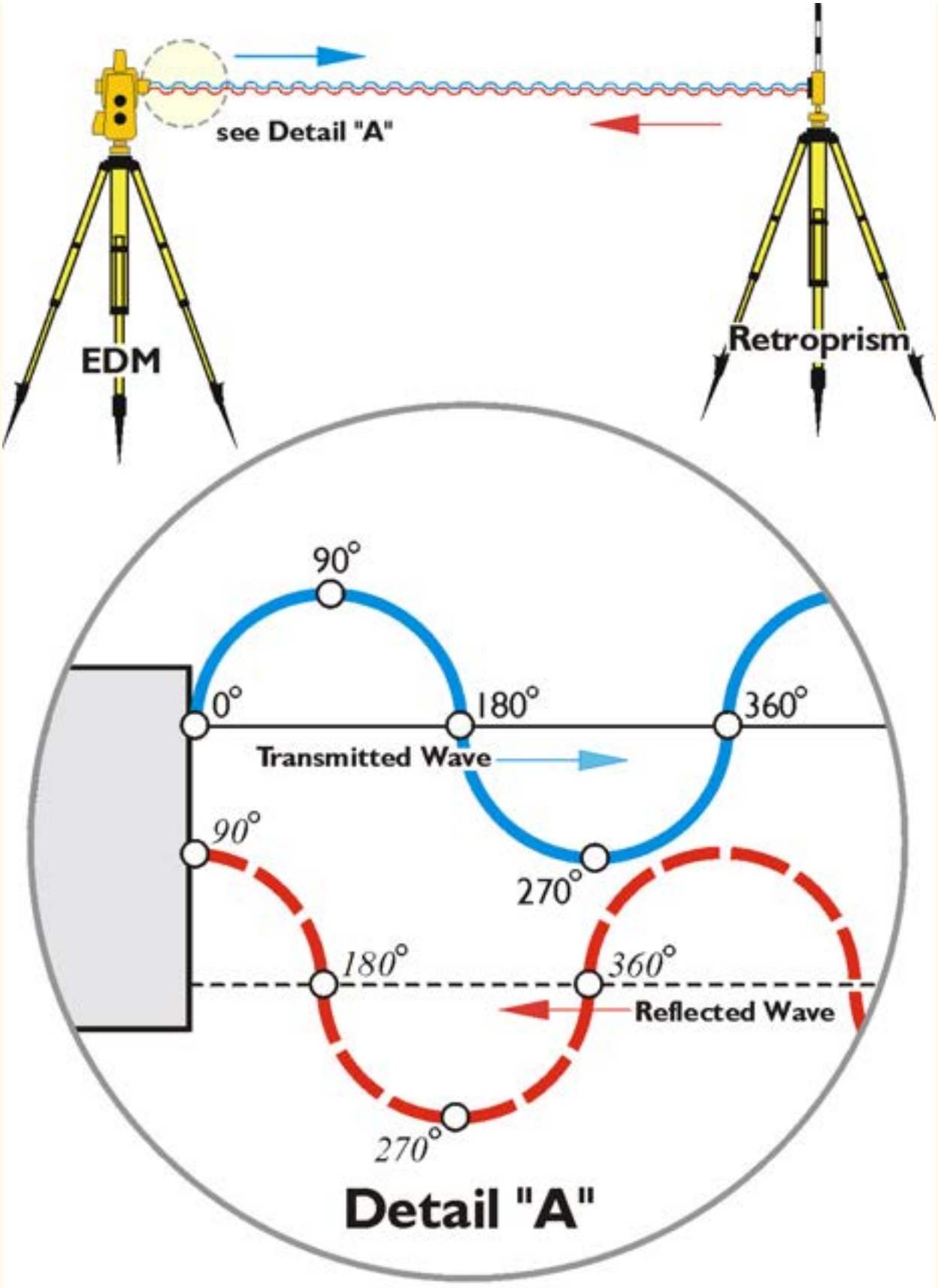
📏 Stable Operation

As long as the oscillator's operation rate is **very stable**:

- **Wavelength length is constant**
- **Elapsed time between wavelengths is same**
- **Phase angles occur at definite distances**
- **Consistent and predictable timing**

> 🎯 **Foundation of Measurements**
>
> The result is a steady series of wavelengths and the foundation of the modulated carrier. This stability ensures that both the length and elapsed time between the beginning and end of every wavelength of the modulation will be the same.

📡 Phase Shift in EDM Measurements



Phase Shift - One Quarter of a Wavelength (90°)

Phase Shift: signal from EDM reflected off of Retroprism showing one quarter phase shift in the wave

Phase Shift: EDM signal reflected off retroprism showing quarter-wavelength phase shift

The image shows the EDM sending out the transmitted wave in **blue**, with phase angles indicated. The signal goes to the retroprism and returns. When it returns, shown in the **dashed red line**, notice the phase angles.

🔍 Phase Comparison Process

1️⃣ Transmission	2️⃣ Reflection	3️⃣ Comparison
EDM sends out modulated carrier wave	Signal reflects off retroprism and returns	Compare returned wave with internal replica

It is clear that the return signal does **not come back exactly in phase** with the transmitted wave. The key element is that the EDM generates another wave that is exactly the same as the wave it transmitted but keeps this additional wave at home for comparison.

📐 Phase Shift Determination

By comparing the returned wave with the exact replica of the transmitted wave, the EDM can determine how much the returned wave is **phase shifted** (out of phase) with the original transmitted wave. This phase shift represents the **fractional part of the measurement**.

🧱 Phase Lock and Distance Calculation

🔒 Phase Relationships

In Phase (Coherent/Phase Locked):

Two waves reach exactly the same phase angle at exactly the same time

Out of Phase (Phase Shifted):

Two waves reach a phase angle at different times

📏 Example Measurement

In the diagram, the returned wave (dashed red) is **out of phase by one-quarter wavelength** compared to the transmitted wave (solid blue).

This phase shift allows precise measurement of the fractional wavelength portion.

📊 Distance Formula

$$\rho = (N\lambda + d) / 2$$

ρ	Nλ	d
Distance	Full wavelengths	Fractional part

Where **N** = the number of full wavelengths the modulated carrier has completed, and **d** = the fractional part of a wavelength that completes the doubled distance. In the example shown, d is three-quarters of a wavelength because it lacks its last quarter.

🔬 Universal Principle

Both EDM and GPS ranging use the method represented in this illustration. Since all measurements will not neatly fit complete wavelengths, the EDM finds the fractional part of its measurement electronically through phase comparison.

📡 GPS Carrier Phase Ranging

In GPS, the measurement of the difference in the phase of the incoming signal and the phase of the internal oscillator in the receiver reveals the small distance at the end of a range. This process is called **carrier phase ranging**.

📶 GPS Application

- **Same phase comparison principle** as EDM
- **Incoming satellite signal** vs. receiver oscillator
- **Measures fractional wavelength** at range end
- **Extremely precise** distance measurement

🎯 Observable

- **Carrier wave itself** is the observable
- **Phase difference measurement**
- **High precision positioning** capability
- **Surveying-grade accuracy**

🚀 Advanced GPS Technique

As the name implies, the **observable** in carrier phase ranging is the carrier wave itself. This technique provides much higher precision than code-based ranging and is essential for surveying and precision applications.

📝 Section Summary

Key Takeaways

This section explained the fundamental concepts of wavelength, phase angles, and phase shift measurements that enable precise distance measurement in both EDM and GPS systems. Understanding phase relationships is crucial for GPS carrier phase ranging applications.

📊 Phase Angles

0°, 90°, 180°, 270°, 360° define points on carrier waves for modulation.

⚙️ Oscillators

Create steady wavelengths by chopping electromagnetic energy at regular intervals.

📡 Phase Ranging

Both EDM and GPS use phase comparison for precise fractional wavelength measurement.

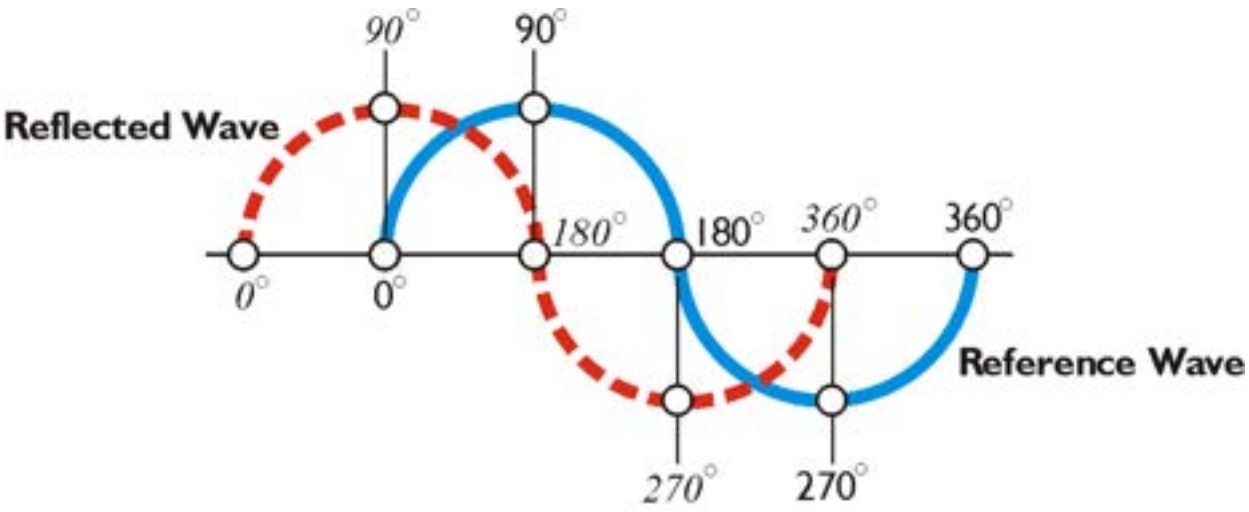
CONTENT

📚 The Integer Ambiguity Problem

Cycle Ambiguity and Integer Problem

GISC 4315 – Satellite Positioning

📊 Phase Shift Measurement Review



Reference and Reflected Waves

Reference and Reflected Waves showing one quarter phase shift

Phase Shift: Reference wave (blue) vs. reflected wave (dashed red) showing quarter-wavelength shift

Both EDM and GPS ranging use the method represented in this illustration. In GPS, the measurement of the difference in the phase of the incoming signal and the phase of the internal oscillator in the receiver reveals the **small distance at the end of a range**.

🔍 What Phase Shift Reveals

- **Fractional wavelength portion** only
- **Small component** of total distance

❓ What Remains Unknown

- **Number of full wavelengths (N)**
- **Complete cycles** between points

- **High precision measurement** of remainder
- **Out-of-phase comparison** with reference

- **Integer portion** of the measurement
- **Cycle ambiguity problem**

🎯 Fundamental Limitation

By comparing the phase of the signal returned from the reflector with the reference wave, an EDM can measure how much the two are out of phase. However, this measurement can only calculate a **small part of the overall distance** - the fractional wavelength portion.

🔢 The Cycle Ambiguity Problem

❓ The Big Unknown: N

Phase measurement only reveals the **fractional wavelength**. The number of **full wavelengths (N)** between transmitter and receiver remains unknown.

$$\text{Distance} = N \times \lambda + \text{fractional part}$$

Where N = unknown number of complete wavelengths

This measurement only discloses the length of a **fractional part of a wavelength** used. This leaves a big unknown, namely the **number of full wavelengths** of the EDM's modulated carrier between the transmitter and the receiver at the instant of the measurement.

🔍 Cycle Ambiguity Symbol

This cycle ambiguity is symbolized by **N**, representing the unknown integer number of complete wavelengths. Fortunately, the cycle ambiguity can be solved in the EDM measurement process.

🔧 EDM Solution: Progressive Wavelengths

Fortunately, the cycle ambiguity can be solved in the EDM measurement process. The key is using **carriers with progressively longer wavelengths**.

📏 Progressive Resolution Method

1️⃣ Submeter

1-meter wavelength
Resolves submeter portion

2️⃣ Meters

10-meter wavelength
Resolves meter component

3️⃣ Tens of Meters

100-meter wavelength
Resolves tens of meters

4️⃣ Hundreds

1000-meter wavelength
Resolves hundreds of meters

🔄 How It Works

- **Send multiple wavelengths** of different lengths
- **Each returns to origin** for comparison
- **Phase comparison** reveals fractional parts
- **Build up total distance** progressively

📊 Example Process

- **100m wavelength:** Determines tens of meters
- **1000m wavelength:** Determines hundreds
- **Progressive refinement** of distance components
- **Complete distance resolution**

🎯 EDM Advantage

This method depends on the fact that the EDM survey can **send out waves of different wavelengths** and have them return to where they came from. That makes comparison of the returned wave with the reference wave possible.

⚠️ GPS Integer Ambiguity Problem

🚫 Why GPS Can't Use EDM's Solution

However, this whole method is **not possible in GPS surveying**.

GPS has only THREE carriers: L1, L2, and L5

They have CONSTANT wavelengths

🔒 GPS Limitations

- **Only three carrier frequencies** (L1, L2, L5)
- **Fixed wavelengths** - cannot vary
- **One-way signal** - no return for comparison
- **Limited progressive resolution**

📊 What GPS Can Do

- **Determine fractional wavelength** from single measurement
- **High precision** for the known portion
- **Phase comparison** with receiver oscillator
- **Requires mathematical solutions** for N

Therefore, while it's possible to determine the **fractional part of the wavelength**, that one small component of the distance, from a single measurement, knowing the **number of full wavelengths between the receiver and satellite is more difficult**.

🎯 Integer Ambiguity Problem

This is known as the **integer ambiguity problem** in GPS. Unlike EDM, GPS cannot easily resolve the unknown number of complete wavelengths, making carrier phase positioning more complex and requiring sophisticated mathematical approaches.

⚖️ EDM vs GPS: Fundamental Differences

✅ EDM Advantages

- **Variable wavelengths** can be transmitted
- **Two-way system** - signals return to origin
- **Reference wave comparison** possible

⚠️ GPS Challenges

- **Fixed wavelengths** (L1, L2, L5 only)
- **One-way system** - no signal return
- **Limited carrier options**
- **Integer ambiguity problem** persists
- **Mathematical solutions** required

- **Progressive resolution** solves cycle ambiguity
- **Complete distance buildup** from components

📝 Section Summary

Key Takeaways

This section explained the fundamental difference between EDM and GPS in handling cycle ambiguity. EDM can resolve the integer wavelength problem using multiple carrier frequencies, while GPS faces the persistent integer ambiguity challenge due to its limited, fixed carrier frequencies.

🔢 Cycle Ambiguity

Unknown number (N) of complete wavelengths between measurement points.

🔧 EDM Solution

Multiple variable wavelengths enable progressive distance resolution.

⚠️ GPS Challenge

Limited to L1, L2, L5 carriers creates persistent integer ambiguity problem.

📚 Two Types of Observables



📊 GPS Observables: Definition and Types

🔍 Observable vs. Observation

The term **"observable"** refers to the signals whose measurement yields the range or distance between satellite and receiver. This distinguishes between the **thing being measured** (the observable) and the **measurement itself** (the observation).

GPS has TWO types of observables

Code observables + Carrier observables

In GPS positioning, the word **observable** is used throughout GPS literature to indicate the signals whose measurement yields the range or distance between the satellite and the receiver. The word is used to draw a distinction between the thing being measured, "the observable" and the measurement, "the observation."

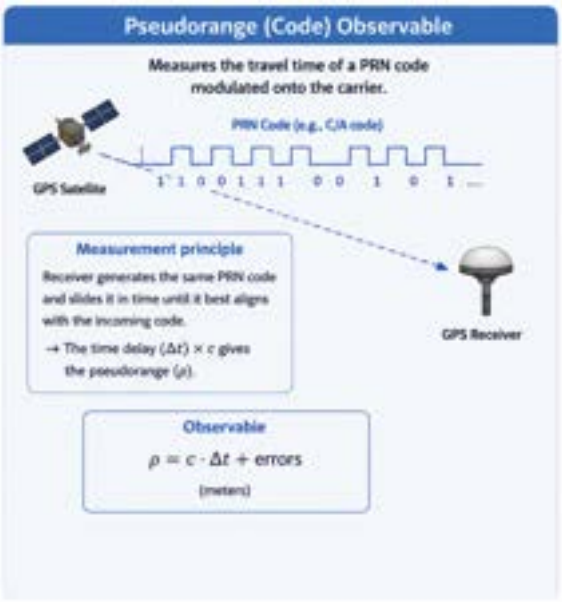
🎯 Two Observable Types

The codes are one type of observable that we've discussed as being modulated onto the carrier. There's another type of observable: the carrier itself without the codes. The carrier is the basis of the techniques used for high-precision GPS surveys.

📈 Visual Comparison: Code vs. Carrier Observables

1️⃣ Pseudorange Code Observable

Pseudorange code observable illustrated as square waves of code states



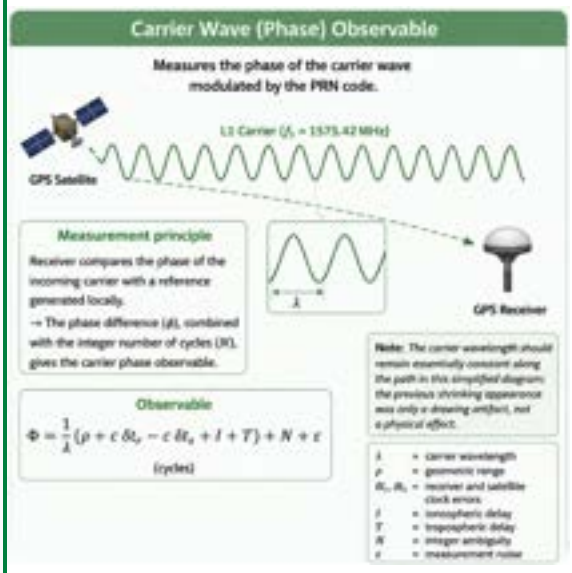
Square waves of code states

📊 Characteristics

- **Digital signal:** Square wave pattern
- **Modulated carrier:** Code imposed on wave
- **Code correlation:** Matched with receiver replica
- **Applications:** Quick positioning, lower accuracy

2️⃣ Carrier Wave Observable

A wave from a GPS Satellite to a GPS Receiver



Constant sine wave, not modulated

📊 Characteristics

- **Pure sine wave:** Unmodulated carrier
- **High precision:** Basis for surveying accuracy
- **Phase measurement:** Fraction of wavelength
- **Applications:** High-precision GPS surveys

⚡ Pseudorange Applications and Processing Strategy

🎯 When to Use Pseudorange Code

⚡ Quick Positioning

When **virtually instantaneous** point positions are required

📊 Lower Accuracy

When **relatively low accuracy** will suffice for the application

🚪 Front Door

As **preliminary step** before carrier phase measurement

🔄 Receiver Processing Strategy

- **Step 1:** Use pseudorange as "front door"
- **Step 2:** Begin position determination with code
- **Step 3:** Switch to carrier for refinement
- **Step 4:** Achieve final high-precision position

⚙️ Code Correlation Process

- **Receive:** Modulated carrier from satellite
- **Generate:** Replica code in receiver
- **Correlate:** Match received vs. replica
- **Calculate:** Time delay = pseudorange

🎯 Modern GPS Receiver Capability

Most GPS receivers used for surveying applications are capable of code correlation (pseudorange determination) AND carrier phase measurement. However, understanding the pseudorange foundation is essential before moving to carrier phase techniques.

🔗 Observable Combinations and Advanced Measurements

🧱 Combining Basic Observables

These basic observables can be **combined in various ways** to generate additional measurements that have certain advantages.

Enhanced Processing Techniques

Code + Carrier = Advanced positioning solutions

🔧 Combination Benefits

- **Improved accuracy:** Leveraging both observables
- **Faster convergence:** Code aids carrier initialization
- **Robustness:** Redundancy in measurements
- **Flexibility:** Application-specific solutions

📊 Context Applications

- **Preliminary step:** Pseudorange leads to carrier
- **Ambiguity resolution:** Code assists phase
- **Quality control:** Cross-validation of results
- **Real-time systems:** Progressive refinement

📝 Section Summary

Key Takeaways

This section introduced the fundamental concept of GPS observables, distinguishing between the two primary types: pseudorange code observables and carrier wave observables. Understanding these observables is essential for comprehending how GPS receivers determine position and the accuracy differences between various measurement techniques.

📡 Observable Types

Code (square waves) and carrier (sine waves) provide different positioning capabilities.

⚡ Processing Strategy

Pseudorange serves as "front door" before switching to carrier for precision.

🔗 Applications

Code for quick positioning; carrier for high-precision surveying applications.

Texas A&M University - Corpus Christi
College of Engineering & Computer Science
GISC 4315, Fall 2026

📚 Spread Spectrum and Code Modulation of L1 GPS Carrier

GPS Signal Modulation and Spread Spectrum

GISC 4315 – Satellite Positioning

📹 Instructor Video Lecture

Understanding GPS Signal Modulation Techniques

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📋 Video Topics Covered

- Carrier wave modulation principles
- Phase modulation vs AM/FM
- Spread spectrum characteristics
- Binary Phase Shift Keying (BPSK)
- Fundamental clock rate (F_0)
- Code chip timing and precision

> **💡 Study Tip**
>
> Focus on understanding how phase modulation enables GPS's spread spectrum advantages.

📻 Carrier Wave Modulation: From Radio to GPS

🔄 Modulation Principles

A carrier wave can be modulated in various ways. Radio stations use modulated carrier waves - AM (amplitude modulated) or FM (frequency modulated). When your radio is tuned to **105 FM**, you're not actually listening to 105 MHz; it's well above human hearing range!

105 MHz = Carrier Wave Frequency

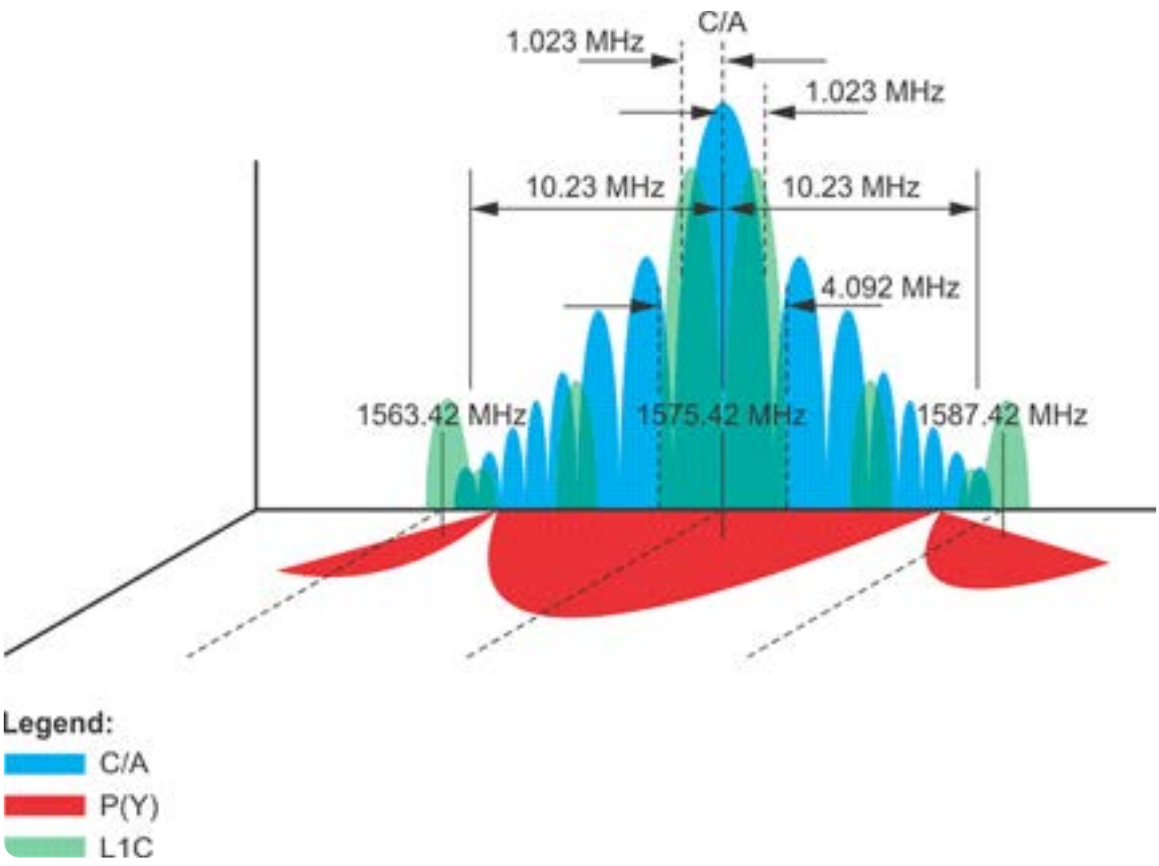
Music and speech come at much slower frequencies

The GPS carriers **L1, L2, and L5** could have been modulated in various ways to carry the binary codes (0s and 1s). However, neither amplitude nor frequency modulation are used in GPS.

🎯 GPS Uses Phase Modulation

It is the **alteration of the phases** of the carrier waves that encodes them. Phase modulation allows them to carry the codes from satellites to receivers.

📊 L1 GPS Signal Structure



L1 GPS Signal Frequency Spectrum

Showing bandwidth distribution of C/A, P(Y), and L1C signals

L1 Carrier	L2 Carrier	L5 Carrier
1575.42 MHz	**1227.60 MHz**	**1176.45 MHz**
Primary GPS frequency	Secondary frequency	Modernized signal

📡 Spread Spectrum: Intentionally Wide Bandwidth

🌊 Spread Spectrum Concept

One consequence of phase modulation is that the signal can occupy a **broader bandwidth** than would otherwise be possible. The GPS signal has a **spread spectrum** because of its intentionally increased bandwidth.

Signal Bandwidth >> Information Bandwidth

GPS signals are much wider than the data they carry

📏 Signal Bandwidth Examples

- **C/A code:** ~2.046 MHz wide
- **P(Y) code:** ~20.46 MHz wide
- **L1C signal:** ~4.092 MHz wide
- **Much wider** than center frequencies

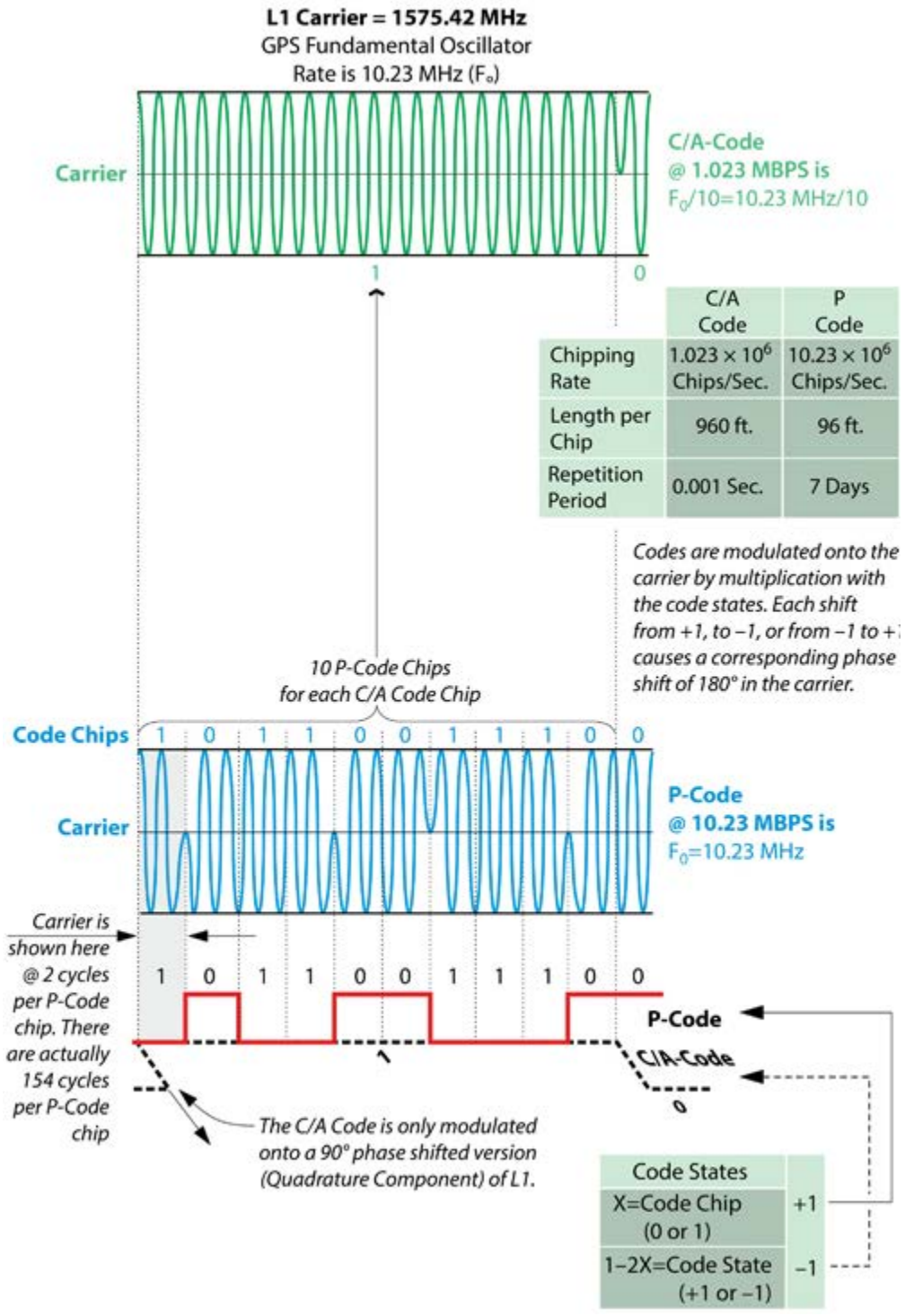
✅ Spread Spectrum Advantages

- **Better signal-to-noise ratio**
- **More accurate ranging**
- **Less interference**
- **Increased security**

⚠️ Trade-off: Signal Power

However, spreading the spectral density of the signal also **reduces its power**. The weakness of the signal makes it difficult to receive undercover.

🔄 Binary Phase Shift Keying (BPSK)



Code Modulation of the L1 Carrier

Demonstrating binary phase shift keying modulation technique

🔧 BPSK Modulation Process

The most commonly used spread spectrum modulation technique is **Binary Phase Shift Keying (BPSK)**. This technique creates the NAV Message, P(Y) code, and C/A code.

Binary Biphase Modulation

0 ↔ 1 switching by 180° phase changes

🔄 Phase Switching Process

- **Code change:** 0 to 1 or 1 to 0
- **Phase flip:** Instantaneous 180° reverse
- **Zero-crossing:** Flip occurs at centerline
- **Each bit** = one "code chip"

💾 Code Chip States

- **0 (zero):** Normal state
- **1 (one):** Mirror image state
- **Binary data:** Encoded as phase shifts
- **Chip timing:** Based on F_0

⏰ Fundamental Clock Rate (F_0) and Signal Timing

🕐 Standard Clock Rate

The rate of all GPS signal components are **multiples of the standard rate** of the oscillators. The standard rate is **10.23 MHz**, known as the fundamental clock rate.

$F_0 = 10.23$ MHz

Fundamental clock rate for all GPS signals

📡 Carrier Frequencies (Multiples of F_0)

- **L1:** 154 × F_0 = 1575.42 MHz
- **L2:** 120 × F_0 = 1227.60 MHz
- **L5:** 115 × F_0 = 1176.45 MHz
- **All based** on fundamental rate

💾 Code Chipping Rates

- **P(Y) code:** 1 × F_0 = 10.23 Mbps
- **C/A code:** 0.1 × F_0 = 1.023 Mbps
- **10.23 chips/microsecond** for P(Y)
- **10:1 ratio** P(Y) to C/A speed

🎯 **Precision Relationship**

Ten P(Y) code chips occur in the time it takes to generate one C/A code chip. This is why a **P(Y) code derived pseudorange is more precise** than a C/A code pseudorange.

📝 Section Summary

Key Takeaways

This section explained how GPS uses phase modulation and spread spectrum techniques to encode binary data onto carrier waves. Understanding BPSK modulation and the fundamental clock rate relationships is essential for comprehending GPS signal structure and timing precision.

📡 Phase Modulation

GPS uses 180° phase shifts to encode binary data instead of AM/FM.

🌊 Spread Spectrum

Wide bandwidth provides advantages but reduces signal power.

⏰ Clock Timing

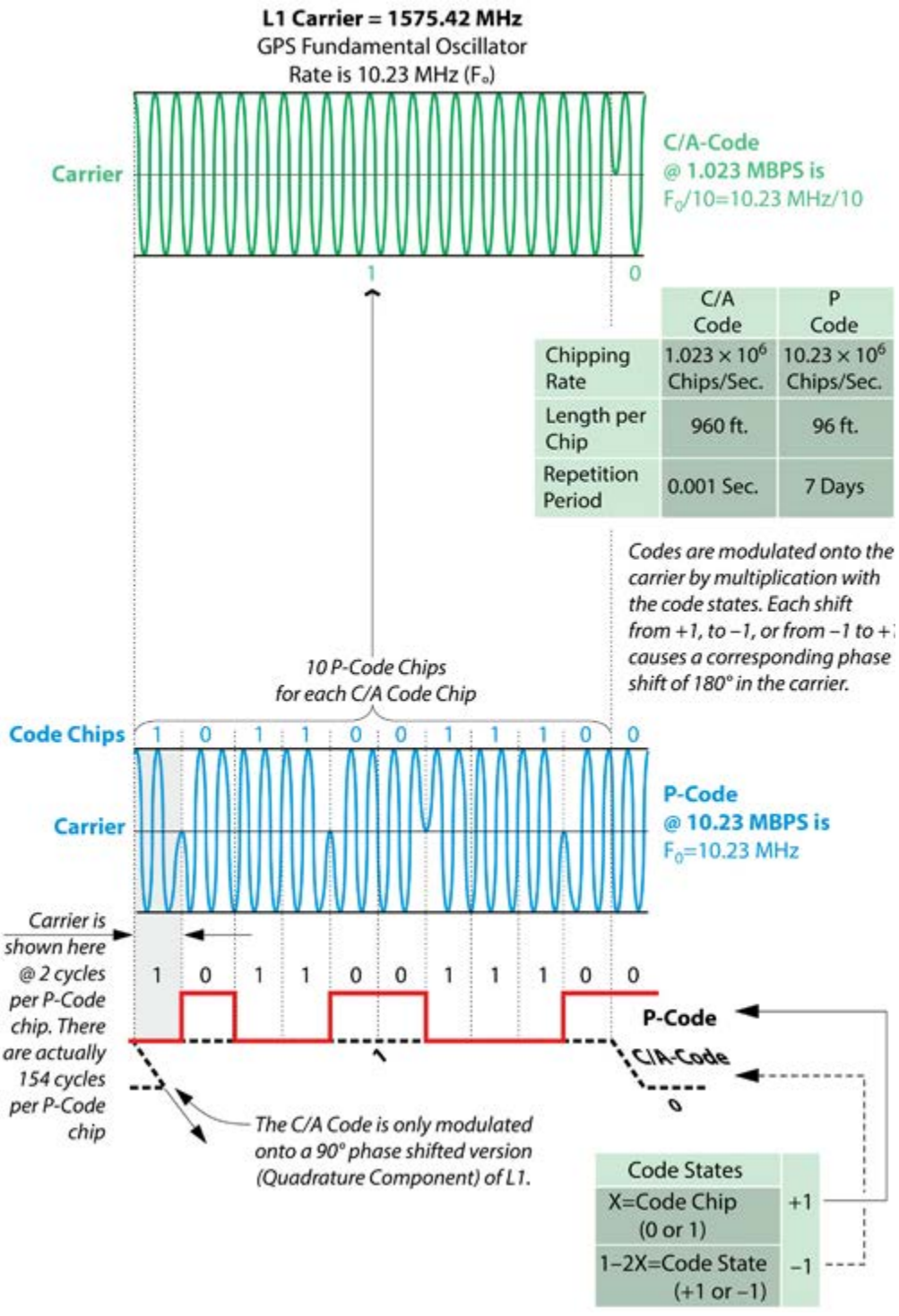
All GPS signals based on F_0 = 10.23 MHz fundamental rate.

CONTENT

📚 More About Code Chips



📊 L1 Signal Diagram: Detailed Component Analysis



Code Modulation of the L1 Carrier

Showing C/A Code (green), P-Code (blue), and phase shift indicators

🔍 Visual Elements in the Diagram

At the top is the **C/A Code in green** and the **P-Code in blue** below it. Several pieces of critical information are displayed in this comprehensive signal structure diagram.

Key Visual Components

Green C/A Code + Blue P-Code + Red Phase Indicators

🎯 Phase Shift Modulation Indicator

One of the things you might notice right away is the **red square wave** from 1 to 0 and 0 to 1 indicated at the bottom. For each 180-degree phase shift, there's this shift from 1 to 0 and back to 1, represented by this red square wave.

🔄 Code Relationships: Quadrature and Phase Alignment

📐 Quadrature Relationship

The **C/A Code is represented by the dashed black square wave** that is turned **90 degrees or in quadrature** to the P-Code. This specific geometric relationship is crucial for signal separation.

C/A Code ⊥ P-Code

90° phase relationship (quadrature)

🎯 Modulation Requirements

- **Phase shifts required:** For code encoding
- **No phase shift:** No modulation occurs
- **No modulation:** No code transmission
- **Essential process:** For GPS functionality

📊 Signal Separation

- **Quadrature:** Enables code separation
- **Same carrier:** Multiple codes on L1
- **Orthogonal:** Prevents interference
- **Receiver:** Can distinguish both codes

⚠️ Critical Principle

If there were **no shift in the phase** of the carriers, they would not be modulated and would not be carrying the codes. Phase modulation is absolutely essential for GPS signal transmission.

⏱️ Code Chipping Rates and Timing Analysis

📊 Chipping Rate Comparison

The code chipping rates are shown on the right-hand side of the diagram. Notice that the **C/A Code chipping rate is 10 times slower** than the P-Code.

P-Code Rate	C/A Code Rate
10.23 Mbps	1.023 Mbps
Fast chipping rate	10× slower rate

📡 L1 Carrier Frequency

- **L1 frequency:** 1575.42 MHz
- **Based on F_0:** 154 × 10.23 MHz
- **Fundamental rate:** 10.23 MHz
- **Multiple relationship:** All rates connected

⚡ Chip Timing Relationship

- **10 P-Code chips:** Per each C/A Code chip
- **Expected ratio:** Exactly as predicted
- **Timing synchronization:** Precise coordination
- **Clock-based:** All from F_0 standard

📏 Physical Code Lengths and Wavelength Analysis

📐 Code Chip Physical Lengths

The physical length difference between code types directly relates to their chipping rates. Faster codes have shorter physical chip lengths.

C/A Code Length	P-Code Length
960 feet	96 feet
Longer physical chip	10× shorter chip

🎯 Speed-Length Relationship

The length of a P-Code chip is **10 times shorter** because the P-Code is **10 times faster**. This inverse relationship between chipping rate and physical chip length is fundamental to GPS ranging precision.

🔒 Code Repetition Periods and Security Implications

🕐 Repetition Period Comparison

The repetition periods of the two codes reveal important security and operational characteristics. Notice the dramatic difference in code repetition timing.

C/A Code Repetition	P-Code Repetition
Very Quick	7 Days
Rapid repetition cycle	Extended cycle period

🔓 C/A Code Characteristics

- **Quick repetition:** Easy to acquire
- **Public access:** Available to all users
- **Initial acquisition:** Entry point for GPS
- **Standard precision:** Meter-level accuracy

🔒 P-Code Security Features

- **7-day cycle:** Extremely long pattern
- **More secure:** Difficult to replicate
- **Classified access:** Precise positioning
- **Enhanced accuracy:** Military applications

🛡️ Security by Design

The P-Code does not repeat for **seven days**, making it more secure as would be expected with a **classified precise code**. This extended repetition period makes unauthorized replication extremely difficult.

📝 Section Summary

Key Takeaways

This detailed analysis of the L1 signal diagram revealed the complex relationships between C/A and P codes, their timing characteristics, physical properties, and security implications. Understanding these relationships is essential for comprehending GPS signal structure and receiver processing.

🔄 Code Relationships

C/A and P codes in quadrature (90°) on same L1 carrier.

⏱️ Timing Precision

10:1 P-Code speed advantage translates to higher precision.

🔒 Security Design

7-day P-Code repetition provides enhanced security features.

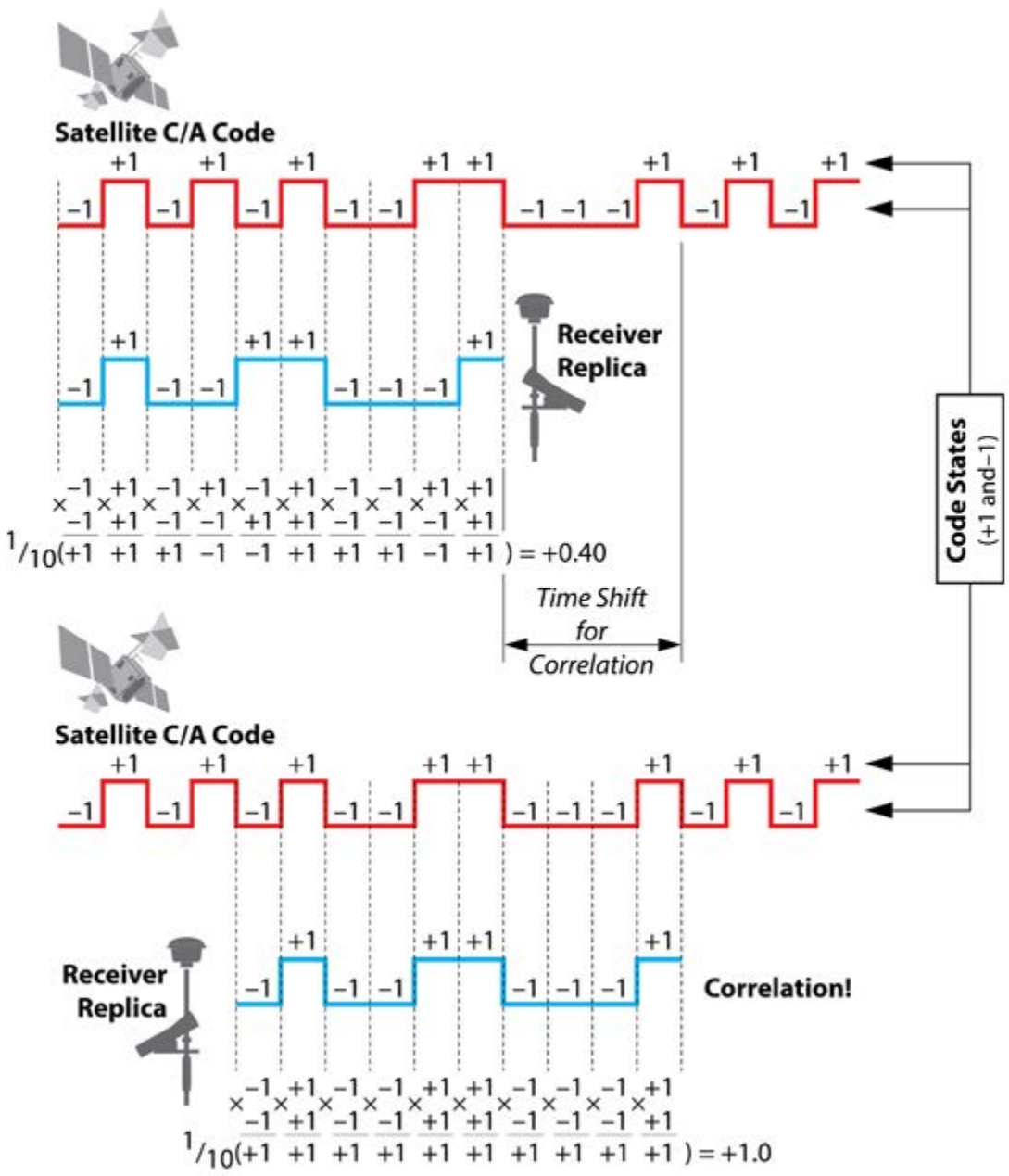
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📚 Code Correlation



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⏱️ Pseudorange Observable: Time Shift Foundation



Code Correlation Process

Demonstrating time shift required for code correlation

📐 Time Shift (dτ) Concept

Strictly speaking, a pseudorange observable is based on a **time shift**. This time shift can be symbolized by **dτ** (d tau), and is the time elapsed between the instant a GPS signal leaves a satellite and the instant it arrives at a receiver.

dτ = Signal Travel Time

From satellite transmission to receiver arrival

🎯 Foundational Principle

The concept can be illustrated by the process of **setting a watch from a time signal heard over a telephone**. This everyday analogy helps understand the fundamental GPS ranging principle.

📞 Propagation Delay: The Telephone Analogy

🕐 Setting a Watch Example

Imagine that a recorded voice said, **"The time at the tone is 3 hours and 59 minutes."** If a watch was set at the instant the tone was **heard**, the watch would be wrong.

❌ Wrong Time

Set watch when tone heard
Includes travel time delay

✅ Correct Time

When tone was broadcast
Actual time at origin

⚡ Electrical Signal Travel

- **Travel time:** Through telephone lines
- **Speed:** Same as electromagnetic energy
- **Speed of light:** Radio signals, electricity
- **Measurable delay:** Distance/speed calculation

📐 Distance Calculation

- **Formula:** Distance = Time × Speed
- **Circuit length:** Elapsed time ÷ speed
- **Measurement principle:** Time reveals distance
- **GPS application:** Same concept in space

🌍 GPS Extension

In GPS, that elapsed time is known as the **propagation delay**, and it is used to measure length. The measurement is accomplished by a combination of codes, similar to the strategy used in EDMs.

🔄 Code Correlation: Matching Process

🗝️ Key and Lock Concept

Where an EDM generates an internal replica of its **carrier wave** to correlate with the signal it receives by reflection, a pseudorange is measured by a GPS receiver using a replica of a portion of the **code that is modulated onto the carrier wave**.

GPS Receiver Generates Code Replica

Compares with incoming satellite code

1️⃣ Generate

Identical codes
One in satellite, one in receiver

2️⃣ Transmit

Satellite sends code
Codes don't initially align

3️⃣ Correlate

Time shift replica
Until codes match perfectly

⏰ Time Shift Reveals Propagation Delay

It is this **time shift** that reveals the propagation delay. The propagation delay is the time it took the signal to make the trip from the satellite to the receiver, $d\tau$. Once the time shift is accomplished, the two codes match perfectly and the signal transit time has been measured.

🔍 Autocorrelation: Code States and Mathematical Matching

💾 Code Chips to Code States Transformation

Actually, lining up the code from the satellite with the replica in the GPS receiver is called **autocorrelation**, and depends on the transformation of code chips into code states.

$$\text{Code State} = 1 - 2x$$

Where x is the code chip value (0 or 1)

🔢 Code State Conversion

- **Chip 0 → State +1:** Normal state
- **Chip 1 → State -1:** Mirror state
- **Binary to bipolar:** Mathematical processing
- **Correlation calculation:** Product summation

📊 Correlation Example

- **10 code states:** Receiver vs satellite
- **Product sum ÷ 10:** Correlation value
- **Perfect match = 1.0:** Successful correlation
- **Mismatch = 0.40:** Needs adjustment

❓ Three Critical Questions in Code Correlation

1️⃣ How does the receiver know when codes DON'T match?

When codes don't match, the sum of the products of each of the receiver's 10 code states with each of the code states from the satellite, when divided by 10, **does not equal 1**.

Example: $0.40 \neq 1.0$ (No correlation)

2️⃣ What does the receiver do when codes don't match?

The receiver makes **iterative small shifts** in both time and frequency until the codes match:

- **Frequency shift:** Accommodate Doppler shift from satellite motion
- **Time shift:** Adjust code timing alignment
- **Iterative process:** Continue until correlation achieved

3️⃣ **How does the receiver know when codes DO match?**

When codes match perfectly, the sum of the products of each code state, divided by 10, **equals exactly 1**.

"Correlation!" - Code fits like a key fits a lock

🎯 **The Result: Distance Measurement**

The time shift for correlation gives the receiver approximately the amount of elapsed time that it took the signal to come from the satellite to the receiver. If the receiver knows the elapsed time and the speed of the signal, it then knows the distance (range) between itself and the satellite.

📝 Section Summary

Key Takeaways

This section explained the fundamental process of code correlation that enables GPS pseudorange measurement. Understanding how receivers generate code replicas, shift them in time and frequency, and achieve mathematical correlation is essential for comprehending GPS ranging principles.

⏱️ `Time Shift (dτ)`
Signal travel time from satellite to receiver enables distance calculation.

🔄 `Code Correlation`
Matching receiver replica with satellite code reveals propagation delay.

🎯 `Perfect Match`
Mathematical correlation = 1.0 indicates successful code alignment.

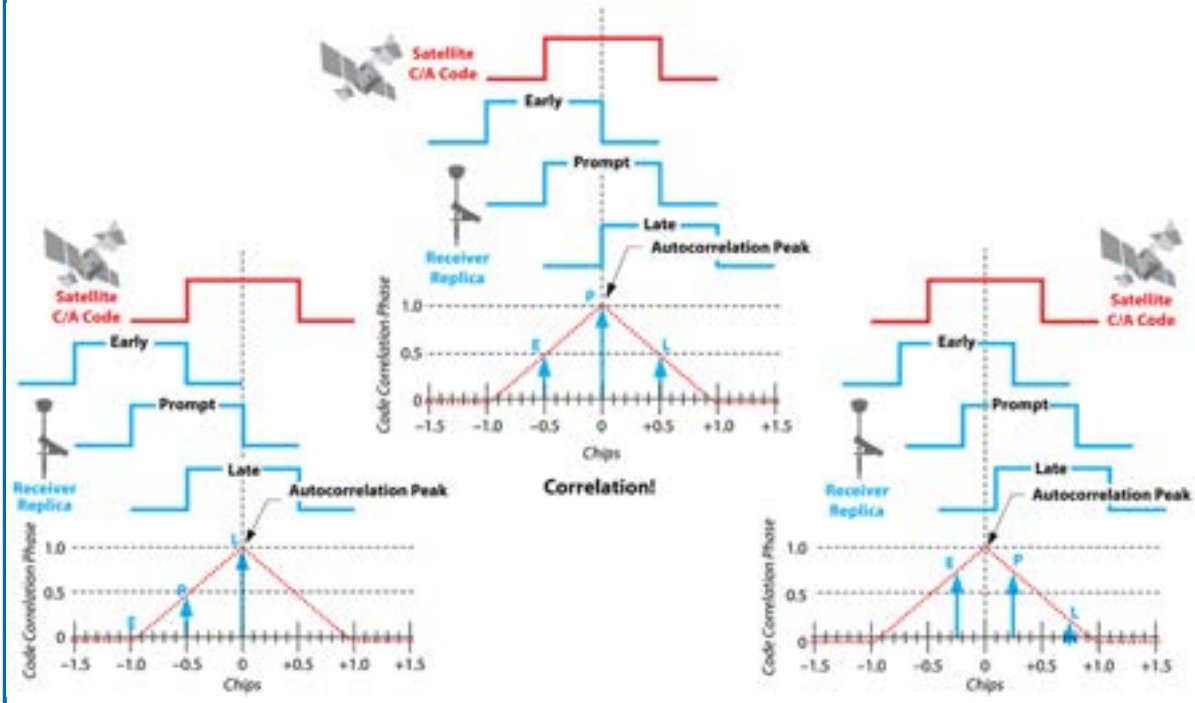
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📚 The Delay Lock Loop

Delay Lock Loop and Signal Tracking

GISC 4315 – Satellite Positioning

🔒 Delay Lock Loop (DLL): Maintaining Code Correlation



Code Correlation: Showing Delay Lock Loop Operation

Demonstrating continuous correlation maintenance through DLL

🔄 DLL Functionality

Once correlation of the two codes is achieved with a **Delay Lock Loop (DLL)**, it is maintained by a correlation channel within the GPS receiver. This continuous process ensures stable signal tracking.

DLL = Continuous Correlation Maintenance

Automatic adjustment to maintain code alignment

🔧 DLL Components

- **Correlation channel:** Dedicated tracking circuit
- **Feedback loop:** Continuous adjustment mechanism
- **Error detection:** Monitors correlation quality
- **Code alignment:** Maintains perfect timing

⚡ DLL Operation

- **Real-time tracking:** Continuous monitoring
- **Dynamic adjustment:** Adapts to signal changes
- **Lock maintenance:** Prevents correlation loss
- **Signal stability:** Ensures consistent tracking

🎯 Critical Function

The DLL is essential for maintaining the precise code correlation initially achieved through the autocorrelation process. Without it, receivers would constantly lose track of satellite signals.

🔐 Signal Lock States: Achieved Lock vs Lost Lock

🎯 Lock Status Terminology

The receiver is sometimes said to have **"achieved lock"** or to be **"locked on"** to the satellites when correlation is successfully maintained through the DLL.

✅ Achieved Lock

Successful correlation
DLL maintaining alignment

❌ Lost Lock

Correlation failure
Signal tracking interrupted

✅ During Lock State

- **Continuous tracking:** Signal logging maintained
- **Stable correlation:** Code alignment preserved

⚠️ Causes of Lock Loss

- **Cycle slip:** Phase tracking discontinuity
- **Signal obstruction:** Physical blockage
- **Multipath interference:** Reflected signals

- **NAV message access:** Data decoding possible
- **Pseudorange updates:** Distance measurements continue

- **Atmospheric disturbances:** Signal degradation

🔄 Continuous Tracking

The receiver can continue to log the signal from the satellite and stay correlated unless it is somehow interrupted. When lock is maintained, reliable positioning data flows continuously.

📡 NAV Message Availability and Clock Corrections

📨 NAV Message Access During Lock

As long as the lock is present, the **NAV message** **is available** to the receiver. Remember that one of its elements is the broadcast clock correction that relates the satellite's onboard clock to GPS time.

Lock State = NAV Message Access

Critical data becomes available during signal tracking

📊 NAV Message Contents

- **Satellite ephemeris:** Orbital position data
- **Clock corrections:** Timing synchronization
- **Almanac data:** Constellation information
- **System health:** Satellite status updates

🕐 Clock Correction Importance

- **Time synchronization:** Satellite to GPS time
- **Error correction:** Clock drift compensation
- **Accuracy maintenance:** Precise timing critical
- **System coordination:** Global time reference

⏰ Timing Synchronization Critical

The broadcast clock correction is essential for relating the satellite's onboard clock to GPS time. Without this correction, pseudorange measurements would contain significant timing errors.

⚠️ Emerging Limitations of the Pseudorange Process

🔍 A Limitation Comes Up

Even with successful code correlation, NAV message access, and proper clock corrections, **a limitation of the pseudorange process comes up**. This limitation affects the accuracy and reliability of GPS measurements.

Perfect Process ≠ Perfect Results

Inherent limitations affect pseudorange accuracy

🎯 Process Success Elements

- **Code correlation:** Successfully achieved
- **Signal lock:** DLL maintaining tracking
- **NAV message:** Data successfully decoded
- **Clock corrections:** Timing synchronized

⚠️ Remaining Challenges

- **Measurement errors:** Various error sources
- **Atmospheric effects:** Signal delays
- **Receiver limitations:** Hardware constraints
- **Systematic biases:** Inherent inaccuracies

🔍 Setting the Stage

While the code correlation and signal tracking processes work effectively, fundamental limitations in the pseudorange measurement process still affect GPS accuracy. Understanding these limitations is crucial for advanced GPS applications.

📝 Section Summary

Key Takeaways

This section covered the critical transition from initial code correlation to continuous signal tracking through the Delay Lock Loop. Understanding how receivers maintain signal lock and access NAV message data sets the foundation for exploring pseudorange measurement limitations.

🔒 Signal Lock

DLL maintains continuous code correlation for stable tracking.

📡 NAV Access

Lock state enables NAV message decoding and clock corrections.

⚠️ Limitations

Process success doesn't eliminate inherent measurement limitations.

CONTENT

📚 Imperfect Oscillators and Clock Corrections



📹 Instructor Video Lecture

GPS Time Differences and Clock Synchronization

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📋 Video Topics Covered

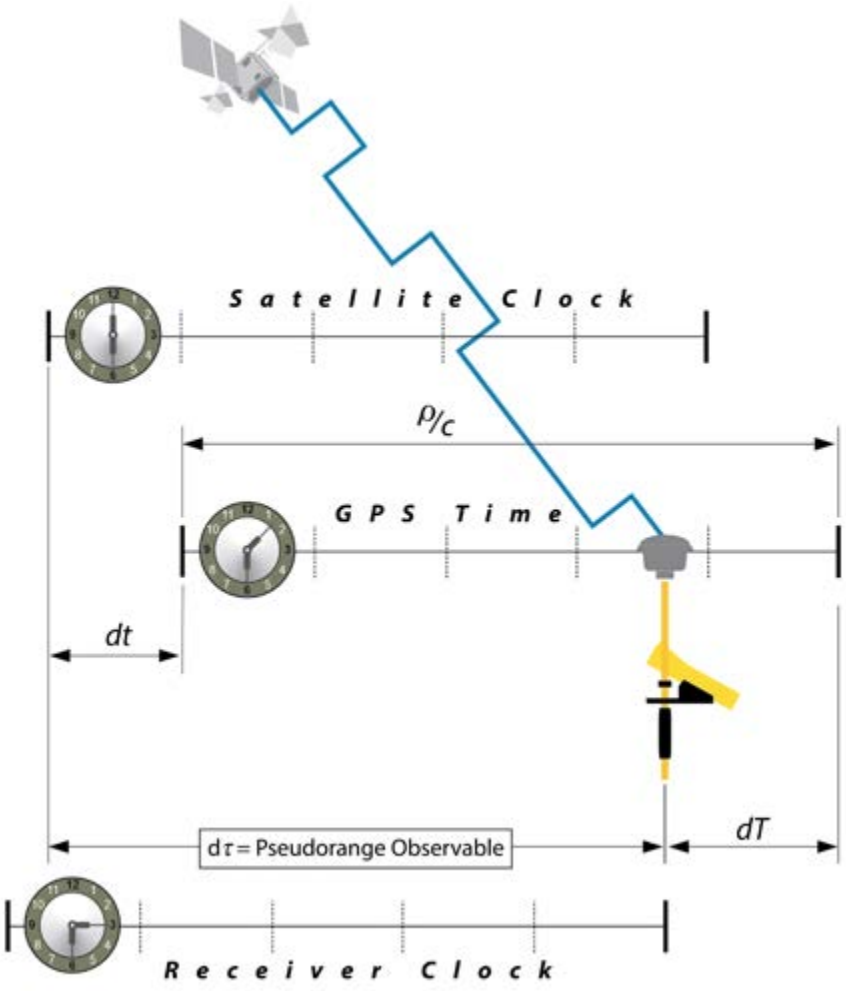
- Clock synchronization challenges in GPS
- Understanding time shift (dτ) measurements
- Satellite vs. receiver clock differences
- Why GPS uses "pseudorange" terminology
- Impact of nanosecond timing on accuracy
- Sources of clock drift and instability

> **💡 Study Tip**
>
> Focus on understanding why perfect clock synchronization is impossible and how this affects all GPS measurements.

🔍 Understanding Clock Synchronization Challenges

⏱️ Time Shift and Clock Errors in GPS Measurements

Time shift contains signal's transit time from satellite to receiver as well as clock errors



Time Differences in GPS Signal Processing

Demonstrating clock synchronization challenges between satellite and receiver

🚫 The Synchronization Problem

One reason the time shift, ***dτ (d tau)***, found in autocorrelation cannot quite reveal the true range, **ρ**, of the satellite at a particular instant is the **lack of perfect synchronization** between the clock in the satellite and the clock in the receiver.

dτ = Signal Transit Time + Clock Errors + Other Errors

Time shift contains multiple error sources

🛰️ Satellite Clock Issues

- **Clock drift:** Gradual deviation from GPS time
- **Temperature effects:** Environmental destabilization
- **Acceleration impacts:** Orbital dynamics influence
- **Radiation exposure:** Space environment effects

📱 Receiver Clock Limitations

- **Quartz oscillators:** Less stable than atomic clocks
- **Temperature sensitivity:** Environmental variations
- **Manufacturing tolerances:** Inherent imperfections
- **Aging effects:** Performance degradation over time

⚠️ Critical Timing Precision

Since the two compared codes are generated directly from the fundamental rate, F_0, of those clocks, and these widely separated clocks cannot be in perfect lockstep, the codes they generate cannot be in perfect sync either.

📏 Pseudorange: The "False Range" Problem

🎯 Why "Pseudorange"?

The difference between the satellite clock's time and GPS time is shown as **dt (d small t)**. The difference in the receiver's clock from GPS time is shown as **dT (d capital T)**.

1 nanosecond ≈ 1 foot of distance error

Timing precision directly affects positioning accuracy

⌚ Clock Error Sources

- **Satellite atomic clocks:** Drift from GPS time standard
- **Receiver quartz clocks:** Inherently less stable
- **Navigation message delays:** Uploaded correction timing
- **Environmental factors:** Temperature, radiation effects

📐 Measurement Challenges

- **Approximate accuracy:** Close but not perfect range
- **Systematic biases:** Consistent error patterns
- **Correction limitations:** Difficult error removal
- **Precision requirements:** Nanosecond-level timing

🔍 Fundamental Limitation

While the pseudorange observable **dτ (d tau)** is intended to measure signal transit time from satellite to receiver, clock discrepancies create errors that are difficult to remove. This is why it's called a *pseudorange* (false range) rather than a true range measurement.

📝 Section Summary

Key Takeaways

Understanding time differences and clock synchronization challenges reveals why GPS measurements are called "pseudoranges" rather than true ranges. The fundamental limitation of imperfect clocks affects all satellite positioning measurements.

⏱️ Clock Errors

Satellite and receiver clocks cannot maintain perfect synchronization.

📏 Pseudorange

Range measurements contain timing errors from multiple sources.

🎯 Precision Impact

Nanosecond timing errors translate to significant distance inaccuracies.

⬅️ Back to Delay Lock Loop

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📚 The Pseudorange Equation

Pseudorange Equation and Error Sources

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📐 The Complete Pseudorange Equation

$$p = \rho + d_\rho + c\,(dt - dT) + d_{ion} + d_{trop} + \varepsilon_{mp} + \varepsilon_p$$

where
- p = the pseudorange measurement
- ρ = the true range
- d_ρ = satellite orbital errors
- c = the speed of light
- dt = the satellite clock offset from GPS Time
- dT = the receiver clock offset from GPS Time
- d_{ion} = ionospheric delay
- d_{trop} = tropospheric delay
- ε_{mp} = multipath
- ε_p = receiver noise

Complete Pseudorange Equation

Showing all error sources that prevent p from equaling ρ

⚠️ The Fundamental Problem

Clock offsets are only one of the errors in pseudoranges. The pseudorange, **p**, and the true range, **ρ**, **cannot be made equivalent** without consideration of clock offsets, atmospheric effects and other biases that are inevitably present.

p = ρ + Clock Errors + Orbital Errors + Atmospheric Delays + Multipath + Noise

Pseudorange equals true range plus multiple error sources

🎯 Key Insight

If the pseudorange was good and complete by itself, these other factors would not need to be considered. But in reality, multiple error sources contaminate every GPS measurement.

🛰️ Satellite and Orbital Error Sources

🌍 Orbital Perturbations

- **Gravitational variations:** Earth's irregular gravity field
- **Solar radiation pressure:** Photon momentum effects
- **Atmospheric drag:** Upper atmosphere interactions
- **Tidal forces:** Sun and moon gravitational pull

⏱️ Clock Error Components

- **Satellite clock offset:** Known from NAV message but imperfect
- **Receiver clock offset:** Generally unknown and unstable
- **Propagation delays:** Signal transit time variations
- **Oscillator imperfections:** Temperature and aging effects

🔬 Technical Detail

The speed of light is constant in a vacuum, but it must be multiplied by the satellite clock offset from GPS time minus the receiver clock offset from GPS time, accounting for propagation delays and imperfect oscillators.

🌍 Atmospheric Delay Effects

☁️ Ionospheric Delay

- **Charged particles:** Free electrons affect signal speed
- **Solar activity dependency:** Variable ionization levels
- **Frequency dependence:** Different delays for L1/L2
- **Time of day variation:** Diurnal ionospheric changes

🌫️ Tropospheric Delay

- **Lower atmosphere:** Layers below ionosphere
- **Weather dependent:** Humidity and pressure effects
- **Elevation angle dependence:** Longer path at low angles
- **Modeling challenges:** Complex atmospheric structure

🌡️ Environmental Impact

Both ionospheric and tropospheric delays vary with atmospheric conditions, satellite elevation angle, and geographic location, making them difficult to predict precisely.

📡 Signal Interference and Receiver Issues

🔄 Multipath Effects

- **Signal reflections:** Bouncing off surfaces before reaching receiver
- **Buildings and terrain:** Urban canyon effects
- **Water surfaces:** Strong reflective properties
- **Distance errors:** Longer path creates false ranges

📻 Receiver Noise

- **Electronic interference:** Internal receiver components
- **Thermal noise:** Temperature-dependent variations
- **Quantization errors:** Digital processing limitations
- **Similar to radio static:** Random signal corruption

🚧 Measurement Challenge

Multipath creates particular difficulties because if you're measuring distances to determine position, any signal bounce introduces errors that are hard to detect and correct.

⚠️ The Reality of GPS Measurements

Why Pseudorange is Just the Starting Point

In short, all these biases contaminate the pseudorange position. So, **while the pseudorange is a good way to get started with a GPS position, it isn't the full answer.**

🎯 Key Understanding

Each error source requires specific correction techniques and modeling approaches to achieve the high accuracy GPS is known for. Understanding these limitations explains why GPS positioning involves complex algorithms beyond simple distance measurements.

📝 Section Summary

Key Takeaways

The complete pseudorange equation reveals why GPS measurements require sophisticated error correction. Each term represents a significant challenge that must be addressed for accurate positioning.

🛰️ Orbital Errors

Gravitational and environmental forces affect satellite positions.

🌍 Atmospheric Delays

Ionosphere and troposphere slow signal propagation.

🔄 Multipath

Signal reflections create false distance measurements.

📻 Receiver Noise

Electronic interference degrades measurement quality.

📚 The Carrier Phase Observable



📏 The One-Percent Rule of Thumb

🎯 Convenient Approximation

The **maximum resolution available in a pseudorange is about 1 percent of the chipping rate** of the code used. This offers a basis to evaluate pseudoranging in general and compare the potentials of P(Y) code and C/A code pseudoranging in particular.

Resolution ≈ 1% of Chipping Rate

Fundamental limit for code-based measurements

🔬 Rule Application

This rule provides a theoretical framework for understanding the precision limitations inherent in different GPS signal structures and helps explain why certain measurement techniques are preferred for high-accuracy applications.

🔢 P(Y) Code Precision Analysis

⚡ P(Y) Code Characteristics

Chip Duration	Chip Rate
0.0978 μs ≈ 0.1 microseconds	**100 nanoseconds** per code chip

1% of 100 ns = 1 nanosecond

Maximum P(Y) code resolution

📐 Distance Conversion

1 nanosecond × speed of light ≈ 30 centimeters ≈ 1 foot

Therefore, the very best P(Y) pseudorange precision is about **1 foot**

📊 C/A Code Precision Comparison

🔄 Relative Performance

Because its chipping rate is **10 times slower**, the C/A code based pseudorange is **10 times less precise** than P(Y) code.

C/A Code Resolution	Practical Distance
10 × 30 cm = 3 meters 1% of C/A chip length	**≈ 10 feet** Using rule of thumb

C/A Code Resolution ≈ 10 feet (3 meters)

10 times less precise than P(Y) code

📡 Carrier Phase: The Path to Higher Precision

🤔 The Critical Question

You might ask, at this point, ***if the pseudorange isn't the full answer, then how do we get the extraordinary accuracies that we depend upon GPS to produce?***

Answer: Carrier Phase Measurements

The key to centimeter-level accuracy

🔑 Precision Breakthrough

The same 1 percent rule of thumb can illustrate the increased precision of the carrier phase observable over the pseudorange, leading to the extraordinary accuracies GPS is known for.

📐 Carrier Wavelength Calculations

🔬 Wavelength Formula

$$\lambda = c_a / f$$

λ = length of each complete wavelength in meters

$\mathbf{c_a}$ = speed of light corrected for atmospheric effects

f = frequency in hertz

📡 L1 Frequency (1575.42 MHz)

$\lambda = (300 \times 10^6 \text{ m/s}) / (1575.42 \times 10^6 \text{ Hz})$

📡 L2 Frequency (1227.60 MHz)

$\lambda = (300 \times 10^6 \text{ m/s}) / (1227.60 \times 10^6 \text{ Hz})$

λ = 19 cm

λ = 24 cm

📏 **Wavelength Summary**

L1 carrier (1575.42 MHz) has a wavelength of approximately **19 cm**, while L2 carrier (1227.60 MHz) has a wavelength of approximately **24 cm**.

🎯 Carrier Phase Precision Advantage

🚀 Superior Resolution

The resolution available from a signal is approximately **1% of its wavelength**. 1% of these wavelengths is approximately **2mm**.

P(Y) Code	C/A Code	Carrier Phase
~1 foot (30 cm)	**~10 feet** (3 meters)	**~2 mm** (millimeter precision!)

Carrier Phase: 1% of ~20 cm ≈ 2 millimeters

Orders of magnitude better than code measurements

⭐ **Preferred Method**

Carrier phase observations are certainly the preferred method for the higher precision work most have come to expect from GPS. This explains how GPS achieves centimeter-level accuracy for surveying and scientific applications.

📝 Section Summary

Key Takeaways

The one-percent rule reveals why GPS evolved from code-based pseudoranges to carrier phase measurements. Understanding these precision limitations explains how GPS achieves its remarkable accuracy for high-precision applications.

📏 Rule of Thumb

Resolution ≈ 1% of signal characteristic length.

🔢 Code Precision

P(Y): ~1 foot, C/A: ~10 feet maximum resolution.

📡 Carrier Phase

Wavelength-based measurements achieve ~2mm precision.

🎯 Applications

Carrier phase enables high-precision GPS surveying.

📡 Module 02 Summary



📡 Module 02: GPS Signal Structure and Processing

Understanding Signal Components and Measurement Techniques

🧭 Module Overview

In this module, we delve deep into the technical foundations of GPS signal structure and processing. You'll explore how GPS satellites generate and transmit their signals, how receivers process these signals to extract positioning information, and the mathematical relationships that enable precise positioning. From the fundamental concepts of trilateration to the sophisticated techniques of carrier phase measurement, this module provides the technical knowledge essential for understanding modern GPS surveying and high-precision applications.

🎯 Learning Objectives - What You've Learned

✅ Module Objectives

By the this point in the module, you should be able to:

- Demonstrate understanding of the basic GPS signal structure and components
- Explain the relationship between GPS positioning and trilateration principles
- Describe the pertinence and structure of the navigation message
- Analyze the structure and function of P(Y) and C/A codes
- Define the creation and modulation of GPS carrier waves
- Identify and differentiate the two primary GPS observables
- Describe the role of autocorrelation in GPS signal processing
- Recognize and apply the pseudorange equation for positioning
- Discuss carrier phase ranging techniques for high-accuracy GPS applications
- Understand clock corrections and timing synchronization in GPS

🔧 Key Technical Concepts

📐 Trilateration and Range Measurement

- **One-way ranging:** Passive measurement from satellite to receiver
- **Trilateration principles:** 3D positioning using distance measurements
- **Geometric relationships:** Converting ranges to coordinates
- **Unknown parameters:** X, Y, Z position plus clock bias
- **Minimum satellites:** Four satellites needed for 3D positioning

📻 GPS Signal Components

- **Carrier frequencies:** L1 (1575.42 MHz), L2 (1227.60 MHz)
- **PRN codes:** C/A code (civilian), P(Y) code (military)
- **Navigation message:** Satellite ephemeris and system data
- **Spread spectrum:** Code division multiple access (CDMA)
- **Signal power:** Extremely weak signals (-125 to -130 dBm)

🔍 GPS Codes and Signal Processing

C/A Code (Coarse/Acquisition)

- 1.023 MHz chip rate
- 1023-bit length, 1 ms period
- Civilian access on L1 frequency
- Initial signal acquisition
- ~1-meter ranging precision

P(Y) Code (Precise/Encrypted)

- 10.23 MHz chip rate
- 7-day repeat cycle
- Military/authorized users
- Higher precision ranging
- ~10-cm ranging precision

📡 Navigation Message and System Information

📊 Ephemeris Data

- Precise satellite orbits
- Valid for 2-4 hours
- Required for accurate positioning
- Broadcast in navigation message

📅 Almanac Data

- Coarse satellite orbits
- Valid for months
- Satellite acquisition aid
- Time to first fix reduction

🕐 GPS Time System

- Atomic time standard
- Week number and time of week
- Leap second information
- Clock correction parameters

📏 GPS Observables and Measurement Techniques

📐 The Two Primary GPS Observables

1️⃣ Code Phase (Pseudorange)

- **Measurement:** Time delay of PRN code
- **Accuracy:** 1-10 meters
- **Equation:** ρ = c·(tr - ts) + clock errors
- **Applications:** Navigation, coarse positioning
- **Ambiguity:** None (unambiguous range)

2️⃣ Carrier Phase

- **Measurement:** Phase of carrier wave
- **Accuracy:** Millimeter to centimeter level
- **Wavelength:** ~19 cm for L1, ~24 cm for L2
- **Applications:** High-precision surveying, RTK
- **Ambiguity:** Integer cycle ambiguity must be resolved

⚙️ Advanced Signal Processing Concepts

🔄 Code Correlation Process

- **Autocorrelation:** Matching received and local PRN codes
- **Time shift detection:** Measuring signal travel time
- **Lock acquisition:** Synchronizing receiver with satellite
- **Code tracking:** Maintaining code alignment

⏰ Delay Lock Loop (DLL)

- **Tracking system:** Maintains code synchronization
- **Early/Late correlators:** Error detection method
- **Feedback control:** Adjusts local code timing
- **Multipath mitigation:** Narrow correlator spacing

🛠️ Clock Corrections and Error Sources

⏱️ Clock Errors

- Satellite clock drift
- Receiver clock bias
- Relativistic effects
- Clock correction polynomials

🌍 Atmospheric Effects

- Ionospheric delay
- Tropospheric delay
- Dual-frequency correction
- Modeling techniques

🔄 Integer Ambiguity

- Carrier phase cycles
- Ambiguity resolution
- LAMBDA method
- Real-time solutions

📊 The Pseudorange Equation

$$\rho = c \cdot \Delta t + c \cdot \delta t_u - c \cdot \delta t_s + I + T + \varepsilon$$

- **ρ:** Measured pseudorange
- **c:** Speed of light
- **Δt:** Signal travel time
- **δtu:** Receiver clock error
- **δts:** Satellite clock error
- **I:** Ionospheric delay
- **T:** Tropospheric delay
- **ε:** Measurement noise

💡 Key Takeaways

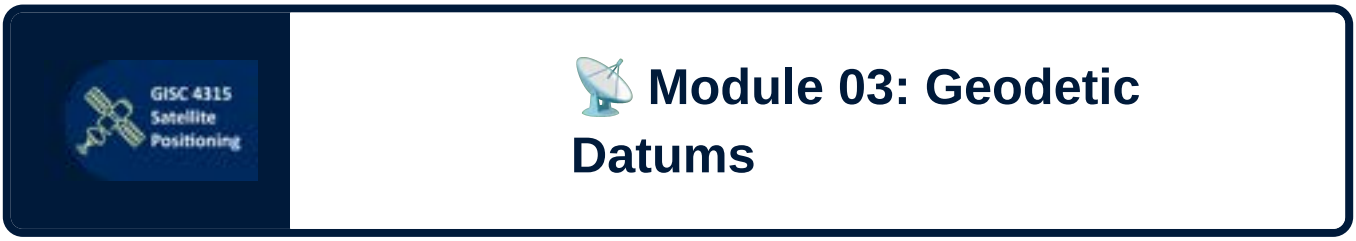
🎯 Bottom Line Up Front

- **GPS signals are complex but precisely structured** with multiple components working together for positioning
- **Two primary observables provide different accuracy levels:** code phase for navigation (~meters) and carrier phase for surveying (~centimeters)
- **Signal processing involves sophisticated correlation techniques** to extract timing information from extremely weak signals
- **Understanding the pseudorange equation is fundamental** to comprehending how GPS positioning calculations work
- **Multiple error sources must be corrected** including clocks, atmosphere, and integer ambiguities for high-precision applications
- **The navigation message provides essential satellite and system information** required for positioning computations

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INTRODUCTION & INSTRUCTIONS

📡 Overview Module 03: Geodetic Datums



🧭 Module Overview

Despite the certainty of the physical surface of the Earth, it remains **notoriously difficult to define in mathematical terms**. The dilemma is illustrated by the ancient struggle to represent its curved surface on flat maps. There have been a whole variety of map projections developed over the centuries that rely on mathematical relationships between positions on the Earth's surface and points on the map. Each projection serves a particular application well, but none of them can represent the Earth without distortion.

As the technology of measurement has improved, the **pressure for greater exactness in the definition of the Earth's shape has increased**.

Plane surveying has traditionally relied on an imaginary flat reference surface, or ***reference frame (datum)***, with Cartesian axes. This rectangular system is used to describe measured positions by ordered pairs, usually expressed in northings and eastings, or *y*- and x- coordinates. Even though we have always known that this assumption of a flat earth is fundamentally unrealistic, it provided, and continues to provide, an adequate arrangement for small areas. The attachment of elevations to such horizontal coordinates somewhat acknowledges the topographic irregularity of the earth, but the whole system is always undone by its inherent inaccuracy as the area grows large. Even with electronic tools that widen the scope and increase the precision of the data, perfection is nowhere in sight.

Module 03 concerns some of the elements of the systems we inevitably use to accommodate the vagaries of assigning coordinates to the results of an **earth-centered-earth-fixed positioning system** based on signals from satellites.

📍 What are Geodetic Datums?

Geodetic datums are fundamental reference systems that define the size, shape, orientation, and positioning of coordinate systems used to locate points on Earth's surface. They serve as the mathematical foundation for all mapping and spatial positioning work.

📚 Module Learning Objectives

At the end of this module, learners will be able to:

- demonstrate understanding of the **basics of geodetic coordinates**
- describe **plane surveying**
- recognize the **structure** of some **geodetic coordinate systems**
- define the **elements of a geodetic datum**
- discuss the **geoid**
- explain the **North American Datum 1983**

📝 Module Instructions

- **Read Chapter 5 in *GPS and GNSS for Land Surveyors*.**
- Review the pages in this module and watch the instructor videos embedded within the module pages.
- After you have reviewed the content for this module, you will complete the assessments and activities to evaluate the learning objectives.

CONTENT

📚 GPS/GNSS and Geodesy

🌍

🗺️ Module 03: Geodetic Reference Frames (Datums)

🧭 Topic Overview

This week will be about geodetic reference frames (datums), because it is not possible to use GPS/GNSS without engaging in geodesy.

Therefore, it is necessary to understand the native format of GPS/GNSS coordinates and how the elements of geodesy influence those coordinates. Geodesy is a complicated subject, but we will be talking about just a few elements of the science.

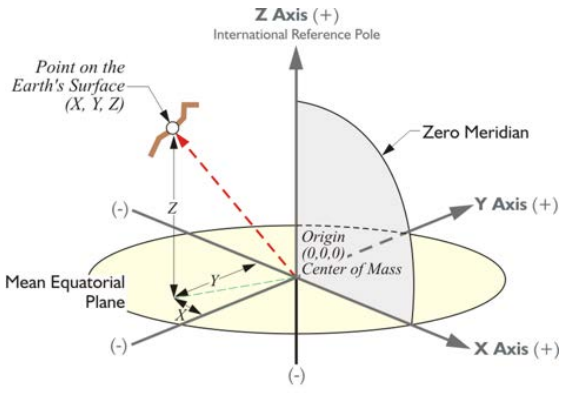
📽️ Instructor Lecture: Geodetic Reference Frames

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📚 What You'll Learn

- **ECEF Coordinates:** Understanding Earth-Centered, Earth-Fixed coordinate systems
- **Polar Motion:** How Earth's rotation affects geodetic measurements
- **Reference Frames:** The importance of datums in GPS/GNSS positioning
- **Coordinate Transformations:** Converting between different geodetic systems

> 💡 **Key Concept**
>
> GPS/GNSS coordinates are natively expressed in ECEF (Earth-Centered, Earth-Fixed) coordinates, making understanding geodetic reference frames essential for accurate positioning.



Three-Dimensional Cartesian Coordinate (ECEF)

Three-Dimensional Cartesian Coordinate (ECEF)

🌐 GPS/GNSS and Geodesy

Today, GPS/GNSS has thrust us into the thick of geodesy, which is no longer the exclusive realm of distant experts. Thankfully, in the age of the microcomputer, the computational drudgery can be handled with software packages. Nevertheless, it is unwise to venture into GPS/GNSS believing that knowledge of the basics of geodesy is, therefore, unnecessary. It is true that GPS/GNSS would be impossible without computers, but blind reliance on the data they generate eventually leads to disaster.



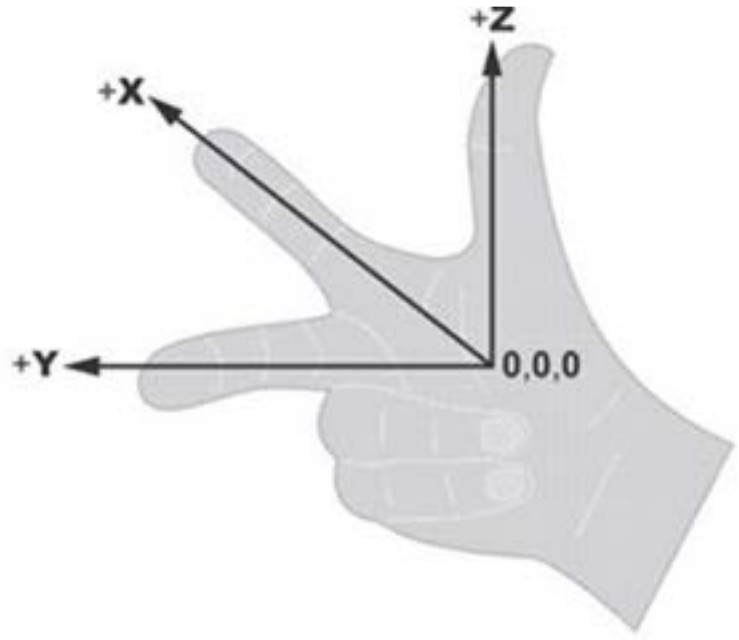
A Hand-Held GPS Receiver

📐 Some Geodetic Coordinate Systems

🔷 Three-Dimensional (3-D) Cartesian Coordinates

A spatial Cartesian system with three axes lends itself to describing the terrestrial positions derived from space-based geodesy. Using three rectangular coordinates instead of two, one can unambiguously define any position on the Earth, or above it for that matter. The three-dimensional Cartesian coordinates (x,y,z) derived from this system are known as Earth-Centered-Earth-Fixed (ECEF) coordinates. It is a right-handed orthogonal system that rotates with and is attached to the Earth, which is why it is called *Earth fixed*.

A three-dimensional Cartesian coordinate system is right-handed if it can be described by the following model: the extended forefinger of the right hand symbolizes the positive direction of the x-axis. The middle finger of the same hand extended at right angles to the forefinger symbolizes the positive direction of the y-axis. The extended thumb of the right hand, perpendicular to them both, symbolizes the positive direction of the z-axis.



Right-Handed System

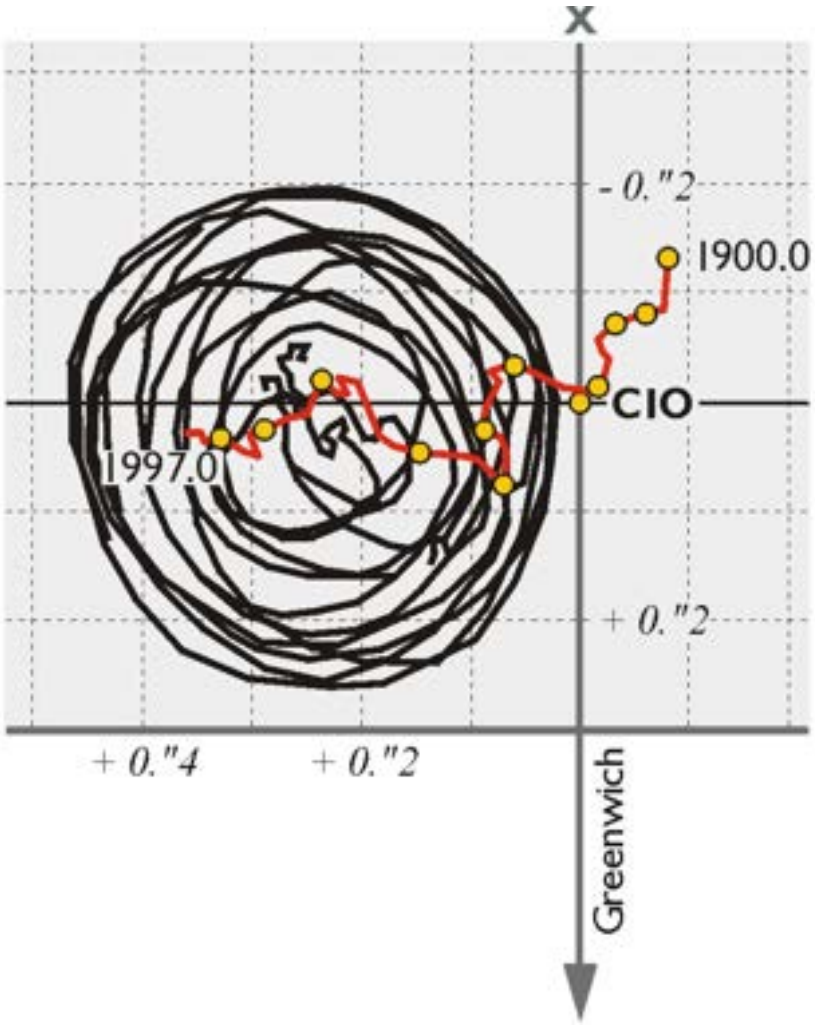
But such a system is only useful if its origin (0,0,0) and its axes (x,y,z) can be fixed to the planet with certainty, something easier said than done.

This illustration shows the three-dimensional Cartesian coordinate system. It is possible to express a point on the Earth's surface in terms of these x, y, and z-coordinates in this Earth-Centered, Earth-Fixed, or ECEF system. This is the native system in which GPS/GNSS coordinates are expressed; obviously, the plane of the equator, the mean equatorial plane, and the zero meridian (aka Prime Meridian), are vital elements of this system.

The usual arrangement is known as the *Conventional Terrestrial Reference System (CTRS)*, and the *Conventional Terrestrial System (CTS)*. The latter name will be used here. The origin is the center of mass of the whole Earth including oceans and atmosphere, the *geocenter*. Since the satellites orbit around the center of mass of the Earth, it is sensible to have the coordinate system derived from satellites with its origin at the center of mass. It can be used to define a position on the Earth's surface or, for that matter, a position above the Earth's surface. It is possible to have an XYZ coordinate in this system of an orbiting satellite. The x-axis is a line from that geocenter through its intersection at the zero meridian, also known as the *International Reference Meridian (IRM)*, with the internationally defined conventional equator. The y-axis is extended from the geocenter along a line perpendicular from the x-axis in the same mean equatorial plane toward 90° East longitude. That means that the positive end of the y-axis intersects the actual Earth in the Indian Ocean. In any case, they both rotate with the Earth around the z-axis, a line from the geocenter through the internationally defined pole known as the International Reference Pole *(IRP)*.

🌀 Polar Motion

The abstract idea of a three-dimensional coordinate system is perfect in the theoretical sense, but when you attach it to the actual, physical Earth, difficulties arise. For example, the Earth's rotational axis just won't hold still. The Earth wobbles so the z-axis of this Earth-Centered, Earth-Fixed three-dimensional Cartesian system is fixed by international agreement. The Earth is constantly moving, of course. While one can say that the Earth has a particular axis of rotation, equator, and zero meridian for an instant, they all change slightly in the next instant. Within all this motion, how do you stabilize the origin and direction of the three axes for the long term? One way is to choose a moment in time and consider them fixed to the Earth as they are at that instant.



Polar Motion

Here is an example of that process of definition. The Earth's rotational axis wanders slightly with respect to the solid Earth in a very slow oscillation called *polar motion*. The largest component of the movement relative to the Earth's crust has a 430-day cycle known as the *Chandler period*. It was named after American Astronomer Seth C. Chandler, who described it in papers in the Astronomical Journal in 1891. Another aspect of polar motion is sometimes called *polar wander*. The conventional terrestrial system of coordinates would be useless if its third axis was constantly wobbling. Originally, an average stable position was chosen for the position of the pole. Between 1900 and 1905, the mean position of the Earth's rotational pole was designated as the *Conventional International Origin (CIO)* and the *z*-axis. This was defined by the *Bureau International de l'Heure (BIH)*. It has since been refined by *the International Earth Rotation Service, (IERS)* using very long baseline interferometry (VLBI) and satellite laser ranging (SLR). It is now placed as it was midnight on New Year's Eve 1983, or January 1, 1984 (UTC). The moment is known as an *epoch* and can be written 1984.0. So, we now use the axes illustrated. The name of the *z*-axis has been changed to the International Reference Pole, IRP epoch 1984, but it remains within 0.005" of the previous definition. It provides a geometrically stable and clear definition of the Earth's surface for the *z*-axis.

ITRF2014
CDOT GOLDEN (CTMC), COLORADO

Antenna Reference Point (ARP) : CDOT GOLDEN CORS ARP

PID =DM5962

ITRF2014 POSITION (EPOCH 2010.0)
Computed in Jun 2019 using data through gpswk 1933.

X = -1287788.532 m	latitude = 39 43 17.51335 N
Y = -4742177.053 m	longitude = 105 11 34.38099 W
Z = 4055414.539 m	height = 1818.969 m

ITRF2014 VELOCITY
Computed in Jun 2019 using data through gpswk 1933.

VX = -0.014 m/yr	northward = -0.0061 m/yr
VY = -0.0001 m/yr	eastward = -0.0142 m/yr
VZ = -0.0047 m/yr	upward = 0.0000 m/yr

Station CTMC

Start Time:	12/14/93 23:24:45 GPS		(727 257085)
Stop Time:	12/14/93 23:55:15 GPS		(727 258915)
Occupation Time:	00:30:30:00		
Measurement Epoch Interval (seconds):	15:00		
Solution Type:	Receiver/Satellite double difference Iono free fixed		
Solution Acceptability:	Passed ratio test		
Baseline Slope Distance Std. Dev. (meters):	17044.376		0.000574
		Forward	*Backward*
Normal Section Azimuth:		179° 04' 38.886169"	359° 04' 47.857511"
		0° 01' 00.456589"	-0° 16' 12.548396"
Baseline Components (meters):	dn -17042.131	de 274.423	du 34.744
Standard Deviations:	dx -6552.297	dy -10264.496	dz -11925.530
	8.437168E-004	1.072974E-003	9.513724E-004
Apostetrior Covariance Matrix:	7.118380E-007		
	7.389287E-007	1.151273E-006	
	-6.141036E-007	-7.690377E-007	9.051094E-004
Variance Ratio Cutoff:	62.1	1.5	
Reference Variance:	0.556		
Observable Count/Rejected RMS:	Iono free phase	451/0	0.006

Differences in ECEF X,Y and Z

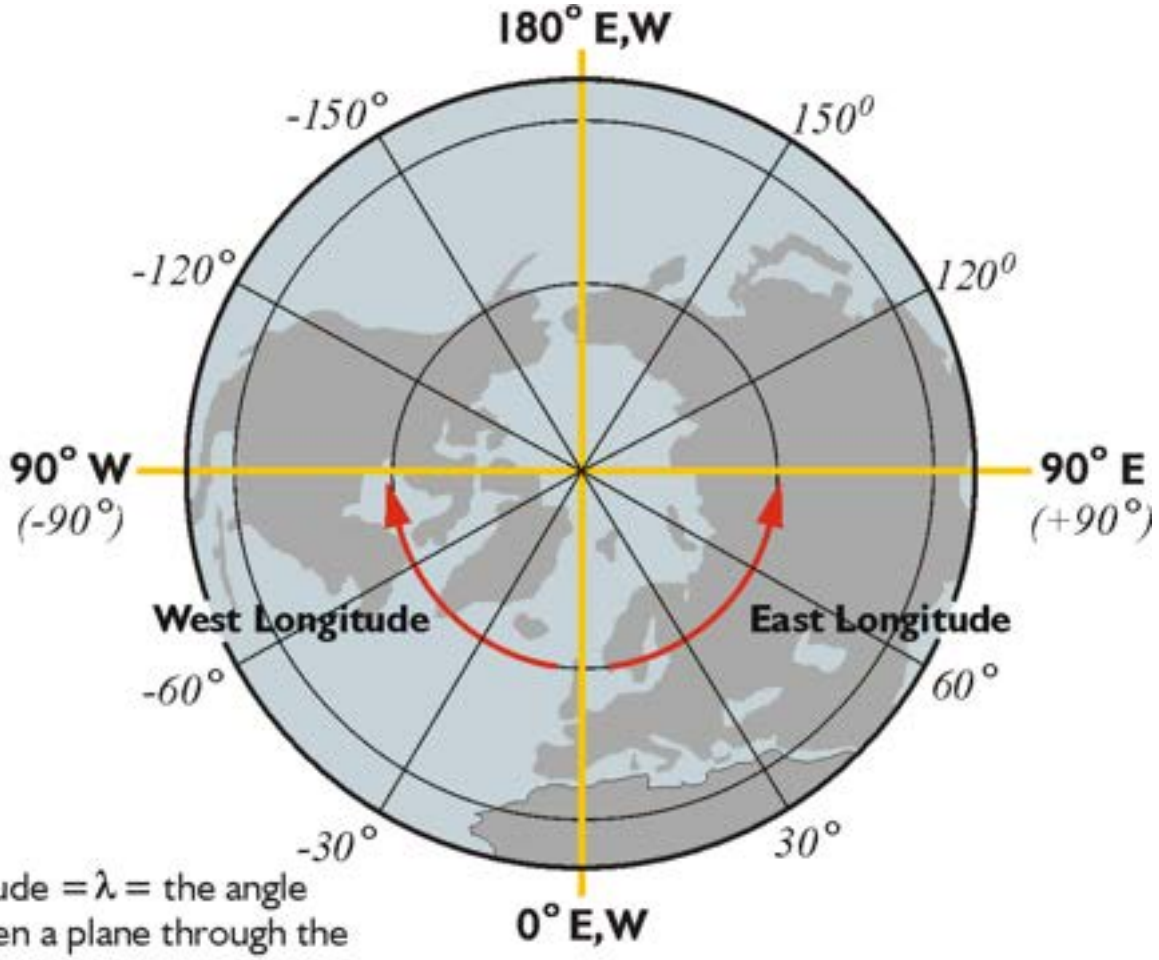
In this three-dimensional right-handed coordinate system, the x-coordinate is a distance from the y-z plane measured parallel to the x-axis. It is always positive from the zero meridian to 90° W longitude and from the zero meridian to 90° E longitude. In the remaining 180°, the X-coordinate is negative. The y-coordinate is a perpendicular distance from the plane of the zero meridian. It is always positive in the Eastern Hemisphere and negative in the Western Hemisphere. The z- coordinate is a perpendicular distance from the plane of the equator. It is always positive in the Northern Hemisphere and negative in the Southern Hemisphere. Here, above, is an example — the position of the station CTMC expressed in three-dimensional Cartesian coordinates of this type expressed in meters, the native unit of the system.

It is important to note that the GPS/GNSS Control Segment generates the position and velocity of the satellites themselves in ECEF coordinates. It follows that most modern GPS/GNSS software provides the GPS/GNSS positions in ECEF, as well. Further, the ends of baselines determined by GPS/GNSS observation are typically given in ECEF coordinates, so that the vectors themselves become the difference between those *x*, *y*, and *z* coordinates. The display of these differences as *DX*, *DY*, and *DZ* is a usual product of these post-processed calculations. This is the way that GPS/GNSS defines vectors. From these coordinates, it is possible to derive any number of other coordinates, i.e., Universal Transfers Mercater System coordinates, State Plane coordinates, latitude, and longitude. All these can be derived from the XYZ coordinates in the three-dimensional Cartesian Earth-Centered, Earth-Fixed system.

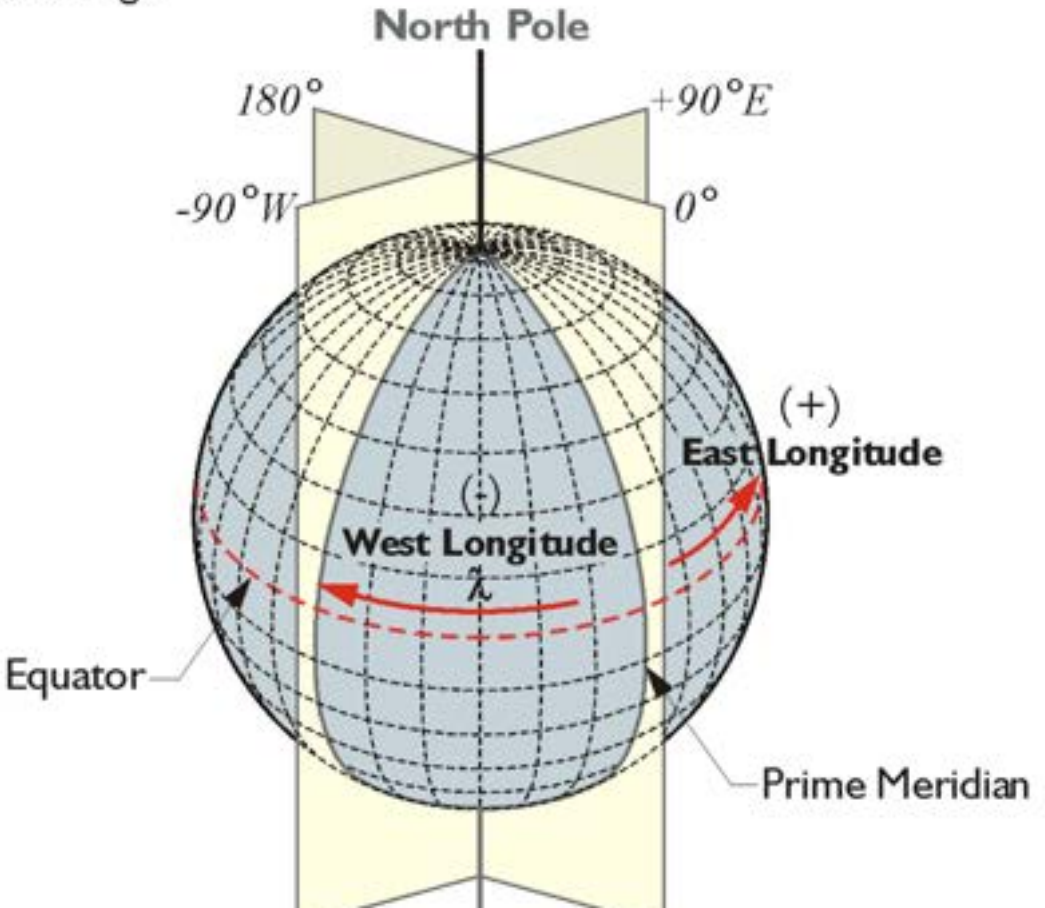
🗺️ Latitude and Longitude

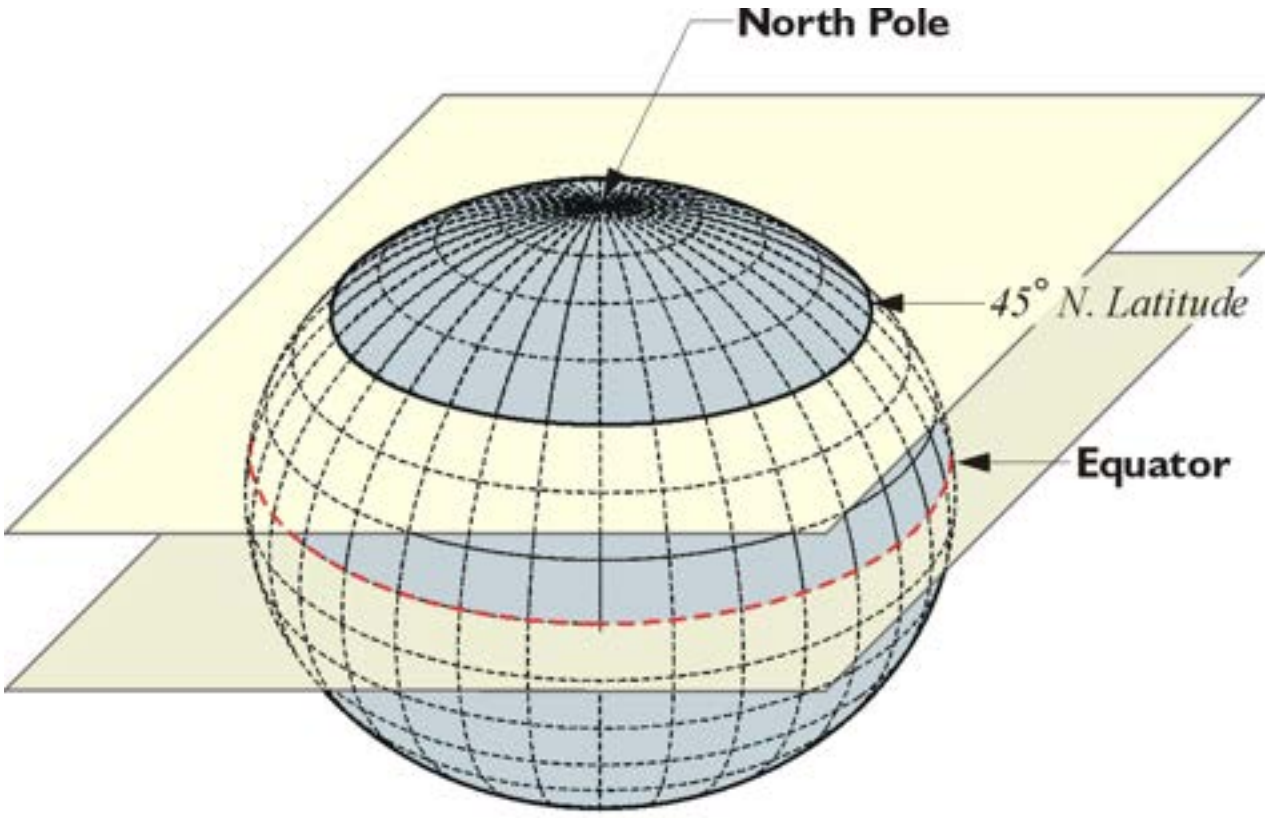
Despite their utility, such 3-D Cartesian coordinates are not the most common method of expressing a geodetic position. Latitude and longitude have been the coordinates of choice for centuries. The designation of these rely on the same two standard lines as 3D Cartesian coordinates: the mean equator and the zero meridian. Unlike them, however, they require some clear representation of the terrestrial surface. In modern practice, latitude and longitude cannot be said to uniquely define a position without a clear definition of the earth itself.

The Prime Meridian is through Greenwich, or approximately through Greenwich. West longitude is negative, is in our neck of the woods. East longitude going the other way around, meeting at the 180 degree longitudinal value. Longitude is usually symbolized by the Greek letter lambda. Latitude is measured from the equator north and south, north being positive and south negative. The plane in the illustration indicates 45 North latitude. There is more than one kind, more than one flavor, of latitude and longitude. In other words, a WGS84 (G1762) latitude and longitude is not the same as a NAD83(2011) latitude and longitude.



Longitude = λ = the angle between a plane through the prime meridian and a plane on the meridian though a point.





Parallels of Latitude

Texas A&M University - Corpus Christi
College of Engineering & Computer Science
GISC 4315 - Satellite Positioning
Fall 2026

📚 Elements of a Geodetic Reference Frame (Datum)

📐 Elements of a Geodetic Reference Frame (Datum)

Understanding Coordinate Systems and Vertical References

🧭 Topic Overview

How can latitude and longitude not define a unique position on the Earth? The reference lines—the mean equator and the zero meridian—are clearly defined, and the units allow for precise measurements. The answer lies in understanding that **there are several categories of latitude and longitude**, and **their definition depends largely on where *"down"* is**. This fundamental concept is crucial for understanding how GPS/GNSS coordinates relate to traditional surveying methods.

📏 Deflection of the Vertical

🎯 What Defines "Down"?

To understand geodetic reference frames, we must first discuss the deflection of the vertical. Consider the astronomical coordinates of station Youghall in North America:

🌟 Astronomical Coordinates

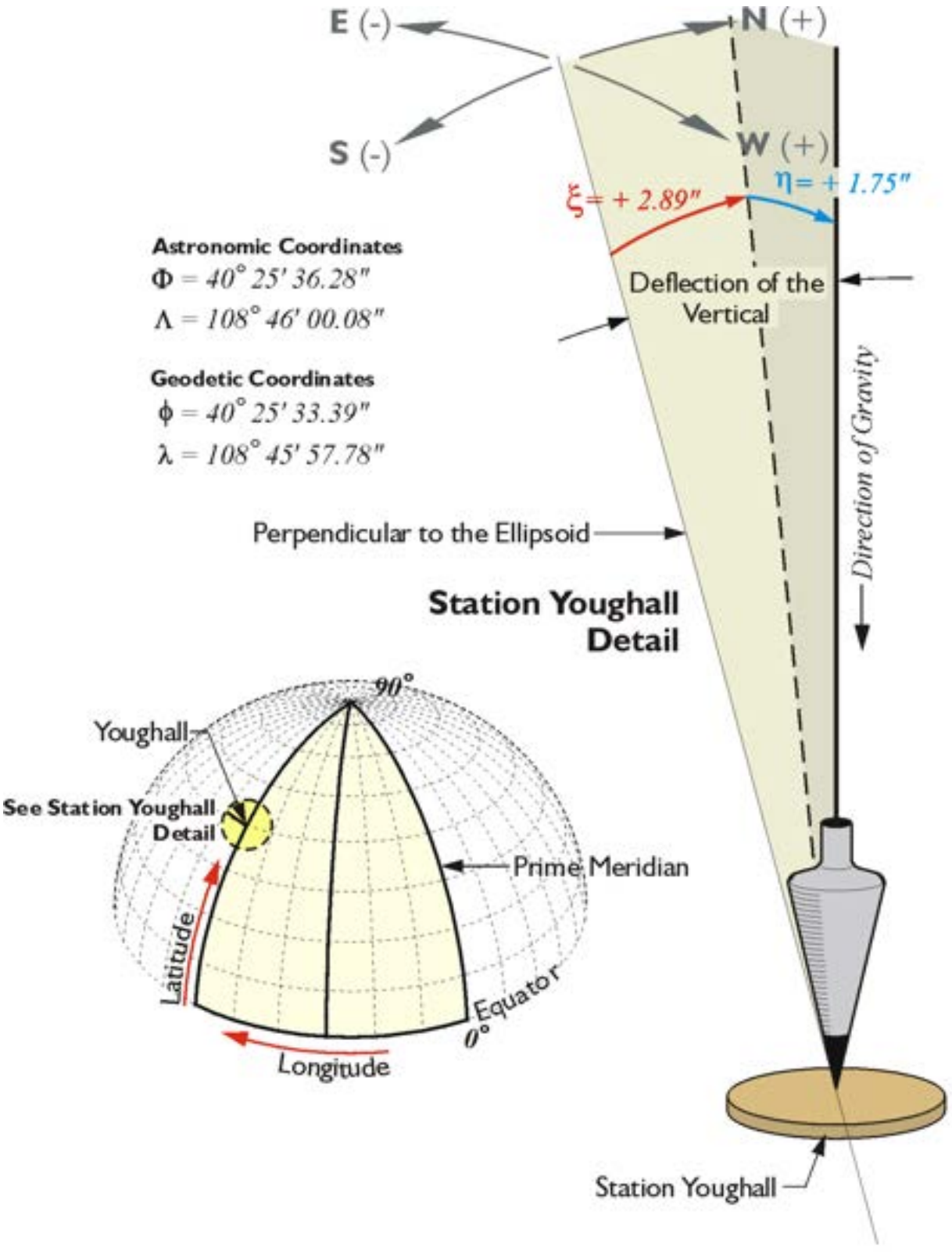
- Latitude: 40° 25' 36.28"
- Longitude: 108° 46' 00.08"
- **Based on:** Direction of gravity (plumb line)

📐 Geodetic Coordinates

- Latitude: 40° 25' 33.39"
- Longitude: 108° 45' 57.78"
- **Based on:** Perpendicular to ellipsoid

These coordinates differ because *"down"* is governed by gravity for astronomical coordinates, while *"down"* is perpendicular to the ellipsoid for geodetic coordinates. The ellipsoid is a smooth,

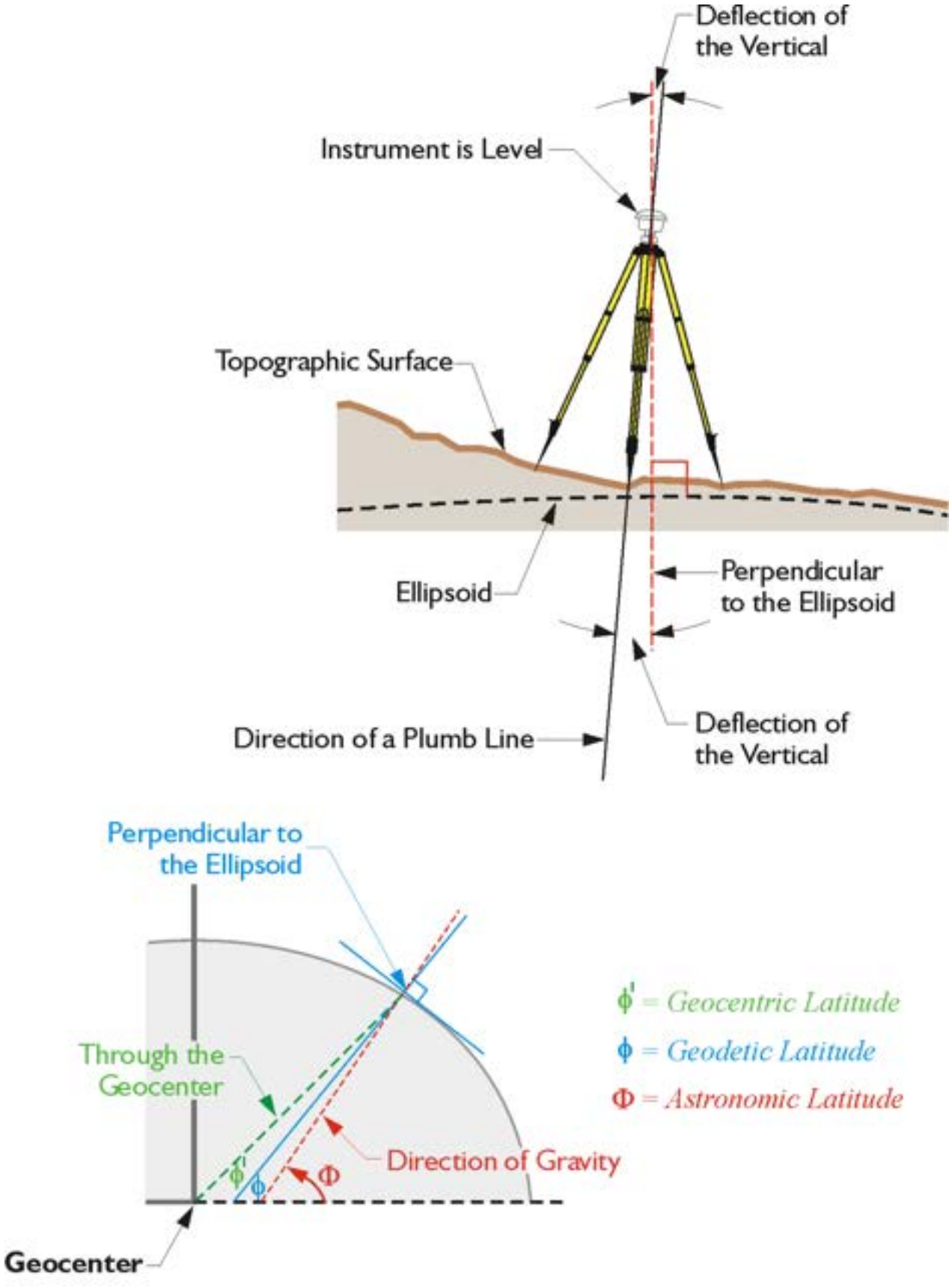
mathematically determined representation of the Earth, free from the irregularities of the actual terrestrial surface and gravity.



Deflection of the Vertical

Three Types of Latitude

📍 Understanding Different Latitude Systems



Different Types of Latitude

In the illustration above, an instrument is set up on the Earth's topographic surface. Notice the three different lines representing different concepts of "vertical":

🔴 **Geodetic Latitude**

Red dashed line: Perpendicular to the ellipsoid at the point. This is the most commonly used latitude system and the basis for modern mapping.

🟢 **Geocentric Latitude**

Green dashed line: From the point on Earth's surface to the geocenter (center of mass of the Earth).

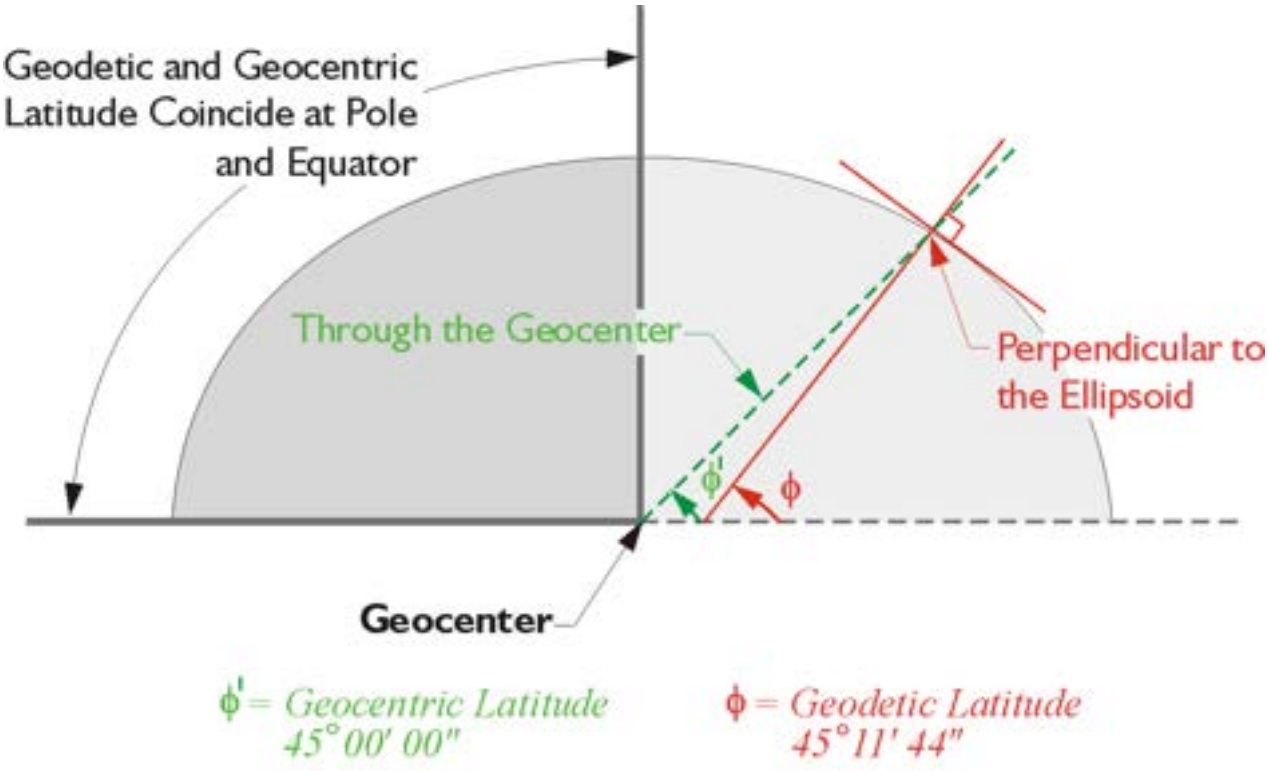
⚫ **Astronomical Latitude**

Black line: Direction of "down" according to a plumb line (gravity). Used in traditional optical surveying.

The difference between the vertical extension of a plumb line and the vertical extension of a line perpendicular to the ellipsoid is called the *deflection of the vertical*. This deflection usually has both north-south and east-west components, affecting both latitude and longitude measurements, as well as azimuths.

🛰️ GPS/GNSS and Coordinate System Relationships

🔄 The Evolution from Astronomical to Geodetic Coordinates



Geodetic and Geocentric Latitudes

GPS/GNSS has fundamentally changed how we determine position coordinates. Before GPS/GNSS, latitude and longitude were determined primarily through astronomical observations using optical instruments carefully leveled with plumb bobs. These instruments, when properly set up, measured astronomical coordinates because they were oriented to gravity.

> 💡 **Key Historical Context**
>
> Traditional surveying instruments set up with plumb bobs and carefully leveled could only measure astronomical latitude and longitude directly. To obtain geodetic coordinates, these measurements required post-observation calculations and corrections by organizations like the National Geodetic Survey.

However, the most commonly used coordinates today are not astronomical latitudes and longitudes, but **geodetic latitudes and longitudes**. GPS/GNSS receivers can display geodetic coordinates immediately because modern computers can complete the necessary calculations with incredible speed.

⚠️ Critical Points to Remember

> 🎯 **Most Important Concept**
>
> **Latitudes and longitudes are dependent upon the reference frame (datum) on which they're measured.** A WGS84 latitude and longitude is NOT the same as a NAD83 latitude and longitude for the same physical point on Earth.

🔧 Instrument Reality

- All instruments are oriented to gravity
- GPS/GNSS antennas follow this rule
- Optical surveying instruments
- Aerial photography cameras
- Even GPS/GNSS satellites themselves

📐 Ellipsoidal Model

- Mathematical representation of Earth
- Not affected by gravity irregularities
- Basis for geodetic coordinates
- Cannot physically set up instruments on it
- Requires mathematical conversion

The fundamental fact remains: instruments are oriented to gravity, but the ellipsoidal model on which geodetic coordinates are based is not. When we speak of latitude and longitude in modern practice, we almost always mean **geodetic latitude and longitude based upon an ellipsoidal model of the Earth**.

💡 Key Takeaways

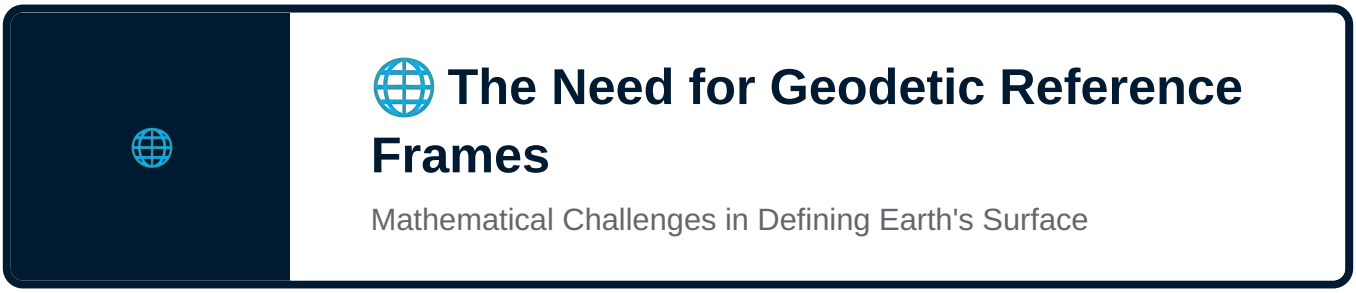
🎯 Bottom Line Up Front

- **Coordinate systems are not universal** - the same physical point has different coordinates in different datums
- **Three types of latitude exist** based on different definitions of "vertical": astronomical, geodetic, and geocentric
- **Deflection of the vertical** explains why astronomical and geodetic coordinates differ for the same point
- **GPS/GNSS changed everything** by providing direct access to geodetic coordinates through rapid computation
- **All physical instruments respond to gravity** while coordinate calculations are based on mathematical models
- **Modern coordinates are geodetic** - based on ellipsoidal models rather than gravity or Earth's center

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CONTENT

📚 Datums (Reference Frames)



🌐 The Need for Geodetic Reference Frames

Mathematical Challenges in Defining Earth's Surface

🧭 Topic Overview

The second part of the answer to our earlier question is this: **if geographic coordinates are to have meaning, they must have a context—a Reference Frame (Datum)**. Despite the physical certainty of Earth's surface, it remains notoriously difficult to define in mathematical terms. This fundamental challenge has driven centuries of innovation in geodesy and continues to influence how we use GPS/GNSS technology today.

📐 The Mathematical Challenge of Earth's Surface



Part of the Earth's Lithosphere

NASA image using Landsat data to texture-map surface created using SRTM elevation data

🌍 The Physical Reality Problem

Despite the certainty of the physical surface of the Earth—the **lithosphere**—it remains notoriously difficult to define in mathematical terms. This dilemma is illustrated by the ancient struggle to represent Earth's curved surface on flat maps.

The challenge is fundamental: *How do you mathematically describe something as complex and irregular as Earth's actual surface?* Every mountain, valley, ocean trench, and building adds to the complexity that cannot be captured in simple mathematical equations.

> ⚠️ **Reality Check**
>
> No modern surveyor would presume to promise a client a high-precision control network with data scaled from a map. The Earth's surface is simply too complex for perfect representation.

🗺️ Map Projections: The Impossible Task

📊 The Historical Struggle

There have been a whole variety of map projections developed over the centuries that rely on mathematical relationships between positions on the Earth's surface and points on the map. This represents humanity's ongoing attempt to solve an impossible mathematical problem.

✅ What Projections Can Do

- Serve particular applications well
- Preserve specific properties (area, distance, or direction)
- Enable mathematical calculations
- Facilitate map-based navigation

❌ What They Cannot Do

- Represent Earth without distortion
- Preserve all geometric properties simultaneously
- Provide perfect accuracy everywhere
- Eliminate the need for geodetic expertise

🎯 The Fundamental Truth

Each projection serves a particular application well, but none of them can represent the Earth without distortion. This is not a limitation of technology—it's a mathematical impossibility.

🎯 The Ever-Increasing Demand for Precision

📈 Technology Drives Accuracy Demands

As the technology of measurement has improved, the pressure for greater exactness in the definition of the Earth's shape has increased dramatically. What was acceptable accuracy for maritime navigation centuries ago is completely inadequate for modern GPS/GNSS applications.

📅 Historical Accuracy

- **18th Century:** Kilometers of uncertainty acceptable

- **19th Century:** Hundreds of meters sufficient
- **Early 20th Century:** Meters adequate for most work
- **Mid-20th Century:** Decimeters required for surveying

🚀 **Modern Requirements**

- **GPS Era:** Centimeter-level accuracy standard
- **RTK Applications:** Millimeter precision needed
- **Scientific Applications:** Sub-millimeter requirements
- **Future Demands:** Even greater precision expected

Even with electronic tools that widen the scope and increase the precision of the data, **perfection is nowhere in sight**. This is not a technological failing—it reflects the fundamental challenge of mathematically defining an infinitely complex physical reality.

💡 The Practical Solution: Geodetic Reference Frames

Since we cannot perfectly define Earth's complex surface, we create simplified mathematical models called **geodetic reference frames** or **datums**. These provide the essential context that makes geographic coordinates meaningful and usable.

🔑 **Key Insight**

Rather than trying to mathematically describe every hill and valley, geodetic reference frames use simplified geometric shapes (primarily ellipsoids) that approximate Earth's overall form. This compromise between accuracy and mathematical tractability makes modern positioning systems possible.

🌍 **What Reference Frames Provide**

- Mathematical consistency
- Standardized coordinate systems
- Reproducible measurements
- Global interoperability
- Computational efficiency

⚖️ **What They Sacrifice**

- Perfect local accuracy
- Complete surface representation
- Absolute physical reality
- Universal applicability
- Elimination of distortions

💡 Key Takeaways

🎯 Bottom Line Up Front

- **Context is everything** - geographic coordinates are meaningless without a reference frame (datum)
- **Earth's surface is too complex** for perfect mathematical representation
- **All map projections involve distortion** - it's mathematically impossible to perfectly represent a curved surface on a flat plane
- **Precision demands have escalated** from kilometers to millimeters as technology advances
- **Reference frames are practical compromises** that balance accuracy with mathematical tractability
- **Perfection remains elusive** despite technological advances - the challenge is fundamental, not technological

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CONTENT

📚 Development of the Ellipsoidal Model



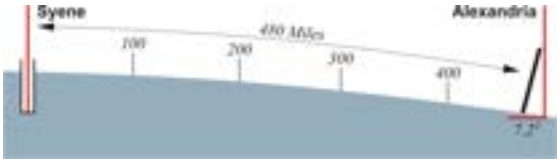
🌍 Historical Development of Earth's Shape Understanding

From Spherical to Ellipsoidal Models

🧭 Topic Overview

The first step in representing positions on the Earth has always been the building of a model of the Earth. **We have been trying to grasp the nature of the planet's size and shape for a long time.** This journey from ancient Greek calculations to Newton's revolutionary insights laid the foundation for modern geodetic reference systems that make GPS/GNSS possible today.

🏛️ Ancient Beginnings: Eratosthenes (200 BC)



Eratosthenes Calculation

📐 The First Scientific Measurement

In 200 BC, Eratosthenes almost got the planet's circumference right, establishing the fundamental idea that it was in fact nearly a sphere. His method was elegantly simple yet remarkably accurate.

🔍 The Method

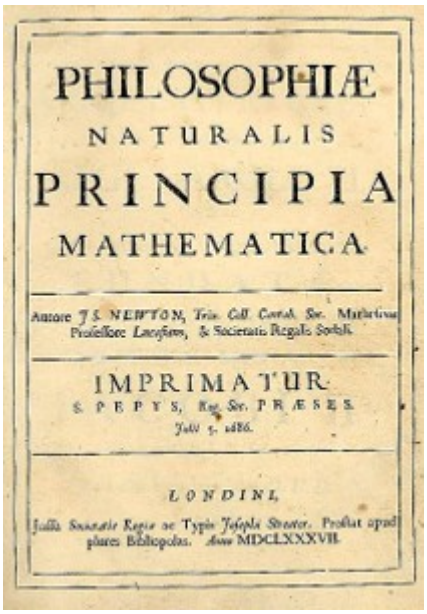
- **Known distance:** Syene was 489 miles south of Alexandria (measured by *rope stretchers*)
- **Key observation:** At noon on midsummer in Syene, no shadows existed—sun was directly overhead
- **Alexandria difference:** On the same day, a pillar cast a shadow forming a 7.2° angle
- **Geometric reasoning:** 7.2° = 1/50th of 360°

📊 The Calculation

Earth's circumference = 50 × 489 miles = ~24,450 miles

Actual circumference: ~24,874 miles - extraordinary accuracy for 200 BC!

🔬 Newton's Revolutionary Insight (1687)


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA
Auctore J.S. NEWTON, Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.
IMPRIMATUR
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.
LONDINI,
Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures Bibliopolas. Anno MDCLXXXVII.

Principia (1687)

🌐 From Sphere to Ellipsoid

Newton's thesis that the Earth was an ellipsoid rather than a sphere came in the first edition of his *Principia* in 1687. This revolutionary idea was not entirely independent—there had been suggestive observations.

🕰️ The Pendulum Clock Evidence

15 years earlier, astronomer **Jean Richer** found that to maintain accuracy of his one-second clock in Cayenne, French Guiana, he had to **shorten its pendulum significantly**.

The clock's pendulum, regulated in Paris, tended to swing more slowly as it approached the equator.

🧠 Newton's Reasoning

Newton attributed this phenomenon to a lessening of the force of gravity. Based on his theoretical work, he proposed:

> *"the earth is higher under the equator than at the poles, and that by an excess of about 17 miles"*

— Philosophiae Naturalis Principia Mathematica, Book III, Proposition XX

⚔️ The Great Scientific Controversy

🏛️ The Cassini Opposition

Obviously, if the Earth is more massive at the equator, bulging out at the equator, the Earth is ellipsoidal—like a squashed ball that bulges out in the middle. Although Newton's model was

supported by some contemporaries, notably Huygens (inventor of Richer's clock), it was attacked by others.



Early Baseline Measurements

👩‍🔬 The Cassini Family's Position

The director of the Paris Observatory, **Jean Dominique Cassini**, took exception to Newton's concept. Even though he had observed the flattening of Jupiter's poles in 1666, neither he nor his son Jacques would accept the same idea for Earth.

🔪 Their "Evidence"

Arc measurements seemed to show the length of one degree of longitude actually *decreased* as it proceeded northward, suggesting the Earth was **prolate** (elongated at the poles) rather than **oblate** (flattened at the poles).

🔬 Arc Measurement Method

Scientists employed arc measurements at various latitudes for geometric verification. They established latitude astronomically at beginning and ending points, then measured north-south cardinal lines to discover the length of one degree of longitude along a meridian arc. One of the most accurate measurements, begun in 1669 by French Abbé J. Picard, was actually used by Newton in formulating his law of gravitation.

🗺️ The Definitive Expeditions (1733-1744)

🌍 Two Expeditions, One Answer

The argument was not resolved until two expeditions between about 1733 and 1744 were completed. They were sponsored by the Paris Académie Royale des Sciences and produced irrefutable proof.

🧊 Arctic Expedition

- **Location:** Near Arctic Circle (66°20' N)
- **Team:** Clairaut and Maupertuis
- **Mission:** Measure meridian arc in Lapland
- **Purpose:** Test degree length at high latitude

🌴 Equatorial Expedition

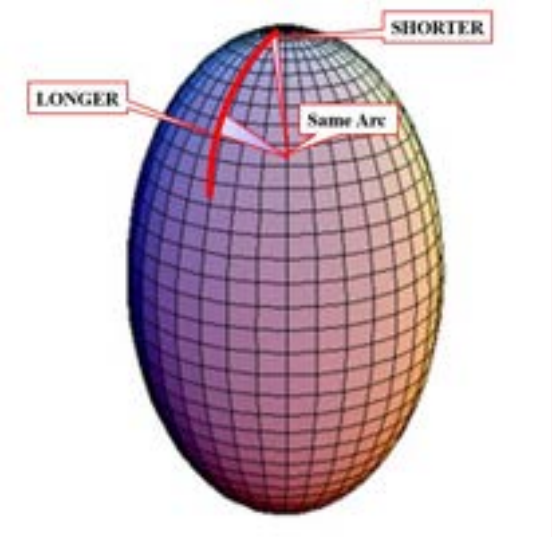
- **Location:** Near equator (01°31' S)
- **Team:** Bouguer and Godin
- **Mission:** Measure arc in what is now Ecuador
- **Purpose:** Test degree length at low latitude

🎯 The Definitive Result

Newton's conjecture was proved correct! The contradictory evidence of Picard's arc measurements was attributed to errors in the measurement of astronomical latitudes.

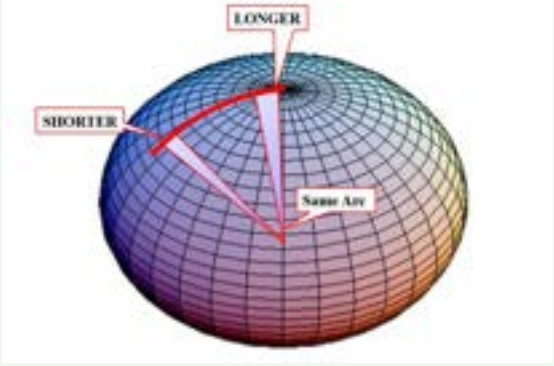
🥚 Prolate vs Oblate: Understanding Earth's True Shape

🔄 The Shape Debate Resolved



❌ Prolate Ellipsoid (Incorrect)

Elongated at poles - what Cassini believed



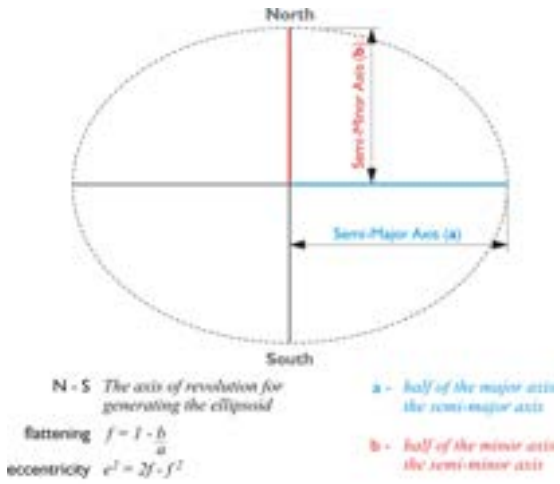
✅ Oblate Ellipsoid (Correct)

Flattened at poles - Newton was right

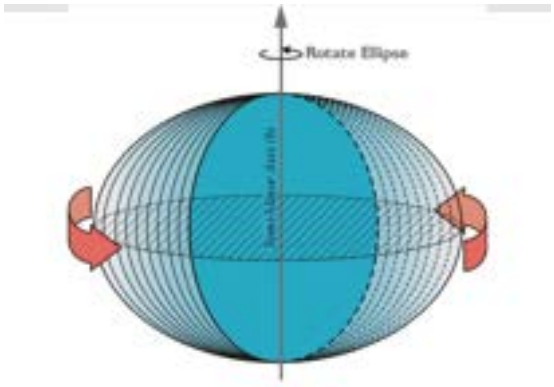
Eventually, these measurements proved Newton correct—the Earth is flattened at the poles and is an oblate ellipsoid.

The Modern Ellipsoidal Model

Defining the Ellipsoid



Ellipse Parameters



3D Ellipsoid

The ellipsoidal model, bulging at the equator and flattened at the poles, has been used ever since as a representation of the general shape of the Earth's surface. Several reference ellipsoids have been established for various regions of the planet.

📏 **Key Parameters**

- **Semi-major axis (a):** Half of the major axis of the ellipse (shown in blue) - the equatorial radius
- **Semi-minor axis (b):** Half of the minor axis of the ellipse - the polar radius
- **Flattening (f):** Measures how much the ellipsoid deviates from a perfect sphere
- **Precise definition:** All reference ellipsoids are defined by their semi-major axis and flattening

🔄 **Creating the 3D Model**

The ellipse can be rotated about the north-south semi-minor axis into the representation of a solid ellipsoid—in this case, a **bi-axial ellipsoid**. It bulges at the equator and is flattened at the poles, providing the mathematical foundation for modern geodetic reference systems.

💡 Key Takeaways

🎯 **Bottom Line Up Front**

- **Ancient wisdom was remarkable** - Eratosthenes calculated Earth's circumference with extraordinary accuracy in 200 BC
- **Newton revolutionized our understanding** by proposing Earth was ellipsoidal, not spherical
- **Scientific controversy drove progress** - the Cassini family's opposition led to definitive expeditions
- **Empirical evidence settled the debate** - Arctic and equatorial measurements proved Newton correct
- **Earth is an oblate ellipsoid** - flattened at poles, bulging at equator
- **Mathematical precision is essential** - ellipsoids are defined by semi-major axis and flattening parameters
- **This foundation enables modern GPS** - understanding Earth's shape is crucial for accurate positioning systems

GISC 4315 - Satellite Positioning
Fall 2026

📚 NAD27

📍 NAD27: The North American Datum of 1927

🧭 Understanding NAD27

The North American Datum of 1927 (NAD27) represents a classic example of how a mathematical ellipsoid is transformed into a practical geodetic datum through careful positioning and orientation. This page explores the **establishment of NAD27 at Meades Ranch, Kansas**, and examines how **regional datums were designed to serve pre-GPS surveying needs**.

📽️ Instructor Lecture: NAD27 Development

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📚 What You'll Learn

- **Clarke 1866 Ellipsoid:** Technical specifications and regional optimization
- **Meades Ranch:** The initial point establishment in Kansas
- **Datum Orientation:** How azimuth and direction complete the system
- **Regional Design:** Why NAD27 was non-geocentric

💡 Key Concept

NAD27 demonstrates how six additional elements transform the Clarke 1866 ellipsoid into a complete geodetic datum: three for center position and three for orientation.
Read more about Meades Ranch





The Monument at the Initial Point of NAD27

🌐 The Role of an Ellipsoid in a Datum

The semimajor axis and flattening can be used to completely define an ellipsoid of revolution. The ellipsoid is revolved around the minor axis. However, in the traditional approach, six additional elements are required if that ellipsoid is to become a geodetic datum: three to specify its center and three more to clearly indicate its orientation around that center. The Clarke 1866 spheroid is one of many reference ellipsoids. Its shape is completely defined by a semimajor axis, a, of 6378.2064 km and a flattening, f, of 1/294.9786982. It is the reference ellipsoid of the datum known as the North American Datum of 1927 (NAD27), but it is not the datum itself.

🔍 **Clarke 1866 Ellipsoid Specifications**

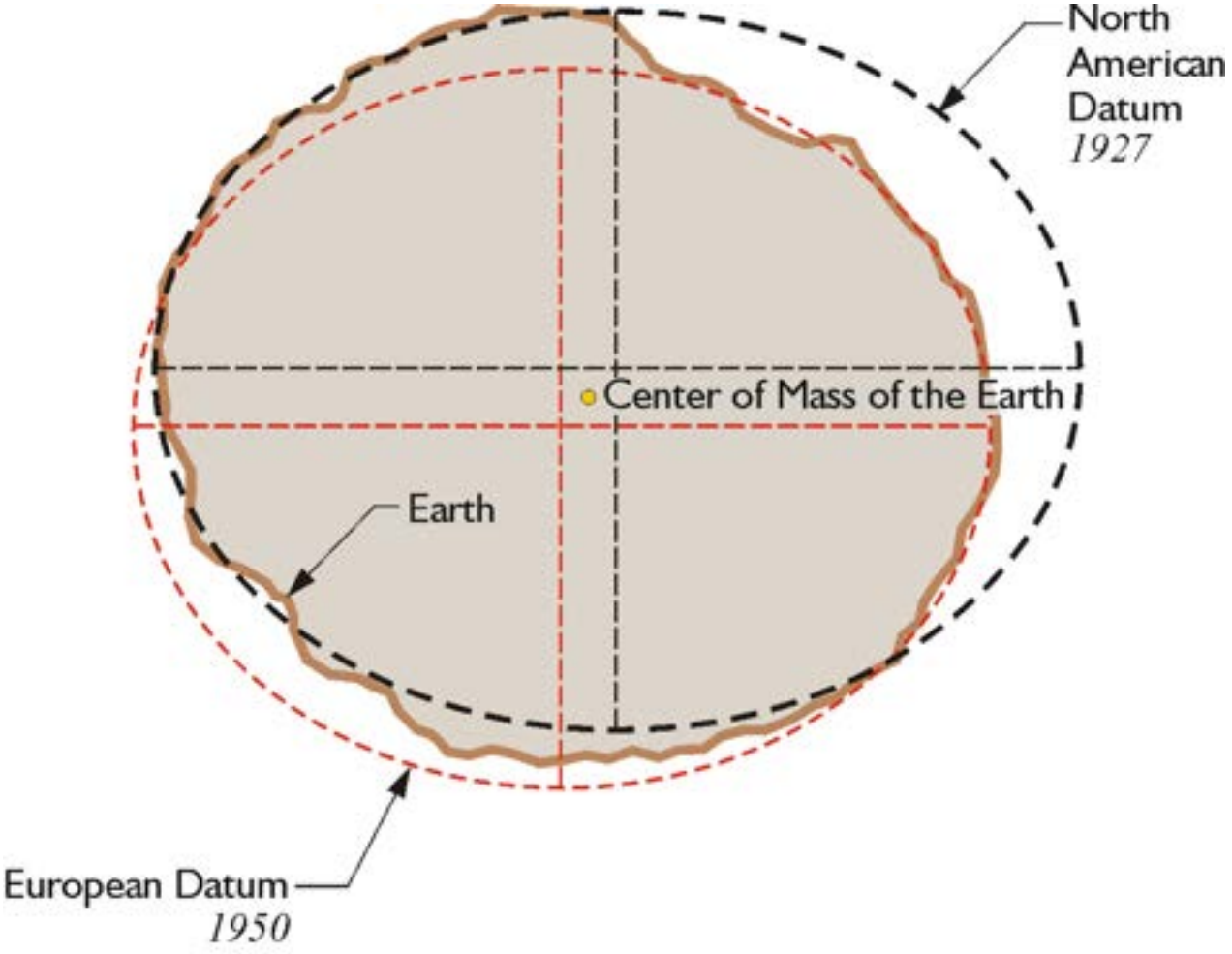
Semimajor axis (a): 6378.2064 km
Flattening (f): 1/294.9786982
Usage: Reference ellipsoid for NAD27

For the Clarke 1866 spheroid to become NAD27, the ellipsoid of reference had to be attached at a point and specifically oriented to the actual surface of the earth. However, even this ellipsoid, which fits North America well, could not conform to that surface perfectly. Therefore, the initial point was chosen near the center of the anticipated geodetic network to best distribute the inevitable distortion between the ellipsoid and the actual surface of the earth as the network extended beyond the initial point. The attachment was established at Meades Ranch, Kansas, 39°13'26".686 Nφ, 98°32'30".506 Wλ, and geoidal height was considered to be zero. Those coordinates were not sufficient, however. The establishment of directions from this initial point was required to complete the orientation.

📍 **Meades Ranch Initial Point**

Location: Meades Ranch, Kansas
Latitude: 39°13'26".686 N
Longitude: 98°32'30".506 W
Geoidal Height: Set to zero

The azimuth from Meades Ranch to station Waldo was fixed at 75°28'09".64 (from South) and the deflection of the vertical set at zero.



Regional Datums

Once the initial point and directions were fixed, the whole orientation of NAD27 was established, including the center of the reference ellipsoid. Its center was imagined to reside somewhere around the center of mass of the earth. However, the two points were certainly not coincident, nor were they intended to be. In short, NAD27 does not use a geocentric ellipsoid.

🔧 Measurement Technology and Datum Selection

In the period before space-based geodesy was tenable, a regional datum was not unusual. The Australian Geodetic Datum 1966, the European Datum 1950, and the South American Datum 1969, among others, were also designed as nongeocentric systems. Achievement of the minimum distortion over a particular region was the primary consideration in choosing their ellipsoids, not the relationship of their centers to the center of mass of the earth. For example, in the Conventional Terrestrial

System (CTS), the 3-D Cartesian coordinates of the center of the Clarke 1866 spheroid as it was used for NAD27 are about X = –4 m, Y = +166 m, and Z = +183 m.

> 🌍 **Examples of Regional Datums**
>
> • **Australian Geodetic Datum 1966**
> • **European Datum 1950**
> • **South American Datum 1969**
> All designed as *nongeocentric* systems optimized for regional accuracy.

This approach to the design of datums was bolstered by the fact that the vast majority of geodetic measurements they would be expected to support were of the classical variety. That is, the work was done with theodolites, towers, and tapes. They were earth-bound. Even after the advent of electronic distance measurement, the general approach involved the determination of horizontal coordinates by measuring from point to point on the earth's surface and adding heights, otherwise known as *elevations*, through a separate leveling operation. As long as this methodological separation existed between the horizontal and vertical coordinates of a station, the difference between the ellipsoid and the true earth's surface was not an overriding concern. Such circumstances did not require a geocentric datum.

CONTENT

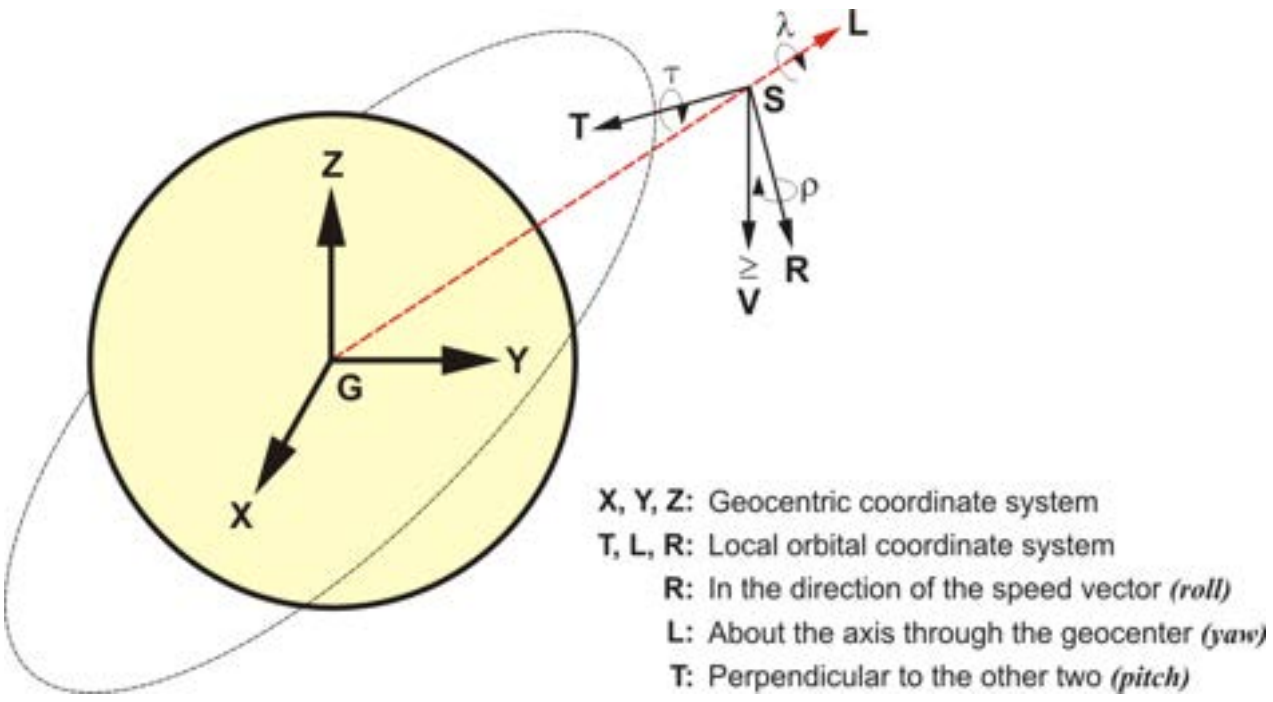
📚 Geocentric Datum

🛰️ 🌍 Geocentric Datum: The Satellite Era

🧭 The Transition to Geocentric Systems

The advent of **satellite geodesy revolutionized how we establish geodetic datums**. Unlike regional datums optimized for specific areas, geocentric datums center on Earth's mass center to support global satellite positioning. This page explores how **space-based observations led to truly global, Earth-centered coordinate systems** essential for GPS/GNSS.

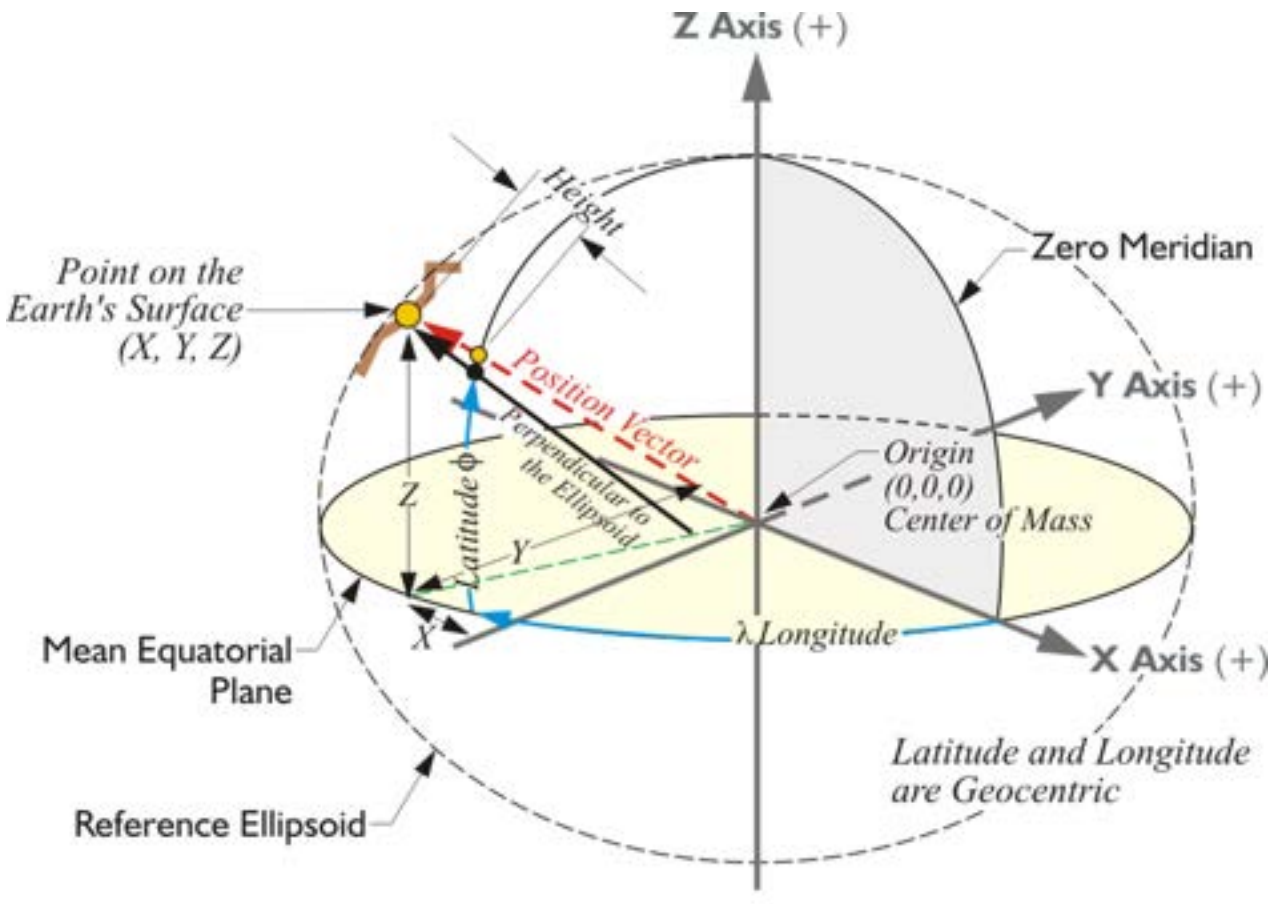
🛰️ Satellite Geodesy and Global Positioning



GPS Orbit

However, as the sophistication of satellite geodesy increased, the need for a truly global, geocentric datum became obvious. The horizontal and vertical information were no longer separate. Since satellites orbit around the center of mass of the earth, a position derived from space-based geodesy can be visualized as a vector originating from that point. So, today, not only are the horizontal and vertical components of a position derived from precisely the same vector, the choice of the coordinate system used to express them is actually a matter of convenience. The position vector can be transformed into the 3-D Cartesian ECEF system, latitude, longitude and height, or virtually any other well-defined coordinate system. However, since the orbital motion and the subsequent position vector derived from satellite geodesy are themselves earth-centered, it follows that the most straightforward representations of that data are earth-centered as well.

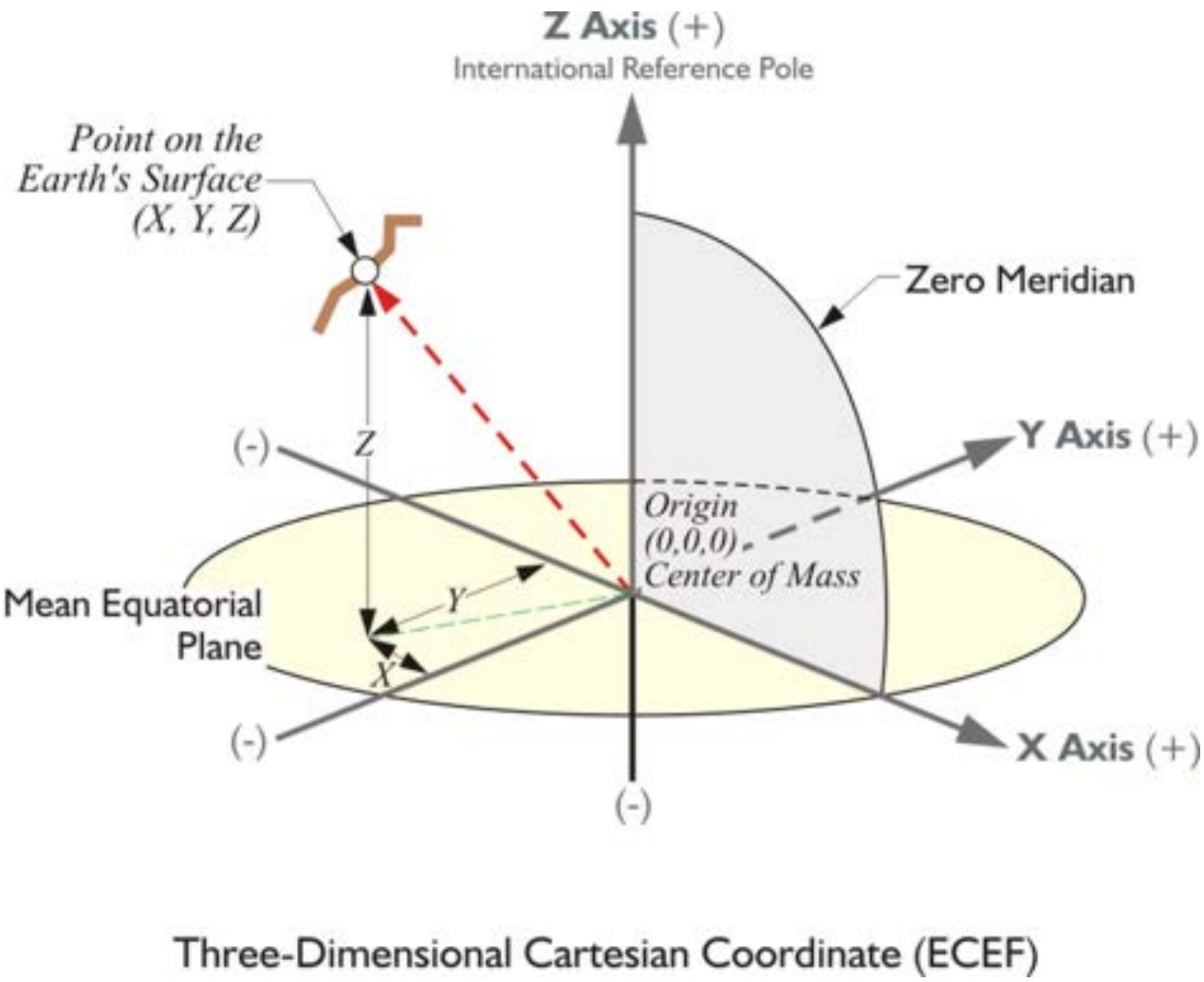
> 🔄 **Coordinate System Flexibility**
>
> A single position vector from Earth's center can be expressed as:
> • **3-D Cartesian ECEF** (X, Y, Z coordinates)
> • **Geodetic coordinates** (latitude, longitude, height)
> • **Any other well-defined system** through mathematical transformation



In this representation, there's a point on the Earth's surface. Its XYZ coordinates are indicated from the center of mass of the Earth in a three-dimensional Cartesian system.

However, there are some additions. For example, you see the longitude, symbolized by lambda, from the Prime Meridian over to a point on the equatorial plane. From there, you see the latitude up to the position of this point as represented on the ellipsoid, the reference ellipsoid of the datum.

Do you remember that an earlier representation of this, the original illustration along this line in the material, had no ellipsoid?



Three Dimensional Cartesian Coordinates

It was simply the XYZ coordinate axes. This is an important distinction. If we're going to have a latitude and longitude derived from an x, y, and z, we have to specify which ellipsoid and which datum (they are not the same thing), is involved. If we are working with the WGS 84 ellipsoid, then we get a latitude and longitude measured along the WGS84 ellipsoid. If we are working with the GRS 80 ellipsoid, we get a different latitude and longitude for exactly the same point on the Earth. If you change the shape of the ellipsoid, the latitude, and longitude changes. Also, the latitude and longitude

of a point in the ITRF2014 (2010.0) reference frame is different than the latitude and longitude for the same point in NAD83 2011 (2010.0).

> ⚠️ **Critical Distinction**
>
> **Same Earth point, different coordinates:**
> • WGS84 ellipsoid → WGS84 lat/lon
> • GRS80 ellipsoid → Different lat/lon
> • ITRF2014 frame → Different lat/lon than NAD83 2011
> *Always specify both ellipsoid AND datum!*

The position vector you see here in red from the center of mass up to the point is different than the black vector that's perpendicular to the ellipsoid. The black vector doesn't go to the center of mass. Also notice that if we're going to go from the position on the ellipsoid up to the actual surface of the Earth, we add a height, an ellipsoidal height. So, we can identify this position on the Earth's surface by an Earth-Centered, Earth-Fixed XYZ coordinate, or we can identify the same point with a geodetic latitude, and longitude, and height. These are just different coordinate systems for expressing the position of the point.

🚀 The Development of a Geocentric Model



The Vanguard Satellite

Satellites have not only provided the impetus for a geocentric datum, they have also supplied the means to achieve it. In fact, the orbital perturbations of man-made near-earth satellites have probably brought more refinements to the understanding of the shape of the earth in a shorter span of time than was ever before possible. For example, the analysis of the precession of Sputnik 2 in the late 1950s showed researchers that the earth's semiminor axis was actually 85 meters shorter than had been previously thought. In 1958, while studying the tracking data from the orbit of Vanguard I, Ann Bailey of the Goddard Spaceflight Center discovered that the planet is shaped a bit like a pear. There is a slight protuberance at the North Pole, a little depression at the South Pole, and a small bulge just south of the equator.

> 🔬 **Key Satellite Discoveries**
>
> **Sputnik 2 (late 1950s):** Earth's semiminor axis 85m shorter than thought
> **Vanguard I (1958):** Ann Bailey discovered Earth's "pear shape"
> • North Pole protuberance
> • South Pole depression
> • Bulge south of equator

These formations and others have been discovered through the observation of small distortions in satellites' otherwise elliptical orbits, little bumps in their road, so to speak. The deviations are caused by the action of Earth's gravity on the satellites as they travel through space. Just as Richer's clock reacted to the lessening of gravity at the equator, and thereby revealed one of the largest features of the earth's shape to Newton, small perturbations in the orbits of satellites, also responding to gravity, reveal details of Earth's shape to today's scientists. The common aspect of these examples is the direct relationship between direction and magnitude of gravity and the planet's form. In fact, the surface that best fits the earth's gravity field has been given a name. It is called the geoid.

> **🌍 The Geoid Concept**
>
> The **geoid** is the surface that best fits Earth's gravity field. Satellite orbital perturbations reveal Earth's true shape by detecting gravitational variations, just as Richer's pendulum clock revealed equatorial bulging to Newton centuries earlier.

📚 The Geoid

🌊 🌍 The Geoid: Earth's Gravity-Defined Surface

🧭 Understanding the Geoid

The **geoid represents Earth's shape as defined purely by gravity**—an equipotential surface where gravitational potential is constant. Unlike the mathematical ellipsoids used in datums, the geoid is a physical reality that influences everything from surveying instruments to satellite orbits. This page explores **why the geoid differs from both sea level and reference ellipsoids**.

📽️ Instructor Lecture: The Geoid Concept

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📚 What You'll Learn

- **Geoid Definition:** Gravity-only equipotential surface concept
- **Sea Level vs. Geoid:** Why mean sea level ≠ geoid
- **Physical Reality:** How surveying instruments relate to the geoid
- **Geoid vs. Ellipsoid:** Physical reality vs. mathematical surface

💡 Key Concept

The geoid is an equipotential surface defined solely by gravity, where gravitational potential is constant everywhere. It's physically real, unlike mathematical ellipsoids.

🌊 Sea Level is Not the Geoid



Sea Level is not the Surface of the Geoid

An often-used description of the geoidal surface involves idealized oceans. Imagine the oceans of the world utterly still, completely free of currents, tides, friction, variations in temperature and all other physical forces, except gravity. Reacting to gravity alone, these unattainable calm waters would coincide with the figure known as the geoid. Admitted by small frictionless channels or tubes and allowed to migrate across the land, the water would then, theoretically, define the same geoidal surface across the continents, too. Of course, the 70% of the earth covered by oceans is not so cooperative, nor is there any such system of channels and tubes.

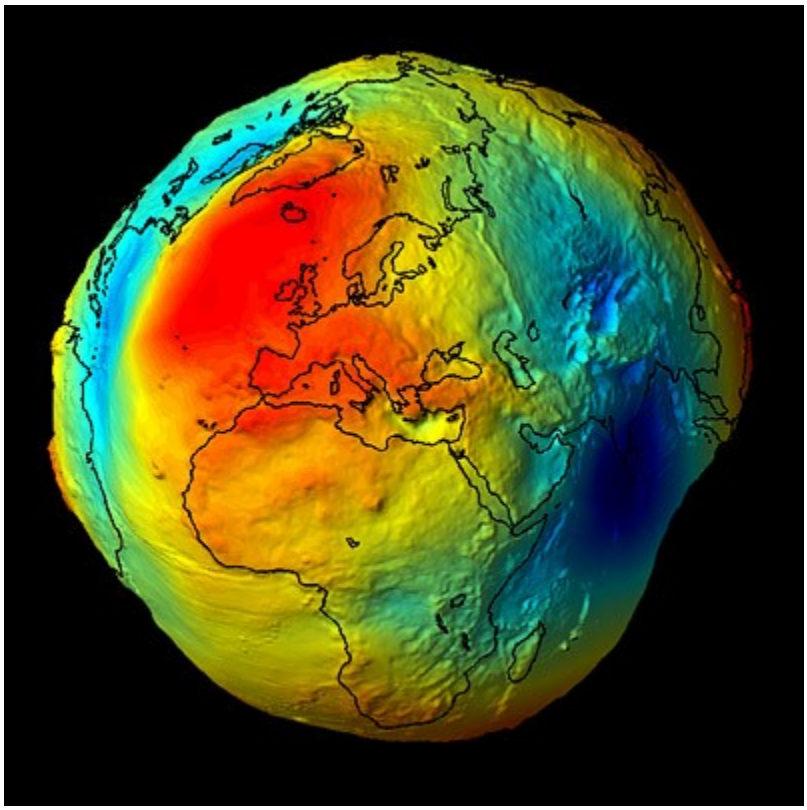
> 🌊 **Idealized Geoid Conditions**
>
> **Theoretical geoidal ocean would be:**
> • Utterly still (no currents)
> • No tides or friction
> • No temperature variations
> • Affected by gravity alone
> • Connected across continents by frictionless channels

In addition, the physical forces eliminated from the model cannot be avoided in reality. The actual oceans of the Earth are affected by temperature, by wave motion, by salinity, and many other aspects that cause variation in their heights. These unavoidable forces actually cause Mean Sea Level to deviate from the geoid. It is a fact frequently mentioned to emphasize the inconsistency of the original definition of the geoid as it was offered by J.B. Listing in 1872. Listing thought of the geoidal surface as equivalent to Mean Sea Level. So even with tidal monitoring, Mean Sea Level is not an indicator of

gravity alone, and that is the concept of the geoid. The geoid is that surface which is affected only by gravity, and it is an equipotential surface, where the gravity potential is always the same.

> 📜 **Historical Note: J.B. Listing (1872)**
>
> J.B. Listing originally defined the geoid as equivalent to Mean Sea Level in 1872. This definition is now known to be inconsistent because real oceans are affected by temperature, waves, salinity, and other forces beyond gravity alone.

⚖️ An Equipotential Surface



And Exaggeration of the Geoid

This somewhat exaggerated image shows what a lumpy surface, Earth would be if we only considered gravity. You can see that gravity isn't consistent across the entire topographic surface of the Earth. At every point it has a magnitude and a direction. In other words, anywhere on the earth, gravity can be described by a mathematical vector. Along the solid earth, such vectors do not have all the *same* direction or magnitude, but one can imagine a surface of constant gravity potential. Such an *equipotential* surface would be level in the true sense. It would coincide with the top of the hypothetical water in the previous example. Mean Sea Level does not define such a figure, nevertheless the geoidal surface is not just a product of imagination.

🔬 **Physical Evidence of the Geoid**

The geoid is real and measurable:
- Vertical axis of leveled surveying instruments is perpendicular to the geoid
- String of a stable plumb bob is perpendicular to the geoid
- Pendulum clocks respond to geoid variations
- Earth-orbiting satellites are affected by the geoid

For example, the vertical axis of any properly leveled surveying instrument and the string of any stable plumb bob are perpendicular to the geoid. Just as pendulum clocks and earth-orbiting satellites, they clearly show that the geoid is a reality.

I hasten to add that the ellipsoid, the nice, smooth, mathematical surface that we use as a reference surface in the definition of a datum, is **different** than the geoid. They are two different surfaces. The geoid is defined entirely by gravity and is a physical reality. The ellipsoid is a purely mathematical imaginary surface.

⚠️ **Critical Distinction**

Geoid vs. Ellipsoid:
Geoid: Physical reality defined by gravity (lumpy, irregular)
Ellipsoid: Mathematical surface used in datums (smooth, regular)
These are completely different surfaces!

📚 Three Figures

📊 Three Figures: Geoid, Ellipsoid, and Topographic Surface

🧭 Understanding the Three Surfaces

Modern geodetic datums involve **three distinct surfaces**: the **geoid** (gravity-defined), the **reference ellipsoid** (mathematical), and the **topographic surface** (physical Earth). Each serves a different purpose in positioning systems, and understanding their relationships is crucial for accurate geodetic work. This page explores why the **ellipsoid serves as the reference surface despite the geoid's physical reality**.

📹 Instructor Lecture: Three Figures in Geodesy

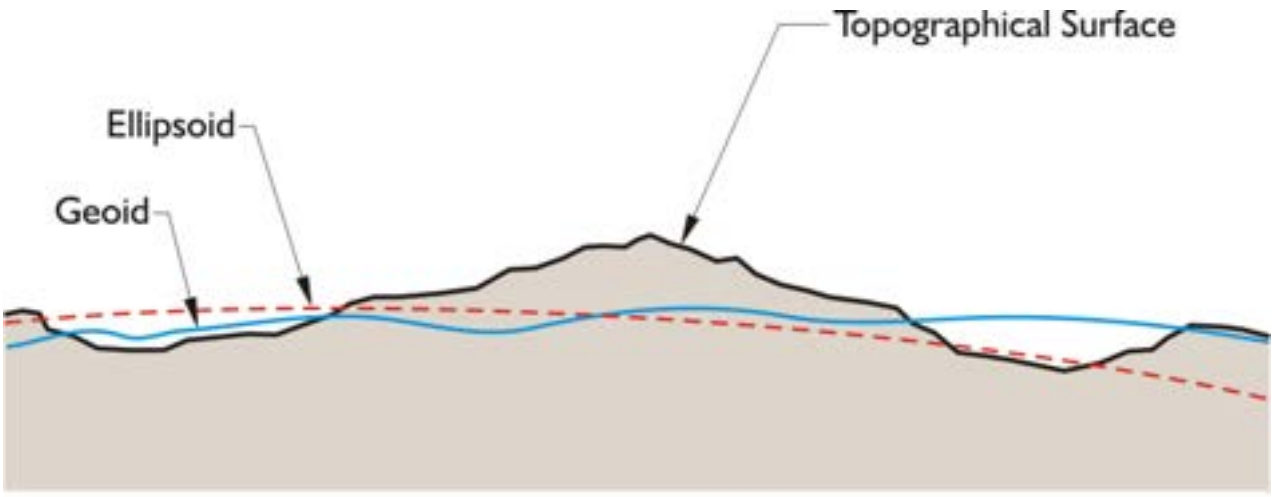
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📚 What You'll Learn

- **Three Surfaces:** Geoid, ellipsoid, and topographic surface roles
- **Surface Properties:** Why each surface has limitations and advantages
- **GRS80 System:** The modern geocentric reference ellipsoid
- **Mathematical vs. Physical:** Why ellipsoids work as reference surfaces

💡 Key Concept

The ellipsoid serves as the reference surface because it can be described mathematically, unlike the irregular geoid or topographic surface.



Three Surfaces

🌍 Topographic Surface

Physical Earth's surface where measurements are made

📐 Reference Ellipsoid

Mathematical surface used for coordinate calculations

🌊 Geoid

Gravity-defined equipotential surface

🔍 Surface Characteristics and Irregularities

The geoid does not precisely follow mean sea level, nor does it exactly correspond with the topography of the dry land. It is irregular like the terrestrial surface. It is bumpy. Uneven distribution of the mass of the planet makes it maddeningly so, because if the solid earth had no internal anomalies of density, the geoid would be smooth and almost exactly ellipsoidal. In that case, the reference ellipsoid could fit the geoid to near perfection and the lives of geodesists would be much simpler. But like the earth itself, the geoid defies such mathematical consistency and departs from true ellipsoidal form by as much as 100 meters in places.

🏔️ Geoid Irregularities

Why the geoid is "bumpy":

- Uneven mass distribution within Earth
- Internal density anomalies
- Departs from ellipsoidal form by up to **100 meters**
- Would be smooth only if Earth had uniform density

Here we have an image of three figures, one is the topographic surface of the Earth that we walk around on, and the next in the red dashed line is the ellipsoid, and the third in the blue wavy line is the geoid. The geoid doesn't follow mean sea level, doesn't correspond with a topographic surface. It's irregular. It does have peaks and valleys. It's bumpy because of the uneven mass of the planet. If the solid Earth didn't have these changes in density, the geoid and the ellipsoid would be the same thing. However, the geoid actually departs from the ellipsoid up to 100 meters in some places.

The figures involved in a latitude, and longitude, and height, are the geoid, the reference ellipsoid, and the Earth itself.

🌐 The Modern Geocentric Datum

Three distinct figures are involved in a geodetic datum for latitude, longitude, and height: the geoid, the reference ellipsoid, and the topographic surface. Due in large measure to the ascendancy of satellite geodesy, it has become highly desirable that they share a common center.

While the level surface of the geoid provides a solid foundation for the definitions of heights and the topographic surface of the earth is necessarily where measurements are made, neither can serve as the reference surface for geodetic positions.

> ❌ **Why Geoid and Topographic Surface Can't Be Reference Surfaces**
>
> **Topographic Surface:** Too irregular from continents to ocean floors
> **Geoid:** Sometimes above, sometimes below Earth's surface
> **Both:** Defy concise geometrical definition
> **Only the ellipsoid:** Can be described completely in mathematical terms

From the continents to the floors of the oceans, the solid earth's actual surface is too irregular to be represented by a simple mathematical statement. The geoid, which is sometimes under, and sometimes above, the surface of the earth, has an overall shape that also defies any concise geometrical definition. But the ellipsoid not only has the same general shape as the earth, but, unlike the other two figures, can be described simply and completely in mathematical terms.

Therefore, a global geocentric system has been developed based on the ellipsoid adopted by the *International Union of Geodesy and Geophysics (IUGG)* in 1979. It is called the *Geodetic Reference System 1980 (GRS80)*. Its semimajor axis, *a*, is 6378.137 km long and is probably within a few meters of the earth's actual equatorial radius. Its flattening, *f*, is 1/298.25722 and likely deviates only slightly from the true value, a considerable improvement over Newton's calculation of a flattening ratio of 1/230. But then, he did not have orbital data from near-earth satellites to check his work.

> 📊 **GRS80 Technical Specifications**
>
> **Geodetic Reference System 1980 (GRS80):**
> • **Adopted by:** International Union of Geodesy and Geophysics (IUGG), 1979

- **Semimajor axis (a):** 6378.137 km
- **Flattening (f):** 1/298.25722
- **Improvement over Newton:** 1/298.25722 vs. Newton's 1/230
- **Advantage:** Based on satellite orbital data

The goal is to have the reference ellipsoid, we use for satellite geodesy, to be geocentric. From the continents to the floors of the ocean, the solid Earth is too irregular to be represented by a simple mathematical statement. The geoid, sometimes under, sometimes above the surface of the Earth, has an overall shape that also defies any concise geometrical definition. However, the ellipsoid not only has the same general shape as the Earth, but unlike the other two figures can be described simply and completely in mathematical terms. That is why we use the ellipsoid as a reference. Therefore, a global geocentric system has been developed based on the ellipsoid adopted by the International Union of Geodesy and Geophysics. It's called the Geodetic Reference System, GRS 80.

CONTENT

📚 Modern Geocentric Datum

🛰️ Modern Geocentric Datum: WGS84 and GRS80

🧭 Contemporary Geodetic Reference Systems

Modern satellite-based positioning requires **geocentric datums that center on Earth's mass center**. This page examines the **two primary geocentric reference systems**: **WGS84** (World Geodetic System 1984) and **GRS80** (Geodetic Reference System 1980), their technical similarities, evolutionary changes, and practical applications in GPS/GNSS positioning.

📊 Two Reference Ellipsoids

Ellipsoid	Semi-major axis	Inverse flattening
GRS80	6,378,137 m	298.257222101
WGS84	6,378,137 m	298.257223563

Two Reference Ellipsoids

🔍 **WGS84 vs. GRS80 Comparison**

Semimajor axis: Identical for both systems
Inverse flattening: Different only at the 6th decimal place
Practical impact: Extremely similar for most applications
Usage: WGS84 for GPS, GRS80 for other geodetic systems

🛰️ WGS84: World Geodetic System 1984

In the table above, you have a comparison between WGS84 and GRS80 ellipsoids. You will notice that the semi major axis is the same, and the inverse flattening is only different when you get out to the sixth place to the right of the decimal. They are very similar. The WGS 84 ellipsoid is the reference of the datum known as the *World Geodetic System 1984 (WGS84)*. In other words, the WGS 84 ellipsoid attached at the center of mass of the Earth is one component of the WGS84 datum, but *please note that while the WGS84 ellipsoid is the reference ellipsoid for the WGS84 datum - it is **not** the datum itself.*

> ⚠️ **Important Distinction**
>
> **WGS84 ellipsoid ≠ WGS84 datum**
> The WGS84 ellipsoid is the *reference surface* for the WGS84 datum system.
> The datum includes the ellipsoid *plus* its positioning and orientation relative to Earth.

The WGS84 datum (reference frame) has been used by the U.S. Military since January 21, 1987. There have been six more incarnations of WGS84 since then. While WGS84 has always been the basis for the GPS Navigation message, the particular version of the datum has changed. As of this writing, the latest version of WGS84 is *WGS84 (G2139)*. The number following the letter G is the number of the GPS week during which the coordinates first were used in the *National Geospatial Intelligence Agency, NGA,* precise ephemeris estimations. Therefore, coordinates provided today by GPS receivers are based in WGS84 (G2139) which is the sixth update to the realization of the WGS84 Reference Frame.

> 📅 **WGS84 Evolution Timeline**
>
> **1987:** Original WGS84 adopted by U.S. Military (Jan 21)
> **1994:** WGS84 (G730) implemented in GPS (June 29)
> **Current:** WGS84 (G2139) - sixth major update
> **GPS Week System:** "G" number indicates GPS week of first use
> **Authority:** National Geospatial Intelligence Agency (NGA)

The original WGS 84 was based on observations from more than 1900 Doppler Stations. It was revised to become WGS84 (G730) to incorporate GPS observations. That realization was implemented in GPS by the *Operational Control Segment, OCS* on June 29, 1994. More GPS based realizations of WGS84 followed. Today, the epoch of WGS84 is (G2139).

🇺🇸 WGS84 and NAD83 Relationship

Most available GPS software can transform those coordinates to a number of other datums as well. The one that is probably of greatest interest in the United States today is the *North American Datum 1983 (NAD83)*. Originally, the difference between WGS84 as originally rolled out in 1987 and NAD83 as first introduced in 1986 coordinates was so small that transformation was unnecessary. That is no longer the case when it comes to NAD83 2011 (2010.0) and WGS84 (G2139) the difference can be ~1 to 2 meters.

📐 WGS84 vs. NAD83 Coordinate Differences

1987 WGS84 vs. 1986 NAD83: Negligible difference (transformation unnecessary)
Modern versions: NAD83 2011 (2010.0) vs. WGS84 (G2139)
Current difference: ~1 to 2 meters
Implication: Coordinate transformation now essential for accuracy

⚖️ Practical Implications

GPS Software: Can transform WGS84 coordinates to multiple datums
U.S. Applications: NAD83 often preferred for domestic work
Survey Accuracy: 1-2 meter differences require proper transformations
Global Work: WGS84 remains standard for international projects

📚 North American Datums



📐 North American Datums: NAD27 and NAD83

🧭 Historical Development of North American Geodetic Systems

The **evolution of North American geodetic datums** represents a fascinating journey from ground-based triangulation networks to satellite-based positioning systems. This page explores the **transition from NAD27**, built through decades of careful triangulation surveys, **to NAD83**, which incorporated early satellite observations and established the foundation for modern GPS applications. Understanding this progression reveals how technological advances **transformed our ability to map and position features across the continent**.

📽️ Instructor Lecture: Evolution of North American Datums

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📚 What You'll Learn

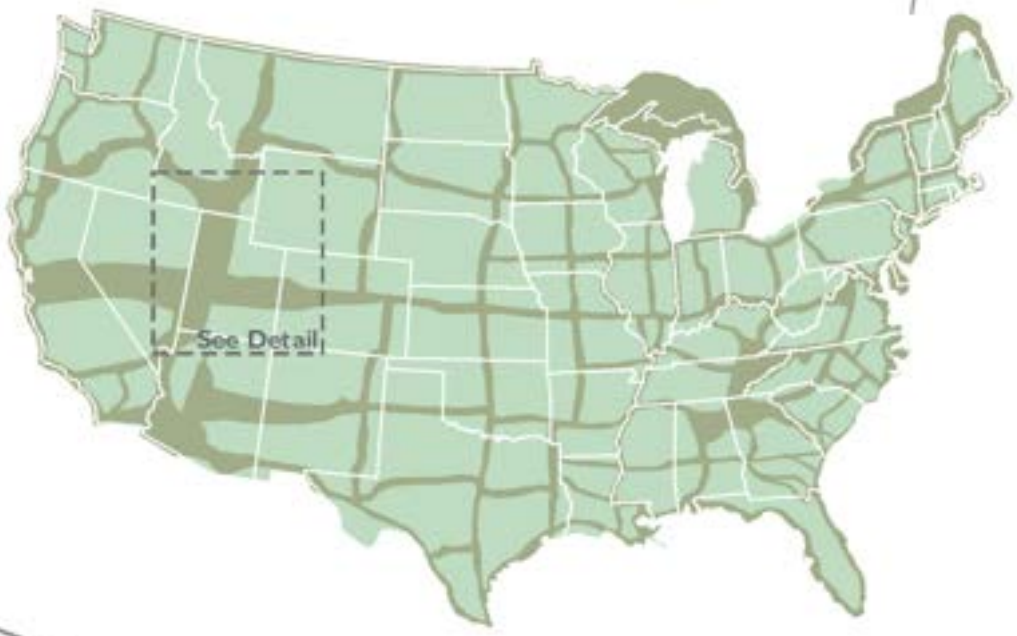
- **NAD27 Foundation:** Triangulation networks and Clarke 1866 ellipsoid
- **Technical Limitations:** Accuracy constraints and local distortions
- **NAD83 Development:** Modern adjustment techniques and satellite integration
- **Coordinate Transformation:** Datum shift implications for mapping

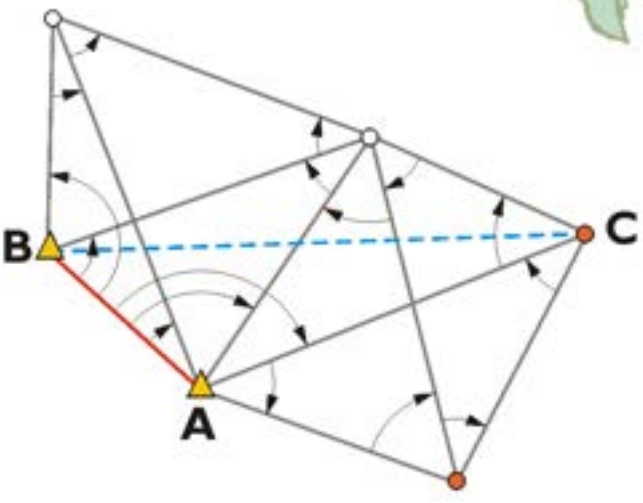
> 💡 **Key Insight**
>
> The transition from **NAD27** to **NAD83** represented more than a technical upgrade—it fundamentally changed how we measure and map North America.

⛰️ NAD27: The Foundation of Triangulation



Detail
Idaho, Utah, Colorado, Wyoming,
Nevada and Arizona





Know Data:
Length of base line AB.
Latitude and longitude of points A and B.
Azimuth of line AB.

Measured Data:
Angles to the new control points.

Computed Data:
Latitude and longitude of point C, and others new points.
Length and azimuth of line C.
Length and azimuth of all other lines.

Triangulation Network Coverage *- Geodetic triangulation stations across the American West*

The Clarke 1866 ellipsoid was the foundation of NAD27, and the blocks that built that foundation were made by geodetic triangulation. After all, an ellipsoid, even one with a clearly stated orientation to the earth, is only an abstraction until physical, identifiable control stations are available for its practical application.

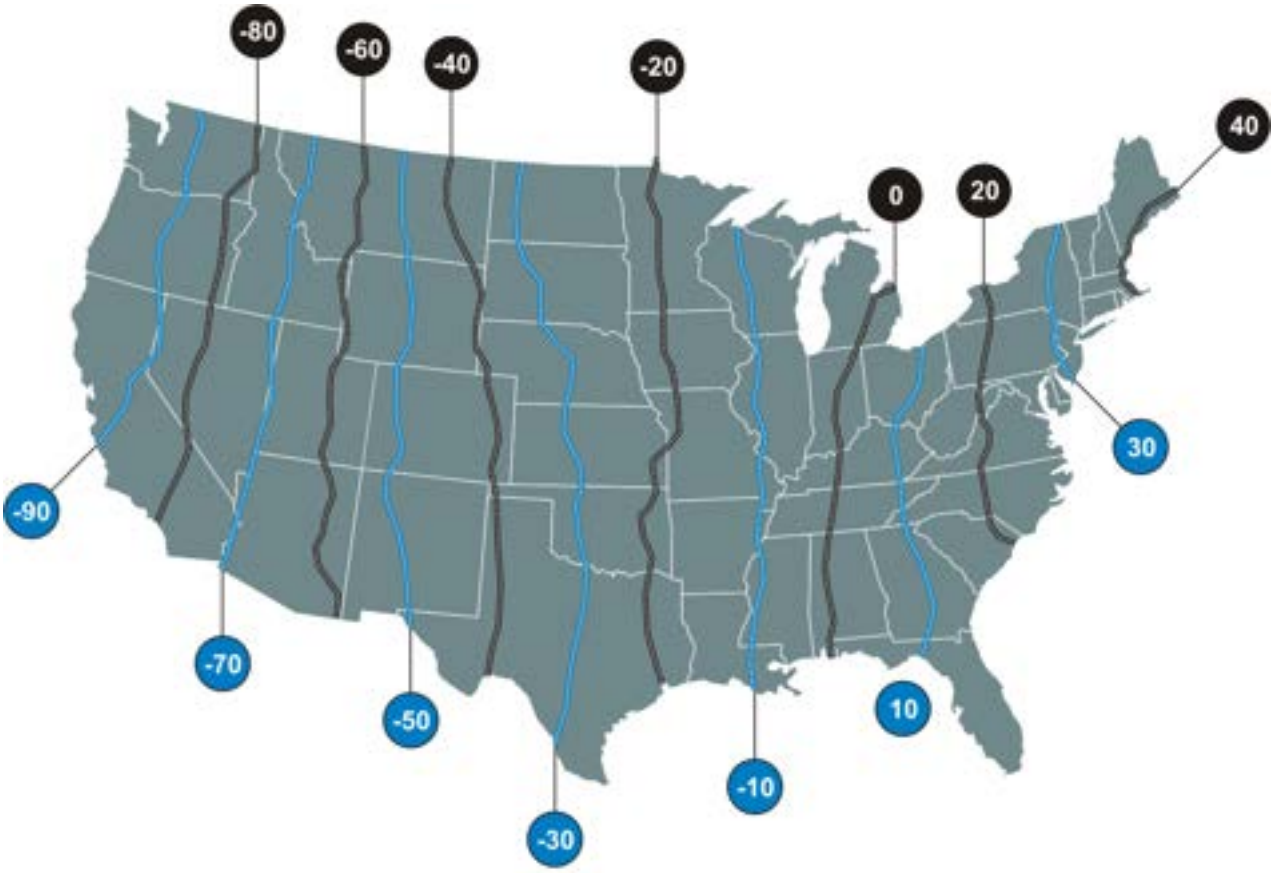
> 🔗 **Triangulation Networks**
>
> **Physical Infrastructure:** Bronze disks set in stone and concrete
> **Coverage:** Tens of thousands of miles of triangulation chains
> **Scope:** Canada to Mexico, coast to coast
> **Applications:** International and state boundaries, mapping, scientific research

During the tenure of NAD27, control positions were tied together by tens of thousands of miles of triangulation and some traverses. Its measurements grew into chains of figures from Canada to Mexico and coast to coast, with their vertices perpetuated by bronze disks set in stone, concrete, and other permanent media.

These tri-stations, also known as brass caps, and their attached coordinates have provided a framework for all types of surveying and mapping projects for many years. They have served to locate international, state, and county boundaries. They have provided geodetic control for the planning of national and local projects, development of natural resources, national defense, and land management.

> ⚠️ **Growing Limitations**
>
> **Accuracy Issues:** Local distortions reaching unacceptable levels
> **Technical Constraints:** Too few baselines and Laplace azimuths
> **Measurement Quality:** Some observations inadequate for modern standards
> **Global Compatibility:** Lack of internationally viable geocentric ellipsoid

But even as application of the nationwide control network grew, the revelations of local distortions in NAD27 were reaching unacceptable levels. The work was excellent, and given the technology of the day, remarkable—but judged by the standards of newer measurement technologies, the quality of some of the observations used in the datum were too low. That, and its lack of an internationally viable geocentric ellipsoid, finally drove its positions to obsolescence.



Datum Shift Visualization *- Coordinate differences between NAD27 and NAD83 across the United States*

> 📐 **Understanding Datum Shifts**
>
> The diagram above illustrates how the same geographic locations have different coordinate values in NAD27 versus NAD83. These shifts vary geographically, with some areas experiencing changes of hundreds of meters.

🛰️ The Development of NAD83

While a committee of the National Academy of Sciences advocated the need for a new adjustment in its 1971 report, work on the new datum, NAD83, did not really begin until after July 1, 1974. Leading the charge was an old agency with a new name.

🏛️ **Organizational Evolution**

1807: Survey of the Coast
1878: U.S. Coast & Geodetic Survey
1899: Coast and Geodetic Survey (C&GS)
Current: National Geodetic Survey (NGS) within NOAA
Authority: U.S. Code, Title 33, USC 883a

NAD83 includes not only the United States, but also Central America, Canada, Greenland, and Mexico. The NGS and the Geodetic Survey of Canada set about the task of attaching and orienting the GRS80 ellipsoid to the actual surface of the earth, as it was defined by the best positions available at the time.

📊 **NAD83 Adjustment Statistics**

Duration: More than 10 years of work
Scope: Entire Western Hemisphere
Classical Observations: 1.7 million weighted observations
EDM Baselines: ~30,000 electronic distance measurements
Astronomic Azimuths: 5,000 measurements
Doppler Stations: 650+ TRANSIT satellite positions
VLBI Vectors: 100+ Very Long Baseline Interferometry observations

It took more than 10 years to readjust and redefine the horizontal coordinate system of North America into what is now NAD83. More than 1.7 million weighted observations derived from classical surveying techniques throughout the Western Hemisphere were involved in the least-squares adjustment. They were supplemented by approximately 30,000 EDM measured baselines, 5000 astronomic azimuths, and more than 650 Doppler stations positioned by the TRANSIT satellite system. Over 100 Very Long Baseline Interferometry (VLBI) vectors were also included. GPS, in its infancy, was not utilized significantly.

🚀 **GPS Integration Planning**

Multi-Agency Coordination: NOAA, NASA, USGS, and Department of Defense
NGS Responsibilities: GPS operation specifications, technology investigation, and crustal motion modeling
Future Integration: Stage set for systematic GPS data infusion as the datum matured

GPS was growing up in the early 80's and some of the agencies involved in its development decided to join forces. NOAA, the National Aeronautics and Space Administration (NASA), the U.S. Geological Survey (USGS), and the Department of Defense coordinated their efforts. As a result, each agency was assigned specific responsibilities. NGS was charged with the development of specifications for GPS operations, investigation of related technologies, and the use of GPS for modeling crustal motion. It was also authorized to conduct its subsequent geodetic control surveys with GPS. So, despite an initial sparseness of GPS data in the creation of NAD83, the stage was set for a systematic infusion of its positions as the datum matured.

Texas A&M University - Corpus Christi
College of Engineering & Computer Science
GISC 4315 - Satellite Positioning
Fall 2026

CONTENT

📚 The International Terrestrial Reference System (ITRS)

🌍 The International Terrestrial Reference System (ITRS)

🧭 Global Geocentric Reference Framework

The **International Terrestrial Reference System** represents the pinnacle of geodetic achievement—a truly geocentric coordinate system that serves as the **global standard for precise positioning**. Unlike earlier regional datums, ITRS **accounts for the dynamic nature of our planet, incorporating tectonic plate motion, crustal deformation, and Earth orientation parameters** into a comprehensive reference framework. This system underlies modern GPS operations and **enables centimeter-level positioning accuracy worldwide**.

🚀 Evolution Beyond NAD83

As **NAD83** has aged, there has been constant improvement in geodesy. When **NAD83** was created, it was intended to be geocentric. It is known that the center of the reference ellipsoid of **NAD83**, **GRS80**, is about **2.24 meters from the true geocenter**.

⚠️ NAD83 Geocentric Limitations

Offset from True Center: ~2.24 meters from Earth's mass center
Original Intent: Designed to be geocentric with 1980s technology
Accuracy Constraints: Limited by available measurement techniques
Need for Improvement: Modern applications require true geocentric positioning

However, there is a reference system that is geocentric. It is known as the International Terrestrial Reference System, ITRS. The reference frame derived from it is the *International Terrestrial Reference Frame (ITRF)*.

🎯 ITRS Technical Specifications

📐 ITRS Fundamental Parameters

Origin: Center of mass of the entire Earth (including oceans and atmosphere)
Unit of Length: The meter (SI standard)
Axis Orientation: Consistent with Bureau International de l'Heure (BIH) at beginning of 1984
Accuracy Level: Centimeter-level precision globally

Its origin is at the center of mass of the whole earth, including the oceans and atmosphere. The unit of length is the meter. The orientation of its axis was established as consistent with that of the IERS's predecessor, Bureau International de l'Heure, BIH, at the beginning of 1984.

🔄 Dynamic Reference Frame

Series of Realizations: First published in 1988, regularly updated
Living System: Continuously refined with new measurements
Global Scope: Hundreds of control stations worldwide
Motion Accounting: Provides both positions and velocities for all stations

🛰️ IERS Global Monitoring Network

Today, the ITRF is maintained by the International Earth Rotation Service (IERS), which monitors Earth Orientation Parameters (EOP) for the scientific community through a global network of observing stations.

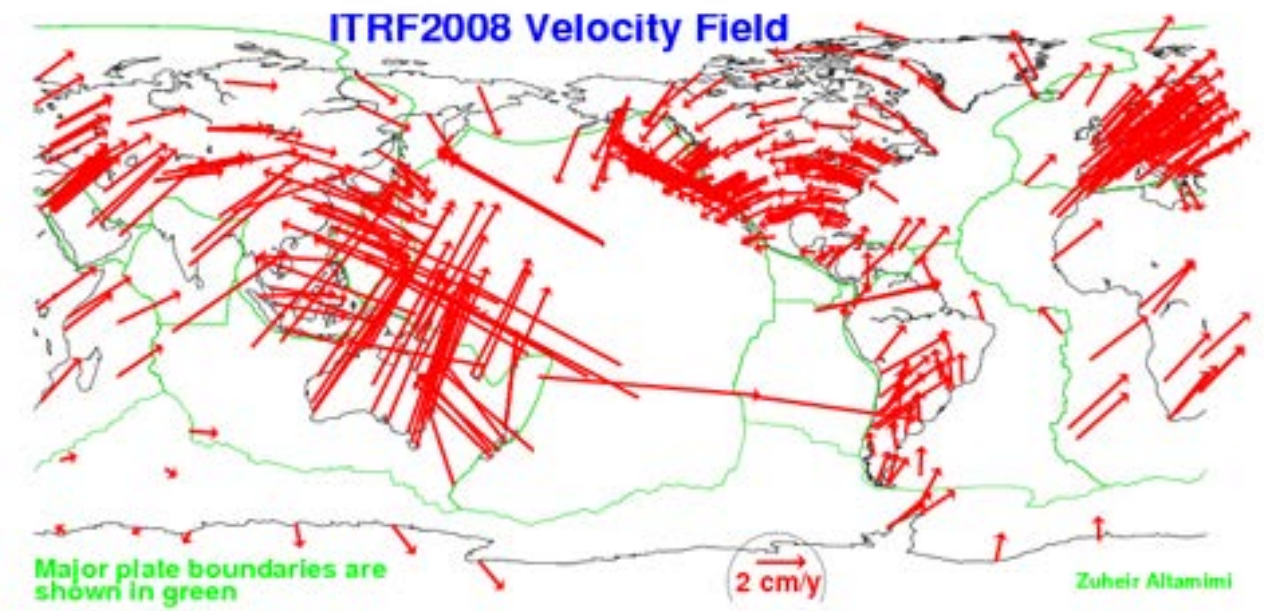
🔬 IERS Measurement Technologies

GPS: Global Positioning System observations
VLBI: Very Long Baseline Interferometry
LLR: Lunar Laser Ranging
SLR: Satellite Laser Ranging
DORIS: Doppler Orbitography and Radiopositioning Integrated by Satellite

This is done with GPS, Very Long Baseline Interferometry (VLBI), Lunar Laser Ranging (LLR), Satellite Laser Ranging (SLR), the Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS), and the positions of the observing stations are considered to be accurate to the centimeter level.

Recognizing that the several hundred control stations worldwide for which it publishes yearly coordinates are actually in motion due to the shifting of approximately 20 tectonic plates worldwide, the IERS also provides velocities for them. The International Terrestrial Reference Frame is actually a series of realizations. The first was in 1988. In other words, it is revised and published on a regular basis.

🌍 Global Tectonic Velocity Field



ITRF Velocity Field *- Global tectonic plate motion vectors from ITRF2008*

📊 **Understanding Velocity Vectors**

Vector Scale: Legend shows 2 centimeters per year
Motion Direction: Arrows indicate tectonic plate movement relative to Earth's center of mass
Global Pattern: ~20 major tectonic plates in constant motion
Measurement Precision: Millimeter-level annual velocity determinations

The vectors that you see on this illustration indicate the motion of the tectonic plates. You can see the legend there, the length of two centimeters per year. It shows the directions that these tectonic plates are in motion relative to the center of mass of the Earth.

🔗 ITRS Applications in Modern Geodesy

GPS Operations: Satellite orbit computation and positioning
Scientific Research: Crustal deformation and earthquake studies
Climate Monitoring: Sea level rise and ice sheet dynamics
Surveying Standards: High-precision mapping and engineering
Navigation Systems: Aviation and maritime positioning
Space Missions: Satellite tracking and interplanetary navigation

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CONTENT

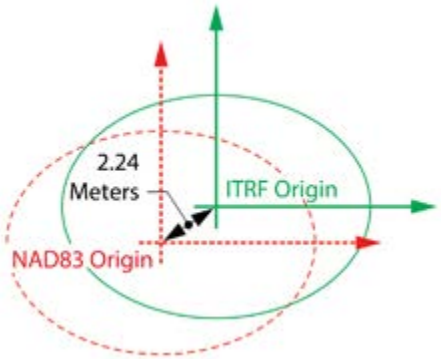
📚 ITRF2014, WGS84 and NAD83

⚖️ ITRF2014, WGS84 and NAD83: Modern Datum Relationships

🧭 Comparing Contemporary Geodetic Reference Systems

Understanding the **relationships between ITRF2014, WGS84, and NAD83** is crucial for modern surveying and GPS applications. While these systems were once nearly identical, technological advances and improved measurements have revealed significant differences—sometimes reaching 1-2 meters. This page explores **how these coordinate systems compare today**, **why their differences have grown**, and the **practical implications for surveyors** working with GPS data across different reference frames.

🎯 Geocentric vs. Non-Geocentric Systems



ITRF is Geocentric and NAD83 is not Geocentric

The North American Datum of 1983 (NAD83) is used everywhere in North America except Mexico. The latest realization of the datum as of this writing is *NAD83 (2011)* epoch 2010.0. This realization in the coterminous United States and Alaska is available through the National *CORS (Continuously*

Operating Reference Stations). Their number of National CORS and Cooperative CORS sites continuously grows with the addition of several new stations each month.

📡 **CORS Network Expansion**

Coverage: Continental United States and Alaska
Current Realization: NAD83 (2011) epoch 2010.0
Growth Pattern: Several new stations added monthly
Station Types: National CORS and Cooperative CORS sites
Accessibility: Provides reference for GPS surveys nationwide

📊 Comparison of ITRF, WGS84, and NAD83

Year	Realization (Epoch)	For all practical purposes equivalent to:
1987	WGS 1984 (ORIG)	NAD83 (1986)
1994	WGS84 (G730)	ITRF91/92
1997	WGS84 (G873)	ITRF94/96
2002	WGS84 (G1150)	ITRF00
2012	WGS (G1674)	ITRF08
2013	WGS (G1762)	Compares to ITRF08 and ITRF2014 within 1cm Root Mean Square (RMS) overall if the epochs are the same

Technical Comparison *- Key parameters of modern reference systems*

As mentioned earlier, in the past we did not have to be concerned with the shift between NAD83 (1986) and WGS84 as introduced in 1987, because the discrepancy easily fell within our overall error budget. NAD83 and WGS84, originally, differed by only a centimeter or two. That is no longer true.

⚠️ **Growing Coordinate Differences**

Historical Accuracy: NAD83 (1986) and WGS84 (1987) differed by only 1-2 cm
Modern Reality: NAD83 (2011) and WGS84 (G1762) differ up to 1-2 meters
Continental US Impact: Differences vary geographically across CONUS
Survey Implications: Exceeds typical survey error budgets

In their new definitions—NAD83 (2011) and WGS84 (G1762)—differ up to one or two meters within the continental United States. On the other hand, ITRF08, ITRF2014 and WGS84 (G1762) are virtually identical if their epochs (moment in time) are the same.

🔄 **Epoch Relationships**

ITRF08 & WGS84 (G1762): Standard epoch 2005.0
ITRF2014: Standard epoch 2010.0
Virtual Identity: ITRF and WGS84 are nearly identical at same epochs
Time Dependency: Coordinates change due to tectonic motion

🌍 Plate Motion and Reference Frame Differences

The typical standard epoch for both are WGS84(G1762) and ITRF08 is 2005.0. The typical standard epoch for both ITRF2014 is 2010.0; NGS has developed a program called Horizontal Time Dependent Positioning (HTDP) to transform positions from one epoch to another.

🛠️ **HTDP Program Capabilities**

Epoch Transformation: Move positions from one date to another
Reference Frame Conversion: Transform between different datum systems
Support Systems: Recent realizations of NAD83, ITRF, and WGS84
Time-Dependent Positioning: Accounts for tectonic motion over time

In other words, this program allows the movement of positions from one date to another, transformation from one reference frame to another and supports the recent realizations of the NAD 83, ITRF and WGS84. This underlines the fact that ITRF and WGS84 systems are global, and their realizations take into account that the Earth is in constant motion due to the shifting of tectonic plates around the world.

🧭 **Fundamental Difference: Global vs. Plate-Fixed**

ITRF & WGS84: Global systems accounting for all tectonic motion
NAD83: Fixed to North American plate, moves with the plate
Motion Rate: NAD83 moves 10-20 mm/year relative to ITRF/WGS84
Continental Impact: Affects all positioning in North America

However, NAD83 is fixed to one plate, the North American plate, and moves with it. Consequently, NAD83, in the continental United States moves approximately 10 to 20 millimeters per year in relation to the realizations of ITRF and WGS84 reference frames.

🔧 The Management of NAD83

Since geodetic accuracy with GPS depends on relative positioning, surveyors continue to rely on NGS stations to control their work, just as they have for generations. Today, it is not unusual for surveyors to find that some NGS stations have published coordinates in NAD83 and others, perhaps needed to control the same project, only have positions in NAD27.

> ⚠️ **Mixed Datum Challenges**
>
> **Project Complexity:** Single projects may require both NAD27 and NAD83 control
> **Transformation Need:** Often desirable to convert NAD27 to NAD83
> **Mathematical Limitation:** No single-step accurate transformation exists
> **Historical Accuracy:** NAD27 coordinates sometimes in error by 1 part in 15,000

In such a situation, it is often desirable to transform the NAD27 positions into coordinates of the newer datum. But, unfortunately, there is no single-step mathematical approach that can do it accurately. The distortions between the original NAD27 positions are part of the difficulty. The older coordinates were sometimes in error as much as 1 part in 15,000.

Problems stemming from the deflection of the vertical, lack of correction for geoidal undulations, low-quality measurements, and other sources contributed to inaccuracies in some NAD27 coordinates that cannot be corrected by simply transforming them into another datum.

🔄 Transformations from NAD27 to NAD83

Nevertheless, various approximate methods are used to transform NAD27 coordinates into NAD83 values. For example, the computation of a constant local translation is sometimes attempted using stations with coordinates in both systems as a guide.

> 🔧 **Transformation Methods**
>
> **Local Translation:** Constant shift using common station coordinates
> **Four-Parameter Model:** Two translations, one rotation, one scale factor
> **3-D Helmert Transformation:** Seven parameters (3 shifts, 1 scale, 3 rotations)
> **Software Solutions:** NGS NADCON program for coordinate conversion

Another technique is the calculation of two translations, one rotation and one scale parameter, for particular locations based on the latitudes and longitudes of three or more common stations. Perhaps

the best results derive from polynomial expressions developed for coordinate differences, expressed in Cartesian or ellipsoidal coordinates using a 3-D Helmert transformation.

However, besides requiring seven parameters (three shift, one scale, and three rotation components) this approach is at its best when ellipsoidal heights are available for all the points involved. Where adequate information is available, software packages such as the NGS program NADCON can provide coordinates.

> 💡 **Best Practice: Return to Original Observations**
>
> **Most Reliable Method:** Use original measurements, not derived coordinates
> **Least-Squares Adjustment:** Incorporate observations into proper adjustment
> **Fundamental Principle:** Coordinates are datum-specific, measurements are absolute
> **Quality Advantage:** NAD83 based on 10× more observations than NAD27

In fact, the only truly reliable method of transformation is not to rely on coordinates at all, but to return to the original observations themselves. It is important to remember, for example, that geodetic latitude and longitude, as other coordinates, are specifically referenced to a given datum (reference frame) and are not derived from some sort of absolute framework.

🎯 High Accuracy Reference Networks (HARN)

Other significant work along this line was accomplished in the state-by-state super-net programs. The creation of *High Accuracy Reference Networks (HARN)* were cooperative ventures between NGS and the states, and often include other organizations as well. The campaign was originally known as *High Precision Geodetic Networks (HPGN)*.

> 📐 **HARN Network Specifications**
>
> **Station Spacing:** 16-62 miles (optimal distribution)
> **Accuracy Target:** 1 part-per-million or better between stations
> **Technology:** Heavy reliance on GPS observations
> **Accessibility:** Vehicle-accessible monuments with good sky visibility
> **Coverage:** State-by-state implementation across the US

A station spacing of not more than about 62 miles and not less than about 16 miles was the objective in these statewide networks. The accuracy was intended to be 1 part-per-million, or better, between stations. In other words, with heavy reliance on GPS observations, these networks were intended to provide extremely accurate, vehicle-accessible, regularly spaced control point monuments with good overhead visibility.

🏆 **HARN Quality Advantages**

Superior Control: More accurate than day-to-day GPS observations
Vector Coherence: Eliminates need to warp vectors to fit inferior control
Statewide Adjustment: Submitted to NGS for network readjustment
Coordinate Shifts: Typical 0.3-1.0m shifts from original NAD83 values
Completion: All state networks concluded by 1998

These stations were intended to provide control superior to the vectors derived from the day-to-day GPS observations that are tied to them. In that way, the HARN points provide the user with a means to avoid any need to warp vectors to fit inferior control.

The original NAD83 adjustment is indicated with a suffix including the year 1986 in parentheses, that is, NAD83 (1986). However, when a newer realization is available, the year in the parentheses will be the year of the adjustment. The most recent realization is NAD83 (2011).

CONTENT

📚 Continuously Operating Reference Stations (CORS)

📊 Continuously Operating Reference Stations (CORS)

🧭 National Reference Station Infrastructure

The **CORS network represents one of the most significant advances in geodetic infrastructure since triangulation surveys**. Beginning in 1992 with just 50 stations for marine navigation, this network has expanded to thousands of continuously operating GPS reference stations providing centimeter-level positioning accuracy nationwide. CORS stations serve as the backbone for modern surveying, supporting everything from precision agriculture to scientific research while **providing real-time access to multiple reference frames including ITRF2014 and NAD83 2011**.

🗺️ National CORS Network Distribution



NGS CORS Map *- Distribution of Continuously Operating Reference Stations across the continental United States*

📈 **CORS Network Growth Timeline**

1992: NGS begins establishing CORS network
Original Purpose: Positioning for navigational and marine needs
1996: Approximately 50 CORS stations operational
Present: Thousands of CORS stations nationwide
Expansion Rate: Continuous growth with state and private partnerships

🎯 CORS Technical Standards and Performance

In about 1992, the NGS began establishing a network of Continuous Operating Reference Stations (CORS) throughout the country. The original idea was to provide positioning for navigational and marine needs. There were about 50 CORS in 1996.

📐 **CORS Accuracy Specifications**

Horizontal Accuracy: 3 centimeters
Vertical Accuracy: 5 centimeters
Standards Compliance: NOAA geodetic requirements
Installation Standards: Rigorous site selection and monumentation
Operation Standards: 24/7 data availability and quality control
Data Distribution: Real-time and post-processing access

Their positional accuracies are 3 cm horizontal and 5 cm vertical. They also must meet NOAA geodetic standards for installation, operation, and data distribution. Today, there are thousands of Continuously Operating Reference Stations (CORS) online.

🛰️ **CORS Data Applications**

Primary Function: Support for carrier phase GPS observations
Data Access: Internet-based postprocessing information
Real-Time Services: Network RTK positioning capabilities
Scientific Research: Crustal deformation and atmospheric studies
Commercial Applications: Precision agriculture, construction, surveying

🏛️ Federal Base Network (FBN)

From 1998 to 2004, NGS introduced another series of observations in each state designed to tie the network to the *Continuously Operating Reference Stations, CORS*. This work resulted in the *Federal Base Network (FBN)*, which is a nationwide network of monumented stations.

🎯 FBN Specifications and Purpose

Development Period: 1998-2004
Station Spacing: Approximately 100 kilometers apart
Coverage: Nationwide network of monumented stations
Special Focus: Particularly dense in crustal motion areas
Integration: Tied to CORS network for enhanced accuracy

These spatial reference positions are among the most precise available and are particularly dense in crustal motion areas. In general, these points are spaced at approximately 100 kilometers apart.

📊 FBN Accuracy Standards

Latitude & Longitude: 1 centimeter accuracy
Ellipsoidal Heights: 2 centimeter accuracy
Orthometric Heights: 3 centimeter accuracy
Quality Level: Among the most precise positions available
Reference Status: Primary control for regional networks

The accuracies intended are: 1 cm-latitudes and longitudes, 2 cm-ellipsoidal heights, and 3 cm-orthometric heights.

🤝 Cooperative Base Network (CBN)

These stations are few compared to the much more numerous *Cooperative Base Network (CBN)*. This is a high-accuracy network of monumented control stations spaced at 25 to 50 km apart throughout the United States and its territories.

🌐 CBN Network Characteristics

Station Density: Much more numerous than FBN
Spacing: 25-50 kilometers apart
Coverage: United States and territories
Management: Created and maintained by state and private organizations

NGS Role: Technical assistance and coordination
Purpose: Densified high-accuracy reference network

The CBN was created and is maintained by state and private organizations with the help of NGS.

🌍 Modern Reference Frame Availability

The Continuously Operating Reference Stations in the NGS network are mostly to provide support for carrier phase observations. Information is available for postprocessing on the Internet.

📡 Current Coordinate Availability

ITRF2014 (2010.0): International Terrestrial Reference Frame coordinates
NAD83 2011 (2010.0): North American Datum coordinates
Data Access: Real-time and post-processing via Internet
Update Frequency: Continuous data streams with regular coordinate updates
Quality Control: Automated monitoring and validation systems

🔬 CORS Network Applications

Scientific Research:
- Crustal deformation monitoring
- Atmospheric studies
- Earthquake research
- Climate change indicators

Commercial Applications:
- Precision agriculture
- Construction and engineering
- Land surveying
- Transportation systems

The coordinates of the CORS are currently available in ITRF2014 (2010.0 and NAD83 2011 (2010.0).

📡 Summary



🗺️ Module 03: Geodetic Datums and Reference Systems

Understanding Earth's Shape and Reference Frameworks

🧭 Module Overview

In this module, we explored the fundamental concepts underlying all geodetic positioning systems—the three distinct figures that define our understanding of Earth's shape and serve as the foundation for satellite-based positioning. You've learned how the geoid, reference ellipsoid, and Earth's physical surface work together to create the mathematical framework that enables GPS and other positioning technologies. This knowledge forms the essential background for understanding coordinate systems, datum transformations, and the precision requirements of modern surveying applications.

🎯 Learning Objectives - What You've Learned

✅ Module Objectives

By this point in the module, you should be able to:

- Distinguish between the three fundamental figures: geoid, reference ellipsoid, and Earth's physical surface
- Explain why the reference ellipsoid serves as the mathematical foundation for geodetic positioning
- Describe the relationship between satellite geodesy and the need for geocentric datums
- Analyze the evolution from local datums (NAD27) to global reference systems (ITRF, WGS84)
- Compare and contrast major geodetic datums including WGS84, GRS80, and NAD83
- Understand the technical specifications and differences between ITRF realizations
- Explain coordinate transformation challenges between different datum systems
- Describe the role of CORS networks in maintaining modern geodetic infrastructure
- Identify the applications and limitations of various reference systems

- Recognize the importance of epoch and plate motion in modern geodetic systems

🔧 Key Technical Concepts

🌍 The Three Fundamental Figures

- **Earth's Physical Surface:** Irregular topography, measurement location
- **The Geoid:** Equipotential surface, height reference, irregular shape
- **Reference Ellipsoid:** Mathematical surface, coordinate reference
- **Common Center:** Essential for satellite geodesy applications
- **Mathematical Definition:** Only ellipsoid can be precisely described

📐 Reference System Evolution

- **Local Datums:** NAD27 with Clarke 1866 ellipsoid
- **Global Systems:** WGS84, ITRF for satellite applications
- **Plate Motion:** ITRS accounts for tectonic movement
- **Epoch Dependencies:** Time-specific coordinate realizations
- **Transformation Methods:** 7-parameter Helmert transformations

🏔️ The Three Distinct Figures in Geodetic Datums

Three distinct figures are involved in a geodetic datum for latitude, longitude and height: the geoid, the reference ellipsoid, and the Earth itself. Due in large measure to the ascendancy of satellite geodesy, it has become highly desirable that they share a common center.

🏔️ Earth's Physical Surface

- Where measurements are made
- Too irregular for mathematical description
- Varies from continents to ocean floors
- Cannot serve as coordinate reference

🌊 The Geoid

- Level surface for height reference
- Sometimes above, sometimes below Earth
- Defies concise geometrical definition
- Irregular, undulating surface

🥚 Reference Ellipsoid

- Same general shape as Earth
- Complete mathematical description
- Suitable for coordinate reference
- Foundation for positioning systems

🛰️ Modern Geocentric Reference Systems

🌐 Global Reference Systems Comparison

🌍 WGS84 & GRS80

- Nearly identical ellipsoids
- Same semimajor axis (6,378,137 m)
- Differ only at 6th decimal place
- WGS84 for GPS applications
- Multiple realizations over time

🔄 ITRF Series

- International Terrestrial Reference Frame
- True geocentric positioning
- Accounts for plate motion
- Regular updates (ITRF2008, ITRF2014)
- Centimeter-level global accuracy

🔗 Datum Relationships and Transformations

📊 Historical vs. Modern Datum Differences

📅 Historical Relationships

- **NAD83 (1986) vs WGS84 (1987):** 1-2 cm difference
- **Error Budget:** Differences within survey tolerances
- **Practical Impact:** Transformations unnecessary
- **Common Usage:** Often used interchangeably

⚠️ Modern Realities

- **NAD83 (2011) vs WGS84 (G1762):** 1-2 meter difference
- **Survey Impact:** Exceeds typical error budgets
- **Transformation Required:** Essential for accuracy
- **Plate Motion:** 10-20 mm/year relative movement

📡 Modern Geodetic Infrastructure

🏗️ CORS Network Hierarchy

📡 CORS Stations

- Thousands nationwide
- 3cm horizontal accuracy

🏛️ Federal Base Network

- ~100 km station spacing
- 1cm lat/lon accuracy

🤝 Cooperative Base Network

- 25-50 km station spacing
- State/private partnerships

- 5cm vertical accuracy
- 24/7 data availability
- Multiple reference frames

- 2cm ellipsoidal heights
- Dense in crustal motion areas
- Tied to CORS network

- High-accuracy monumentation
- Densified reference network
- NGS coordination

🧱 Why the Ellipsoid is Essential

While the level surface of the geoid provides a solid foundation for the definitions of heights and the topographic surface of the Earth is necessarily where measurements are made, neither can serve as the reference surface for geodetic positions.

❌ Why Not Earth's Surface?

- Too irregular from continents to ocean floors
- Cannot be represented by simple mathematical statement
- Solid Earth's actual surface defies geometric definition

❌ Why Not the Geoid?

- Sometimes under, sometimes above Earth's surface
- Overall shape defies concise geometrical definition
- Irregular undulations prevent mathematical modeling

✅ Why the Ellipsoid Works

But the ellipsoid not only has the same general shape as the Earth, but, unlike the other two figures, can be described simply and completely in mathematical terms. This mathematical precision enables coordinate computations, transformations, and the precise positioning that satellite systems require.

💡 Key Takeaways

🎯 Bottom Line Up Front

- **Three figures define geodetic systems:** Earth's surface (measurement location), geoid (height reference), and ellipsoid (coordinate reference)
- **Only the ellipsoid provides mathematical precision** necessary for coordinate calculations and satellite positioning systems
- **Satellite geodesy requires geocentric systems** where all three figures share a common center at Earth's mass center
- **Modern datum differences can exceed survey tolerances** requiring careful attention to reference frame selection and transformation
- **CORS networks provide the infrastructure** for accessing multiple reference frames with centimeter-level accuracy

- **Understanding these fundamentals is essential** for comprehending satellite-based positioning and its applications

🔮 Looking Ahead

A good understanding of these ideas is essential to clear comprehension of satellite based positioning. The next lesson will focus on the most usual manifestations of GPS coordinates in the United States, State Plane Coordinates and UTM coordinates. We will also have a bit of discussion on an often misunderstood topic—heights.

INTRODUCTION & INSTRUCTIONS

📡 Overview Module 04: State Plane Coordinates and Heights



📐 Module 04: State Plane Coordinates & Heights

🧭 Module Overview

A stake driven into the ground holds a clear position, but it is **awfully hard for its coordinates to be so certain**, even if the figures are precise. Coordinate systems must and do evolve as measurement technology improves. That has certainly been the case with the advent of GPS/GNSS.

That is reflected in the commonly used systems of **State Plane Coordinates and also in heights**. It may come as a bit of surprise, but the height, or elevation, that is most easily derived from satellite positioning is almost certainly not the sort of height you are after. Also, did you know that **the string of a plumb bob is actually very slightly curved**?

We will discuss these issues and the concepts at their foundation, exploring how **coordinate systems evolve with advancing measurement technology** and why understanding the relationship between different height systems is crucial for accurate geospatial positioning.

📐 What are State Plane Coordinates?

State Plane Coordinates are standardized coordinate systems used across the United States that project the curved Earth's surface onto flat planes for each state or region. These systems bridge the gap between geodetic positioning and practical surveying applications, making GPS/GNSS measurements compatible with traditional mapping workflows.

📚 Module Learning Objectives

At the successful completion of this lesson, students should be able to:

- explain the **basics of State Plane Coordinates**
- describe **NAD83 positions and plane coordinates**
- identify **map projections**
- define **map distortion**
- differentiate between **SPCS27 and SPCS83**
- describe **scale and distance in State Plane Coordinates**
- explain the **basics of heights**
- identify **ellipsoidal heights**
- recognize **orthometric heights**
- discuss the **evolution of the vertical datum in North America**
- recognize **the geoid**

📝 Module Instructions

- **Read Chapter 5 in *GPS and GNSS for Land Surveyors*.**
- Review the pages in this module and watch the instructor videos embedded within the module pages.
- After you have reviewed the content for this module, you will complete the assessments and activities to evaluate the learning objectives.

CONTENT

📚 State Plane Coordinates

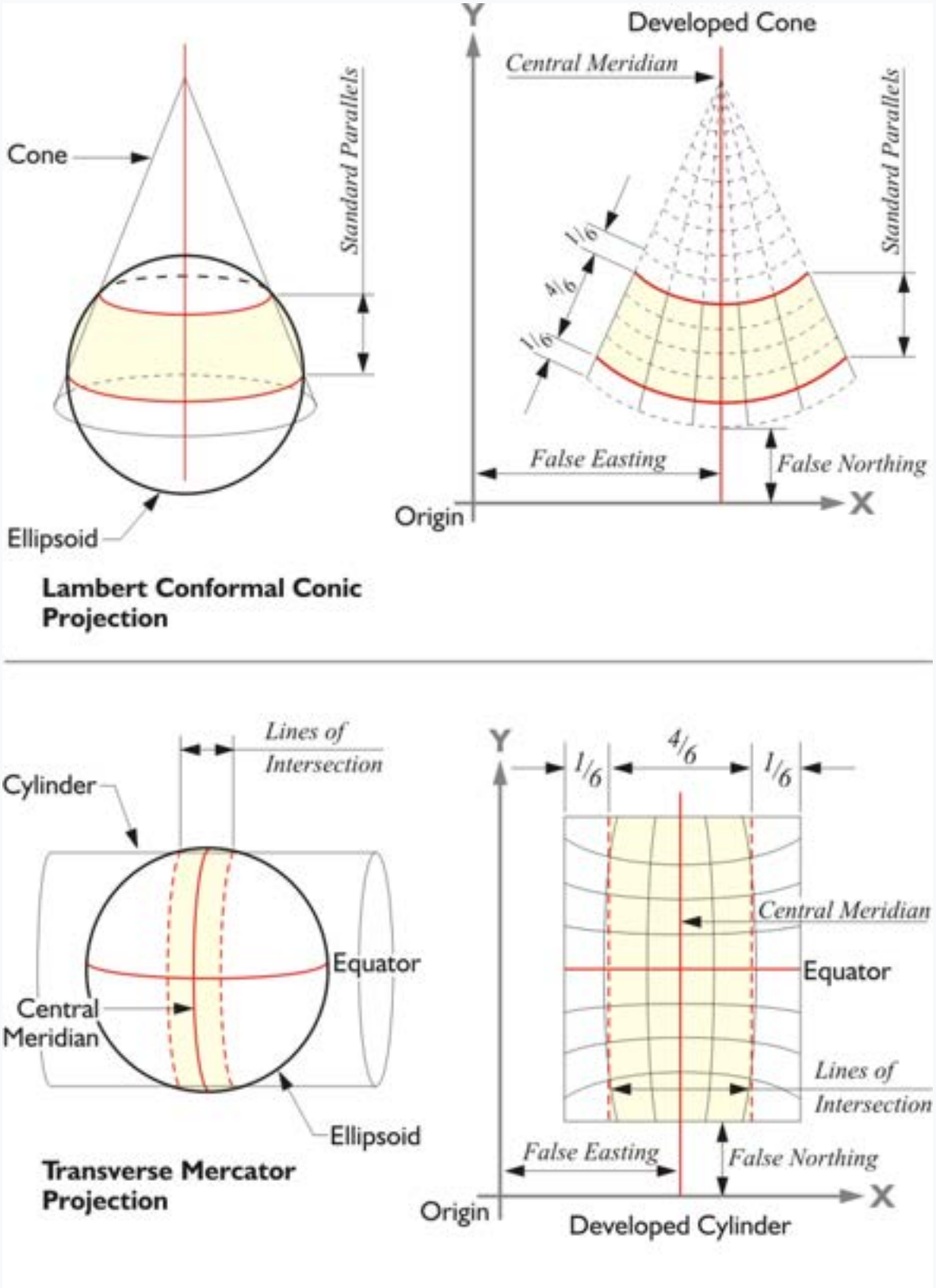
Module 04: State Plane Coordinates & Heights

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📍 The Foundation of Plane Coordinates

Why Use Flat Coordinate Systems?

This time, we'll be talking about **two coordinate systems**. These coordinate systems that we're going to discuss are **plane coordinate systems based upon the fiction that the earth is flat**, which, of course, immediately introduces distortion. However, much of GIS work—and GPS/GNSS work as well—is done based upon this presumption. So, it is worthwhile to spend some time discussing how the distortions are handled and some of the elements.



Plane Coordinates

Lambert Conformal Conic Projection (top) and Transverse Mercator Projection (bottom)

📐 State Plane Coordinates

- **Grid every state** into rectangular coordinate systems
- Based on **Transverse Mercator**, **Oblique Mercator**, and **Lambert Conic** projections
- Include **Puerto Rico and U.S. Virgin Islands**
- Use **northings and eastings** (y- and x-coordinates)
- Enable **plane trigonometry calculations**

🗺️ Universal Transverse Mercator (UTM)

- Adopted by **IUGG** (International Union of Geodesy and Geophysics)
- Used by **NATO and military organizations** worldwide
- Useful for **planning work over large areas**
- Global coverage with **60 zones**
- Based on **Transverse Mercator projection**

🔍 **Why Plane Coordinates Persist**

Despite the fact that the assumption of a flat Earth is fundamentally wrong, **calculation of areas, angles and lengths using latitude and longitude can be complicated**, so plane coordinates persist because they are convenient. The calculations can be done with plane trigonometry. GIS coordinate systems typically are plane. As long as the extent of the coverage of the coordinate system is limited, the curvature aspect—while it leads to distortion—can be managed.

⚖️ Understanding Map Distortions

Managing the Inevitable Distortions

When we project the **curved Earth's surface onto flat maps**, distortions are unavoidable. However, different projection methods allow us to control and minimize specific types of distortion depending on our application needs.

📏 Distance Distortion

How lengths are affected:

- **Scale factors** vary across the map
- Some projections preserve **distances along certain lines**
- **Local scale** vs. **principal scale** differences
- Critical for **surveying applications**

📐 Angular Distortion

How angles and shapes change:

- **Conformal projections** preserve local angles
- **Shape preservation** at small scales
- Important for **navigation and mapping**
- **Right angles remain perpendicular**

📊 Area Distortion

How areas are affected:

- **Equal-area projections** preserve relative sizes
- Critical for **statistical mapping**
- **Population density** and resource analysis
- **Cannot be conformal** simultaneously

⚡ **Key Insight: Projection Selection**

It's when the flat map, the flat coordinate system, **extends beyond a limited area that the distortion can get out of hand**. Therefore, the projection of points from the Earth's surface onto a reference ellipsoid and finally onto flat maps is still viable when properly managed within appropriate geographic extents.

📏 Height Systems: More Complex Than You Think

Three Different Ways to Measure Height

We'll be discussing **ellipsoidal heights**, **geoidal heights**, and **orthometric heights**. Understanding the relationships between these height systems is crucial for accurate GPS/GNSS positioning work.

📡 Ellipsoidal Heights

What GPS gives you directly

- Height above the **reference ellipsoid**
- **Mathematical surface**, not physical
- Used in **geodetic calculations**
- **Not the elevation you usually want**

🌍 The Geoid

Earth's "true" shape

- **Equipotential surface** of Earth's gravity field
- Sea level extended **under continents**
- **Undulates** relative to the ellipsoid
- The surface water would form **if at rest**

🌊 Orthometric Heights

Height above mean sea level

- What we typically call **"elevation"**
- Referenced to the **geoid**
- Used on **topographic maps**
- What **surveyors and engineers expect**

⚠️ **Important Relationship**

Ellipsoidal Height = Orthometric Height + Geoid Height

To convert GPS ellipsoidal heights to the elevations you need, you must know the local geoid height (geoid separation).

🔧 Practical Applications

Working with Coordinate Systems in Practice

GPS/GNSS surveys performed for **local projects and mapping are frequently reported** in the plane coordinates of State Plane or UTM systems. Understanding how to work with these systems is essential for modern geospatial professionals.

✅ Advantages of Plane Coordinates

- **Simple calculations** using plane trigonometry
- **Compatible with GIS systems**
- **Easy area and distance computation**
- **Familiar rectangular grid system**
- **Integration with existing mapping**

⚠️ Limitations to Consider

- **Distortion increases** with distance from projection center
- **Scale factors** must be applied for accurate measurements
- **Zone boundaries** create discontinuities
- **Not suitable for global applications**
- Height conversion often required

🔍 Did You Know?

The string of a plumb bob is actually **very slightly curved**! This is because it aligns with the local gravity vector, which is influenced by Earth's mass distribution and doesn't point exactly toward the center of the reference ellipsoid.

CONTENT

📚 Map Projection



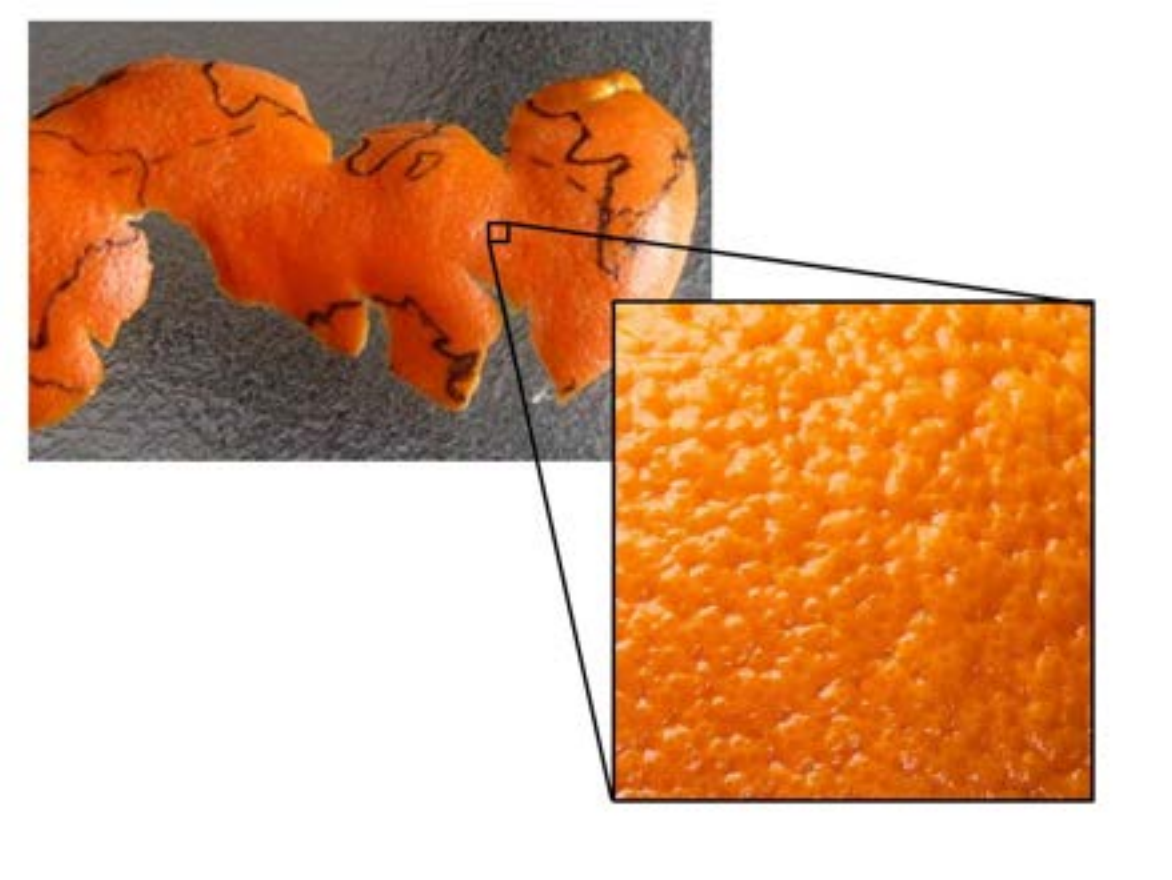
Module 04: Understanding Map Projections

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🍊 The Orange Peel Analogy

Understanding the Fundamental Challenge

State Plane Coordinate Systems are built on **map projections**. Map projection means representing a portion of the actual Earth on a plane. Done for hundreds of years to create paper maps, it continues, but map projection today is most often really a **mathematical procedure done in a computer**. Nevertheless, even in an electronic world, **it cannot be done without distortion**.



Orange Peel Analogy

Small sections can be flattened with minimal distortion

The problem is often illustrated by trying to **flatten part of an orange peel**. The orange peel stands in for the surface of the earth. To minimize the amount of natural distortion that comes with projecting curved data to a flat surface, we can focus in on a small area of interest. You can imagine that it is impossible to flatten a grapefruit skin without stretching or tearing, but a sugar ant sized piece of paper would sit flat on that grapefruit with no bending tearing or stretching.

Distortion Becomes Severe with Larger Areas

As the area increases, tearing and stretching become inevitable

⚠️ **The Fundamental Problem**

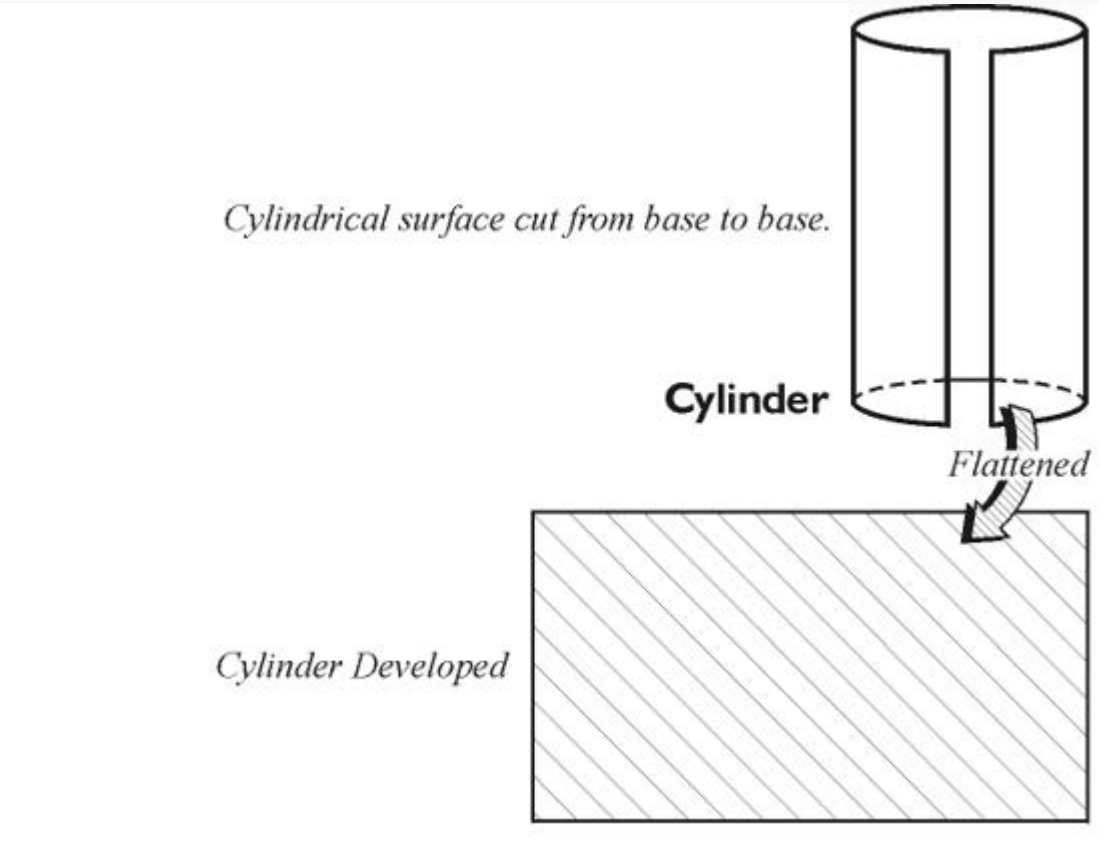
However, when the portion of the orange peel gets larger, problems appear. Suppose the whole orange peel is involved; as the center is pushed down, **the edges tear or stretch, or both**. And

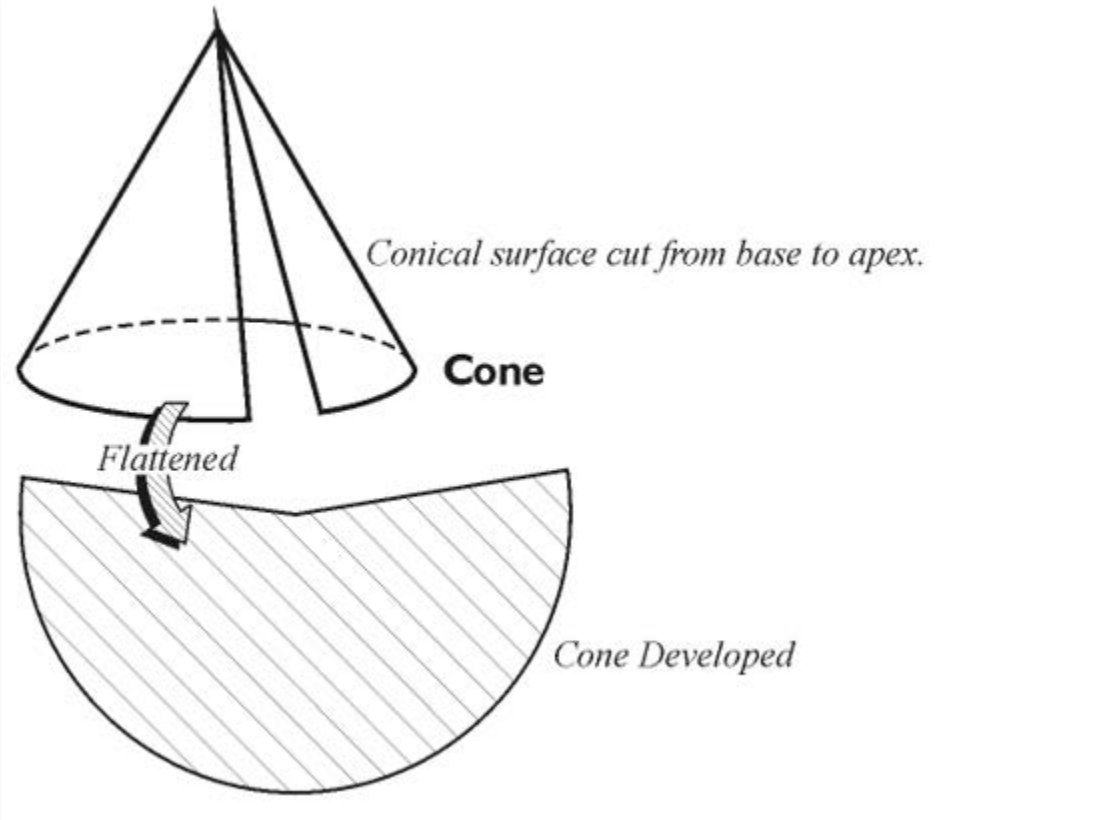
if the peel gets even bigger, the tearing gets more severe. So, if a map is drawn on the orange before it is peeled, the map gets distorted in unpredictable ways when it is flattened.

📐 Developable Surfaces: The Solution

Using Surfaces That Can Be Flattened

These are the problems that a map projection needs to solve to be useful. The first problem is the surface of an ellipsoid, like the orange peel, is ***non-developable***. In other words, flattening it inevitably leads to distortion. So, a useful map projection ought to start with a surface that is ***developable***, a surface that may be flattened without all that unpredictable deformation.





Developable Surfaces

Cylinders and cones can be cut and flattened without distortion

🥤 Cylindrical Projections

- Used in **Mercator projections**
- Used in **Transverse Mercator projections**
- Cylinder can be **cut and flattened perfectly**
- **Most common in UTM systems**
- Good for **east-west trending areas**

🎯 Conical Projections

- Used in **Lambert Conic projections**
- Cone can be **cut along an element and flattened**
- **Most common in State Plane systems**
- Good for **north-south trending areas**
- Excellent for **mid-latitude regions**

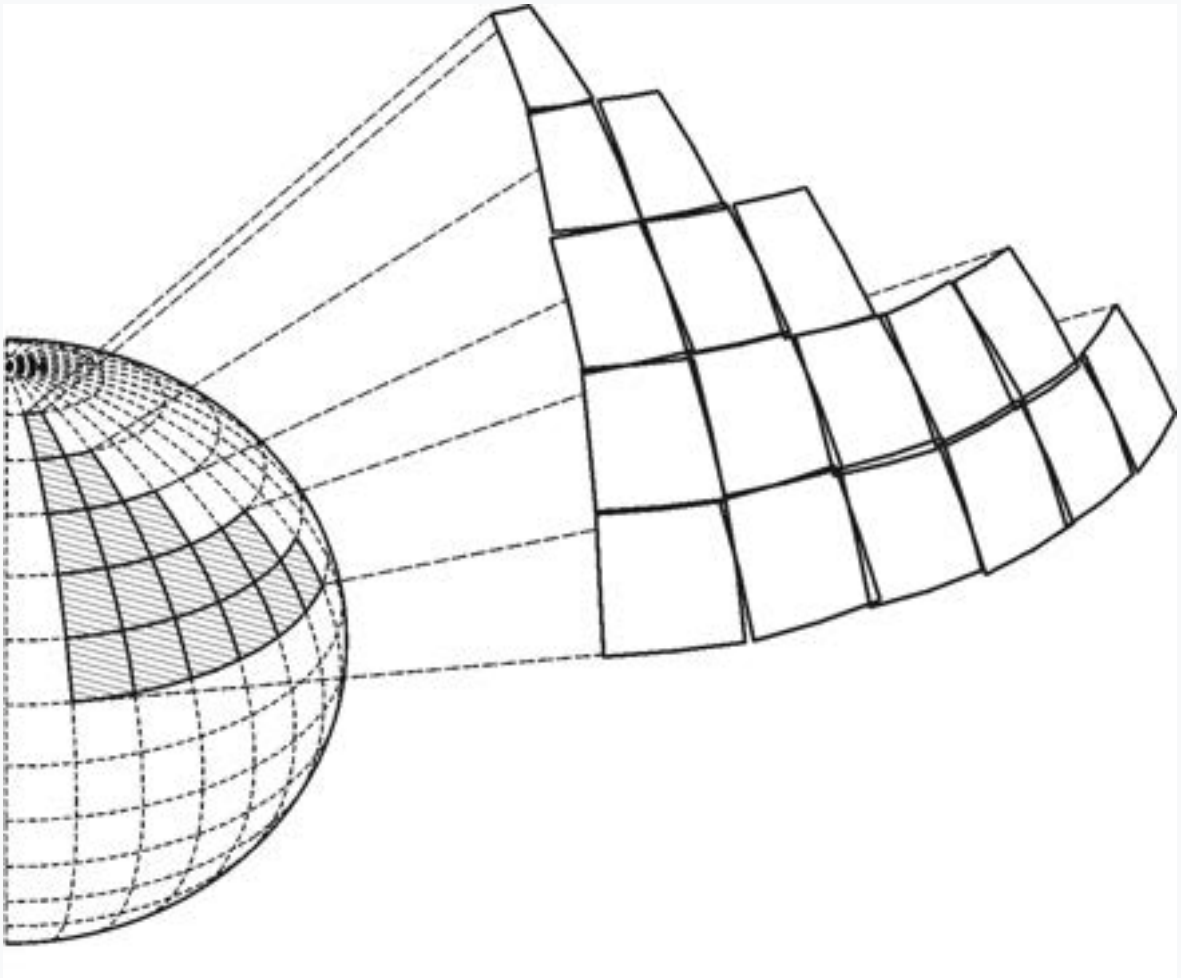
🔍 Key Insight

A paper cone or cylinder both illustrate this idea nicely. They are illustrations only, models for thinking about the issues involved. If a right circular cone is cut perpendicularly from the base to its apex, **the cone can then be made completely flat without trouble**. The same may be said of a cylinder cut perpendicularly from base to base.

📍 Local Coordinate Systems

The Simplest Case: Tangent Planes

Or one could use the simplest case, a surface that is already developed. **A flat piece of paper is an example**. If the center of a flat plane is brought tangent to the earth, a portion of the planet can be mapped on it, that is, it can be projected directly onto the flat plane.



Local Coordinate Systems

Multiple tangent planes projected onto the globe

✅ Advantages of Local Systems

- **Simple Cartesian coordinates**
- **Convenient and easy to use**
- **Little manipulation of field observations**

❌ Problems with Local Systems

- **Don't match when brought together**
- **Overlaps and gaps** between systems
- **Different scales and orientations**

- **Negligible distortion for small areas**
- Based on idea that **small sections are nearly flat**

- **Becomes untenable as area grows**
- Limited to **very small extents**

In fact, this is the typical method for establishing an **independent local coordinate system**. These simple Cartesian systems are convenient and satisfy the needs of small projects. The method of projection, onto a simple flat plane, is based on the idea that a small section of the earth, as with a small section of the orange mentioned previously, **conforms so nearly to a plane that distortion on such a system is negligible**.

🚧 When Local Systems Fail

The Need for Comprehensive Systems

Subsequently, local tangent planes have been long used. Such systems demand little if any manipulation of the field observations, and the approach has merit **as long as the extent of the work is small**. It's convenient. It's easy. It can satisfy small projects fairly well, but **the larger each of the planes grows, the more untenable it becomes**.

> ⚠️ **Critical Problem**
>
> Another difficulty arises when the local coordinate systems are brought together—as you see in this image—they simply **don't match**. There are overlaps, there are gaps. The scale of each of the local systems are not the same, nor is their orientation.

In other words, a local tangent coordinate system can be useful in a very limited extent, as long as one stays within the zone that it intends to cover. When you bring in another coordinate system from another direction, **trouble ensues**.

📈 As Areas Grow Larger

The reduction of observations becomes more complicated and must take account of:

- **The actual shape of the earth**
- **The ellipsoid**
- **The geoid**
- **Geographical coordinates** (latitude and longitude)

🎯 The Solution

At that point, surveyors and engineers rely on **map projections** to:

- **Mitigate the situation**
- **Limit the troublesome distortion**
- Provide **manageable distortion levels**
- Cover **entire states or large portions**

🏛️ Enter State Plane Coordinate Systems

A Well-Designed Solution

This is one of the reasons **State Plane coordinate systems were devised**. A well-designed map projection can offer the convenience of working in plane Cartesian coordinates and while not eliminating the inevitable distortion **keep it at manageable levels**. They cover an entire state or a fairly good portion of the state with a zone or zones.

🎯 **State Plane System Benefits**

- **Convenience of plane Cartesian coordinates**
- **Manageable distortion levels** across large areas
- **Consistent system** across entire states
- **No gaps or overlaps** between adjacent areas
- **Standardized** for professional use
- **Compatible with GPS/GNSS** positioning

💡 **Key Takeaway**

State Plane coordinate systems represent the optimal balance between the convenience of flat, rectangular coordinates and the need to minimize distortion across large geographic areas. They solve the fundamental problems that arise when trying to bring multiple local coordinate systems together.

CONTENT

📚 Distortion

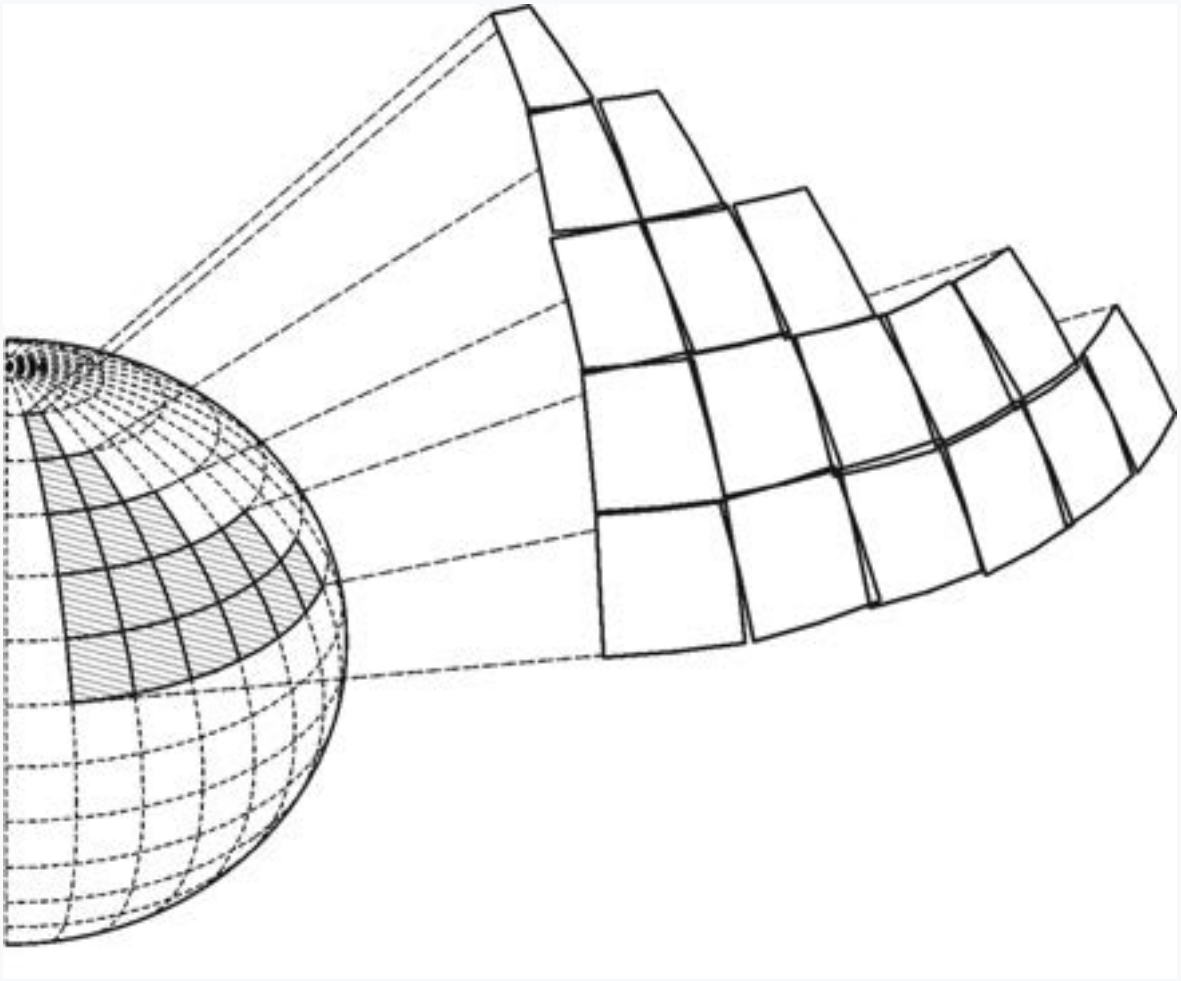
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🚫 The Fundamental Problem with Local Systems

Why Multiple Local Planes Cannot Be Joined

Here is a bit more on that idea. The design of **local tangent plane projections must accommodate some awkward facts**. For example, while it would be possible to imagine mapping a considerable portion of the earth using a large number of small individual planes, like facets of a gem, **it is seldom done** because when these planes are brought together they cannot be edge-matched accurately.



Edge Matching Difficulty

Local tangent planes cannot be accurately joined at their borders

⚠️ The Core Issue

They **cannot be joined properly along their borders**. And the problem is unavoidable because the planes, tangent at their centers, inevitably depart more and more from the reference ellipsoid at their edges, and **the greater the distance between the ellipsoidal surface and the surface of the map** on which it is represented, **the greater the distortion** on the resulting flat map.

📏 Distortion at the Worst Possible Location

Where Accuracy Matters Most

This is true of all methods of map projection. Therefore, one is faced with the **daunting task of joining together a mosaic of individual maps along their edges where the accuracy of the representation is at its worst**.

📐 Distortion Distribution

- **Center of plane:** Minimal distortion (tangent point)
- **Moving outward:** Distortion increases progressively
- **Edges of plane:** Maximum distortion
- **Junction points:** Where accuracy is worst

🔗 Edge Matching Challenge

- **Need to join** where distortion is greatest
- **Cannot achieve** seamless connections
- **Accuracy compromised** at critical boundaries
- **Cumulative errors** across multiple junctions

⚠️ Even Perfect Distortion Matching Won't Help

Even if one could overcome the problem by making the distortion, however large, **the same on two adjoining maps**, another difficulty would remain. This highlights that distortion is not the only problem with local systems.

🎯 Coordinate System Incompatibility

When Every System is Unique

Typically, **each of these planes has a unique coordinate system**. The orientation of the axes, the scale, and the rotation of each one of these individual local systems will not be the same as those elements of its neighbor's coordinate system.

🧭 Orientation Issues

Axis Direction Problems:

- Each system has **different axis orientations**
- **North direction** varies between systems
- **Grid rotation** differs for each plane

📏 Scale Variations

Measurement Problems:

- **Different scale factors** for each system
- **Unit inconsistencies** between adjacent areas
- **Distance calculations** become problematic

🔄 Rotation Problems

Angular Issues:

- **Each system rotated** differently
- **Bearing measurements** incompatible
- **Angular relationships** cannot transfer

🚫 **Inevitable Results**

Subsequently, there are **gaps and overlaps between adjacent maps**, and their attendant coordinate systems, because there is **no common reference system**.

🧭 The Direction Problem

When North is Not Constant

Even if it was possible to accommodate, let's say, a GIS coordinate system with a local coordinate system, it wouldn't be terribly desirable because of the problems of scale and distortion as mentioned but also **all problems of direction**.

🌍 Geographic Reality

Obviously, north changes as one goes from meridian to meridian around the earth. This must be accommodated by any plane coordinate system.

Key Insight: Each meridian has a slightly different orientation relative to any fixed plane coordinate system.

🎯 System Requirements

Any comprehensive coordinate system must handle:

- **Meridian convergence**
- **Grid azimuth vs. geodetic azimuth**
- **Consistent directional references**
- **Angular relationship preservation**

⚖️ The Final Verdict on Local Systems

Limited but Valid Applications

So, the idea of a **self-consistent, local map projection** based on small, flat planes tangent to the earth, or tangent to the reference ellipsoid, is convenient, but only for small projects that have no need to be related to adjoining work.

✅ **When Local Systems Work**

- **Small, isolated projects**
- **No need to connect** to adjacent work
- **Self-contained** within single zone

❌ **When They Fail**

- **Significant area coverage** needed
- **Multiple projects** must connect
- **Regional or state-wide** applications

- **Entirely adequate** for limited scope
- **Simple and convenient** to implement

- **GIS integration** requirements
- **Standardization** across organizations

🔍 **The Bottom Line**

In that case there is **no need to venture outside the bounds** of a particular local system, it can be entirely adequate. But if a significant area needs coverage, **another strategy is needed**.

CONTENT

📚 Decreasing Distortion

Module 04: Decreasing Distortion

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📐 Reducing Distortion: The Eternal Challenge

Moving Beyond Simple Tangency

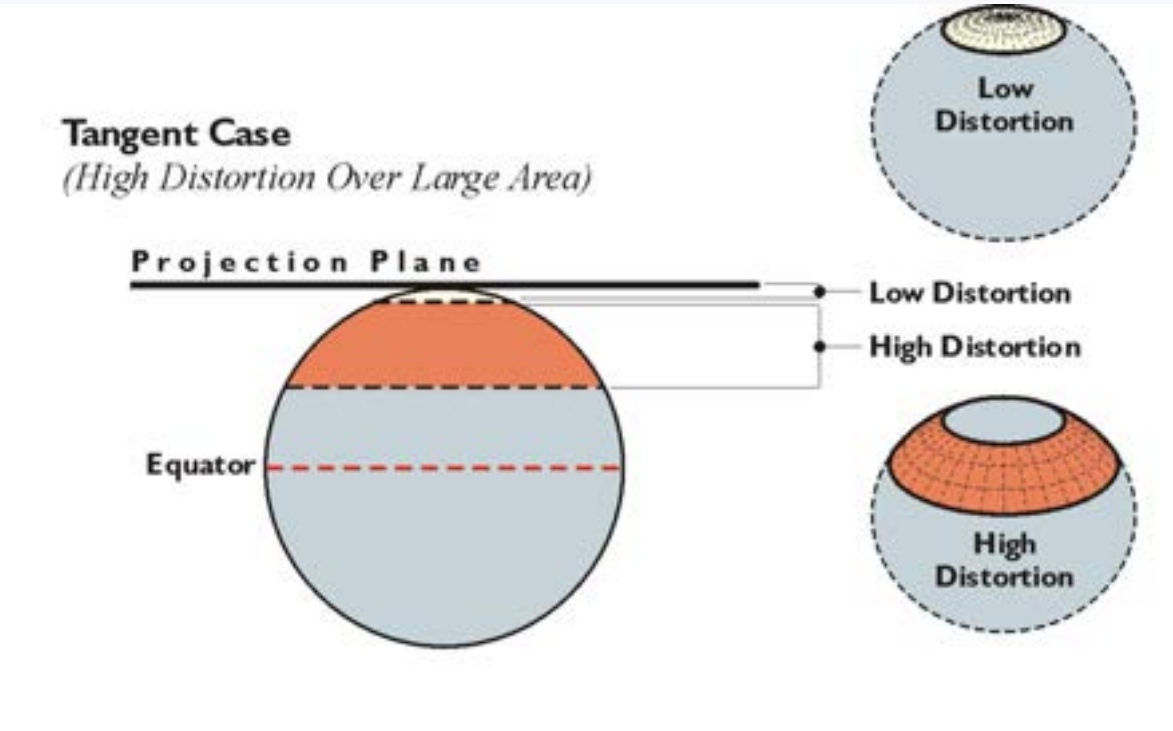
Decreasing distortion is a constant and elusive goal in map projection. It can be done in several ways. Most involve reducing the distance between the map projection surface and the ellipsoidal surface. One way this is done is to move the mapping surface from tangency with the ellipsoid and make it actually cut through it. This strategy produces what is known as a **secant projection**.

A secant projection is one way to shrink the distance between the map projection surface and the ellipsoid. Thereby, the area where distortion is in an acceptable range on the map can be effectively increased.

📊 Tangent vs. Secant: Visual Comparison

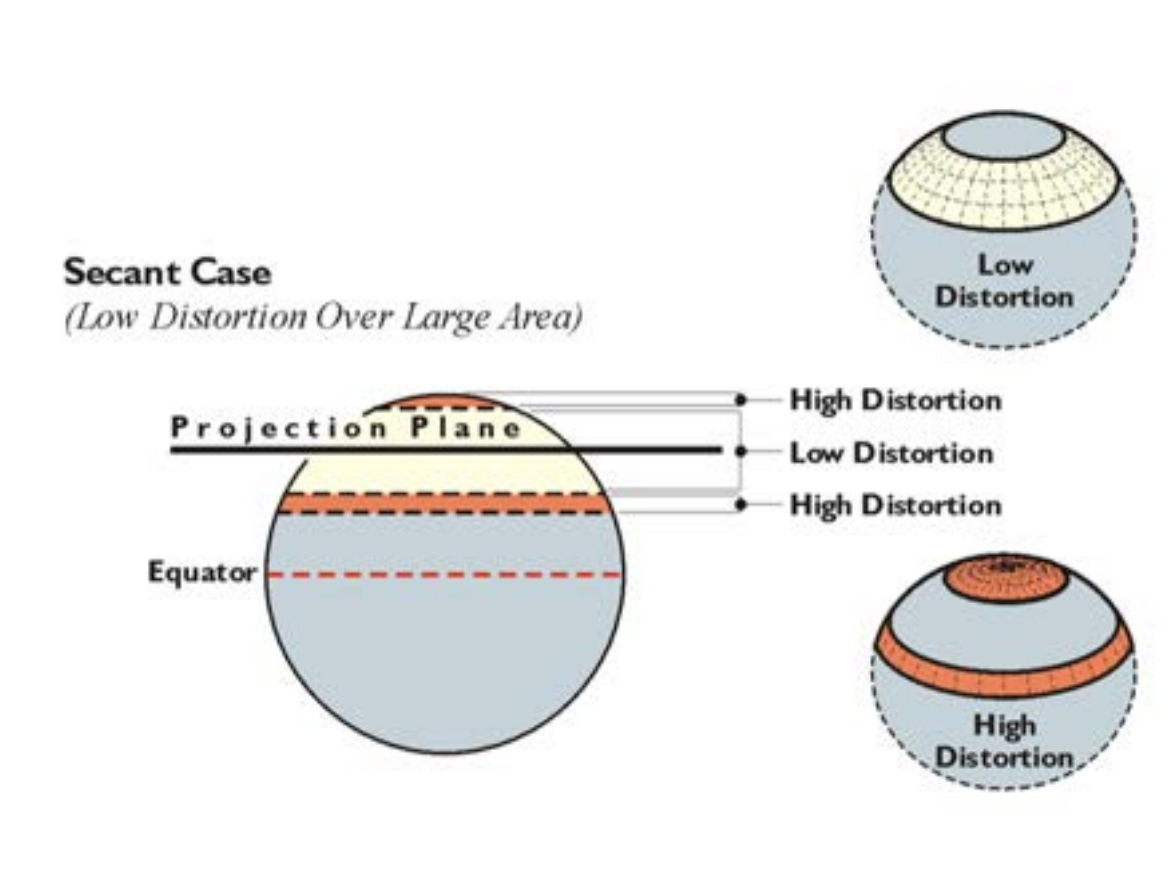
Understanding the Difference

The illustrations here are intended to show how the tangent case distorts a larger area than does the secant case.



Tangent Case

High distortion over large area



Secant Case

Low distortion over large area

📍 Tangent Case Problems

The tangent on the top shows a flat projection plane touching the Earth at one point. This is the approach used in polar protections (North Pole, South Pole).

- The area in **orange has high distortion**
- The low distortion, the area in **light beige, is much smaller**
- **Single point of contact** with Earth
- **Large areas of unacceptable distortion**

✂️ Secant Case Advantages

However, if the projection plane cuts the Earth rather than just touching it at one point, there's a larger area of relatively low distortion.

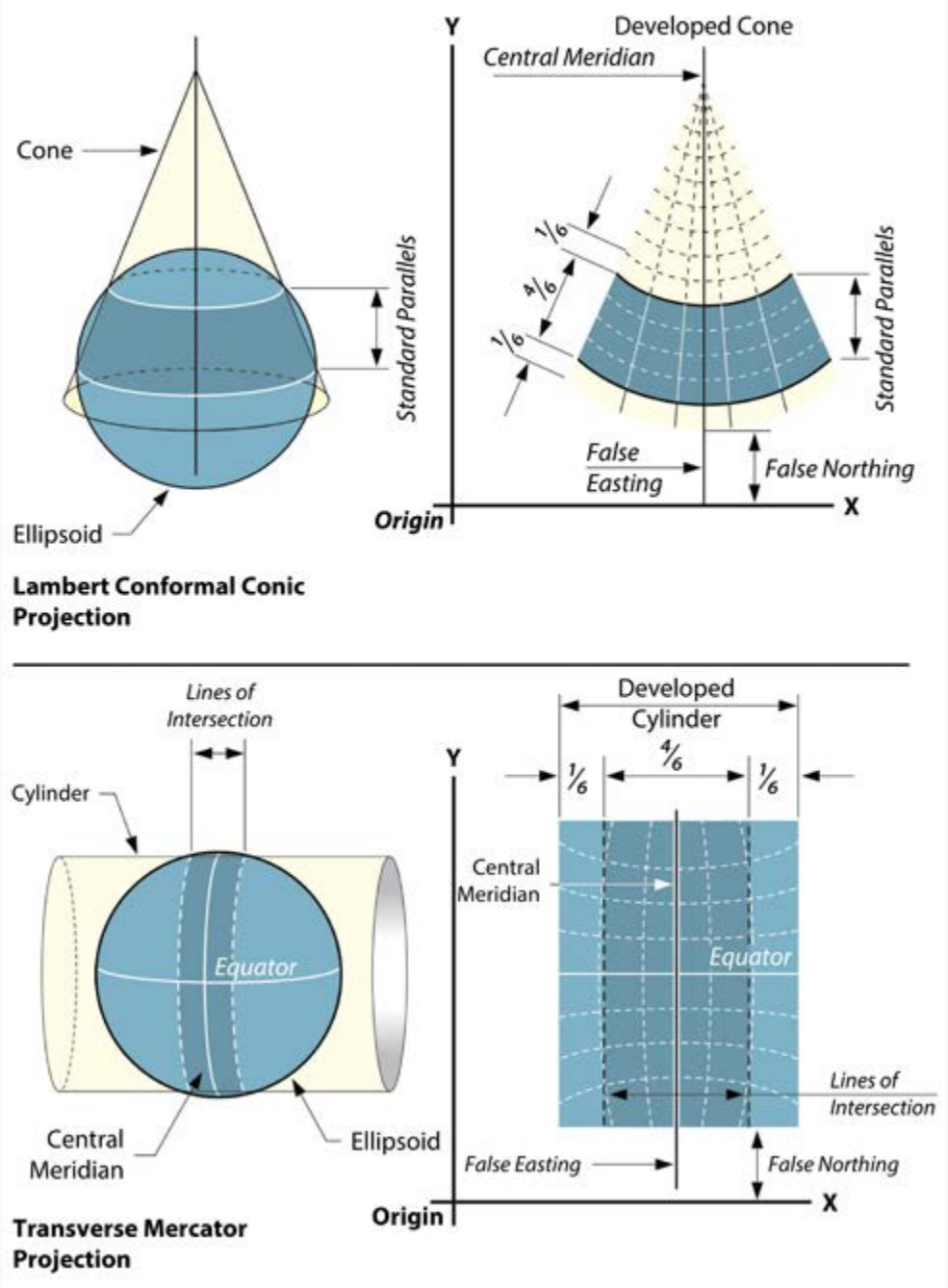
- The **less the distance between the plane and the Earth, the less the distortion**
- Areas closest to plane have **less distortion**
- **Lower distortion over larger area** than tangent case
- **Multiple contact lines** with Earth

> 🎯 **Key Principle**
>
> (The areas on the Earth closest to the plane have less distortion than the areas farther away from the plane.) The secant case has lower distortion over a larger area than does the tangent case.

🗺️ Why State Plane Systems Use Secant Projections

Lines of Exact Scale

This is why State Plane coordinate systems in the United States use secant projections. In the case of Lambert projection, there are two parallels of latitude where the mapping plane cuts the Earth. In the case of Transverse Mercator, there are two approximately north-south lines that are not meridian of longitude. In both cases, these are **lines of exact scale**.



Secant Projections

Lambert Conformal Conic and Transverse Mercator in secant configuration

🎯 Lambert Conic Secant

- **Two standard parallels** of latitude
- Cone **cuts through the Earth** at these lines
- **Exact scale** along both parallels

📐 Transverse Mercator Secant

- **Two approximately north-south lines**
- Lines are **not meridians of longitude**
- **Exact scale** along both lines

- Best for **east-west trending** areas
- Minimal distortion **between the parallels**

- Best for **north-south trending** areas
- Minimal distortion **between the lines**

🏗️ The Advantage of Curved Surfaces

Why Cones and Cylinders Work Better

Both cones and cylinders have an advantage over a flat map projection plane. They are curved in one direction and can be designed to follow the curvature of the area to be mapped in that direction.

🌐 Matching Earth's Curvature

The curved nature of cones and cylinders allows them to **conform more closely to the Earth's shape** in one dimension, reducing the fundamental mismatch between the curved Earth and the flat map.

🗺️ Multiple Zones for Complete Coverage

The Zone System Approach

Also, if a large portion of the ellipsoid is to be mapped, several cones or several cylinders may be used together in the same system to further limit distortion. In that case, each cone or cylinder defines a ***zone*** in a larger coverage. This is the approach used in State Plane Coordinate systems.

Generally, one cone or one cylinder is not sufficient to cover an entire state (though there are exceptions). Usually a collection of cones or cylinders, one for each zone, is used. That's the approach that was used in the original design of most State Plane coordinate systems in this country.

🎯 Zone System Benefits

- **Each zone optimized** for its specific area
- **Distortion minimized** within each zone
- **Complete state coverage** achieved
- **Seamless transitions** between adjacent zones
- **Practical compromise** between accuracy and usability

📏 Design Considerations

- **State shape and extent** determine zone number
- **Geographic orientation** influences projection choice
- **Each zone** has its own parameters
- **Standardized approach** across the nation
- **Historical design** still serves modern needs

🔍 **Historical Success**

The original design approach has proven so effective that it continues to serve as the foundation for modern GPS/GNSS coordinate transformations and GIS applications across the United States.

📝 The Power of Strategic Cutting

From Concept to Implementation

The transition from tangent to secant projections represents a fundamental advance in cartographic thinking. By strategically cutting through the Earth rather than merely touching it, mapmakers achieved the optimal balance between accuracy and coverage that makes State Plane coordinate systems possible.

📚 Secant Projections

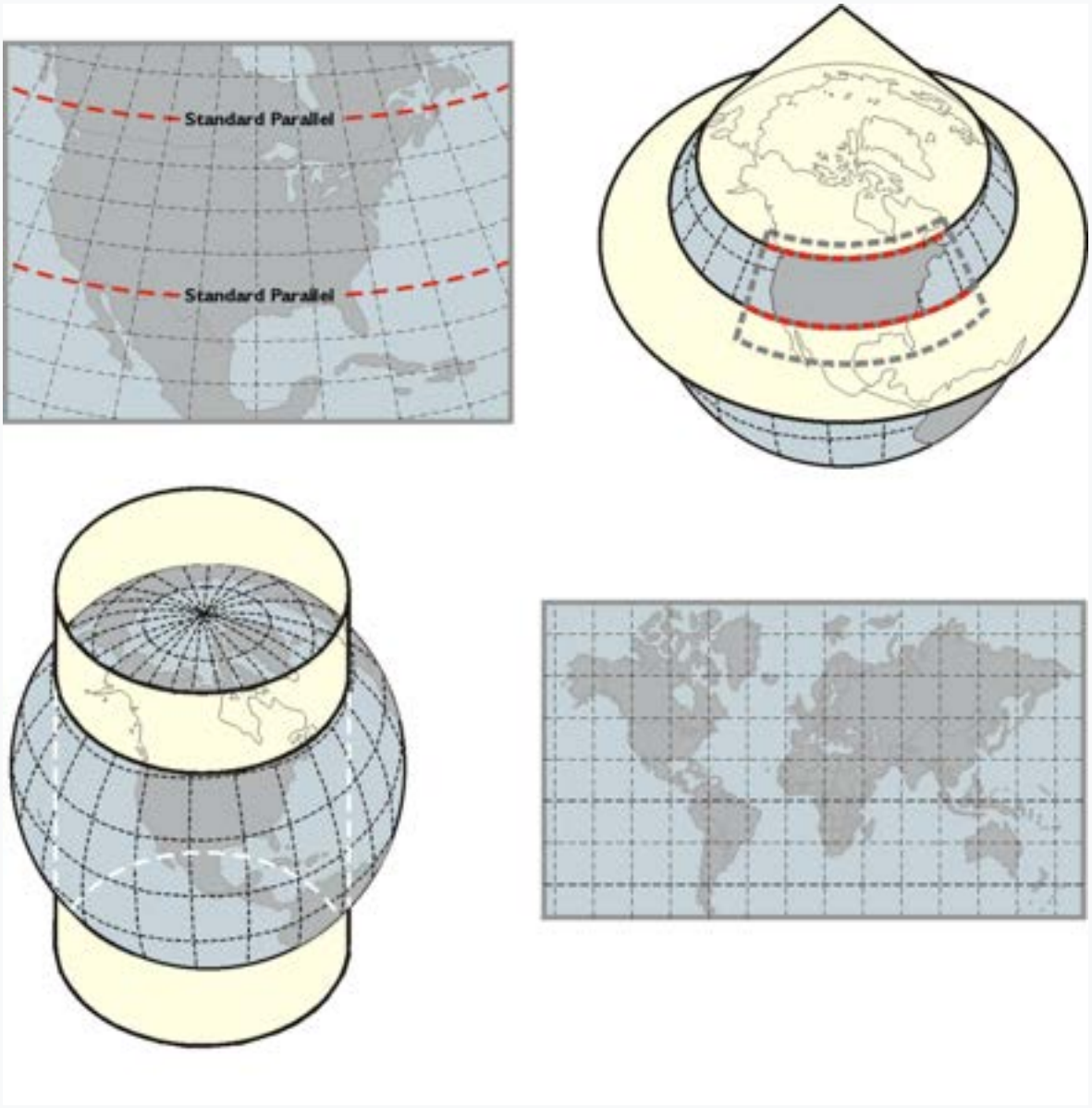
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How Secant Projections Work

Intersecting the Ellipsoid

As mentioned, when a conic or a cylindrical map projection surface is made secant, it intersects the ellipsoid, and the map is brought close to its surface. For example, the conic and cylindrical projections shown in the illustration cut through the ellipsoid. The map is projected both inward and outward onto it and there is no distortion on the ***two lines of exact scale***, standard lines.



Secant Projections

Conic and cylindrical surfaces cutting through the ellipsoid

Key Concept

They are created along the small circles where the cone and the cylinder intersect the ellipsoid. They are called ***small circles*** because they do not describe a plane that goes through the center of the Earth, as do the previously mentioned great circles.

Understanding Distance Relationships

Above, Below, and On the Standard Lines

Between the standard lines, the map is under the ellipsoid, and outside of the standard lines, the map is above it. That means that between the standard lines, a distance from one point to another is actually longer on the ellipsoid than it is shown on the map, and outside the standard lines, a distance on the ellipsoid is shorter than it is on the map.

On Standard Lines

Perfect Scale Match:

- **No distortion** whatsoever
- **Exact scale** representation
- **Map distance = Earth distance**
- These are the **reference lines**

Between Standard Lines

Map Under Ellipsoid:

- **Earth distance > Map distance**
- Map shows distances **shorter** than reality
- **Negative scale factor**
- **Compressed representation**

Outside Standard Lines

Map Above Ellipsoid:

- **Earth distance < Map distance**
- Map shows distances **longer** than reality
- **Positive scale factor**
- **Expanded representation**

Lines of Exact Scale

Why They Matter

Any length that is measured along a standard line is the same on the ellipsoid and on the map, which is why another name for standard parallels is ***lines of exact scale***.

Perfect Accuracy

These lines represent the only locations on the entire map where measurements can be made with **perfect accuracy**—no scale corrections are needed along these lines.

Different Projections, Different Lines

Lambert vs. Transverse Mercator

They are called standard parallels in the case of a Lambert projection that is illustrated at the top here. The lower picture is Transverse Mercator projection and the lines of exact scale neither meridians of longitude nor parallels of latitude, but are rather **complex curves**.

Lambert Conic Projection

Standard Parallels:

- **Two parallels of latitude**
- **East-west running lines**
- **Constant latitude values**
- **Straight lines on the map**
- Best for **east-west trending** areas

Transverse Mercator Projection

Complex Curves:

- **Not meridians of longitude**
- **Not parallels of latitude**
- **Complex curved lines**
- **Approximately north-south** trending
- Best for **north-south trending** areas

📚 Conformal

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🎯 The Fundamental Goal

From Ellipsoid to Flat Map

Ultimately, the goal is very straightforward relating each position on one surface, the reference ellipsoid, to a corresponding position on another surface as faithfully as possible and then flattening that second surface to accommodate Cartesian coordinates. In fact, the whole procedure is in the service of moving from geographic to Cartesian coordinates and back again. These days, the complexities of the mathematics are handled with computers. Of course, that was not always the case.

Two Projections

Lambert Conformal Conic Projection and Transverse Mercator Projection

📊 Coordinate Transformation

The entire process serves one primary purpose: **moving from geographic to Cartesian coordinates and back again** with maximum fidelity and minimal distortion.

📐 Conformal Projections: Preserving Shape

Orthomorphic Properties

Map projections in which shape is preserved are known as ***conformal*** or ***orthomorphic***. Orthomorphic means right shape. In a conformal projection, the angles between intersecting lines and curves retain their original form on the map.

📏 Angle Preservation

Short Lines (Under 10 Miles):

- A **45° angle on the ellipsoid** remains a **45° angle on the map**
- **Perfect angular preservation**
- **Local shape consistency**
- Critical for **surveying applications**

⚖️ Scale Consistency

Isotropic Scaling:

- **Scale is the same in all directions** from a point
- This characteristic **preserves the angles**
- **No directional distortion** at any given point
- Enables **accurate local measurements**

⚠️ Long Lines Limitation

On long lines, angles on the ellipsoid are not exactly the same on the map projection. Nevertheless, **the change is small and systematic**.

✅ Benefits for State Plane Users

Why Conformal Properties Matter

These aspects were certainly a boon for all State Plane Coordinate users.

🎯 Practical Advantages

- **Surveying calculations** remain accurate for local work
- **Angular measurements** are preserved
- **Shape integrity** maintained for small areas
- **Compatible with traditional** surveying methods
- **Predictable distortion patterns** for corrections

🗺️ The Three Original Projections

State Plane Coordinate System Foundations

Actually, all three of the projections that were used in the designs of the original State Plane Coordinate Systems were conformal. Each system was originally based on the North American Datum 1927, NAD27. Along with the Oblique Mercator projection, which was used on the panhandle of Alaska, the two primary projections were the Lambert Conic Conformal Projection and the Transverse Mercator projection.

🎯 Lambert Conic Conformal

- **Primary projection** for many states
- **Best for east-west** trending areas
- **Two standard parallels**
- **Excellent for mid-latitudes**

📐 Transverse Mercator

- **Second primary** projection
- **Best for north-south** trending areas
- **Complex standard lines**
- **Rotated Mercator** concept

🔄 Oblique Mercator

- **Special case** projection
- **Alaska panhandle** only
- **Oblique orientation**
- **Unique geographic** requirements

📅 Evolution of Datums

From NAD27 to Modern Systems

Today, they're in North American Datum of 1983, NAD83 2011 (2010.0).

📊 Original Foundation

NAD27 Era:

- **1927 datum** foundation
- **Original State Plane** system design
- **Pre-GPS technology**
- **Ground-based triangulation** networks

🛰️ Modern Evolution

NAD83 2011:

- **2010.0 epoch** reference
- **GPS/GNSS compatible**
- **Enhanced accuracy**
- **Same projection principles**

> 🔄 **Continuity and Progress**
>
> While the underlying datum has evolved from NAD27 to modern NAD83 realizations, the fundamental conformal projection principles remain unchanged, ensuring continuity while incorporating improved accuracy from satellite-based positioning systems.

📚 Projection Design

Module 04: Projection Design

GISC 4315

🚧 The Problem: 1932 North Carolina

When Long Route Surveys Failed

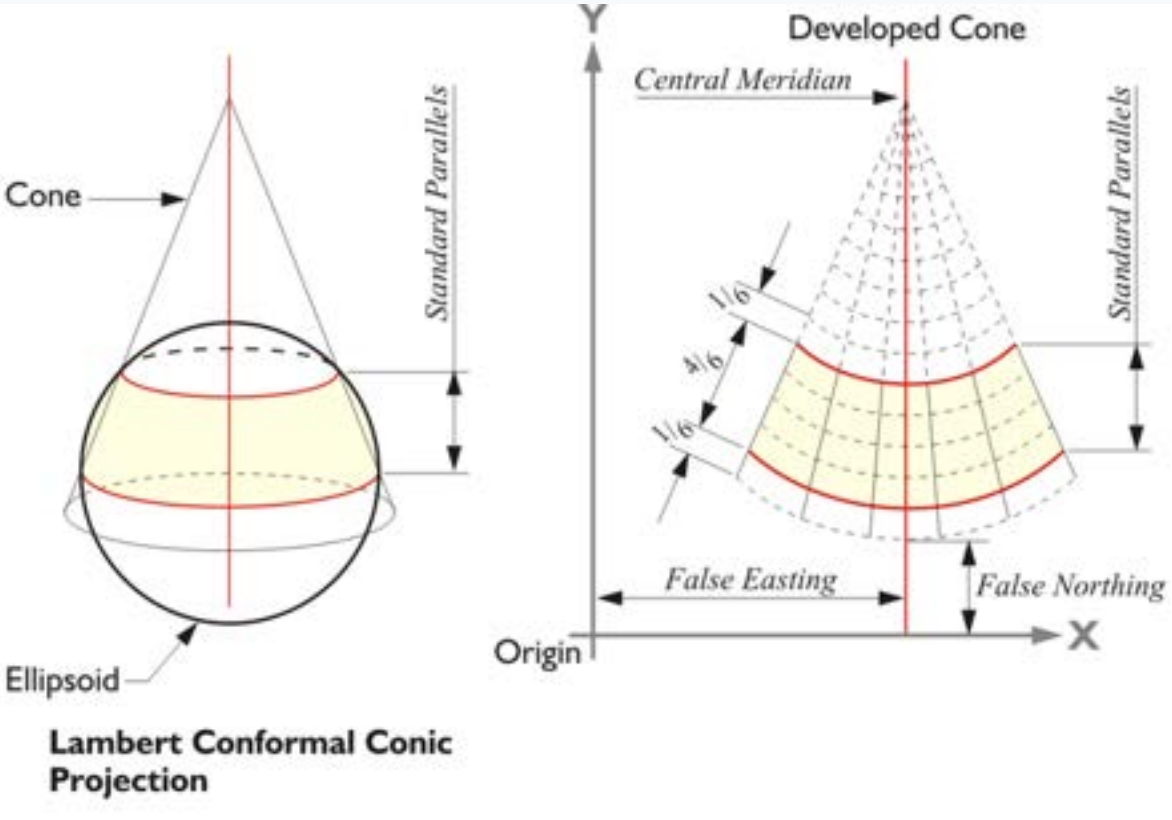
In 1932, two engineers in North Carolina's highway department, O.B. Bester and George F. Syme, appealed to the then Coast & Geodetic Survey (C&GS, now NGS) for help. They had found that the stretching and compression inevitable in the representation of the curved earth on a plane was so severe over long route surveys that they could not check into the C&GS geodetic control stations across a state within reasonable limits.

> ⚠️ **The Critical Issue**
>
> Highway surveyors could not maintain acceptable accuracy when trying to connect their long route surveys to the existing geodetic control network because of excessive distortion in their coordinate systems.

💡 The Proposal and Solution

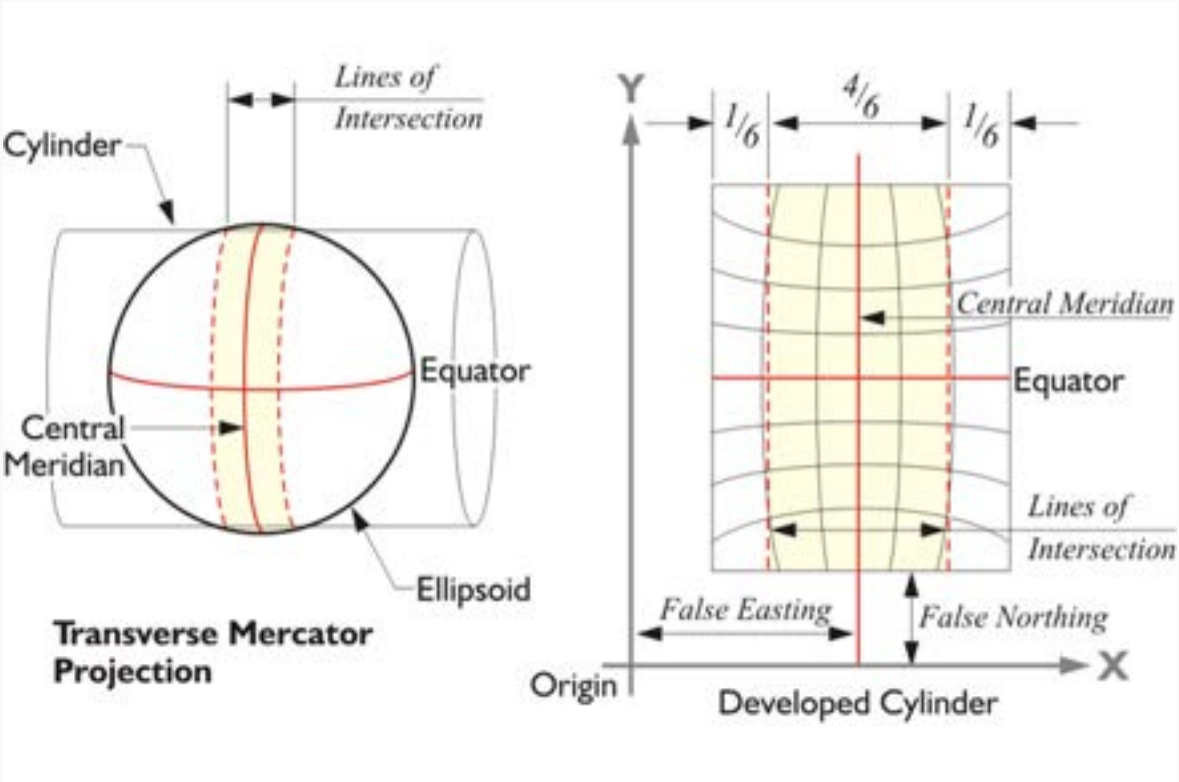
Dr. Oscar Adams' Innovation

The engineers suggested that a plane coordinate grid system be developed that was mathematically related to the reference ellipsoid, but could be utilized using plane trigonometry rather than laborious geodetic calculations because in those days such calculations required sharp pencils, logarithmic tables and lots of midnight oil. Dr. Oscar Adams of the Division of Geodesy, assisted by Charles Claire, designed the first State Plane Coordinate System to mediate the problem. It was based on a map projection called the ***Lambert Conformal Conic Projection***.



A Conic Secant Projection

The foundation of Lambert Conformal Conic systems



A Cylindrical Secant Projection

The foundation of Transverse Mercator systems

⚖️ The Inevitable Trade-offs

Perfect Maps Are Impossible

Adams realized that it was possible to use this map projection and allow one of the four elements of area, shape, scale, or direction to remain virtually unchanged from its actual value on the earth, but not all four. On a perfect map projection, all distances, directions, and areas could be conserved. They would be the same on the ellipsoid and on the map. Unfortunately, it is not possible to satisfy all of these specifications simultaneously, at least not completely.

There are inevitable choices. It must be decided which characteristic will be shown the most correctly, but it will be done at the expense of the others. And there is no universal best decision. Still, a solution that gives the most satisfactory results for a particular mapping problem is always available.

📏 Scale	📐 Shape	📊 Area	🧭 Direction
Distance accuracy	Angular preservation	Size relationships	Bearing accuracy

🎯 Adams' Choice

Adams decided to preserve **shape** (conformality) as the most important characteristic for surveying applications, accepting the trade-offs in the other three properties.

🗺️ The Two Primary Projections

Lambert Conic vs. Transverse Mercator

On the Lambert Conformal Conic Projection, parallels of latitude are arcs of concentric circles, and meridians of longitude are equally spaced straight radial lines, and the meridians and parallels intersect at right angles. The axis of the cone is imagined to be a prolongation of the polar axis. The parallels are not equally spaced because the scale varies as you move north and south along a meridian of longitude.

Adams decided to use this map projection in which shape is preserved based on a developable cone. The Transverse Mercator projection is based on a cylindrical mapping surface much like that illustrated here. However, the axis of the cylinder is rotated so that it is perpendicular with the polar axis of the ellipsoid. Unlike the Lambert Conic projection, the Transverse Mercator represents meridians of longitude as curves rather than straight lines on the developed grid.

🎯 Lambert Conformal Conic

- **Parallels:** Arcs of concentric circles
- **Meridians:** Equally spaced straight radial lines
- **Intersections:** Right angles
- **Cone axis:** Prolongation of polar axis
- **Spacing:** Parallels not equally spaced

📐 Transverse Mercator

- **Base:** Cylindrical mapping surface
- **Orientation:** Cylinder axis perpendicular to polar axis
- **Meridians:** Curves (not straight lines)
- **Grid:** Developed cylindrical surface
- **Rotation:** Rotated from standard Mercator

⚠️ Important Distinction

The Transverse Mercator projection is not the same thing as the ***Universal Transverse Mercator system (UTM)***. UTM was originally a military system that covers the entire earth and differs significantly from the Transverse Mercator system used in State Plane Coordinates.

🎯 Adams' Strategic Design

Zones and Secant Projections

In using these projections as the foundation of the State Plane Coordinate systems, Adams wanted to have the advantage of conformality and also cover each state with as few ***zones*** as possible. A zone in this context is a belt across the state that has one Cartesian coordinate grid with one origin and is projected onto one mapping surface.

One strategy that played a significant role in achieving that end was Adams's use of ***secant*** projections in both the Lambert and Transverse Mercator systems. Using a single secant cone in the Lambert projection and limiting the extent of a zone, or belt, across a state to about 158 miles, approximately 254 km, Dr. Adams limited the distortion of the length of lines. Not only were angles preserved in the final product, but also there were no radical differences between the length of a measured line on the Earth's surface and the length of the same line on the map projection. In other words, the scale of the distortion was small.

> 📏 **Zone Definition**
>
> A **zone** is a belt across the state that has one Cartesian coordinate grid with one origin and is projected onto one mapping surface.

📊 The Mathematical Precision

1 Part in 10,000 Accuracy

He placed 4/6th of the map projection plane between the standard lines, 1/6th outside at each extremity. The distortion was held to 1 part in 10,000. A maximum distortion in the lengths of lines of 1 part in 10,000 means that the difference between the length of a 2-mile line on the ellipsoid and its representation on the map would be about 1 foot.

📐 Zone Layout	📏 Edge Areas	🎯 Accuracy
4/6th between standard lines	**1/6th at each extremity**	**1 part in 10,000**
Main coverage area	Outside standard lines	Maximum distortion

📏 **Practical Example**

A maximum distortion of 1 part in 10,000 means that the difference between the length of a **2-mile line on the ellipsoid** and its representation on the map would be about **1 foot**.

🎯 Historical Context and Purpose

Designed for Its Time

State Plane Coordinates were created to be the basis of a method that approximates geodetic accuracy more closely than the then commonly used methods of small-scale plane surveying. Today, surveying methods can easily achieve accuracies far beyond those prevalent in those days, but the State Plane Coordinate systems were designed in a time of generally lower accuracy and efficiency in surveying measurement.

⏰ **Historical Perspective**

The system was designed when surveying technology was far less sophisticated than today, but the fundamental approach remains sound and continues to serve modern GPS/GNSS applications effectively.

🏆 Rapid Success and National Adoption

From North Carolina to the Nation

The original State Plane Coordinate System (SPCS) was so successful in North Carolina, similar systems were devised for all the states in the Union within a year or so. The system was successful because, among other things, it overcame some of the limitations of mapping on a horizontal plane while avoiding the imposition of strict geodetic methods and calculations. It managed to keep the distortion of the scale ratio under 1 part in 10,000 and preserved conformality. It did not disturb the familiar system of ordered pairs of Cartesian coordinates, and it covered each state with as few zones as possible whose boundaries were constructed to follow county lines. County lines were generally used, so that those relying on State Plane Coordinates could work in one zone throughout a jurisdiction.

✅ **Why It Succeeded**

- **Overcame horizontal plane** limitations
- **Avoided strict geodetic** calculations

🗺️ **Practical Design**

- **Few zones per state**
- **Boundaries follow county lines**

- **Maintained 1:10,000** accuracy
- **Preserved conformality**
- **Familiar Cartesian coordinates**

- **Single zone per jurisdiction**
- **Administrative convenience**
- **National standardization**

🎯 **Key Success Factor**

County lines were generally used, so that those relying on State Plane Coordinates could **work in one zone throughout a jurisdiction**, eliminating the complications of zone boundaries within administrative areas.

📝 A Foundation That Endures

From 1932 to GPS Era

The vision of O.B. Bester, George F. Syme, and Dr. Oscar Adams in 1932 created a coordinate system framework so robust that it continues to serve as the foundation for modern GPS/GNSS applications nearly a century later. Their careful balance of mathematical rigor, practical usability, and administrative convenience established principles that remain valid in today's satellite-based positioning world.

CONTENT

📚 State Plane Coordinate Zones 1983, False Eastings and Scale

Module 04: State Plane Coordinate Zone 1983 False Eastings and Scale

GISC 4315

🗺️ Evolution from SPCS27 to SPCS83

Zone Boundary Changes and FIPS Numbering

In several instances, the boundaries of State Plane Coordinate Zones today, ***SPCS83***, the State Plane Coordinate System based on NAD83 2011 (2010.0) and its reference ellipsoid GRS80, differ from the original zone boundaries. The foundation of the original State Plane Coordinate System, SPCS27 was NAD27 and its reference ellipsoid Clarke 1866.

State Plane Coordinate Zones 1983

Current SPCS83 zone boundaries across the United States

As mentioned earlier, NAD27 geographical coordinates, latitudes and longitudes, differ significantly from those in NAD83 2011 (2010.0). In fact, conversion from geographic coordinates, latitude and longitude, to grid coordinates, ***y*** and ***x***, and back is one of the three fundamental conversions in the State Plane Coordinate system. It is important because the whole objective of the SPCS is to allow the user to work in plane coordinates, but still have the option of expressing any of the points under consideration in either latitude and longitude or State Plane Coordinates without significant loss of accuracy.

📊 SPCS27 Foundation

- **Based on NAD27**
- **Clarke 1866** reference ellipsoid

🗺️ SPCS83 Evolution

- **Based on NAD83 2011 (2010.0)**
- **GRS80** reference ellipsoid

- **Original zone boundaries**
- **No FIPS numbering**

- **Modified zone boundaries**
- **FIPS numbering system**

🔄 **Migration Opportunity**

Therefore, when geodetic control was migrated from NAD27 to NAD83 2011 (2010.0), the State Plane Coordinate System had to go along. When the migration was undertaken in the 1970s, it presented an opportunity for an overhaul of the system. Many options were considered, but in the end, just a few changes were made. One of the reasons for the conservative approach was the fact that **37 states had passed legislation supporting the use of State Plane Coordinates**.

🔢 FIPS Numbering and Zone Changes

Federal Information Processing Standards

Nevertheless, some zones got new numbers, and some of the zones changed. The zones are numbered in the SPCS83 system known as ***FIPS***. FIPS stands for ***Federal Information Processing Standard***, and each SPCS83 zone has been given a FIPS number. These days, the zones are often known as ***FIPS zones***. SPCS27 zones did not have these FIPS numbers.

📋 **FIPS System**

Federal Information Processing Standard provides standardized numbering for all SPCS83 zones, creating consistency across the national coordinate system that was absent in the original SPCS27.

📉 Zone Consolidations and Scale Changes

Abandoning the 1:10,000 Standard

As mentioned earlier, the original goal was to keep each zone small enough to ensure that the scale distortion was 1 part in 10,000 or less, but when the SPCS83 was designed, that scale was not maintained in some states. In five states, some SPCS27 zones were eliminated altogether and the areas they had covered were consolidated into one zone or added to adjoining zones. In three of those states, the result was one single large zone.

🗺️ Single Zone States

Three states consolidated to one zone:

- **South Carolina:** 2 zones → 1 (FIPS 3900)
- **Nebraska:** 2 zones → 1 (FIPS 2600)
- **Montana:** 3 zones → 1 (FIPS 2500)

🏛️ California Changes

Zone elimination and consolidation:

- **Eliminated zone 7**
- **Added area to FIPS zone 0405** (formerly zone 5)

🏝️ Territory Changes

Other consolidations:

- **Puerto Rico & Virgin Islands:** 2 zones → 1 (FIPS 5200)
- **Michigan:** Three Transverse Mercator zones eliminated

⚠️ Scale Standard Compromise

Therefore, because the area covered by these single zones has become so large, **they are not limited by the 1 part in 10,000 standard**. This represents a significant departure from the original design philosophy in favor of administrative convenience.

📐 Coordinate System Setup

False Eastings, False Northings, and Positive Coordinates

In both the Transverse Mercator and the Lambert Conic projection, the positions of the axes are similar in all SPCS zones. As you can see in the illustration, each zone has a central meridian. These central meridians are true meridians of longitude near the geometric center of the zone. Please note that the central meridian is not the ***y***-axis. If it were the ***y***-axis negative coordinates would result. To avoid them, the actual ***y***-axis is moved far to the west of the zone itself.

State Plane Projections

Coordinate system setup for Lambert and Transverse Mercator

In the old SPCS27 arrangement, the ***y***-axis was 2,000,000 feet west from the central meridian in the Lambert Conic projection and 500,000 feet in the Transverse Mercator projection. In the SPCS83 design, those constants have been changed. The most common values are 600,000 meters for the Lambert Conic and 200,000 meters for the Transverse Mercator. However, there is a good deal of variation in these numbers from state to state and zone to zone. In all cases, however, the ***y***-axis is still far to the west of the zone and there are no negative State Plane Coordinates.

📊 SPCS27 Values

False Easting (Y-axis offset):

- **Lambert Conic:** 2,000,000 feet west
- **Transverse Mercator:** 500,000 feet west
- **Unit:** U.S. survey foot

🗺️ SPCS83 Values

False Easting (most common):

- **Lambert Conic:** 600,000 meters west
- **Transverse Mercator:** 200,000 meters west
- **Unit:** Meter
- **Note:** Variation by state/zone

No negative coordinates, because the ***x***-axis, also known as the baseline, is far to the south of the zone. Where the ***x***-axis and ***y***-axis intersect is the origin of the zone, and that is always south and west of the zone itself. This configuration of the axes ensures that all State Plane Coordinates occur in the first quadrant and are, therefore, always positive.

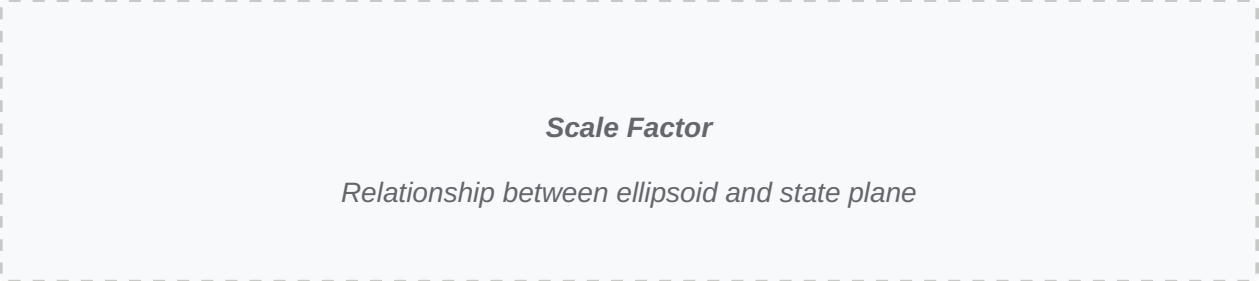
> 📐 **Coordinate System Design**
>
> The origin is placed **south and west** of every zone, ensuring all coordinates fall in the **first quadrant** and remain positive, eliminating the need to work with negative coordinate values.

It is important to note that the fundamental unit for SPCS27 was the U.S. survey foot, but "the U.S. survey foot will be phased out as part of the modernization of the National Spatial Reference System (NSRS). From this point forward, the international foot will be simply called the foot." https://www.nist.gov/pml/us-surveyfoot. The fundamental unit for SPCS83, it is the meter.

📏 Scale Factors: Theory and Practice

Understanding the K Factor

This brings us to the scale factor, also known as the K factor and the projection factor. It was this factor that the original design of the State Plane Coordinate system sought to limit to 1 part in 10,000. As implied by that effort, scale factors are ratios that can be used as multipliers to convert ellipsoidal lengths, also known as geodetic distances, to lengths on the map projection surface, also known as grid distances, and vice versa. Please notice that the geodetic distance is the distance on the ellipsoid of reference, not the distance measured on the surface of the earth.



So, the geodetic length of a line, on the ellipsoid, multiplied by the appropriate scale factor will give you the grid length of that line on the state plane (the map). And the grid length multiplied by the inverse of that same scale factor would bring you back to the geodetic length again. There's another factor that will get you from topographic surface of the earth —where the measurement was made— down to the ellipsoid. However, at the moment we're talking about the scale factor.

Here in this image, you see a state plane, and a horizontal line is indicated between the bases of the red arrows. Please notice that between the standard lines, the scale is too small on the state plane. And outside the standard lines, the scale is too large on the state plane. So, the line between the bases of those two red arrows on the ellipsoid of reference will be projected inward from the ellipsoid to the state plane. As it's projected inward —the line shortens. That means that between the intersection of standard lines, the grid (state plane) is under the ellipsoid. In that area, a distance from one point to another is longer on the ellipsoid than on the state plane.

📍 Between Standard Lines

Scale too small:

- **Grid under ellipsoid**
- **Minimum scale factor**
- **Typical minimum: 0.9999**

📏 On Standard Lines

Perfect scale:

- **Scale factor = 1.0000**
- **Lines of exact scale**
- **No distortion**

📍 Outside Standard Lines

Scale too large:

- **Grid above ellipsoid**
- **Maximum scale factor**
- **Typical maximum: 1.0001**

This means that right in the middle of the State Plane coordinate systems zone, the scale is at its minimum. In the middle, a typical minimum State Plane coordinate scale factor is not less than 0.9999. Outside of the standard lines, the grid (state plane) is above the ellipsoid where the distance from one point to another is shorter on the ellipsoid than it is on the state plane. There, at the edge of the zone, a maximum typical State Plane coordinate system scale factor is generally not more than 1.0001.

🗺️ Scale Factor Behavior by Projection

Lambert Conic vs. Transverse Mercator Patterns

The projection used most on states that are longest from east to west is the Lambert Conic. In this projection, the scale factor for east-west lines is constant. In other words, the scale factor is the same all along the line. One way to think about this is to recall that the distance between the ellipsoid and the map projection surface does not change east to west in that projection. On the other hand, along a north-south line, the scale factor is constantly changing on the Lambert Conic. And it is no surprise then to see that the distance between the ellipsoid and the map projection surface is always changing along the north to south line in that projection.

But looking at the Transverse Mercator projection, the projection used most on states longest north to south, the situation is exactly reversed. In that case, the scale factor is the same all along a north-south line, and changes constantly along an east-west line.

🎯 Lambert Conic (East-West States)

- **East-West lines:** Constant scale factor
- **North-South lines:** Constantly changing scale factor
- **Distance to ellipsoid:** Constant east-west, varies north-south
- **Best for:** States wider than tall

📐 Transverse Mercator (North-South States)

- **North-South lines:** Constant scale factor
- **East-West lines:** Constantly changing scale factor
- **Distance to ellipsoid:** Constant north-south, varies east-west
- **Best for:** States taller than wide

📊 Design Strategies and Examples

Zone Width and Distortion Management

Both the Transverse Mercator and the Lambert Conic used a secant projection surface and originally restricted the width to 158 miles. These were two strategies used to limit scale factors when the State Plane Coordinate systems were designed. Where that was not optimum, the width was sometimes made smaller, which means the distortion was lessened. As the belt of the ellipsoid projected onto the map narrows, the distortion gets smaller.

🏛️ Connecticut Example

Narrow State Benefits:

- **Width:** Less than 80 miles north-south
- **Zones:** Only one zone needed
- **Boundaries:** 1 part in 40,000 (4x better)
- **Middle:** 1 part in 79,000 (8x better)

🇹🇽 Texas Compromise

Large State Challenges:

- **Scale factor:** Slightly worse than 1:10,000
- **Coverage:** Complete with five zones
- **Principle:** As few zones as possible
- **Boundaries:** Follow county lines

For example, Connecticut is less than 80 miles wide north to south. It has only one zone. Along its northern and southern boundaries, outside of the standard parallels, the scale factor is 1 part in 40,000, a fourfold improvement over 1 part in 10,000. And in the middle of the state, the scale factor is 1 part in 79,000, nearly an eightfold increase. On the other hand, the scale factor was allowed to get a little bit smaller than 1 part in 10,000 in Texas. By doing that, the state was covered completely with five zones. And among the guiding principles in 1933 was covering the states with as few zones

as possible and having zone boundaries follow county lines. Still it requires ten zones and all three projections to cover Alaska.

🧮 Scale Factor Calculations

From Tables to Equations

When SPCS 27 was current, scale factors were interpolated from tables published for each state. In the tables for states in which the Lambert Conic projection was used, scale factors change north–south with the changes in latitude. In the tables for states in which the Transverse Mercator projection was used, scale factors change east-west with the changes in ***x***-coordinate. Today, scale factors are not interpolated from tables for SPCS83. For both the Transverse Mercator and the Lambert Conic projections, they are calculated directly from equations. There are also several software applications that can be used to automatically calculate scale factors for particular stations. They can be used to convert latitudes and longitudes to State Plane Coordinates. Given the latitude and longitude of the stations under consideration, part of the available output from these programs is typically the scale factors for those stations.

📊 SPCS27 Method

- **Interpolated from tables**
- **Lambert:** Changes with latitude (N-S)
- **Transverse Mercator:** Changes with x-coordinate (E-W)
- **Published for each state**

🗺️ SPCS83 Method

- **Calculated directly from equations**
- **Both projections** use mathematical formulas
- **Software applications** available
- **Automatic calculation** with lat/lon input

To illustrate the use of these factors, consider a line with a length on the ellipsoid of 130,210.44 feet, a bit over 24 miles. That would be its geodetic distance. Suppose that the scale factor for that line was 0.9999536, then the grid distance along the line would be:

*Geodetic Distance * Scale Factor = Grid Distance*

130,210.44 *ft.* * 0.999953617 = 130,204.40 *ft.*

The difference between the longer geodetic distance and the shorter grid distance here is a little more than 6 feet. That is actually better than 1 part in 20,000; please recall that the 1 part in 10,000 ratio was originally considered the maximum. Distortion lessens, and the scale factor approaches 1 as a line nears a standard parallel. Please also recall that on the Lambert projection, an east–west line, that is a line that follows a parallel of latitude, has the same scale factor at both ends and throughout. However, a line that bears in any other direction will have a different scale factor at each end. A

north–south line will have a great difference in the scale factor at its north end compared with the scale factor of its south end.

📍 Practical Example: Stormy to Seven

Scale Factor Calculation Between Two Points

Where K is the scale factor for a line, K_1 is the scale factor at one end of the line and K_2 is the scale factor at the other end of the line. Scale factor varies with the latitude in the Lambert projection. For example, suppose the point at the north end of the 24-mile line is called Stormy and has a geographic coordinate of:

Stormy (North End):

37°46'00.7225"

103°46'35.3195"

and at the south end the point is known as Seven with a geographic coordinate of:

Seven (South End):

37°30'43.5867"

104°05'26.5420"

Scale Factor Between Stormy and Seven

Illustration of scale factor variation along a line

The scale factor for point Seven is 0.99996113 and the scale factor for point Stormy is 0.99994609. It happens that point Seven is further south and closer to the standard parallel than is point Stormy, and it therefore follows that the scale factor at Seven is closer to 1. It would be exactly 1 if it were on the standard parallel, which is why the standard parallels are ***called lines of exact scale***. The typical scale factor for the line is the average of the scale factors at the two end points:

📊 **Scale Factor Analysis**

- **Seven (South):** 0.99996113 (closer to standard parallel)
- **Stormy (North):** 0.99994609 (farther from standard parallel)

- **Line scale factor:** Average of endpoints

Deriving the scale factor at each end and averaging them is the usual method for calculating the scale factor of a line. The average of the two is sometimes called K_m.

However, it can also be done by the precise method $K = K_1 + K_2 + 4K_m/6$.

Calculation Methods

Simple method: Average the scale factors at both ends
Precise method: $K = (K_1 + K_2 + 4K_m)/6$

CONTENT

📹 State Plane Coordinates by Dave Doyle

State Plane

GISC 4315

📹 Educational Video

State Plane Coordinates by Dave Doyle

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CONTENT

📚 Elevation Factor



Module 04: Elevation Factor

GISC 4315

📍 The Missing Piece: From Earth Surface to Grid

Why Scale Factor Alone Isn't Enough

But, that is not the whole story when it comes to reducing distance to the State Plane Coordinate grid. Measurement of lines must always be done on the topographic surface of the Earth, and not on the ellipsoid. Therefore, the first step in deriving a grid distance must be moving a measured line from the Earth to the ellipsoid. The scale factor moves us from the ellipsoid to the state plane (aka grid). However, as mentioned here, the measurements are not made on the ellipsoid. They are made on the surface of the Earth.

Elevation Factor

Moving measurements from Earth's surface to the ellipsoid

If we're going to measure the line from one point to another — from station Boulder to station Peak as shown in the illustration— that distance would be measured on the surface of the Earth. To move from the surface of the Earth down to the ellipsoid, we need to have what's known as an ***elevation factor***.

🔍 **Key Insight**

In the illustration, you may have noticed the approximate radius of the Earth, **20,906,000 feet**. It is shown from the center of mass of the Earth up to the ellipsoid. It is an approximation used in creating this elevation factor.

🧮 Calculating the Elevation Factor

Average Method and Combined Factor

Typically, the elevation factor for a line is the average of the elevation factors at each end of the line. For example, the average of the elevation factor at station Boulder (h_1), and the elevation factor at station Peak (h_2) is the elevation factor for the line (h_{Avg}). That would give the elevation factor for the line Boulder-Peak. Once the elevation factor is available, it is multiplied with the scale factor. The result is known as the **combined factor**. The combined factor will move the line from the topographic surface of the Earth to the state plane (aka grid).

📊 The Complete Process

Step 1: Elevation Factor moves from Earth surface to ellipsoid
Step 2: Scale Factor moves from ellipsoid to state plane
Result: Combined Factor = Elevation Factor × Scale Factor

🌊 Historical Context: Sea Level Factor vs. Elevation Factor

Why Names Matter

In other words, converting a distance measured on the topographic surface to a geodetic distance on the reference ellipsoid is done with another ratio that is also used as a multiplier. Originally, this factor had a rather unfortunate name. It used to be known as the ***sea level factor*** in SPCS27. It was given that name because as you may recall that when NAD27 was established using the Clarke 1866 reference ellipsoid, the distance between the ellipsoid and the geoid was declared to be zero at Meades Ranch in Kansas.

That meant that in the middle of the country the ***sea level*** surface, the geoid, and the ellipsoid were coincident by definition. And since the Clarke 1866 ellipsoid fit the United States quite well, the separation between the two surfaces, the ellipsoid and geoid, only grew to about 12 meters anywhere in the country. With such a small distance between them, many practitioners at the time took the point of view that, for all practical purposes, the ellipsoid and the geoid were in the same place. And that place was called sea level. Hence, reducing a distance measure on the surface of the Earth to the ellipsoid was said to be reducing it to sea level.

📊 NAD27 Era

Clarke 1866 Ellipsoid:

- **Zero separation** at Meades Ranch, Kansas

🗺️ NAD83 Reality

GRS80 Ellipsoid:

- **Certainly not** the same as Mean Sea Level

- **Good fit** for United States
- **Maximum 12 meters** geoid-ellipsoid separation
- **"Sea level factor"** terminology made sense

- **Up to 53 meters** geoid-ellipsoid separation
- **Dramatically improved** measurement technology
- **"Ellipsoid factor"** more accurate term

⚠️ **Why the Old Name Was Misleading**

Today, that idea and that name for the factor are misleading because, of course, the GRS80 ellipsoid on which NAD83 is based is certainly not the same as Mean Sea Level. The separation between the geoid and ellipsoid can grow as large as 53 meters. And technology by which lines are measured has improved dramatically. Therefore, in SPCS83, the factor for reducing a measured distance to the ellipsoid is known as the ***ellipsoid factor***.

In any case, both the old and the new name can be covered under the name the elevation factor.

🧱 Mathematical Definition

The Earth Radius Ratio

Regardless of the name applied to the factor, it is a ratio. The ratio is the relationship between an approximation of the Earth's radius and that same approximation with the mean ellipsoidal height of the measured line added to it. For example, consider station Boulder and station Peak illustrated in the figure above.

📍 **Boulder Station**

- N39°59'29.1299"
- W105°15'39.6758"

🏔️ **Peak Station**

- N40°01'19.1582"
- W105°30'55.1283"

The distance between these two stations is 72,126.21 feet. This distance is sometimes called the ***ground distance***, or the ***horizontal distance at mean elevation***. In other words, it is not the slope distance but rather the distance between them corrected to an averaged horizontal plane, as is common practice. For practical purposes, then, this is the distance between the two stations on the topographic surface of the Earth. On the way to finding the grid distance Boulder to Peak, there is the interim step, calculating the geodetic distance between them, that is the distance on the ellipsoid. We need the elevation factor, and here is how it is determined.

📊 **Key Measurements**

- **Ground distance:** 72,126.21 feet
- **Distance type:** Horizontal distance at mean elevation

- **Not:** Slope distance
- **Goal:** Convert to geodetic distance on ellipsoid

📐 Elevation Factor Calculation

Step-by-Step Calculation

The ellipsoidal height of Boulder, h_1, is 5,437 feet. The ellipsoidal height of Peak, h_2, is 9,099 feet. The approximate radius of the Earth, traditionally used in this work, is 20,906,000 feet. The elevation factor is calculated:

$$Elevation\ Factor = \mathrm{R}/[\mathrm{R} + \mathrm{h(average)}]$$

$$Elevation\ Factor = 20{,}906{,}000ft./(20{,}906{,}000+7{,}268ft.)$$

$$Elevation\ Factor = 20{,}906{,}000ft\ ./\ 20{,}913{,}268ft.$$

$$Elevation\ Factor = 0.99965247$$

Elevation Factor Formula

🏔️ Boulder Height	⛰️ Peak Height	🌍 Earth Radius
h_1 = 5,437 feet	h_2 = 9,099 feet	R = 20,906,000 feet

This factor then is the ratio used to move the ground distance down to the ellipsoid, down to the geodetic distance.

$$Ground\ Distance\ Boulder\ to\ Peak = 72{,}126.21ft.$$

$$Geodetic\ Distance = Ground\ Distance * Elevation\ Factor$$

$$Geodetic\ Distance = 72{,}126.21 * 0.99965247$$

$$Geodetic\ Distance = 72{,}101.14ft.$$

Ground Distance to Geodetic Distance Conversion

🔍 **Understanding the Ratio**

The elevation factor is the relationship between the Earth's radius and that same radius plus the mean ellipsoidal height of the measured line. This accounts for the fact that measurements are made above the ellipsoid surface.

🔬 Technical Refinements

Improving Calculation Precision

It is possible to refine the calculation of the elevation factor by using an average of the actual radial distances from the center of the ellipsoid to the end points of the line, rather than the approximate 20,906,000 feet. In the area of stations Boulder and Peak, the average ellipsoidal radius is actually a bit longer, but it is worth noting that within the continental United States such variation will not cause a calculated geodetic distance to differ significantly.

📊 Standard Method

- **Uses approximation:** 20,906,000 feet
- **Consistent across CONUS**
- **Adequate for most applications**
- **Widely accepted practice**

🎯 Refined Method

- **Uses actual radial distances**
- **Average of endpoint radii**
- **Slightly more accurate**
- **Minimal practical difference** in CONUS

📍 **Regional Variation**

In the Boulder-Peak area, the average ellipsoidal radius is actually a bit longer than the standard approximation, but the difference is not significant enough to materially affect calculated geodetic distances for most surveying applications.

📝 Completing the Coordinate Transformation

The Essential Bridge

The elevation factor represents the crucial bridge between real-world measurements made on Earth's topographic surface and the mathematical models used in State Plane coordinate systems. Without

> this factor, the carefully designed scale factors and projections would be meaningless, as they depend on distances referenced to the ellipsoid rather than the actual measurement surface.

CONTENT

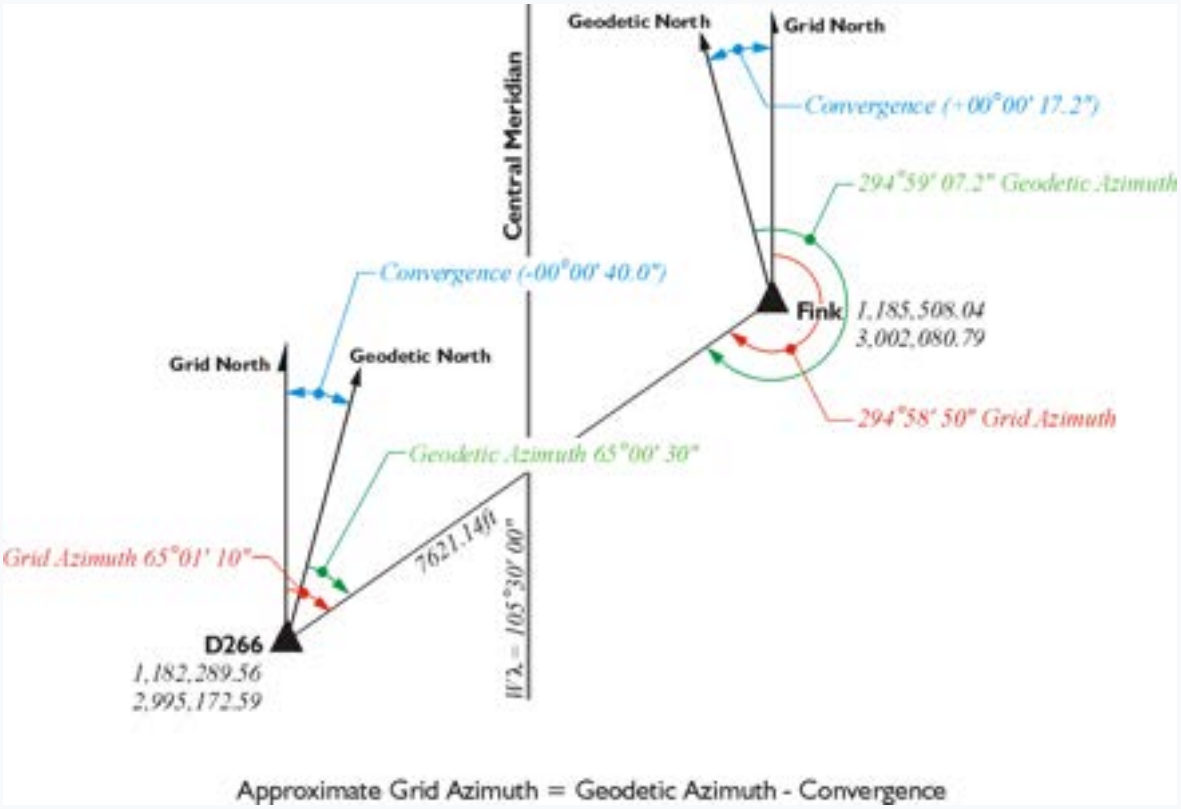
📚 Grid and Geodetic Azimuths



📍 Central Meridians and Grid North

The Foundation of Direction in State Plane Systems

You will find that every zone in a State Plane coordinate system has a central meridian. In Colorado, the central meridian is 105 degrees, 30 minutes West Longitude, as shown here, on the left you have station D266 and on the right, station Fink. Notice that, in both cases, that grid north is exactly parallel with the central meridian. This is the convention for any plane coordinate system, so that north is always up, and always the same direction. All the norths are parallel in the plane systems; this is, of course, unrealistic.



Grid and Geodetic Azimuths

Colorado example showing convergence at stations D266 and Fink

🗺️ Grid System Convention

- **Grid north** is exactly parallel with central meridian
- **North is always up** and same direction
- **All norths are parallel** in plane systems
- **Consistent orientation** across the zone

🌍 Real Earth Reality

- **Meridians converge** on Earth's surface
- **Parallel grid north** is unrealistic
- **True north varies** by location
- **Convergence creates** directional differences

📐 Colorado Example

Central Meridian: 105°30' West Longitude
Stations: D266 (west) and Fink (east)
Grid North: Parallel to 105°30'W at both stations

📐 Understanding Convergence

The Difference Between Grid North and Geodetic North

Meridians converge on the surface of the Earth. Therefore, there is a value, shown here in **blue**, called convergence. Convergence is the difference between geodetic north and grid north. You see over here on the right at station Fink, the convergence is plus 17.2 arcseconds. All the convergence angles east of the Central Meridian are positive. On the left, at station D266, the convergence is minus 40.0 arcseconds. All the convergence angles west of the Central Meridian are negative.

📍 Station D266 (West)

Position: West of Central Meridian

- **Convergence:** -40.0 arcseconds
- **Sign:** Negative (west of central meridian)
- **Meaning:** Grid north is west of geodetic north

📍 Station Fink (East)

Position: East of Central Meridian

- **Convergence:** +17.2 arcseconds
- **Sign:** Positive (east of central meridian)
- **Meaning:** Grid north is east of geodetic north

🔍 Convergence Rule

East of Central Meridian: Positive convergence angles
West of Central Meridian: Negative convergence angles
Definition: Convergence = Geodetic North - Grid North

🎯 Grid Azimuth vs. Geodetic Azimuth

The Mathematical Proof of Convergence

Please note that the sum of the grid azimuths (in **red**) at station Fink and station D266 is 360 degrees. This is a direct result of the grid north lines being parallel with each other. However, the sum of the geodetic azimuths (in **green**) at station Fink and station D266 is not 360 degrees. This is a direct result of the geodetic north lines converging. On the Earth they are not parallel with each other.

🔴 Grid Azimuths

Key Characteristic:

- **Sum of opposite azimuths = 360°**
- **Grid north lines are parallel**
- **Consistent across the zone**
- **Based on plane coordinate system**

🟢 Geodetic Azimuths

Key Characteristic:

- **Sum of opposite azimuths ≠ 360°**
- **Geodetic north lines converge**
- **Varies by location on Earth**
- **Based on Earth's actual shape**

- **Mathematically predictable**

- **Reflects true meridian convergence**

📊 **Mathematical Demonstration**

The difference in azimuth sums provides direct mathematical proof of meridian convergence. Grid azimuths sum to exactly 360° because grid north lines are artificially parallel, while geodetic azimuths sum to something other than 360° because they reflect the true convergence of meridians on Earth's curved surface.

🌍 The Fundamental Reality

Grid Convenience vs. Earth Reality

The bottom line is this. On the State Plane coordinate grid, north is always parallel with the central meridian, but at the points on the Earth, north is along the meridian that passes through them. They are certainly not parallel with one another and certainly not parallel with the central meridian.

🗺️ State Plane Grid Reality

- **North is always parallel** with central meridian
- **Consistent direction** throughout zone
- **Simplified calculations**
- **Cartesian coordinate convenience**
- **Mathematical uniformity**

🌍 Earth's Physical Reality

- **North follows** the meridian through each point
- **Meridians are not parallel**
- **Convergence varies** by location
- **True geodetic relationships**
- **Reflects Earth's curvature**

⚖️ **The Essential Trade-off**

State Plane coordinate systems deliberately sacrifice the physical reality of meridian convergence in favor of the mathematical convenience of parallel grid lines. This trade-off enables the use of simple Cartesian coordinates and plane trigonometry, but requires understanding convergence angles to relate grid directions to true Earth directions.

🧱 Practical Applications

When Convergence Matters

Understanding the relationship between grid and geodetic azimuths is crucial for accurate surveying work, especially when:

🎯 Critical Applications

- **Boundary surveys** requiring legal accuracy
- **Large-scale projects** covering significant areas
- **GPS/GNSS integration** with traditional surveys
- **Astronomical observations** for azimuth determination
- **Navigation applications**

📐 Correction Process

- **Calculate convergence angle** for location
- **Apply correction** to grid azimuth
- **Convert between** grid and geodetic directions
- **Account for location** relative to central meridian
- **Use appropriate sign** (+ east, - west)

> 🔄 **Conversion Relationship**
>
> **Geodetic Azimuth = Grid Azimuth + Convergence**
> Where convergence is positive east of the central meridian and negative west of it.

📝 The Direction Dilemma Resolved

Bridging Grid and Earth

The concept of convergence elegantly solves the fundamental directional challenge in State Plane coordinate systems: how to maintain the computational convenience of parallel grid lines while preserving the ability to relate measurements to true Earth directions. This systematic approach to handling the difference between mathematical convenience and physical reality exemplifies the thoughtful design principles underlying the entire State Plane coordinate framework.

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CONTENT

📚 Universal Transverse Mercator

Module 04: Universal Transverse Mercator System

GISC 4315

🗺️ UTM: Global Convenience vs. Local Precision

Large Area GIS Applications

A plane coordinate system that is convenient for GIS work over large areas is the Universal Transverse Mercator (UTM) system. UTM with the Universal Polar Stereographic system covers the world in one consistent system. It is four times less precise than typical State Plane Coordinate systems, with a scale factor that reaches 0.9996. A Universal Transverse Mercator zone embraces a much larger portion of the earth than does a state plane coordinate zone. When you get a larger bite, a larger portion of the earth, the scale factor is less attractive. Yet, the ease of using UTM and its worldwide coverage makes it very attractive for work that would otherwise have to cross many different SPCS zones.

UTM Zones for the United States

Conterminous United States spans UTM zones 10 to 19

✅ UTM Advantages

- **Worldwide coverage** in one consistent system
- **Large area convenience** for GIS work
- **Avoids crossing** many different SPCS zones
- **Standardized globally**
- **Used on USGS quad sheets**

⚠️ Precision Trade-offs

- **Four times less precise** than State Plane
- **Scale factor reaches 0.9996**
- **Larger zones = larger distortions**
- **Less attractive scale factors**

For example, nearly all ***National Geospatial-Intelligence Agency (NGA)*** topographic maps, ***U.S. Geological Survey (USGS)*** quad sheets, and many aeronautical charts show the UTM grid lines.

🏛️ Historical Development: Post-WWII International Cooperation

Military Coordination Necessity

It is often said that UTM is a military system created by the U.S. Army; but several nations, and the North Atlantic Treaty Organization (NATO), played roles in its creation after World War II. At that time, the goal was to design a consistent coordinate system that could promote cooperation between the military organizations of several nations. Before the introduction of UTM, allies found that their differing systems hindered the synchronization of military operations. Conferences were held on the subject from 1945 to 1951, with representatives from Belgium, Portugal, France, and Britain, and the outlines of the present UTM system were developed. By 1951, the U.S. Army introduced a system that was very similar to that currently used.

🤝 International Collaboration

Timeline: 1945-1951 conferences
Participants: Belgium, Portugal, France, Britain, USA, NATO
Goal: Consistent coordinate system for military cooperation
Problem Solved: Differing systems hindered synchronized operations

🔢 Zone Structure and Numbering System

60 Zones Starting at the International Date Line

The UTM projection divides the world into 60 zones that begin at longitude 180°, the International Date Line. Zone 1 is from 180° to the 174° W longitude. The conterminous United States is within UTM zones 10 to 19. Here is a convenient way to find the zone number for a particular longitude. Consider west longitude negative and east longitude positive, add 180° and divide by 6. Any answer greater than an integer is rounded to the next highest integer, and you have the zone. For example, Denver, Colorado is near 105° W. Longitude, –105°.

🧱 Zone Number Calculation

$$-105° + 180° = 75°$$

75° / 6 = 12.50

Round up to 13

Therefore, Denver, Colorado, is in UTM Zone 13.

📋 Zone Calculation Formula

Step-by-step process:

1. Convert west longitude to negative, east longitude to positive
2. Add 180° to the longitude value
3. Divide the result by 6
4. Round any decimal up to the next highest integer

🌍 Global Zone Coverage and Grid Letters

Worldwide Grid System

All UTM zones have a width of 6° of longitude. From north to south, the zones extend from 84° N latitude to 80° S latitude. Originally the northern limit was at 80° N latitude and the southern 80° S latitude. On the south, the latitude is a small circle that conveniently traverses the ocean well south of Africa, Australia, and South America. However, 80° N latitude was found to exclude parts of Russia and Greenland and was extended to 84° N latitude.

World UTM Zones

Global coverage with 60 zones and alphabetic grid letters

You will notice letters of the alphabet along the right edge of the illustration. There are a few letters missing. You see C,D,E, F, G,H —but there is no I, because it's too close to the number 1 and could be confused— J,K, L, M, N —but there is no O, it could be confused for the number zero— P,Q, R, S,T U, V, W, X, there is no Y, — and finally there is Z. I point this out because there are GPS/GNSS systems that use these letter designations along with the number of the UTM zone to identify a particular quadrangle. For example, C21 would be the square that you see just immediately above 21 in the Southern Hemisphere. This is a useful method of referring to a particular quadrangle in a particular UTM zone. The designations are also used by the military.

📏 Zone Dimensions

- **Width:** 6° of longitude each
- **Total zones:** 60 worldwide
- **North limit:** 84° N latitude
- **South limit:** 80° S latitude
- **Coverage:** Nearly global

🔤 Grid Letters

- **Missing I:** Too close to number 1
- **Missing O:** Could be confused with zero
- **Missing Y:** Not used in system
- **Example:** C21 = Grid C, Zone 21
- **Used by:** Military and GPS/GNSS systems

🌐 Hemisphere Origins and Duplicate Coordinates

Two Origins Per Zone

Unlike any of the systems previously discussed, every coordinate in a UTM zone occurs twice, once in the Northern Hemisphere and once in the Southern Hemisphere. This is a consequence of the fact that there are two origins in each UTM zone. The origin for the portion of the zone north of the equator is moved 500 km west of the intersection of the zone's central meridian and the equator. This arrangement ensures that all of the coordinates for that zone in the Northern Hemisphere will be positive. The origin for the coordinates in the Southern Hemisphere for the same zone is 500 km west of the central meridian, as well. But in the Southern Hemisphere, the origin is not on the equator, it is 10,000 km south of it, close to the South Pole. This orientation of the origin guarantees that all of the coordinates in the Southern Hemisphere are in the first quadrant and are positive.

In other words, the intersection of each zone's central meridian with the equator defines its origin of coordinates. In the southern hemisphere, each origin is given the coordinates:

🗺️ UTM Origin Coordinates

Northern Hemisphere

Easting = X_o = 500,000 meters
Northing = Y_o = 0 meters

(at the equator)

Southern Hemisphere

Easting = X_o = 500,000 meters
Northing = Y_o = 10,000,000 meters

(10,000 km south of equator)

⚠️ **Important Consequence**

Every coordinate in a UTM zone **occurs twice**—once in the Northern Hemisphere and once in the Southern Hemisphere. This requires careful attention to hemisphere designation when working with UTM coordinates.

🧊 Polar Coverage: Universal Polar Stereographic

Completing Global Coverage

The UTM zones nearly cover the earth, except the Polar Regions, which are covered by two azimuthal polar zones called the Universal Polar Stereographic (UPS) projection.

Universal Polar Stereographic Projection

UPS covers polar regions not handled by UTM zones

🌐 **Complete System**

UTM + UPS = Global Coverage
UTM handles 84°N to 80°S, while UPS projections cover the polar regions beyond these limits, creating one comprehensive worldwide coordinate system.

🔧 Technical Foundation

Secant Transverse Mercator Projection

The foundation of the 60 UTM zones is a secant Transverse Mercator projection, very similar to those used in some State Plane Coordinate systems. The central meridian of the zones is exactly in the middle. For example, in Zone 1, from 180° W to the 174° W longitude, the central meridian is 177° W longitude, so each zone extends 3 degrees east and west from its central meridian.

One UTM Zone

Secant Transverse Mercator projection with 3° extent each side of central meridian

The UTM secant projection gives approximately 180 kilometers between the lines of exact scale where the cylinder intersects the ellipsoid. The scale factor grows from 0.9996 along the central meridian of a UTM zone to 1.00000 at 180 km to the east and west. Please recall that SPCS zones are usually more limited in width, ~158 miles, and, therefore, have a smaller range of scale factors than do the UTM zones. In State Plane coordinates, the scale factor is usually no more than 1 part in 10,000. In UTM coordinates, it can be as large as 1 part in 2500. The reference ellipsoids for UTM coordinates vary.

🎯 UTM Scale Factors

- **Central meridian:** 0.9996
- **At ±180 km:** 1.00000 (exact scale)
- **Maximum distortion:** 1 part in 2,500
- **Lines of exact scale:** ~180 km apart

📊 SPCS Comparison

- **Zone width:** ~158 miles (narrower)
- **Maximum distortion:** 1 part in 10,000
- **Scale factor range:** Smaller variation
- **Precision:** Four times better than UTM

⚖️ The Precision Trade-off

UTM zones cover larger areas than State Plane zones, resulting in larger scale factor variations and lower overall precision. This trade-off provides global consistency and convenience at the cost of local accuracy.

📝 Global Standardization Achievement

Universal Coverage with Managed Precision

The Universal Transverse Mercator system represents a remarkable achievement in international cooperation and technical standardization. By accepting reduced precision compared to local systems like State Plane coordinates, UTM provides the global consistency essential for worldwide mapping, military operations, and large-scale GIS applications. The system's enduring success demonstrates the value of thoughtful engineering compromises in creating universally applicable coordinate frameworks.

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📚 Heights



Module 04: Heights

GISC 4315

📍 The Third Dimension of Position

Beyond Latitude and Longitude

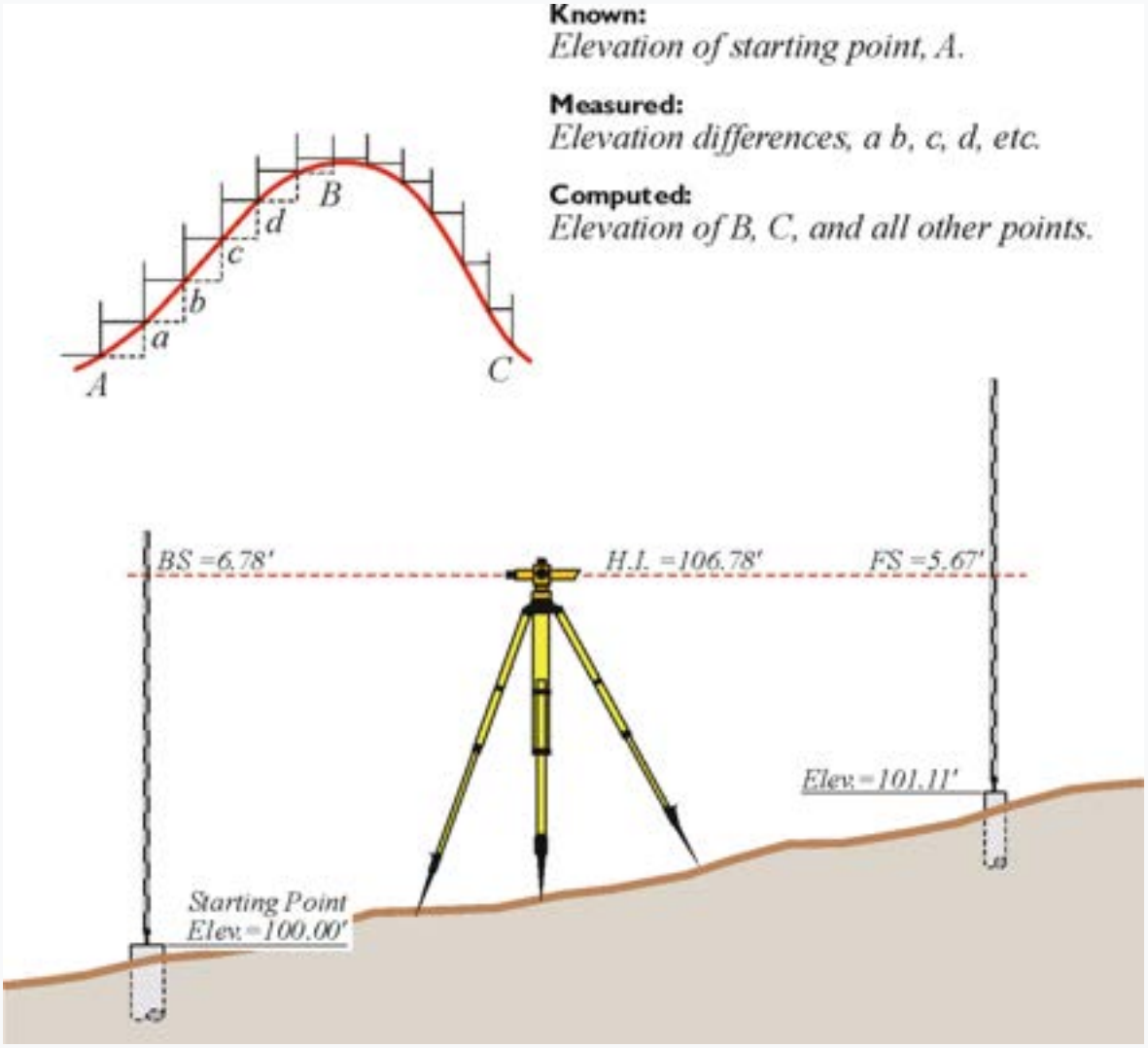
A point on the earth's surface is not completely defined by its latitude and longitude. In such a context, there is, of course, a third element, that of height. Surveyors have traditionally referred to this component of a position as its elevation. One classical method of determining elevations is spirit leveling. A level, correctly oriented at a point on the surface of the earth, defines a line parallel to the geoid at that point. Therefore, the elevations determined by level circuits are orthometric; that is, they are defined by their vertical distance above the geoid as it would be measured along a plumb line.

> 🎯 **Key Concept**
>
> However, **orthometric elevations are not directly available** from the geocentric position vectors derived from GPS/GNSS measurements. This fundamental difference creates important considerations for modern surveying applications.

📐 Spirit Leveling: The Classical Method

Gravity-Based Height Determination

In a normal course of things, a coordinate —YX, or northing easting, or latitude and longitude— is not fully a definition of a position. We've discussed also Earth-Centered, Earth-Fixed XYZ coordinates, but it is typical for a horizontal coordinate to be given a height, or elevation. This third element, was most often originally determined by spirit leveling or, in some cases, by a technique known as trigonometric leveling.



A Spirit Level Run

Traditional method for determining orthometric elevations

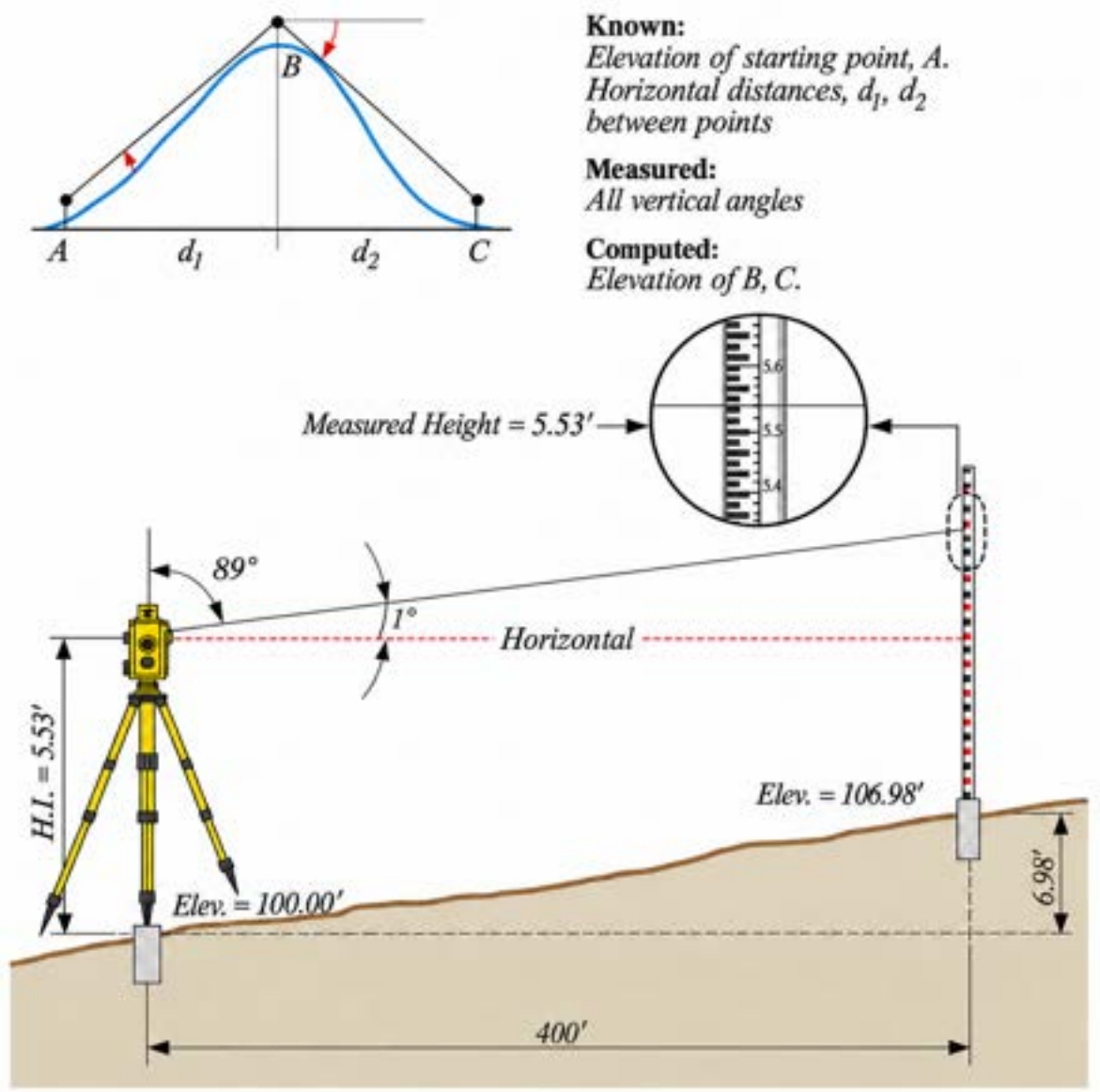
🔬 **Spirit Leveling Principles**

A level, correctly oriented at a point on the surface of the earth, **defines a line parallel to the geoid** at that point. This creates elevations that are **orthometric**—defined by their vertical distance above the geoid as measured along a plumb line.

📐 Trigonometric Leveling: An Alternative Approach

Angular Measurements for Height Determination

Despite the surveying method that was used in the past, it was based on optical instruments oriented to gravity, therefore oriented to the geoid. Therefore, the heights were based upon gravity.



Trigonometric Leveling

Using angular measurements and distances to determine elevations

📏 Spirit Leveling

- **Direct measurement** of height differences
- **Bubble level instruments**
- **Horizontal sight lines** parallel to geoid
- **High accuracy** for elevation determination
- **Sequential measurements** over long distances

📐 Trigonometric Leveling

- **Angular measurements** to determine heights
- **Theodolites and total stations**
- **Vertical angles** and horizontal distances
- **Faster over long distances**
- **Less accurate** than spirit leveling

🌍 Common Foundation

Both methods shared a crucial characteristic: they were **based on optical instruments oriented to gravity, therefore oriented to the geoid**. This means all traditional height measurements were fundamentally gravity-based.

🏛️ The Legacy of Gravity-Based Heights

Historical Infrastructure of Elevation Data

There is a long legacy of benchmarks, recorded heights, and archives that are based upon this sort of methodology of determining heights. And so, this is what it follows that these values are generally what we think of as elevations —we think of orthometric heights based upon gravity.

🏛️ Historical Infrastructure

- **Long legacy of benchmarks**
- **Recorded heights** in archives
- **Established survey networks**
- **Decades of accumulated data**
- **Standardized procedures**

🌊 Orthometric Heights

- **Based upon gravity**
- **Referenced to the geoid**
- **What we think of as "elevations"**
- **Familiar to surveyors**
- **Used in traditional mapping**

📊 What We Call "Elevation"

When most people think of elevation, they're thinking of **orthometric heights based upon gravity**. This creates a fundamental challenge when working with GPS/GNSS systems, which provide ellipsoidal heights rather than the gravity-based elevations that traditional surveying has established.

⚠️ The GPS/GNSS Challenge

Incompatible Height Systems

The challenge arises because GPS/GNSS measurements provide ellipsoidal heights (heights above the mathematical reference ellipsoid), while traditional surveying infrastructure is built on orthometric heights (heights above the geoid, which approximates mean sea level).

📡 GPS/GNSS Heights

- **Ellipsoidal heights**
- **Mathematical reference surface**
- **Geocentric positioning**
- **Not directly usable** for traditional applications
- **Require conversion** to orthometric

🌊 Traditional Survey Heights

- **Orthometric heights**
- **Gravity-based reference**
- **Geoid surface**
- **What users expect** as "elevation"
- **Extensive benchmark network**

🔄 Integration Challenge

Modern surveying must bridge these two height systems: the traditional gravity-based orthometric heights that form our elevation infrastructure, and the ellipsoidal heights provided by GPS/GNSS technology. This requires understanding geoid models and height conversion procedures.

📝 Bridging Traditional and Modern Height Systems

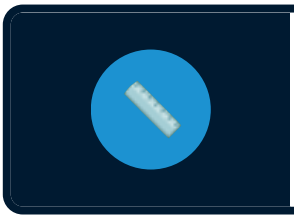
Understanding the Foundation

The traditional gravity-based methods of height determination—spirit leveling and trigonometric leveling—created the elevation infrastructure that still forms the backbone of surveying and mapping today. Understanding these methods and their relationship to the geoid is essential for modern practitioners who must integrate GPS/GNSS ellipsoidal heights with the orthometric height system that users expect and regulatory frameworks require.

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CONTENT

📚 Ellipsoidal Heights



Module 04: Ellipsoidal Heights

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A point on the Earth's surface is not completely defined by its latitude and longitude. There is, of course, a **third element: height**. Surveyors have traditionally referred to this component of a position as its elevation, determined through classical methods like spirit leveling.

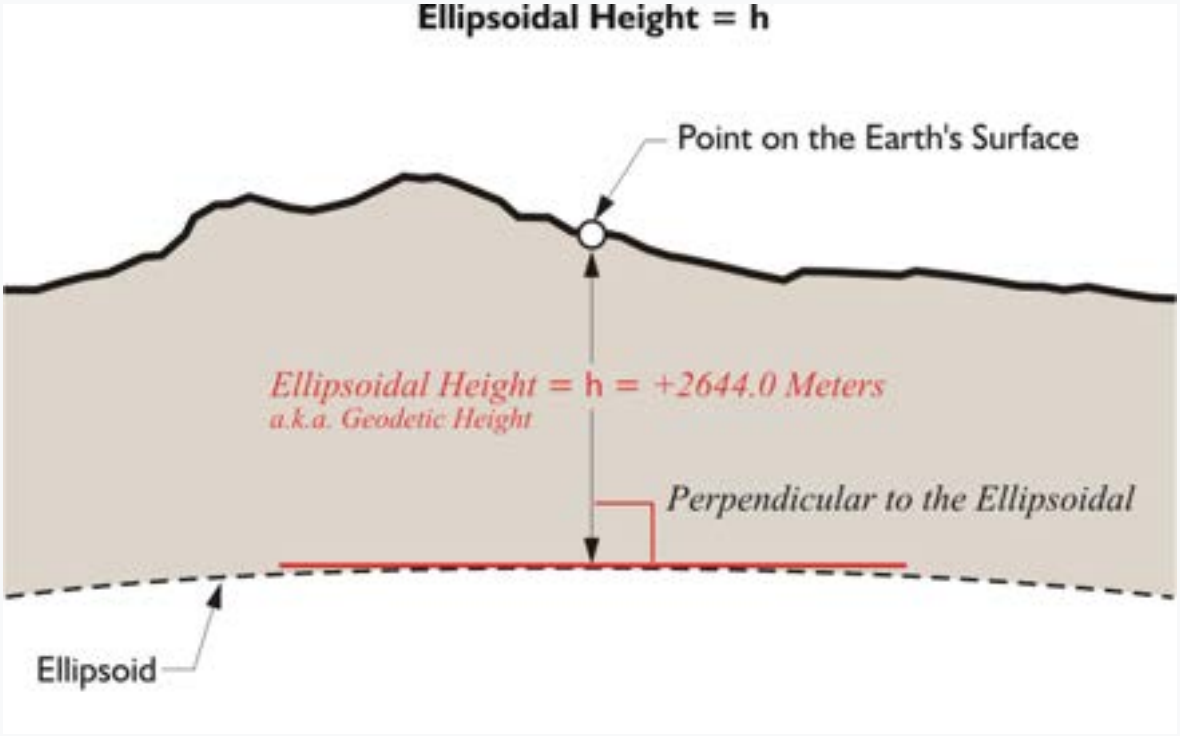
However, the **orthometric elevations from traditional surveying are not directly available** from GPS/GNSS measurements. This fundamental difference creates important considerations for modern surveying applications, requiring us to understand the relationship between gravity-based heights and ellipsoidal heights.

This module explores the traditional methods of height determination and examines the challenges of integrating GPS/GNSS technology with established elevation systems.

📍 The Third Dimension of Position

Beyond Latitude and Longitude

A point on the earth's surface is not completely defined by its latitude and longitude. In such a context, there is, of course, a third element, that of height. Surveyors have traditionally referred to this component of a position as its **elevation**. One classical method of determining elevations is spirit leveling.



Ellipsoidal Heights vs Earth Surface

The distance between the reference ellipsoid and Earth's surface

A level, correctly oriented at a point on the surface of the earth, defines a line parallel to the geoid at that point. Therefore, the elevations determined by level circuits are **orthometric**; that is, they are defined by their vertical distance above the geoid as it would be measured along a plumb line.

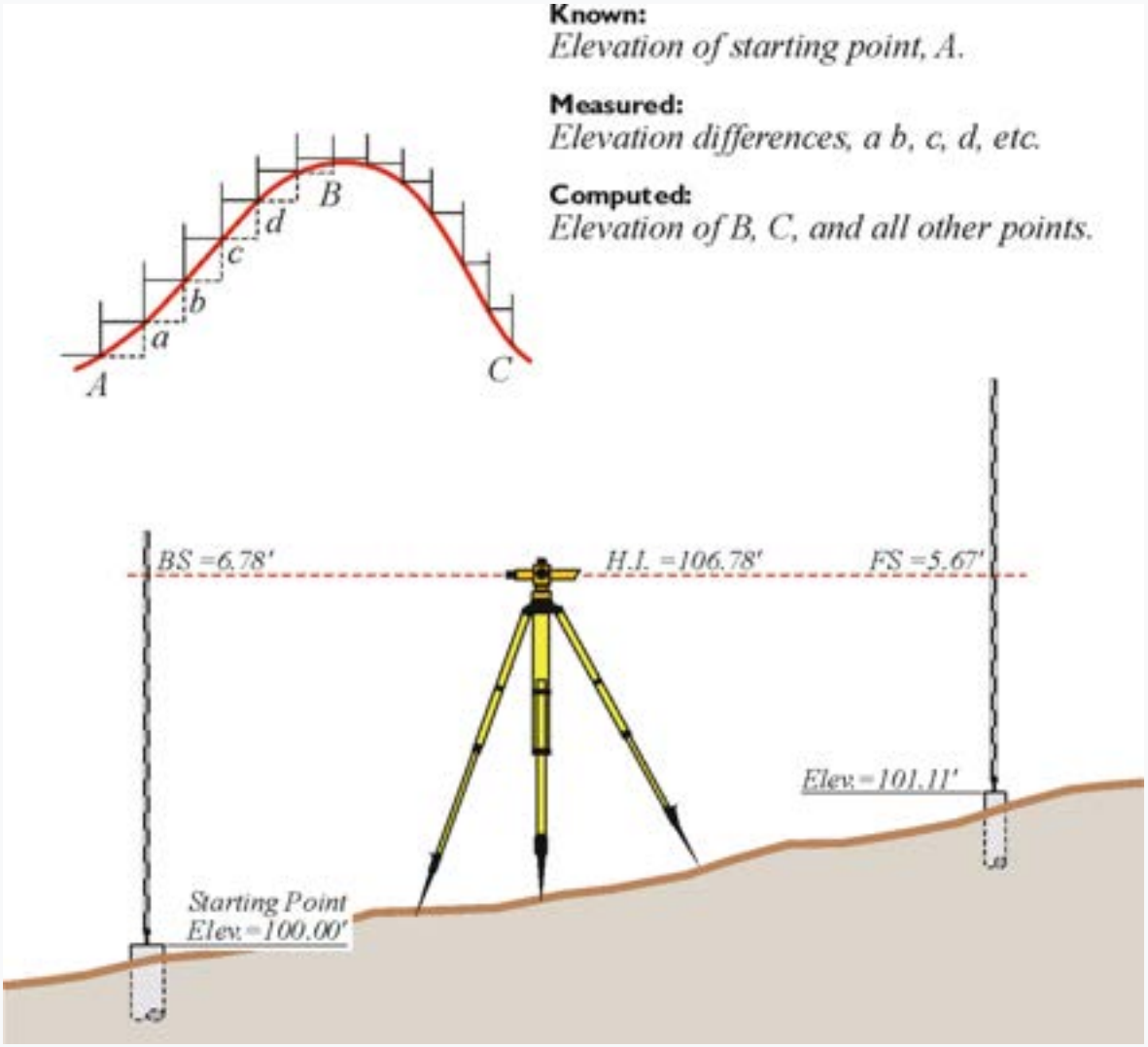
> 🎯 **Key Concept**
>
> However, **orthometric elevations are not directly available** from the geocentric position vectors derived from GPS/GNSS measurements. This fundamental difference creates important considerations for modern surveying applications.

📐 Spirit Leveling: The Classical Method

Gravity-Based Height Determination

In a normal course of things, a coordinate —YX, or northing easting, or latitude and longitude— is not fully a definition of a position. We've discussed also Earth-Centered, Earth-Fixed XYZ coordinates, but it is typical for a horizontal coordinate to be given a height, or elevation. This third element, was

most often originally determined by **spirit leveling** or, in some cases, by a technique known as trigonometric leveling.



A Spirit Level Run

Traditional method for determining orthometric elevations

🔬 Spirit Leveling Features

- **Direct measurement** of height differences
- **Bubble level instruments**
- **Horizontal sight lines** parallel to geoid
- **High accuracy** for elevation determination
- **Sequential measurements** over long distances

🌊 Orthometric Properties

- **Based upon gravity**
- **Referenced to the geoid**
- **Parallel to equipotential surfaces**
- **What we think of as "elevations"**
- **Foundation of traditional surveying**

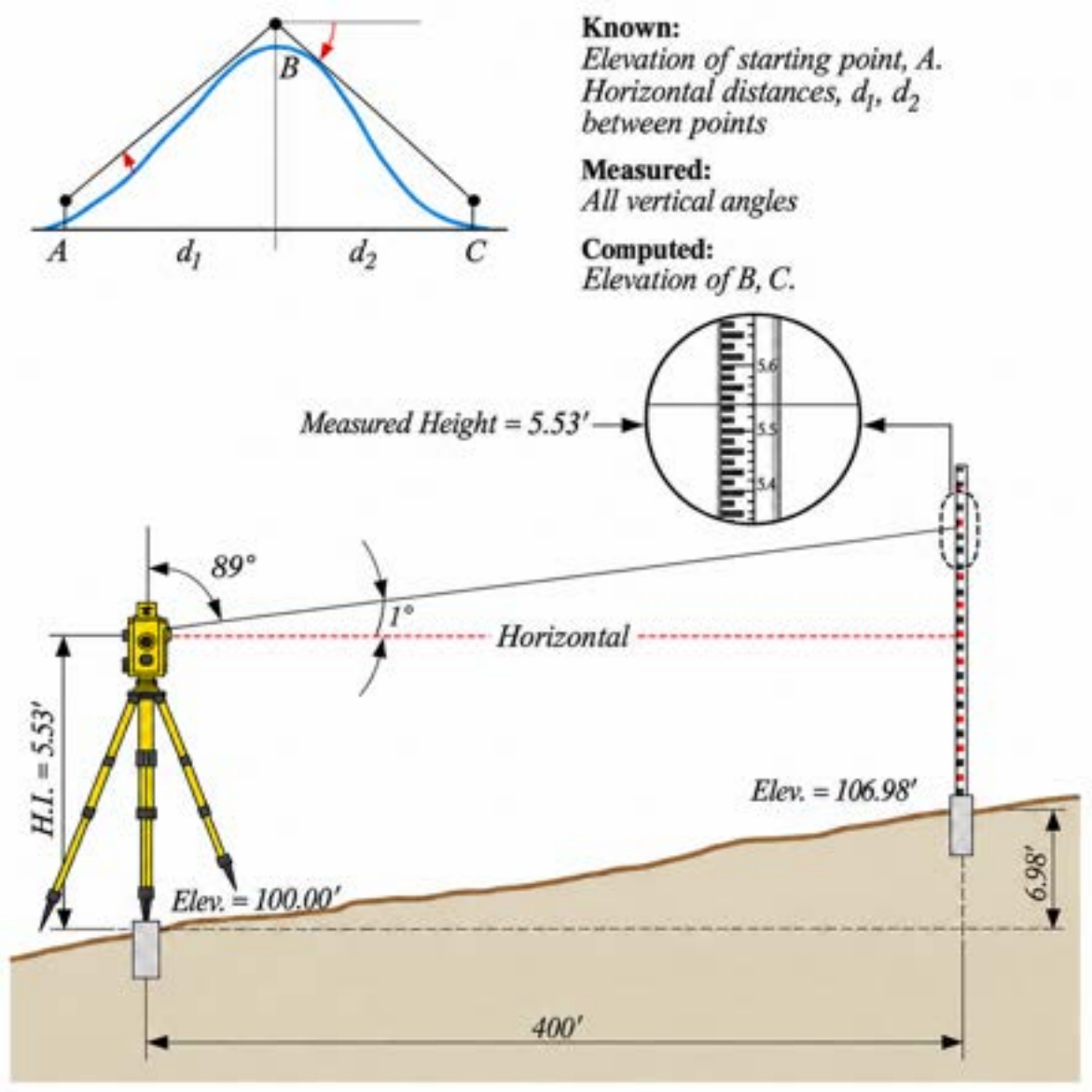
🔬 **Spirit Leveling Principles**

A level, correctly oriented at a point on the surface of the earth, **defines a line parallel to the geoid** at that point. This creates elevations that are **orthometric**—defined by their vertical distance above the geoid as measured along a plumb line.

📐 Trigonometric Leveling: An Alternative Approach

Angular Measurements for Height Determination

Despite the surveying method that was used in the past, it was based on optical instruments oriented to gravity, therefore oriented to the geoid. Therefore, the heights were based upon gravity.



Trigonometric Leveling

Using angular measurements and distances to determine elevations

📏 Spirit Leveling

- **Direct measurement** of height differences
- **Bubble level instruments**
- **Horizontal sight lines** parallel to geoid
- **High accuracy** for elevation determination
- **Sequential measurements** over long distances

📐 Trigonometric Leveling

- **Angular measurements** to determine heights
- **Theodolites and total stations**
- **Vertical angles** and horizontal distances
- **Faster over long distances**
- **Less accurate** than spirit leveling

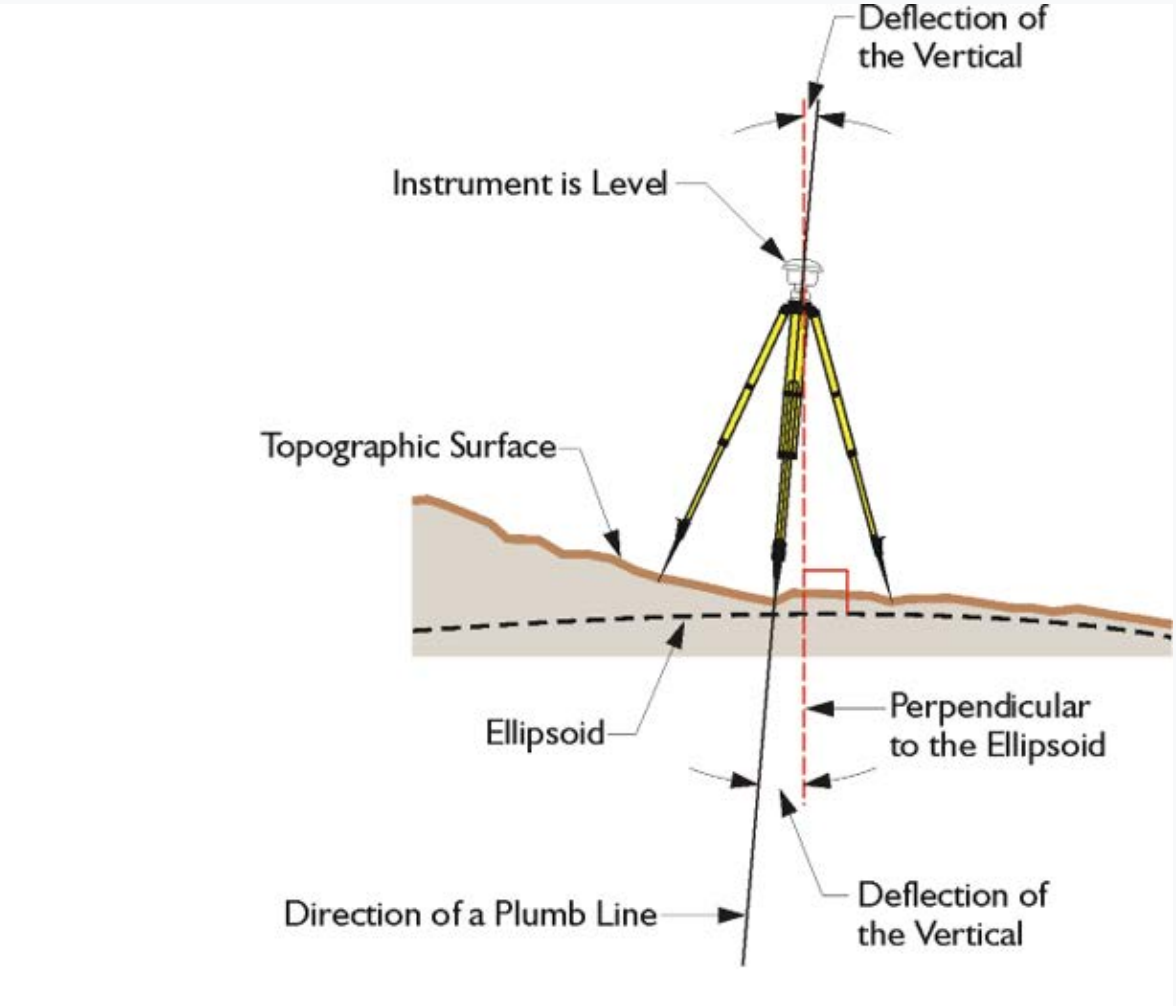
🌍 **Common Foundation**

Both methods shared a crucial characteristic: they were **based on optical instruments oriented to gravity, therefore oriented to the geoid**. This means all traditional height measurements were fundamentally gravity-based.

🌊 Ellipsoidal Heights: The GPS Challenge

Understanding Ellipsoidal Heights

Modern geodetic datums rely on the surfaces of geocentric ellipsoids to approximate the surface of the Earth. But the actual surface of the Earth does not coincide with these nice smooth surfaces, even though that is where the points represented by the coordinate pairs lay.



The Deflection of the Vertical

Showing how gravity direction differs from ellipsoid perpendicular

📐 Ellipsoidal Height Characteristics

- **Perpendicular to ellipsoid surface**
- **Not measured along gravity**
- **Mathematical reference surface**
- **What GPS provides directly**
- **Symbol: h (small h)**

🎯 Key Properties

- **Positive:** Earth surface above ellipsoid
- **Negative:** Earth surface below ellipsoid
- **Cannot be measured directly** with traditional instruments
- **Readily available** from GPS/GNSS
- **Varies with datum changes**

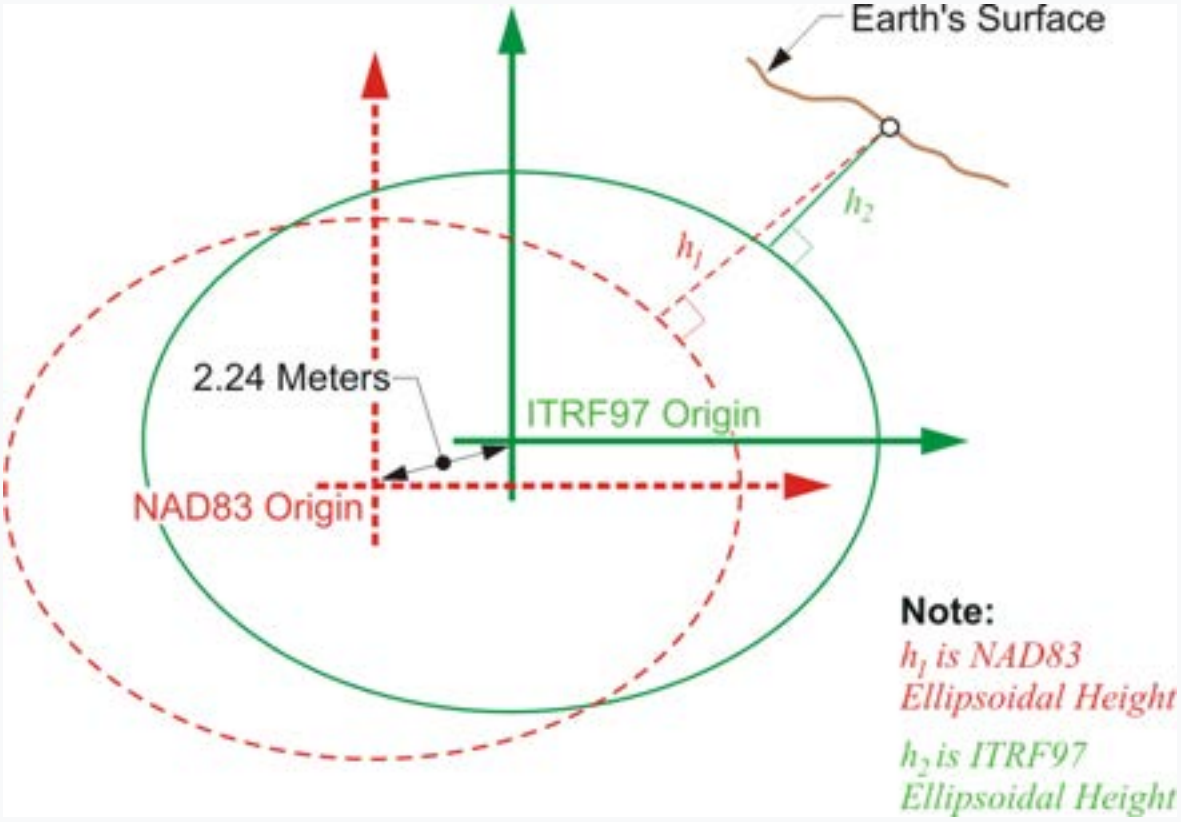
🛰️ GPS Reality

The ellipsoid height, **not the orthometric height, is what is determined directly from a GPS/GNSS measurement**. GPS satellites orbit around the center of mass of the Earth and have no concept of where the surface of the Earth is.

🏛️ The Legacy of Gravity-Based Heights

Historical Infrastructure of Elevation Data

There is a long legacy of benchmarks, recorded heights, and archives that are based upon this sort of methodology of determining heights. And so, this is what it follows that these values are generally what we think of as elevations —we think of orthometric heights based upon gravity.



A Shift to the Geocenter

Showing difference between datum origins affecting ellipsoidal heights

🏛️ Historical Infrastructure

- **Long legacy of benchmarks**
- **Recorded heights** in archives
- **Established survey networks**
- **Decades of accumulated data**
- **Standardized procedures**

🌊 Orthometric Heights

- **Based upon gravity**
- **Referenced to the geoid**
- **What we think of as "elevations"**
- **Familiar to surveyors**
- **Used in traditional mapping**

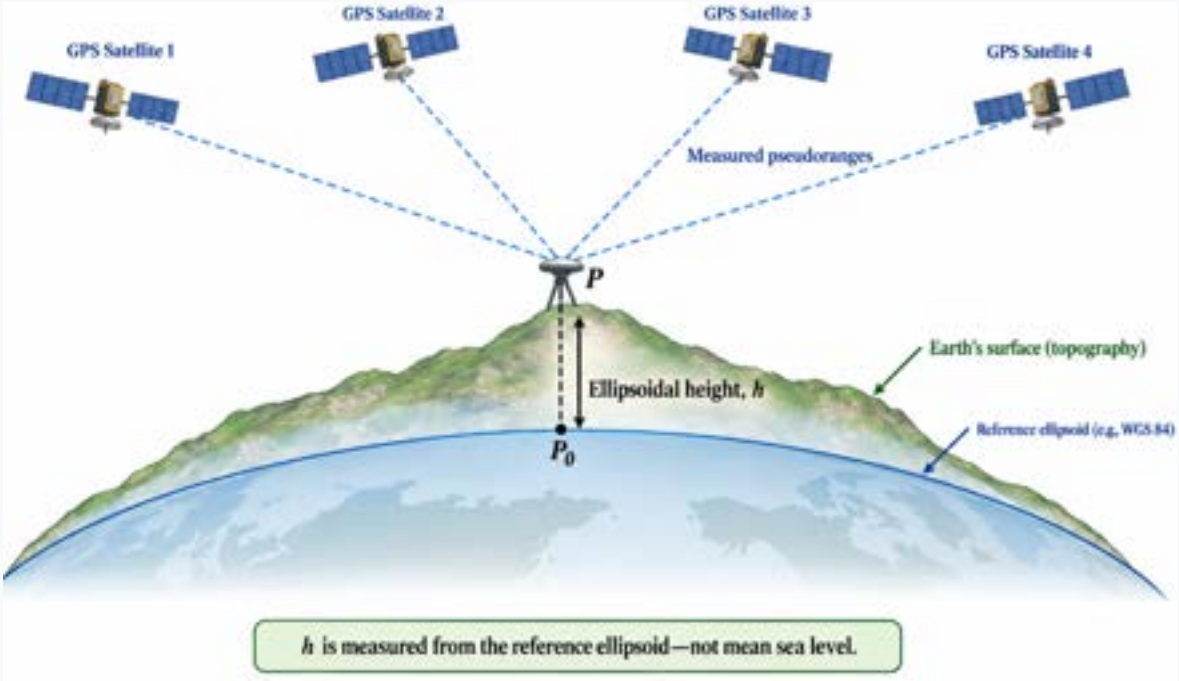
📊 What We Call "Elevation"

When most people think of elevation, they're thinking of **orthometric heights based upon gravity**. This creates a fundamental challenge when working with GPS/GNSS systems, which provide ellipsoidal heights rather than the gravity-based elevations that traditional surveying has established.

⚠️ The Integration Challenge

Bridging Incompatible Height Systems

The challenge arises because GPS/GNSS measurements provide ellipsoidal heights (heights above the mathematical reference ellipsoid), while traditional surveying infrastructure is built on orthometric heights (heights above the geoid, which approximates mean sea level).



GPS Positioning Fundamentals

How GPS determines ellipsoidal heights from satellite signals

📡 GPS/GNSS Heights

- **Ellipsoidal heights (h)**
- **Mathematical reference surface**
- **Geocentric positioning**

🌊 Traditional Survey Heights

- **Orthometric heights (H)**
- **Gravity-based reference**
- **Geoid surface**

- **Not directly usable** for traditional applications
- **Require conversion** to orthometric

- **What users expect** as "elevation"
- **Extensive benchmark network**

🔄 **Integration Formula**

The relationship between height systems: **h = H + N**
Where: h = ellipsoidal height, H = orthometric height, N = geoid height (undulation)

🎯 Heights Summary

Understanding the Foundation

The traditional gravity-based methods of height determination—spirit leveling and trigonometric leveling—created the elevation infrastructure that still forms the backbone of surveying and mapping today. Understanding these methods and their relationship to the geoid is essential for modern practitioners who must integrate GPS/GNSS ellipsoidal heights with the orthometric height system that users expect and regulatory frameworks require.

🏛️ Traditional Methods

- Spirit leveling
- Trigonometric leveling
- Gravity-based
- Orthometric heights

🛰️ GPS/GNSS System

- Satellite positioning
- Ellipsoidal heights
- Mathematical reference
- Geocentric system

🔄 Integration

- Geoid models
- Height conversions
- h = H + N
- Modern surveying

CONTENT

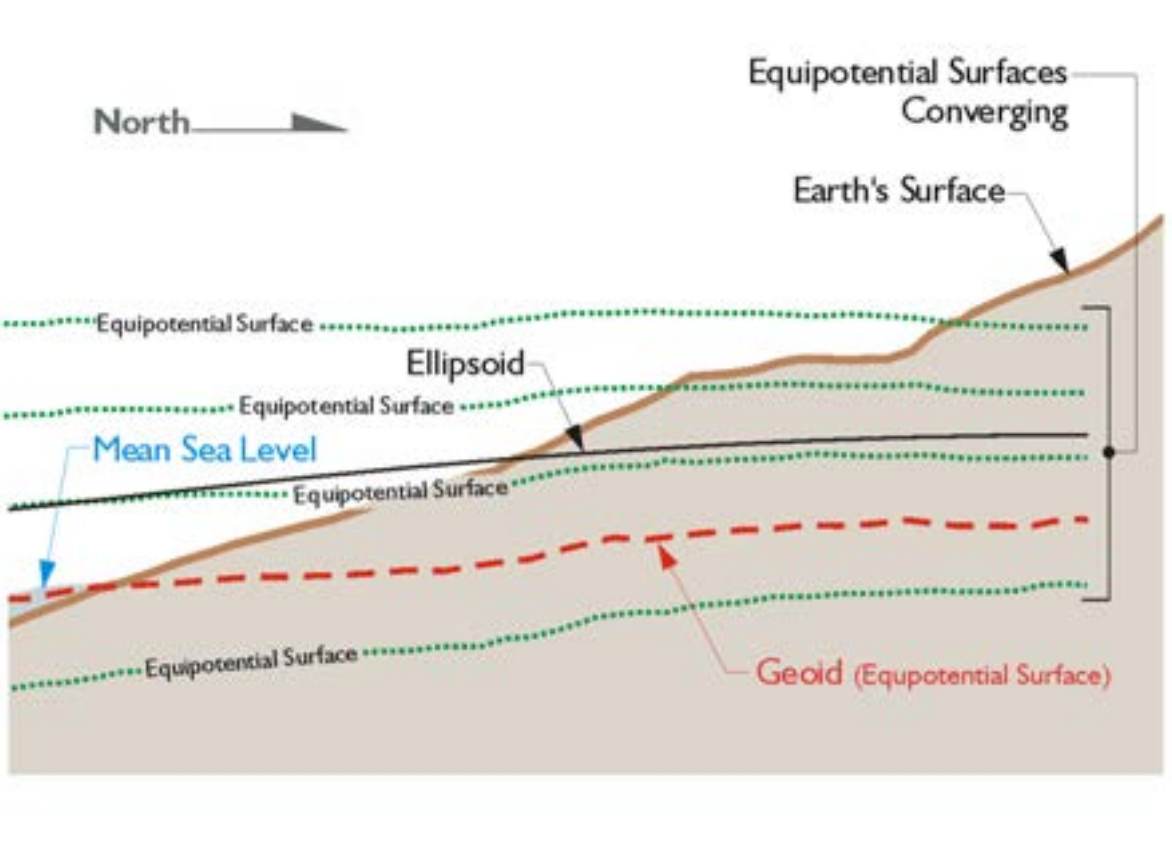
📚 The Geoid



🌍 Understanding Equipotential Surfaces

Gravity and Potential Energy

Any object in the earth's gravitational field has **potential energy** derived from being pulled toward the Earth. Quantifying this potential energy is one way to talk about height, because the amount of potential energy an object derives from the force of gravity is related to its height.



Ellipsoid-Geoid-Mean Sea Level with Several Equipotential Surfaces
Showing the relationship between different reference surfaces

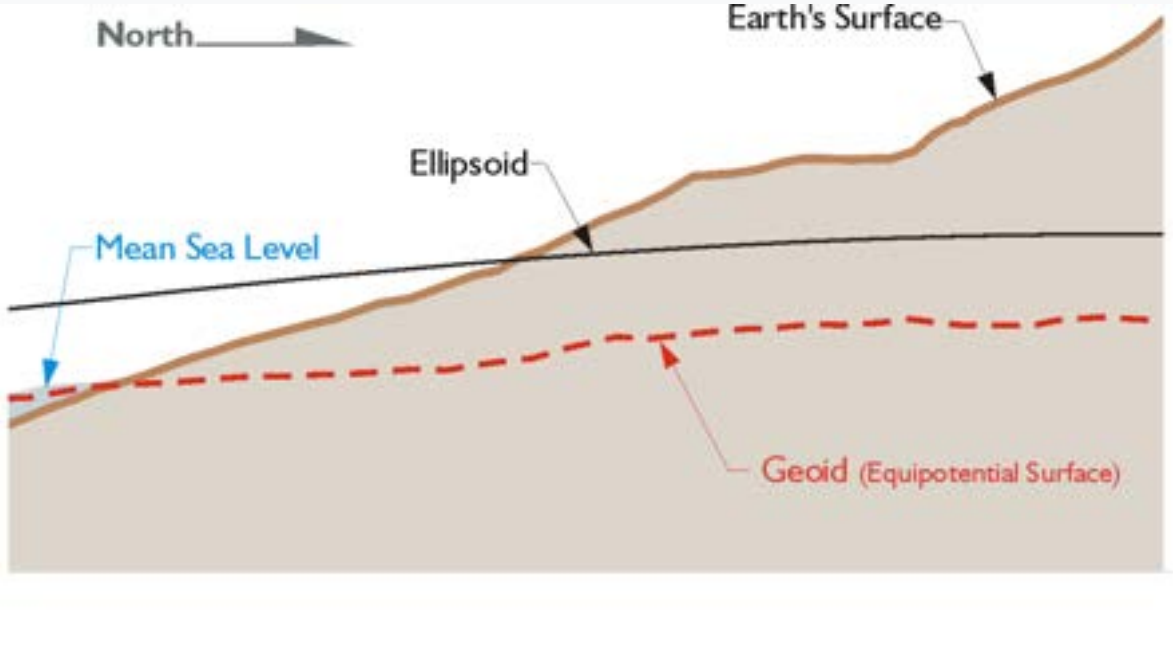
There are an infinite number of points where the potential of gravity is always the same. They are known as **equipotential surfaces**. Mean Sea Level itself is not an equipotential surface at all, of course. Forces other than gravity affect it, forces such as temperature, salinity, currents, wind, and so forth.

> 🎯 **Key Concept**
>
> The **geoid, on the other hand, is defined by gravity alone**. The geoid is the particular equipotential surface arranged to fit Mean Sea Level as well as possible, in at least squares sense.

🌊 Geoid vs. Mean Sea Level

Different Reference Surfaces

The geoid and the ellipsoid are not the same. Remember that the legacy heights determined by optical instruments in the past were always relative to the geoid, because, of course, the instruments were oriented to gravity. However, GPS/GNSS heights are related directly at least to the ellipsoid — two different surfaces.



Ellipsoid-Geoid-Mean Sea Level

Comparison of different reference surfaces used in surveying

🌊 The Geoid

- **Defined by gravity alone**
- **Equipotential surface**
- **Approximates Mean Sea Level**
- **Reference for orthometric heights**
- **Irregular and lumpy surface**

🌊 Mean Sea Level

- **Affected by multiple forces**
- **Temperature and salinity effects**
- **Currents and wind influence**
- **Deviates from geoid**
- **Up to 1-2 meters difference**

🔍 Important Distinction

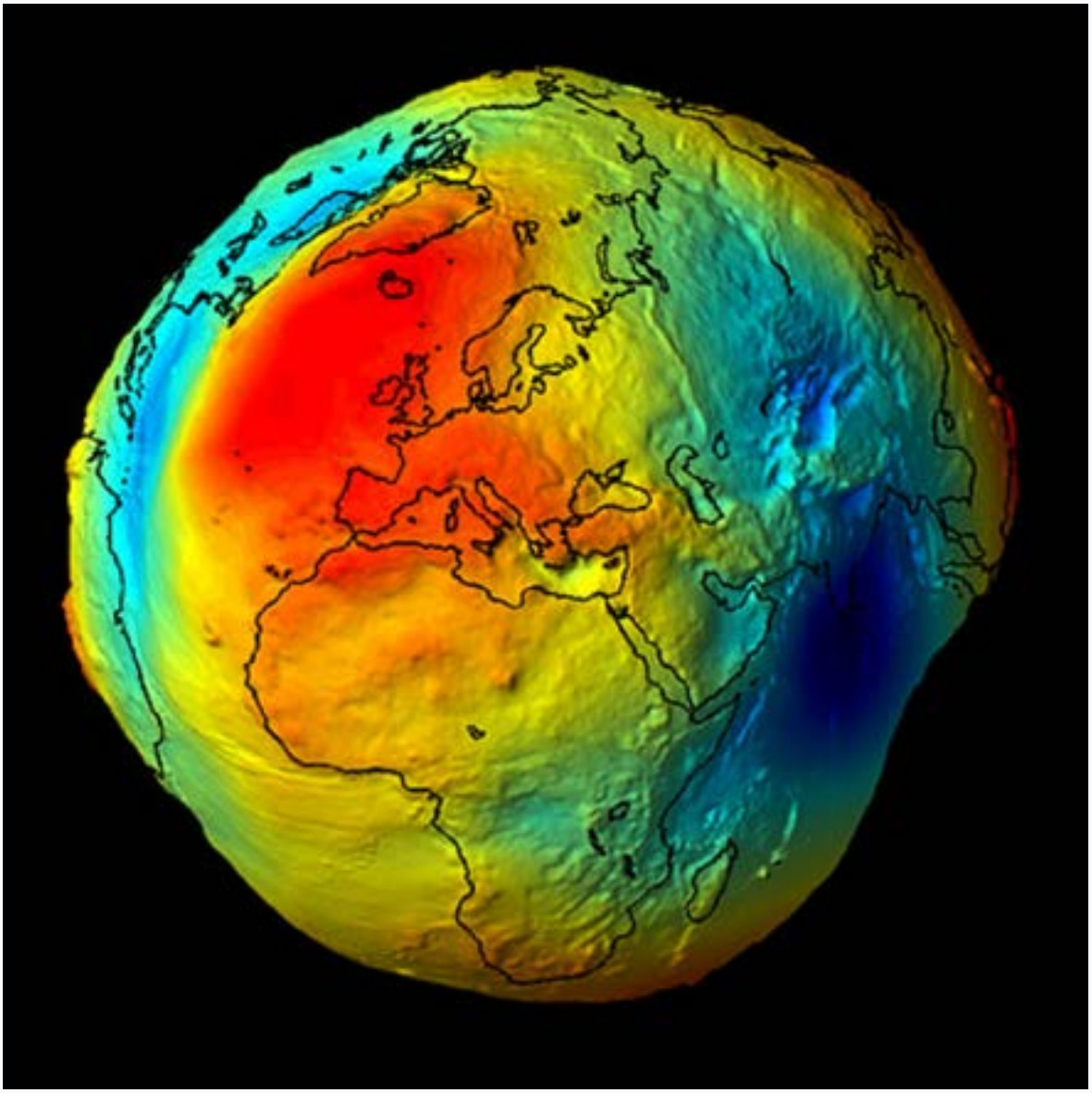
While there is a relationship between Mean Sea Level and the geoid, they are not the same. They could be the same if the oceans of the world could be utterly still, completely free of currents, tides, friction, variations in temperature, and all other physical forces, except gravity.

However, these unavoidable forces actually cause Mean Sea Level to deviate up to 1, even 2, meters from the geoid.

🏔️ The Lumpy Geoid

Gravity Variations Create Irregularities

The geoid is completely is not smooth and continuous. It is **lumpy**, because gravity is not consistent across the surface of the earth. At every point, gravity has a magnitude and a direction, but these vectors do not all have the *same* direction or magnitude.



Global Geoid Variations

Showing the "lumpy" nature of the geoid surface worldwide

🌍 Causes of Geoid Variations

- **Uneven mass distribution** in Earth
- **Density variations** affect gravity
- **Non-spherical Earth shape**
- **Geological structures** influence gravity
- **Topographic features** create variations

📏 Global Geoid Separations

- **Maximum high:** +85 meters (west of Ireland)
- **Maximum low:** -106 meters (south of India)
- **Total variation:** ~191 meters globally
- **Reference:** GRS80 ellipsoid
- **Irregular surface** follows gravity

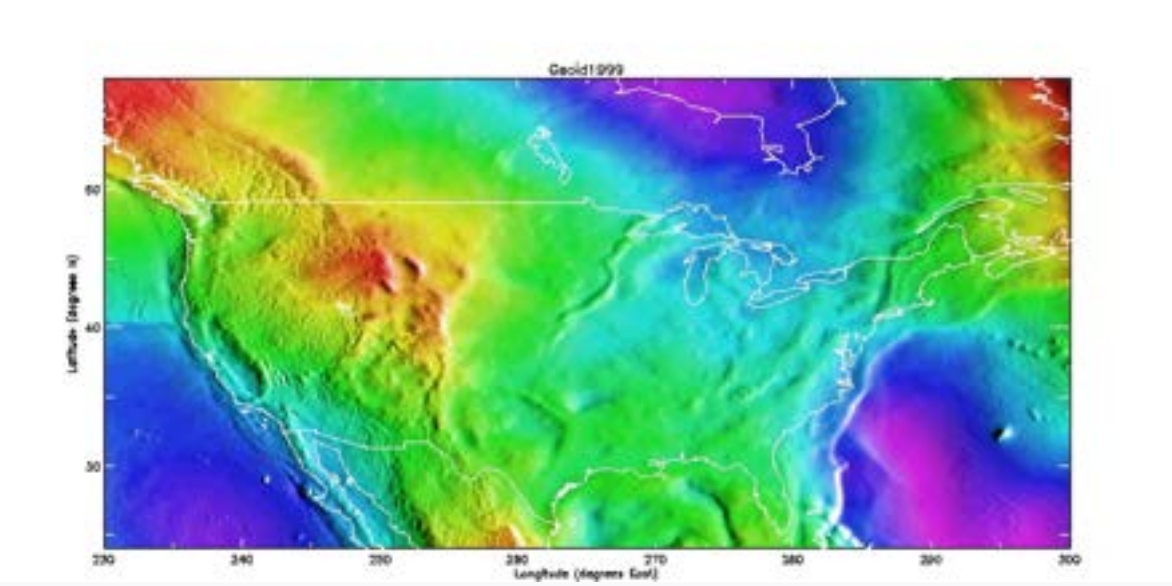
🔬 **Scientific Reality**

Some parts of the earth are denser than others. Where the earth is denser, there is more gravity, and the fact that the earth is not a sphere also affects gravity. The geoid undulates with the uneven distribution of the mass of the earth and has all the irregularity that the attendant variation in gravity implies. In fact, the separation between the lumpy surface of the geoid and the smooth GRS80 ellipsoid worldwide varies from about +85 meters west of Ireland to about -106 meters, the latter in the area south of India near Ceylon.

🇺🇸 The Geoid in the United States

CONUS Geoid Variations

In the conterminous United States, sometimes-abbreviated *CONUS*, the distances between the geoid and the GRS80 ellipsoid, known as *geoid heights,* are less. They vary from about –8 meters to about –53 meters.



The Geoid in the Coterminous United States

Showing geoid height variations across the continental US

🌍 Global Range

- +85 to -106 meters
- 191-meter total range
- Extreme variations
- Complex patterns

🇺🇸 CONUS Range

- -8 to -53 meters
- 45-meter total range
- More moderate variations
- Regional patterns

📊 Significance

- Critical for surveying
- Height conversions
- GPS applications
- Mapping accuracy

📏 Geoid Heights (Undulations)

These separations between the geoid and the ellipsoid are called **geoid heights** or **geoid undulations**. They represent the fundamental correction needed to convert between GPS ellipsoidal heights and traditional orthometric elevations.

🎯 Understanding the Geoid's Role

Bridging Traditional and Modern Systems

Understanding the geoid is essential for modern surveying because it represents the bridge between traditional gravity-based height systems and GPS/GNSS ellipsoidal heights. The geoid's irregular surface reflects the true gravitational field of the Earth, making it the natural reference for elevations as we experience them.

🏛️ Traditional System

- Gravity-based heights
- Referenced to geoid
- Orthometric elevations
- What surveyors expect

🌊 The Geoid

- Gravity equipotential surface
- Irregular and lumpy
- Approximates Mean Sea Level
- Bridge between systems

🛰️ GPS System

- Ellipsoidal heights
- Mathematical reference
- Smooth surface
- Requires conversion

🔄 The Conversion Key

The geoid height (N) is the critical value needed for the fundamental height conversion equation: **h = H + N**, where h is ellipsoidal height, H is orthometric height, and N is geoid height. Understanding geoid variations is essential for accurate surveying work.

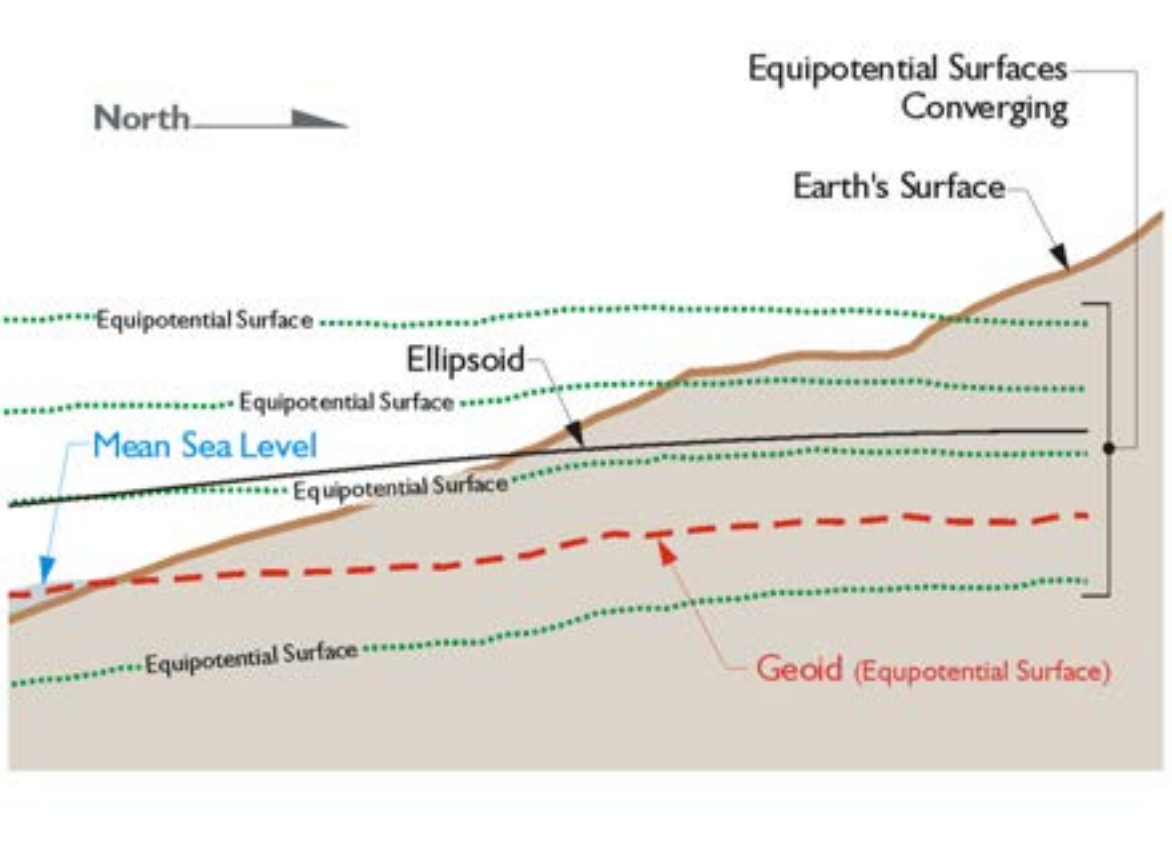
📚 Height Conversion

🔄 Height Conversion

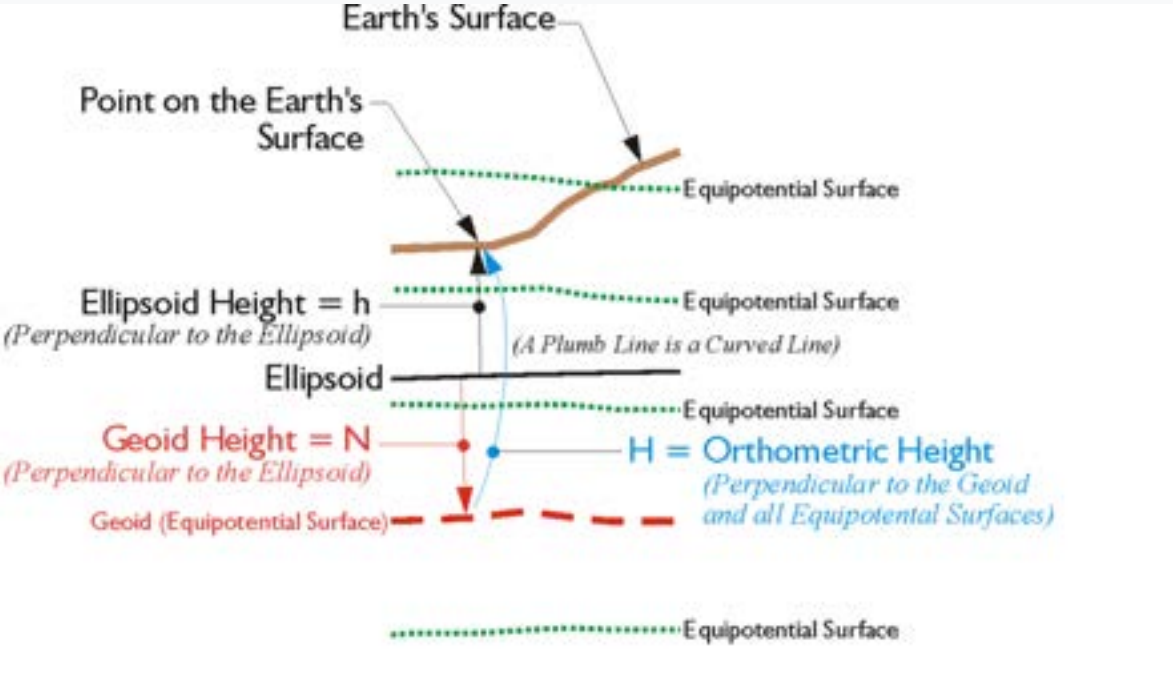
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📐 Understanding Height System Relationships

Visualizing the Three Reference Surfaces



Ellipsoid-Geoid-Topographic Surface with Equipotential Surfaces



Height Conversion

📏 Geoid Height (N) - The Key to Conversion

Understanding Geoid Height Measurements

A geoidal height is the distance measured along a line perpendicular to the ellipsoid of reference to the geoid. Also, as you can see in the illustration, these geoid heights are negative. They are usually symbolized, *N*. If the geoid is above the ellipsoid, N is positive; if the geoid is below the ellipsoid, N is negative. It is negative here because the geoid is underneath the ellipsoid throughout the conterminous United States. In Alaska, it is the other way around; the ellipsoid is underneath the geoid and N is positive.

🇺🇸 Continental United States (CONUS)

- **Geoid is below ellipsoid**
- **N is negative**
- **Range: -8 to -53 meters**
- **Ellipsoidal heights < orthometric heights**

🏔️ Alaska

- **Ellipsoid is below geoid**
- **N is positive**
- **Opposite relationship**
- **Different conversion behavior**

📐 **Measurement Direction**

Geoid height (N) is measured along a line **perpendicular to the ellipsoid** from the ellipsoid to the geoid. The sign indicates which surface is higher at that location.

📊 The Three Height Types

Defining Each Height System

Please recall that an ellipsoid height is symbolized, *h*. The ellipsoid height is also measured along a line perpendicular to the ellipsoid of reference, but to a point on the surface of the Earth.

However, an orthometric height, symbolized, *H*, is measured along a plumb line from the geoid to a point on the surface of the Earth.

📏 Ellipsoidal Height (h)

- **Perpendicular to ellipsoid**
- **To Earth's surface**
- **What GPS/GNSS provides**
- **Mathematical reference**
- **Not gravity-based**

🌊 Orthometric Height (H)

- **Along plumb line**
- **From geoid to surface**
- **Traditional "elevation"**
- **Gravity-based reference**
- **What surveyors expect**

🔄 Geoid Height (N)

- **Ellipsoid to geoid**
- **Perpendicular to ellipsoid**
- **Positive or negative**
- **Regional variations**
- **Conversion factor**

🔍 **Key Difference**

Ellipsoidal height (h) is measured perpendicular to the mathematical ellipsoid, while orthometric height (H) is measured along the curved plumb line that follows gravity's direction.

⚖️ The Conversion Formula

From GPS Heights to Traditional Elevations

By using the formula,

$$H = h - N$$

one can convert an ellipsoidal height, h, derived say from a GPS/GNSS observation, into an orthometric height, H, by knowing the extent of geoid-ellipsoid separation, also known as the geoidal height, N, at that point. The ellipsoid height of a particular point is actually smaller than the orthometric height throughout the conterminous United States.

🔄 Conversion Process

- **Start with:** GPS ellipsoidal height (h)
- **Obtain:** Local geoid height (N)
- **Calculate:** H = h – N
- **Result:** Orthometric elevation (H)

🇺🇸 CONUS Example

- **GPS height (h):** 100.00 m
- **Geoid height (N):** -30.00 m
- **Calculation:** H = 100 – (-30)
- **Elevation (H):** 130.00 m

🎯 Practical Application

This formula allows GPS users to convert satellite-derived heights into the traditional elevations that surveyors, engineers, and mapping professionals expect and regulatory frameworks require.

📐 Plumb Line Curvature

Why the Simple Formula Works

The formula H = h – N does not account for the fact that the plumb line along which an orthometric height is measured is curved, as you see in the figure above. Curved, because it is perpendicular with each and every equipotential surface through which it passes. The equipotential surfaces are not parallel with each other. They converge toward the pole because the Earth is oblate therefore, the plumb line must curve to maintain perpendicularity with them all.

🌍 Why Plumb Lines Curve

- **Equipotential surfaces** not parallel
- **Converge toward poles** (oblate Earth)
- **Plumb line maintains perpendicularity**
- **Must curve** through all surfaces
- **Follows gravity's direction**

📊 Negligible Impact

- **Maximum deviation:** ~1 minute of arc
- **Only in extreme cases**
- **Height difference:** negligible
- **6+ miles high:** only 1mm difference
- **Formula remains valid**

✅ **Practical Conclusion**

This deviation of a plumb line from the perpendicular to the ellipsoid reaches about 1 minute of arc in only the most extreme cases. Therefore, any height difference that is caused by the curvature is negligible. It would take a height of over 6 miles for the curvature to amount to even 1 mm of difference in height.

CONTENT

📹 Vetical Datums by Dave Doyle

Vertical Datums

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📹 Educational Video

Vertical Datums by Dave Doyle

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Dave Doyle brings extensive experience in surveying and geodesy, providing valuable perspective on how these fundamental concepts are applied in professional practice and why understanding vertical datums is essential for accurate positioning work.

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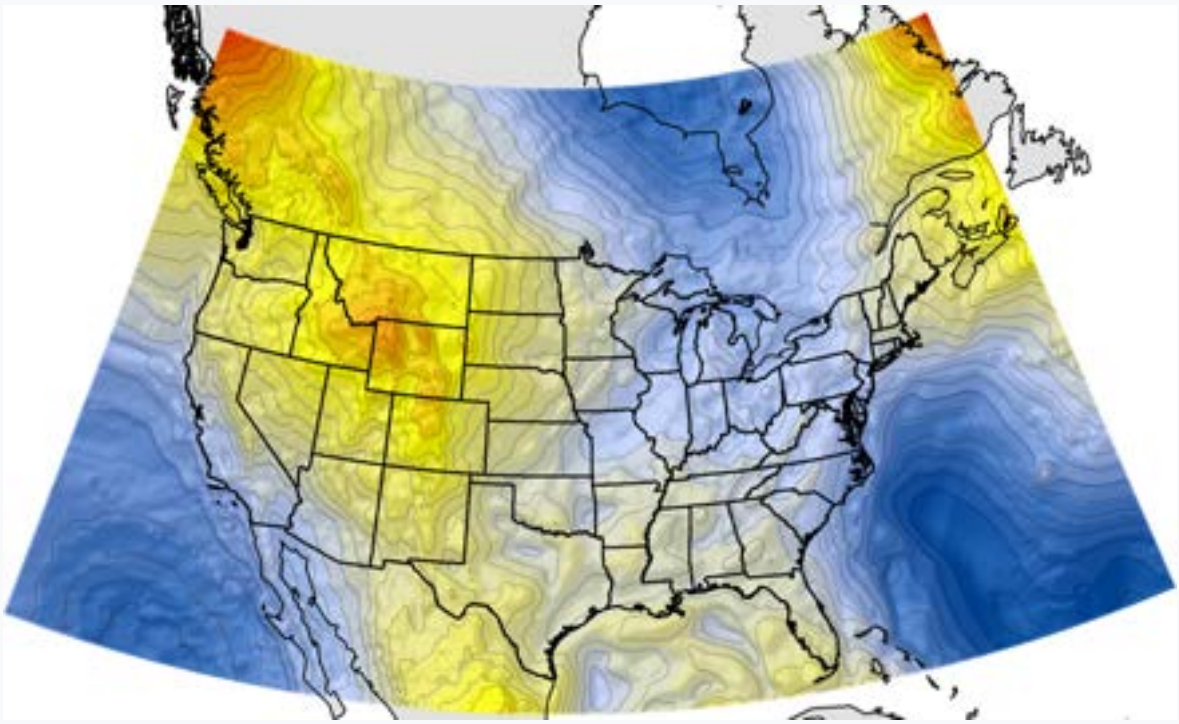
📚 Geoidal Models

Geoidal Models

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🌍 Evolution of Geoid Mapping

Quarter-Century of Improvements



Geoid 12A
Modern high-resolution geoid model for the United States

Major improvements have been made over the past quarter-century or so in mapping the geoid on both national and global scales. And because there are large complex variations in the geoid related to both the density and relief of the earth, geoid models and interpolation software have been developed to support the conversion of GPS elevations to orthometric elevations.

🎯 Purpose of Geoid Models

Geoid models provide the critical *N* values (geoidal heights) needed to convert GPS ellipsoidal heights to orthometric elevations using the formula H = h – N.

📈 GEOID90 - The Beginning (1991)

First NGS Geoid Program

For example, in early 1991, NGS presented a program known as **GEOID90**. This program allowed a user to find *N*, the geoidal height, in meters for any NAD83 latitude and longitude in the United States.

📊 GEOID90 Specifications

- **Computation:** End of 1990
- **Data source:** Over 1 million gravity observations
- **Grid resolution:** 3' × 3' (lat × lon)
- **Accuracy:** About 10 cm
- **Coverage:** United States

🎯 Key Features

- **First comprehensive model**
- **NAD83 coordinate system**
- **Nationwide coverage**
- **User-friendly program**
- **Foundation for future models**

⚡ GEOID93 - Major Data Expansion

Five-Fold Increase in Gravity Data

The GEOID90 model was computed at the end of 1990, using over a million gravity observations. It was followed by the **GEOID93** model. It was computed at the beginning of 1993, using more than 5 times the number of gravity values used to create GEOID90. Both provided a grid of geoid height values in 3 minutes of latitude by 3 minutes of longitude grid with an accuracy of about 10 cm.

📊 GEOID90

- 1990 computation
- 1+ million observations
- 3' × 3' grid
- ~10 cm accuracy

📈 GEOID93

- 1993 computation
- 5+ million observations
- 3' × 3' grid
- ~10 cm accuracy

🚀 Improvement

- **500% more data**
- Better accuracy
- Enhanced coverage
- More reliable results

🎯 GEOID96 - Higher Resolution

Improved Grid Resolution

Next, the **GEOID96** model resulted in a gravimetric geoid height grid in a 2 minutes of latitude by 2 minutes of longitude grid.

📐 **Resolution Enhancement**

GEOID96 improved spatial resolution from 3' × 3' to **2' × 2'**, providing more detailed geoid height information for precise GPS height conversions.

🌎 GEOID99 - Expanded Coverage

First Hybrid Model with GPS Data

GEOID99 covered the conterminous United States, and it includes U.S. Virgin Islands, Puerto Rico, Hawaii, and Alaska. The grid is 1 degree of latitude by 1 degree of longitude, and it is the first to combine gravity values with GPS ellipsoid heights on previously leveled benchmarks.

🗺️ **Coverage Expansion**

- **Conterminous United States**
- **U.S. Virgin Islands**
- **Puerto Rico**
- **Hawaii**
- **Alaska**

🚀 **Innovation**

- **First hybrid model**
- **Gravity + GPS data**
- **Previously leveled benchmarks**
- **1° × 1° grid resolution**
- **Improved accuracy**

📊 **NGS Accuracy Assessment**

According to NGS, "When comparing the GEOID99 model with GPS ellipsoid heights in the NAD 83 reference frame and leveling in the NAVD 88 datum, it is seen that GEOID99 has

roughly a **4.6 cm absolute accuracy (one sigma)** in the region of GPS on benchmark coverage."

⏭️ Continued Evolution

Modern Geoid Models

GEOID03 superseded the previous models for the continental United States. It was followed by **GEOID09**, and the current geoid model in place is known as **GEOID18**.

📅 GEOID03

- Superseded previous models
- Continental US focus
- Enhanced methodology
- Improved accuracy

📈 GEOID09

- Further improvements
- Updated data sources
- Better algorithms
- Higher precision

🎯 GEOID18

- **Current model**
- Latest technology
- Best available accuracy
- Comprehensive coverage

🚀 GRAV-D: The Future of Vertical Datums

Gravity for the Redefinition of the American Vertical Datum

NGS has an ongoing project known gravity mapping project known as **GRAV-D**, "The gravity-based vertical datum resulting from this project will be accurate at the **2 cm level** where possible for much of the country. The proposal is official policy for NGS and is included in the NGS 10 year plan. The project is currently underway and actively collecting gravity data across the United States and its holdings."

🎯 GRAV-D Goals

- **2 cm accuracy target**
- **Gravity-based vertical datum**
- **Nationwide coverage**
- **Future datum foundation**
- **Revolutionary precision**

📊 Project Status

- **Official NGS policy**
- **10-year plan inclusion**
- **Currently underway**
- **Active data collection**
- **US and territories**

🔗 **Learn More**

For detailed information about the GRAV-D project: https://geodesy.noaa.gov/GRAV-D/

CONTENT

📹 2022 Datum Change by Dave Doyle

2022 Datum Change

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📽️ Educational Video

2022 Datum Change by Dave Doyle

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🚀 Critical Knowledge

Understanding the 2022 datum change is essential for modern geospatial professionals. This transformation represents a fundamental shift in how coordinate systems work in the United States

INTRODUCTION & INSTRUCTIONS

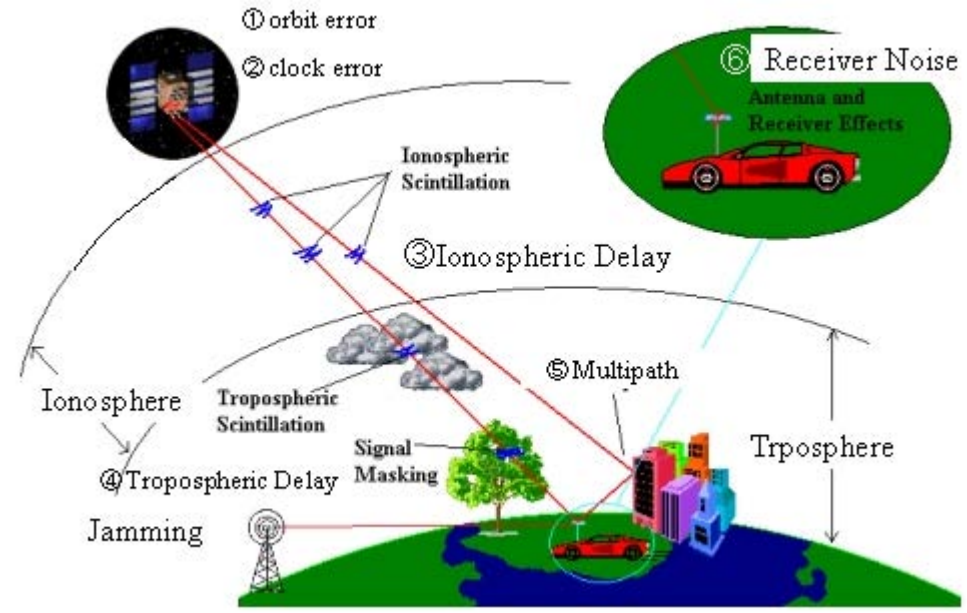
📡 Overview Module 05: GPS Error Sources



📖 Overview

Understanding GPS Error Sources

Biases are fascinating. The frequency standards, the clocks in GPS satellites, run faster than the clocks in GPS receivers, and the clocks in the GPS receivers are much less sophisticated than the clocks in the satellites. During the trip through the ionosphere, the information on the GPS signal appears to slow, and the ionosphere affects L2 more than L1. You might find it interesting that when a GPS signal reflects off a surface before it reaches your receiver, this is called multipath, and, believe it or not, it actually has something in common with a billiard ball (as a billiard ball often bounces off several surfaces before finding its way to the pocket). In this lesson, we will also talk about how GPS surveying is done.



Common Errors on GPS Signal

Major error sources affecting GPS positioning accuracy

🕐 Clock Errors

- **Satellite clocks** run faster
- **Receiver clocks** less sophisticated
- **Frequency standards** differ
- **Timing precision** critical

🌍 Atmospheric Effects

- **Ionospheric delays**
- **L2 affected more than L1**
- **Signal appears to slow**
- **Variable conditions**

🎱 Multipath

- **Signal reflections**
- **Like billiard balls** bouncing
- **Surface interactions**
- **Path distortions**

🎯 Lesson Focus

This lesson explores these fascinating error sources and examines how GPS surveying techniques address these challenges to achieve high-precision positioning.

- **Read Chapter 2 in *GPS and GNSS for Land Surveyors*.**
- Review the pages in this module and watch the instructor videos embedded within the module pages.

🎯 Learning Objectives

What You Will Learn

At the successful completion of this lesson, students should be able to:

📊 Error Analysis

- **demonstrate** biases and solutions
- **explain** the error budget
- **explain** the biases in the observation equations
- **describe** user equivalent range error

🕐 Clock Biases

- **identify** the satellite clock bias, dt
- **recognize** the receiver clock bias, dT
- **describe** the orbital bias

🌍 Atmospheric Effects

- **define** the ionospheric effect, d_{ion}
- **explain** the tropospheric effect, d_{trop}
- **identify** multipath

📐 Processing Techniques

- **recognize** differencing
- **differentiate** between classifications of positioning solutions
- **discuss** relative and autonomous positioning
- **recognize** the benefits of single, double, and triple differencing

💡 Learning Approach

These objectives build from understanding individual error sources to mastering the advanced processing techniques that make high-precision GPS surveying possible.

🔍 Key Error Categories

Understanding Error Sources

The learning objectives organize GPS errors into logical categories that build understanding from basic concepts to advanced processing techniques used in professional surveying.

📐 Processing

- Differencing methods

🛰️ Applications

- Relative positioning

⚡ System Errors

- Clock biases (dt, dT)
- Orbital errors
- Error budgets
- Range errors

🌍 Environmental

- Ionosphere (d_{ion})
- Troposphere (d_{trop})
- Multipath effects
- Signal delays

- Solution classifications
- Processing benefits
- Error mitigation

- Autonomous methods
- GPS surveying
- Precision techniques

🎯 Professional Relevance

Understanding these error sources and mitigation techniques is essential for professional GPS surveying, where centimeter-level accuracy depends on properly addressing each type of bias and error.

CONTENT

📚 The Error Budget



Understanding Range Biases

The understanding and management of errors is indispensable for finding the true geometric range ρ between a satellite and a receiver from either a pseudorange or carrier phase observation.

$$p = \rho + d_{\rho} + c(dt - dT) + d_{ion} + d_{trop} + \varepsilon_{mp} + \varepsilon_{p} \text{ (pseudorange)}$$

$$\Phi = \rho + d_{\rho} + c(dt - dT) + N\lambda - d_{ion} + d_{trop} + \varepsilon_{m\Phi} + \varepsilon_{\Phi} \text{ (carrier phase)}$$

Both equations include environmental and physical limitations called *range biases*.

Error Sources and Components

Atmospheric errors are among the biases; two are the ionospheric effect, d_{ion}, and the tropospheric effect, d_{trop}. Other biases, clock errors symbolized by *(dt-dT)* and receiver noise, ε_{ρ} and ε_{φ}, multipath, $\varepsilon_{m\rho}$ and $\varepsilon_{m\varphi}$, and orbital errors, d_{ρ}, are unique to satellite surveying methods. As you can see, each of these biases comes from a different source. They are each independent of one another, but they

combine to obscure the true geometric range. The objective here is to discuss each of them separately.

Here, we see the formula of the pseudorange error budget on the upper portion. As it indicates, p is the pseudorange, measurement equals ρ, rho, the true range between the GPS receiver and satellite. However, as you can see, there are many more elements, errors, or biases that contaminate the pseudorange— the satellite orbital errors, the ephemeris errors, etc.

The time difference *(dt-dT)* is between the satellite clock offset, and the receiver clock offset from GPS Time, as was mentioned in the previous lesson. There is also the ionospheric delay, the attenuation, of the signal as it passes through the ionosphere. Distinct from these biases are multipath and receiver noise. These two are distinct because, unlike the others, it is difficult to deal with them in a statistical way. Nevertheless, all are part of the error budget.

Directly below the pseudorange formula, you see the carrier phase formula. It also equals true range symbolized by capital phi, Φ. It includes errors that are very similar to those in the pseudorange formula. But one is included that is obviously different. It is symbolized by the capital N followed by the letter lambda, λ. This is the integer ambiguity. It doesn't occur in the pseudorange measurement.

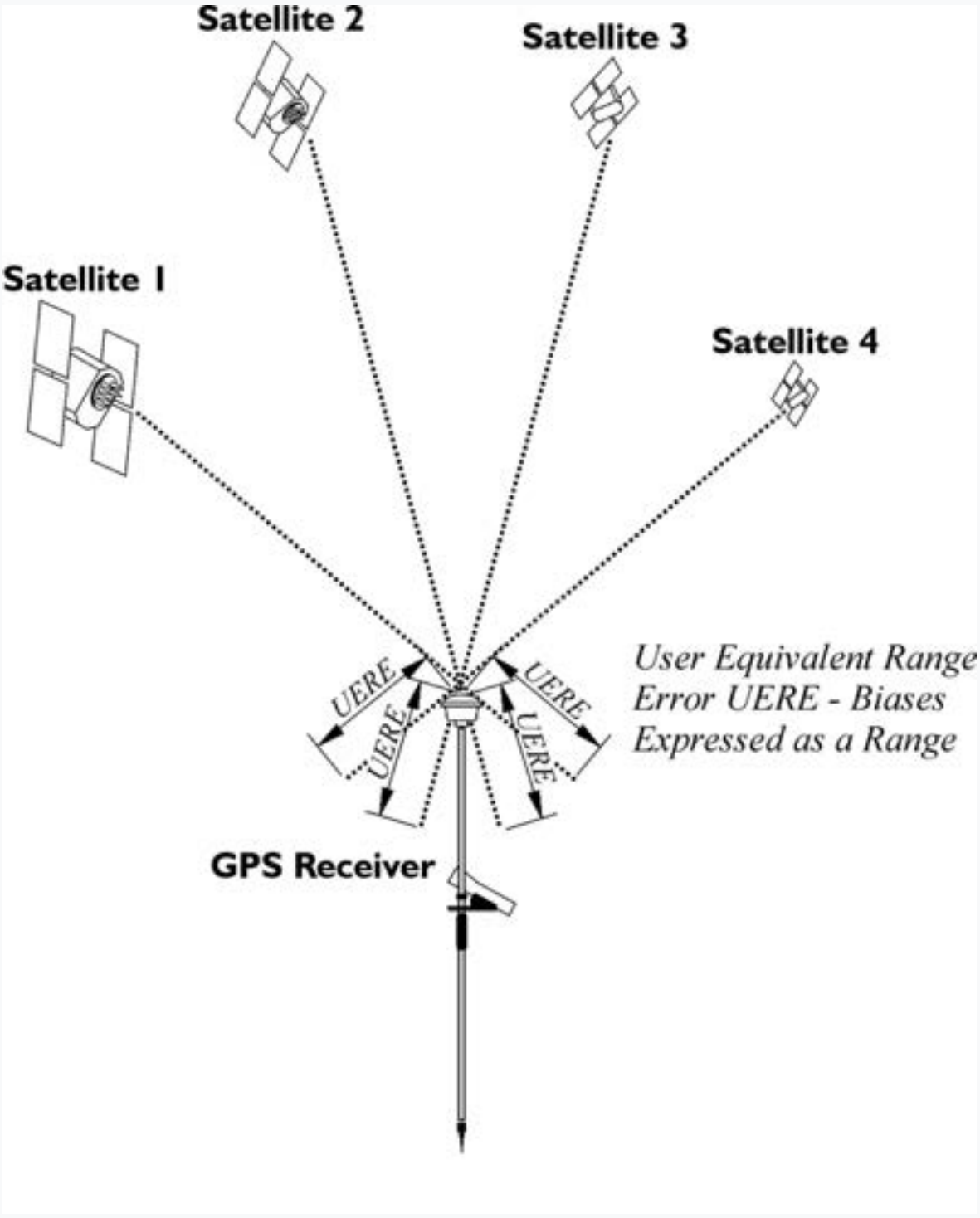
The management of the errors shown in the formulas, the biases, is indispensable for finding the true range from either a pseudorange or a carrier phase observation. In this lesson, we're going to try to understand the source of these errors so that they can be dealt with.

CONTENT

📚 User Equivalent Range Error



Understanding UERE



User Equivalent Range Error

The summary of the total error budget affecting a pseudorange from the user's point of view is called the *user equivalent range error*, *UERE*. This expression, often used in satellite surveying literature, is the square root of the sum of the squares of the individual biases. See figure above. The UERE is the

ranging error between a satellite and the user's receiver. It can be shown as a distance that includes both the signal in space ranging error (SISRE) and the user equipment error (UEE).

Baseline-Dependent vs Independent Errors

Some of the biases that make up the UERE such as those attributable to the atmosphere d_{ion}, d_{trop} and satellite orbits d_{ρ} increase and decrease with the length of the baselines between receivers. Differential correction techniques can often limit the effect of these errors. Others included in the UERE, such as those due to receiver noise, ε_{ρ} and ε_{φ}, multipath, $\varepsilon_{m\rho}$ and $\varepsilon_{m\varphi}$, do not increase and decrease with the length of the baselines between receivers.

CONTENT

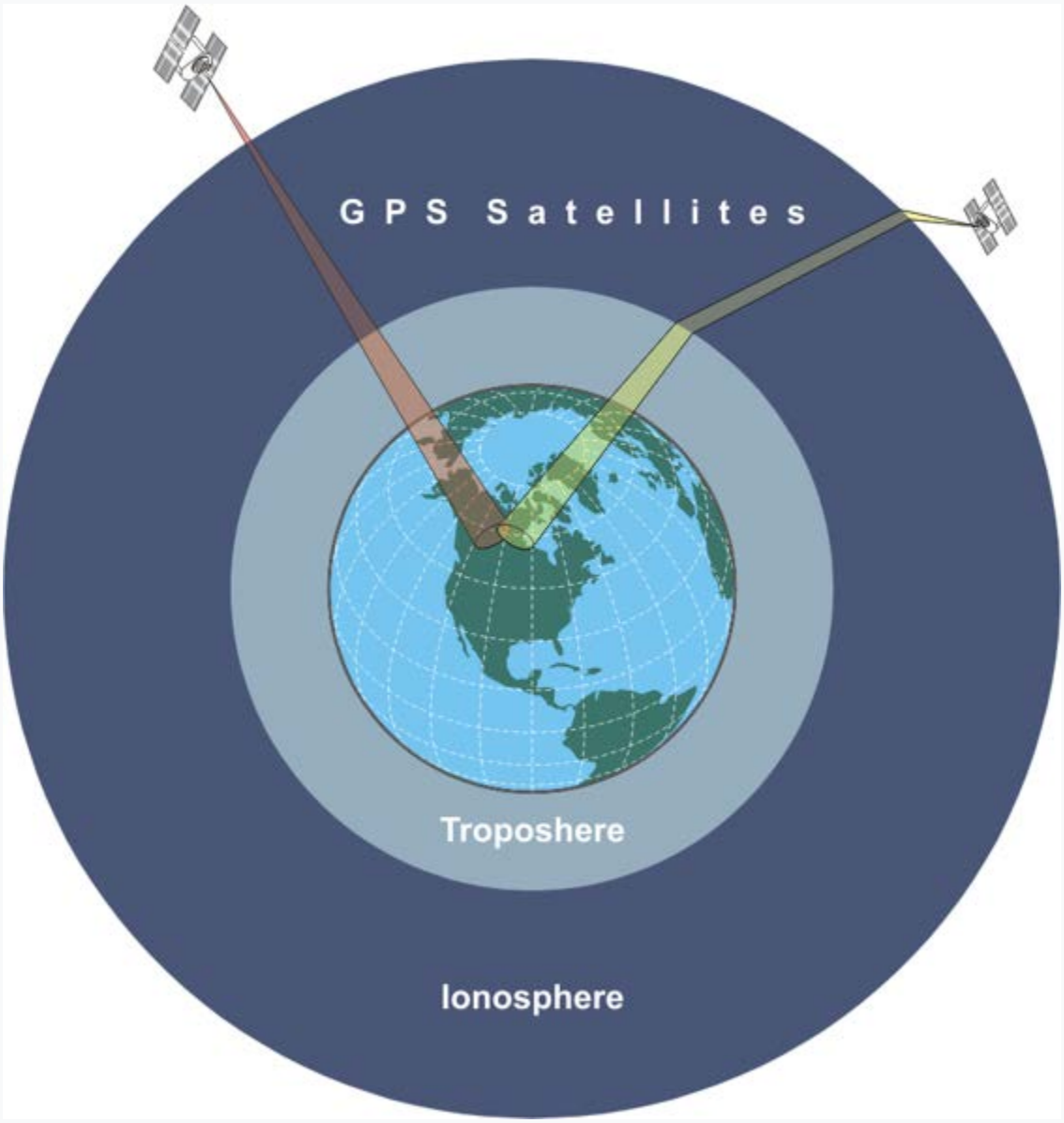
📚 The Ionospheric Effect



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The Ionospheric Effect, d_{ion}

The Earth's Ionosphere is variable



Imaging the Earth's Ionosphere

One of the largest errors in GPS positioning is attributable to the atmosphere. The long, relatively unhindered travel of the GPS signal through the virtual vacuum of space changes as it passes through the earth's atmosphere. Through both refraction and diffraction, the atmosphere alters the apparent speed and, to a lesser extent, the direction of the signal. This causes an apparent delay in the signal's transit from the satellite to the receiver.

Ionized Plasma

The ionosphere is ionized plasma comprised of negatively charged electrons which remain free for long periods before being captured by positive ions. It is the first part of the atmosphere that the signal encounters as it leaves the satellite. The magnitude of these delays is determined by the state of the ionosphere at the moment the signal passes through, so it's important to note that its density and stratification varies. The sun plays a key role in the creation and variation of these aspects. Also, the daytime ionosphere is rather different from the ionosphere at night.

Ionosphere and the Sun

Earth's Orbit

When gas molecules are ionized by the sun's ultraviolet radiation, free electrons are released. As their number and dispersion varies, so does the electron density in the ionosphere. This density is often described as *total electron content* or TEC, a measure of the number of free electrons in a column through the ionosphere with a cross-sectional area of 1 square meter: 10^{16} is one TEC unit. The higher the electron density, the larger the delay of the signal, but the delay is by no means constant.

The ionospheric delay changes slowly through a daily cycle. It is usually least between midnight and early morning, and most around local noon or a little after. During the daylight hours in the midlatitudes, the ionospheric delay may grow to be as much as five times greater than it was at night, but the rate of that growth is seldom more than 8 cm per minute. It is also nearly four times greater in November, when the earth is nearing its *perihelion*, its closest approach to the sun, than it is in July near the earth's *aphelion*, its farthest point from the sun. The effect of the ionosphere on the GPS signal usually reaches its peak in March, about the time of the vernal equinox.

Ionospheric Stratification

Atmospheric Model

For ease of reference, the ionosphere can be said to extend from 50 kilometers to 1,000 kilometers above the earth's surface. It can be divided into the mesosphere and thermosphere, et cetera that are themselves composed of layers: D, E, and F. Neither the boundaries between these regions, nor the upper layer of the ionosphere, can be defined strictly. Here are some general ideas on the subject. The lowest detectable layer, the D region, extends from about 50 km to 90 km. It has almost no effect on GPS signals and virtually disappears at night. The E region, also a daytime phenomenon, is between 90 km and 120 km. Its effect on the signal is slight, but it can cause the signal to scintillate. The layer that affects the propagation of electromagnetic signals the most is the F region. It extends from about 120km to 1000km The F region contains the most concentrated ionization in the atmosphere. In the daytime, the F layer can be further divided into F1 and F2. F2 is the most variable. F1, the lower of the two, is most apparent in the summer. These two layers combine at night. Above, the F layer is fully ionized. It is sometimes known as the photosphere or the H region.

The ionosphere is not homogeneous and unchanging. It is in constant flux. Therefore, it's impossible to have a correction that's static. Also its behavior in one region of the earth is liable to be unlike its behavior in another. For example, ionospheric disturbances can be particularly harsh in the polar regions. But the highest TEC values and the widest variations in the horizontal gradients occur in the band of about 60° of *geomagnetic latitude*. That band lies 30° north and 30° south of the earth's magnetic equator.

Satellite Elevation and Group vs Phase Delay

The severity of the ionosphere's effect on a GPS signal depends on the amount of time that signal spends traveling through it. A signal originating from a satellite near the observer's horizon must pass through a larger amount of the ionosphere to reach the receiver than does a signal from a satellite near the observer's zenith. In other words, the longer the signal is in the ionosphere, the greater the ionosphere's effect on it.

Group and Phase Delay

The ionosphere is *dispersive*, which means that the apparent time delay contributed by the ionosphere depends on the frequency of the signal. During the signal's trip through the ionosphere, this dispersive property causes the codes, the modulations on the carrier wave, to be affected

differently than the carrier wave itself. The P code, the C/A code, the Navigation message, and all the other codes appear to be delayed, or slowed, affected by what is known as the *group delay*. But the carrier wave, itself, appears to speed up in the ionosphere. It is affected by what is known as the *phase delay*. It may seem odd to call an increase in speed a delay. It is sometimes called *phase advancement*. In any case, it is governed by the same properties of electron content as the group delay, phase delay just increases negatively. Please note that the algebraic sign of d_{ion} is negative in the carrier phase equation and positive in the pseudorange equation. In other words, a range from a satellite to a receiver determined by a code observation will be a bit too long and a range determined by a carrier observation will be a bit too short.

Different Frequencies Are Affected Differently

Another consequence of the dispersive nature of the ionosphere is that the attenuation for a higher frequency carrier wave is less than it is for a lower frequency wave. That means that L1, 1575.42 MHz, is not affected as much as L2, 1227.60 MHz, and L2 is not affected as much as L5, 1176.45MHz. This fact provides one of the greatest advantages of a dual-frequency receiver over the single-frequency receivers. The separations between the L1 and L2 frequencies (347.82 MHz), the L1 and L5 frequencies (398.97 MHz) and even the L2 and L5 frequencies (51.15 MHz) are large enough to facilitate estimation of the ionospheric group delay. Therefore, by tracking all the carriers, a multiple-frequency receiver can model and remove, not all, a portion of the ionospheric bias. There are now several possible combinations, L1/L2, L1/L5 and L2/L5. It is even possible to have a triple frequency combination to help ameliorate this bias.

The frequency dependence of the ionospheric effect is described by the following expression (Klobuchar 1983 in Brunner and Welsch, 1993):

$$v = (40.3 / cf^2) \bullet TEC$$

Where:

v = the ionospheric delay
c = the speed of light in meters per second
f = the frequency of the signal in Hz
TEC = the quantity of free electrons per square meter

As the formula illustrates, the time delay is inversely proportional to the square of the frequency; in other words, the higher the frequency, the less the delay. For example, the ionospheric delay at

1227.60 (L2) is 65% larger than at 1575.42 MHz (L1), and at 1176.45 MHz (L5) it is 80% larger than 1575.42 (L1).

Broadcast Correction

A predicted total UERE is provided in each satellite's Navigation message as the user range accuracy (URA), but it is minus ionospheric error. To help remove some of the effect of the ionospheric delay on the range derived from a single frequency receiver, there is an ionospheric correction available in another part of the Navigation message, subframe 4. However, this broadcast correction should not be expected to remove more than about three-quarters of the error, which is most pronounced on long baselines. Where the baselines between the receivers are short, the effect of the ionosphere can be small, but as the baseline grows, so does the significance of the ionospheric bias.

CONTENT

📚 The Satellite Clock

Satellite Clock Bias

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The Satellite Clock Bias, dt



Cesium Frequency Standard (Clock)

One of the largest errors can be attributed to the satellite clock bias. It can be quite large, especially if the broadcast clock correction is not used by the receiver to bring the time signal acquired from a satellite's onboard clock in line with GPS time. As time is a critical component in the functioning of GPS, it is important to look closely at the principles behind this bias.

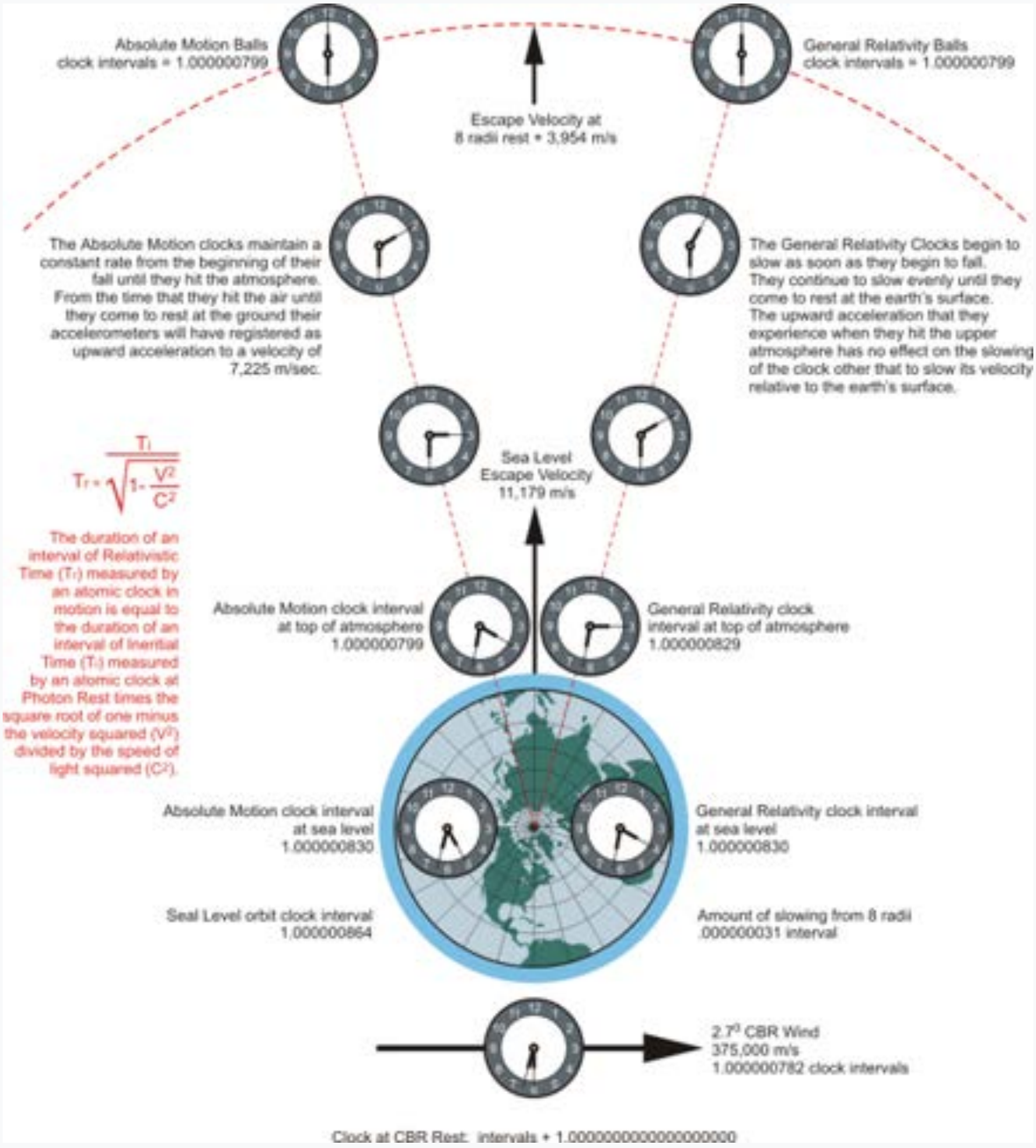
The onboard satellite clocks are independent of one another. While GPS time itself is designed to be kept within one microsecond, 1 µsec or one-millionth of a second, of UTC, excepting leap seconds, the satellite clocks can be allowed to drift up to a millisecond, 1 msec or one-thousandth of a second, from GPS time. They are allowed to drift within this tolerance because the Control Segment doesn't want to constantly tweak the clocks, as this would cause the clocks to deteriorate more rapidly. Although the GPS satellites have been robust, I think it's fair to say that the clocks are one of their more fragile components. Instead, we have the broadcast clock correction in the Navigation

Message. This is the correction that the Control Segment provides to the receiver to use to bring the satellite clock in line with GPS time.

Relativistic Effects on the Satellite Clock

Albert Einstein's special and general theories of relativity apply to the clocks involved here. At 3.874 kilometers per second, the clocks in the GPS satellites are traveling at great speed, and that makes the clocks on the satellites appear to run slower than the clocks on earth by about 7 microseconds a day. However, this apparent slowing of the clocks in orbit is counteracted by the weaker gravity around them. The weakness of gravity makes the clocks in the satellites appear to run faster than the clocks on earth by about 45 microseconds a day. Therefore, on balance, the clocks in the GPS satellites in space appear to run faster by about 38 microseconds a day than the clocks in GPS receivers on earth. So, to ensure the clocks in the satellites will actually produce the correct fundamental frequency of 10.23 MHz in space, their frequencies are set to 10.22999999543 MHz before they are launched into space.

There is yet another consideration, the eccentricity of the orbit of GPS satellites. With an eccentricity of 0.02, this effect on the clocks can be as much as 45.8 nanoseconds. Fortunately, the offset is eliminated by a calculation in the GPS receiver itself; thereby avoiding what could be ranging errors of about 14 meters. The receiver is moving, too, of course, so an account must be made for the motion of the receiver due to the rotation of the earth during the time it takes the satellite's signal to reach it. This is known as the Sagnac effect, and it is 133 nanoseconds at its maximum. Luckily, these relativistic effects can be accurately computed and removed from the system.



Relativistic Effects on the Satellite Clock

CONTENT

📚 The Receiver Clock Bias, dT

Receiver Clock Bias

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The Receiver Clock Bias, *dT*



An Oven-Controlled Quartz Crystal Oscillator (OCXO) on a Board

The third-largest error, which can be caused by the receiver clock, is its oscillator. Both a receiver's measurement of phase differences and its generation of replica codes depend on the reliability of this internal frequency standard.

Typical Receiver Clocks

GPS receivers are usually equipped with quartz crystal clocks, which are relatively inexpensive and compact. They have low power requirements and long life spans. For these types of clocks, the frequency is generated by the piezoelectric effect in an oven-controlled quartz crystal disk, a device sometimes symbolized by *OCXO*. Their reliability ranges from a minimum of about 1 part in 10^8 to a maximum of about 1 part in 10^{10}, a drift of about 0.1 nanoseconds in 1 second. Even at that, quartz clocks are not as stable as the atomic standards in the GPS satellites and are more sensitive to temperature changes, shock, and vibration. In other words, they're relatively stable, but they certainly aren't as sophisticated as those that are found in the GPS satellite. They're not atomic standards, nor need they be.

Obviously, the receivers that we use would be tremendously expensive if the oscillators —if the clocks that were in them— had to be of that great stability. It really isn't necessary for a GPS receiver clock to be wonderful, because we are solving for time. There are four unknowns (x, y, z, and time) and, therefore, four observations to make the solution. Still, we can't get along without an oscillator in the receiver. It is necessary for producing the replica code, for example. The replica code needs to match the incoming signals from the satellites. So, obviously, a receiver clock is necessary, but it doesn't need to be anything like an atomic standard.

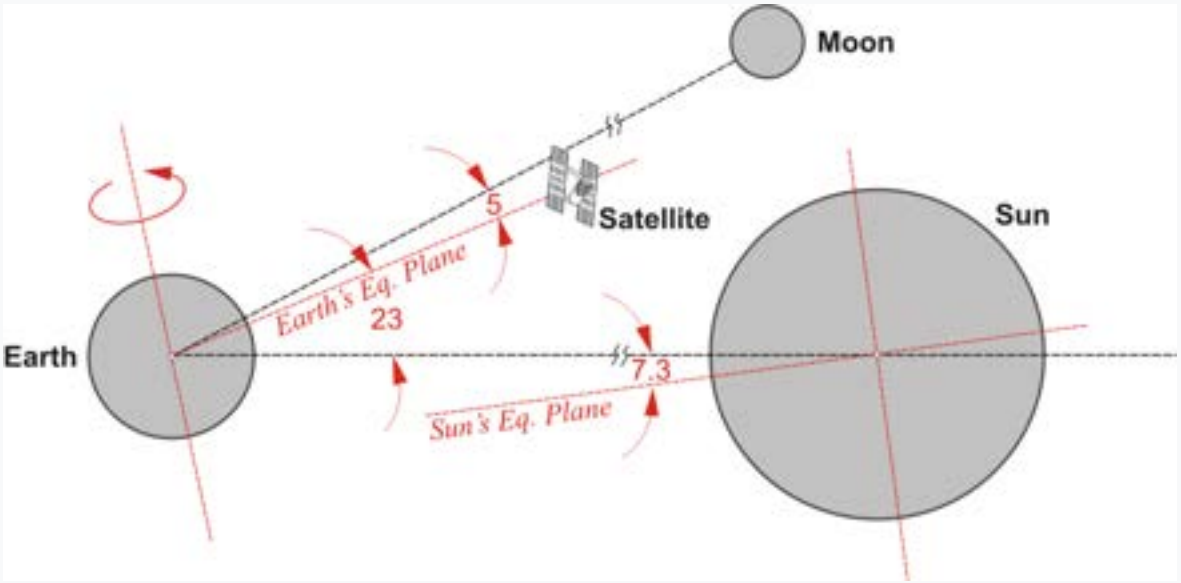
CONTENT

📚 The Orbital Bias

Orbital Bias

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Forces Disturbing the Satellite's Orbit



The Sources of Some Forces Disturbing the Satellite's Orbit

Orbital bias has the potential to be the fourth-largest error. It is addressed in the broadcast ephemeris.

The orbital motion of GPS satellites is not only a result of the earth's gravitational attraction, there are several other forces that act on the satellite. The primary disturbing forces are the non-spherical nature of the earth's gravity (the earth is not equally dense in all its portions), tidal forces, the attractions of the sun and the moon, and solar radiation pressure as the satellite moves from darkness to light. All in all, it is a rather bumpy road for the satellite. The best model of these forces is the actual motion of the satellites themselves, and the government facilities distributed around the world, known collectively as the Control Segment or the *Operational Control System (OCS)*, track them for that reason, among others. The modeling of the orbit by the Control Segment is good, but

they can upload the ephemerides only so frequently. There is always a certain amount of age in the ephemerides and that means that the position of the satellite expressed in its ephemeris at the moment of observation cannot be perfect. So orbital bias could be thought of as the error in the broadcast ephemeris.

CONTENT

📚 The Control Segment

The Control Segment

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The Control Segment

The GPS Control Segment

There are government tracking and uploading facilities distributed around the world. These facilities not only monitor the L-band signals from the GPS satellites and update their Navigation Messages, but also track the satellite's health, their maneuvers, and many other things, even battery recharging. Taken together, these facilities are known as *the Control Segment*.

Master Control Station

The Master Control Station (MCS), once located at Vandenberg Air Force Base in California, now resides at the Consolidated Space Operations Center (CSOC) at Schriever (formerly Falcon) Air Force Base near Colorado Springs, Colorado, and has been manned by the 2nd Space Operations Squadron *(2SOPS)*, since 1992. There is an alternate MCS in at Vandenberg Tracking Station in California.

The 2SOPS squadron controls the satellites' orbits. For example, they maneuver the satellites from the highly eccentric orbits into which they are originally launched to the desired mission orbit and spacecraft orientation. They monitor the state of each satellite's onboard battery, solar, and propellant systems. They resolve satellite anomalies, activate spare satellites, and control Selective Availability (SA) and Anti-Spoofing (A/S). They dump the excess momentum from the *wheels*, the series of gyroscopic devices that stabilize each satellite. With the continuous constellation tracking data available and aided by Kalman filter estimation to manage the noise in the data, they calculate and

update the parameters in the Navigation message (ephemeris, almanac, and clock corrections) to keep the information within limits, because the older it gets, the more its veracity deteriorates. This process is made possible by a persistent two-way communication with the constellation that includes both monitoring and uploading, accomplished through a network of ground antennas and monitoring stations.

Monitoring Stations and the L-AII Initiative

The data that feeds the MCS comes from monitoring stations. These stations track the entire GPS constellation. In the past, there were limitations. There were only six tracking stations. It was possible for a satellite to go unmonitored for up to two hours each day. It was clear that the calculation of the ephemerides and the precise orbits of the constellation could be improved with more monitoring stations in a wider geographical distribution. It was also clear that if one of the six stations went down, the effectiveness of the Control Segment could be considerably hampered. These ideas, and others, led to a program of improvements known as *the Legacy Accuracy Improvement Initiative*, *L-AII*. During this initiative from August 18 to September 7 of 2005, six *National Geospatial Intelligence Agency, NGA*, stations were added to the Control Segment. This augmented the information forwarded to the MCS with data from Washington, D.C., England, Argentina, Ecuador, Bahrain, and Australia. With this 12-station network in place, every satellite in the GPS constellation was monitored almost continuously from at least three stations when it reached at least 5° above the horizon.

Today, there are 6 Air Force and the 11 National Geospatial-Intelligence Agency (NGA) monitoring stations. The monitoring stations track all the satellites; in fact, every GPS satellite is tracked by at least 3 of these stations all the time. The monitoring stations collect range measurements, atmospheric information, satellite's orbital information, clock errors, velocity, right ascension, and declination and send them to the MCS. They also provide pseudorange and carrier phase data to the MCS. The MCS needs this constant flow of information. It provides the basis for the computation of the almanacs, clock corrections, ephemerides, and other components that make up the Navigation message. The new stations also improve the geographical diversity of the Control Segment, and that helps with the MCS isolation of errors, for example, making the distinction between the effects of the clock error from ephemeris errors. In other words, the diagnosis and solution of problems in the system are more reliable now because the MCS has redundant observations of satellite anomalies with which to work. Testing has shown that the augmented Control Segment and subsequent improved modeling has improved the accuracy of clock corrections and ephemerides in the Navigation Message substantially, and may contribute to an increase in the accuracy of real-time GPS of 15% or more.

Ground Antennas for Uploading

However, once the message is calculated, it needs to be sent back up to the satellites. Some of the stations have ground antennas for uploading. Four monitoring stations are collocated with such antennas. The stations at Ascension Island, Cape Canaveral, Diego Garcia, and Kwajalein upload navigation and program information to the satellites via S-band transmissions. The station at Cape Canaveral also has the capability to check satellites before launch.

Control Segment Modernization

The modernization of the Control Segment has been underway for some time, and it continues. In 2007, the Launch/Early Orbit, Anomaly Resolution, and Disposal Operations mission (LADO) PC-based ground system replaced the mainframe based Command-and-Control System (CCS). Since then, LADO has been upgraded several times. It uses Air Force Satellite Control Network (AFSCN) remote tracking stations only, not the dedicated GPS ground antennas to support the satellites from spacecraft separation through checkout, anomaly resolution, and all the way to end of life disposal. It also helps in the performance of satellite movements and the presentation of telemetry simulations to GPS payloads and subsystems. Air Force Space Command (AFSPC) accepted the LADO capability to handle the most modern GPS satellites at the time, the Block IIF. Another modernization program is known as the Next Generation Operational Control System or OCX. OCX will facilitate the full control of the new GPS signals like L5, as well as L2C, L1C, M code and the coming GPS III program.

CONTENT

📚 The Tropospheric Effect, dtrop

Tropospheric Effect

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GPS Receiver and two satellites: one at Zenith, one showing tropospheric refraction decreases with height

🌍 Troposphere

The troposphere is that part of the atmosphere closest to the earth. It extends from the surface to about 9 km over the poles and about 16 km over the equator, but in this work, it will be combined with the tropopause and the stratosphere, as it is in much of GPS literature. Therefore, the following discussion of the tropospheric effect will include the layers of the earth's atmosphere up to about 50 km above the surface. While the troposphere is no more cooperative than the ionosphere in terms of being consistent, the troposphere and the ionosphere are by no means alike in their effect on the satellite's signal. The troposphere is refractive, its refraction of a GPS satellite's signal is not related to its frequency. The refraction is tantamount to a delay in the arrival of a GPS satellite's signal. It can also be conceptualized as a distance added to the range the receiver measures between itself and the satellite. The troposphere is part of the electrically neutral layer of the earth's atmosphere, meaning it is not ionized. The troposphere is also nondispersive for frequencies below 30 GHz or so. Therefore, L1, L2, and L5 are equally refracted. This means that the range between a receiver and a satellite will be shown to be a bit longer than it actually is. However, as it is in the ionosphere, density affects the severity of the delay of the GPS signal as it travels through the troposphere. For example, when a satellite is close to the horizon, the delay of the signal caused by the troposphere is maximized. The tropospheric delay of the signal from a satellite at zenith, directly above the receiver, is minimized.

📐 Satellite elevation and tropospheric effect

The situation is analogous to atmospheric refraction in astronomic observations; the effect increases as the energy passes through more of the atmosphere. The difference in GPS is that it is the delay, not the angular deviation caused by the changing density of the atmosphere, that is of primary interest. The GPS signal that travels the shortest path through the troposphere will be the least delayed by it. In other words, there is less tropospheric delay when the satellite is at the zenith of the receiver.

Modeling. Modeling the troposphere is one technique used to reduce the bias in GPS data processing, and it can be up to 95% effective. However, the residual 5% can be quite difficult to remove. Several a priori models have been developed, for example, the Saastamoinen model and the Hopfield models, which perform well when the satellites are at reasonably high elevation angles. However, it is advisable to limit GPS observations to those signals above 15% or so to ameliorate the effects of atmospheric delay.

💧 The dry and wet components of refraction

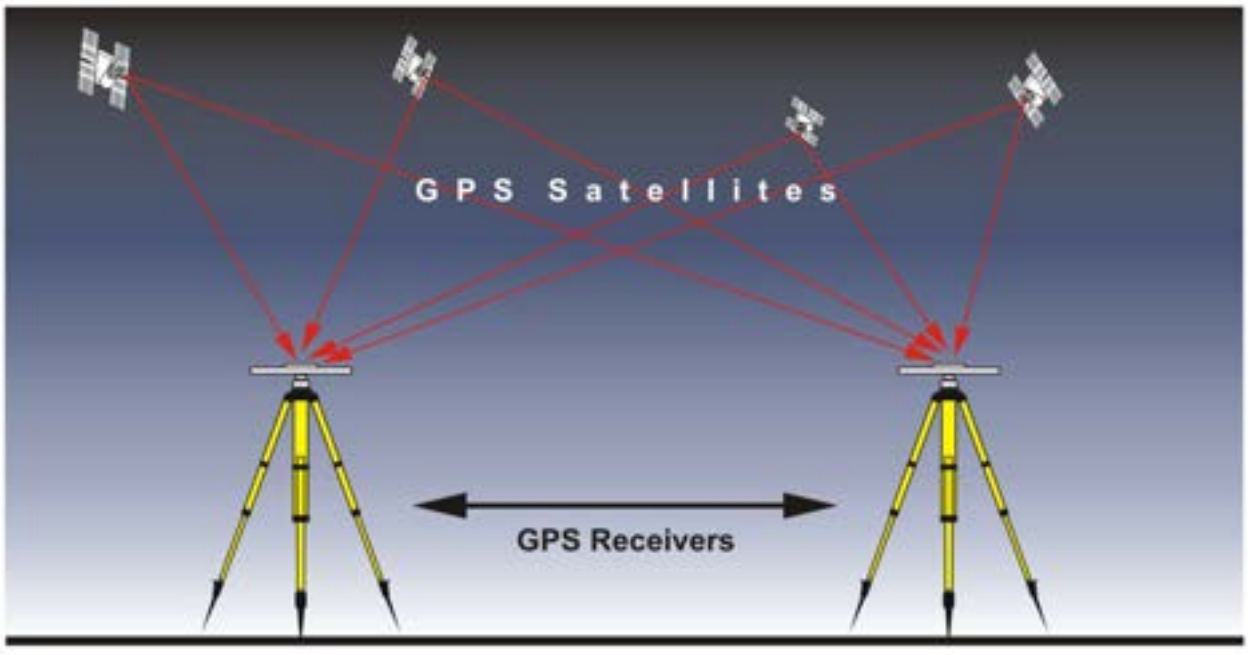
Refraction in the troposphere has a dry component and a wet component. The dry component which contributes most of the delay, perhaps 80% to 90%, is closely correlated to the atmospheric pressure. The dry component can be more easily estimated than the wet component. It is fortunate that the dry component contributes the larger portion of range error in the troposphere, because the size of the delay attributable to the wet component depends on the highly variable water vapor distribution in the atmosphere. Even though the wet component of the troposphere is nearer to the Earth's surface, measurements of temperature and humidity are not strong indicators of conditions on the path between the receiver and the satellite. While instruments that can provide some idea of the conditions along the line between the satellite and the receiver are somewhat more helpful in modeling the tropospheric effect, the high cost of sending water vapor radiometers and radiosondes aloft generally restricts their use to only the most high-precision GPS work. In most cases, this aspect must remain in the purview of mathematical modeling; such calculations include a hydrostatic model with corrections and a horizontal gradient component. It is important to recognize that the index of tropospheric refraction decreases as height increases.

CONTENT

📚 Atmosphere and Baseline Length

A Baseline

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Two GPS receivers each receiving signals from each of 4 GPS satellites

A Baseline

📏 Receiver spacing and the atmospheric biases

Receiver spacing and the atmospheric biases

There are other practical consequences of the atmospheric biases. As mentioned earlier, the character of the atmosphere is never homogeneous; therefore, the importance of atmospheric modeling increases as the distance between GPS receivers grows. Consider a signal traveling from one satellite to two receivers that are close together. That signal would be subjected to very similar

atmospheric effects, and, therefore, atmospheric bias modeling would be less important to the accuracy of the measurement of the **relative** distance between them. But a signal traveling from the same satellite to two receivers that are far apart may pass through levels of atmosphere quite different from one another. In that case, atmospheric bias modeling would be more important. In other words, the importance of atmospheric correction increases as the *differences* in the atmosphere through which the GPS satellite signal must pass to reach the receivers increase. Such differences can generally be related to baseline length.

CONTENT

📚 Multipath

📶 Multipath, $\varepsilon_{m\rho}$ and $\varepsilon_{m\varphi}$

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📡 Multipath Signal Effects

Reflected Multipath Signal (ex. reflecting off a window) and Lost Signal (Diffusion, hitting trees)

Multipath, $\varepsilon_{m\rho}$ and $\varepsilon_{m\varphi}$

Multipath

Multipath is an uncorrelated error. It is a range delay that has a different symbol in the two equations we talked about at the top. It is symbolized by $\varepsilon_{m\rho}$ in the pseudorange equation and $\varepsilon_{m\varphi}$ in the carrier phase equation. As the name implies, it is the reception of the GPS signal via multiple paths rather than from a direct line of sight. It occurs when part of the signal from the satellite reaches the receiver after one or more reflections or scattering from the ground, a building, or another object. These reflected signals can interfere with the signal that reaches the receiver directly from the satellite and cause the correlation peak to become skewed. It is simple in principle. Since we are measuring distances to the satellites, and that is the foundation of the positions in GPS, a signal that bounces from an obstruction before it reaches the antenna of the GPS receiver causes a problem. If there's a bend in the signal, that upsets the distance, the range, that the receiver measures to the satellite. As illustrated here, the obstruction can be buildings, or really any reflective surface. The signal can reflect more than once. It can reflect from the ground. Multipath is one of the main reasons ground planes are used in geodetic work.

Since carrier phase multipath is based on a fraction of the carrier wavelength, and code multipath is relative to the chipping rate, the effect of multipath on pseudorange solutions is orders of magnitude larger than it is in carrier phase solutions. However, multipath in carrier phase is harder to mitigate than multipath in pseudoranges.

The high frequency of the GPS codes tends to limit the field over which multipath can contaminate pseudorange observations. Once a receiver has achieved lock, that is, its replica code is correlated with the incoming signal from the satellite, signals outside the expected chip length can be rejected. Generally speaking, multipath delays of less than one chip, those that are the result of a single reflection, are the most troublesome.

Fortunately, there are factors that distinguish reflected multipath signals from direct, line-of-sight, signals. For example, reflected signals at the frequencies used for L1, L2, and L5 tend to be weaker and more diffuse than the directly received signals. Another difference involves the circular polarization of the GPS signal. The polarization is actually reversed when the signal is reflected. Reflected, multipath signals *become Left Hand Circular Polarized, LHCP*, whereas the signals received directly from the GPS satellites are *Right Hand Circular Polarized, RHCP*. RHCP means that it rotates clockwise when observed in the direction of propagation. However, in the face of multiple reflections, the attribute is not determinative. In any case, these characteristics allow some multipath signals to be identified and rejected at the receiver's antenna.

📡 GPS Antennas



Choke ring, 5 satellites and a GPS Receiver, and ground plane

GPS Antennas

Antenna Design and Multipath

GPS antenna design can play a role in minimizing the effect of multipath. Ground planes, usually a metal sheet, are used with many antennas to reduce multipath interference by eliminating signals from low elevation angles. Generally, larger ground planes, multiple wavelengths in size, have a more stabilizing influence than smaller ground planes. However, such ground planes do not provide much protection from the propagation of waves along the ground plane itself. When a GPS signal's wave

front arrives at the edge of an antenna's ground plane from below, it can induce a surface wave on the top of the plane that travels horizontally. Another way to mitigate this problem is the use of a choke ring antenna. Choke ring antennas, based on a design first introduced by the Jet Propulsion Laboratory (JPL), can reduce antenna gain at low elevations. This design contains a series of concentric circular troughs that are a bit more than a quarter of a wavelength deep. A choke ring antenna can prevent the formation of these surface waves. But neither ground planes nor choke rings remove the effect of reflected signals from above the antenna very effectively. There are signal processing techniques that can reduce multipath. A widely used strategy is the 15° cutoff or *mask angle*. This technique calls for tracking satellites only after they are more than 15° above the receiver's horizon. Careful attention in placing the antenna away from reflective surfaces, such as nearby buildings, water, or vehicles, is another way to minimize the occurrence of multipath. The ground plane, seen on the right in the illustration, helps eliminate the bounce from the ground underneath the antenna. However, waves can almost crawl across the top of that ground plane and so there is the choke ring antenna, shown on the left in the illustration, can prevent these surface waves from propagating across the ground plane. I've mentioned the mask angle more than once. It is effective in ameliorating multipath. However, there really is no way to eliminate multipath entirely. It can be minimized by good location, away from reflective surfaces, and by uses of these sorts of antennas and also from software and the receiver itself. But, as far as I know, there is no way to absolutely eliminate the effect of multipath in all cases.

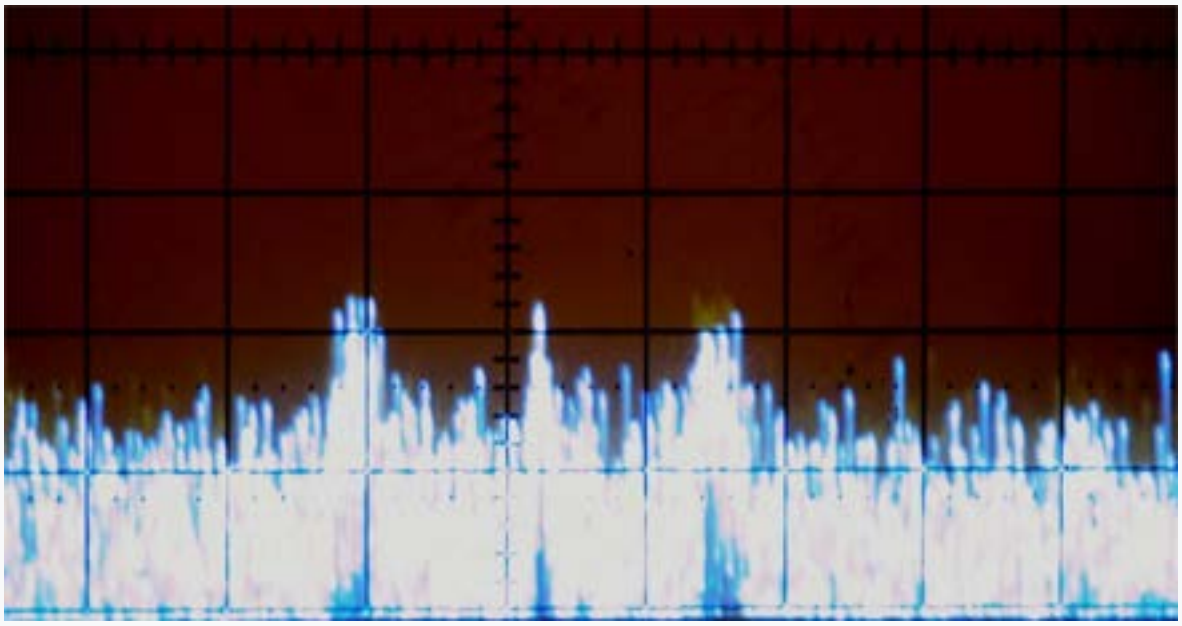
📚 Receiver Noise

Receiver Noise, ε_ρ and ε_φ

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📈 Receiver Noise, ε_ρ and ε_φ

Receiver Noise, ε_ρ and ε_φ



Graph of receiver noise

Noise Floor

Receiver noise is directly related to thermal noise, dynamic stress, and so on in the GPS receiver itself. Receiver noise is also an uncorrelated error source, meaning the effects of both multipath and the receiver noise are not related to the length of the baseline between GPS receivers. They are uncorrelated in that regard. The effects of receiver noise on carrier phase measurements symbolized

by ε_{φ}, like multipath, are small when compared to their effects on pseudorange measurements, ε_{ρ}. Generally speaking, the receiver noise error is about 1% of the wavelength of the signal involved. In code solutions, the size of the error is related to chip width. For example, the receiver noise error in a C/A code solution might be about an order of magnitude more than it might be in a P(Y) code solution, and it would be even less in carrier phase solutions. Receiver noise is inevitable, and it must be considered, but it is a relatively small contributor to the GPS error budget, in most cases.

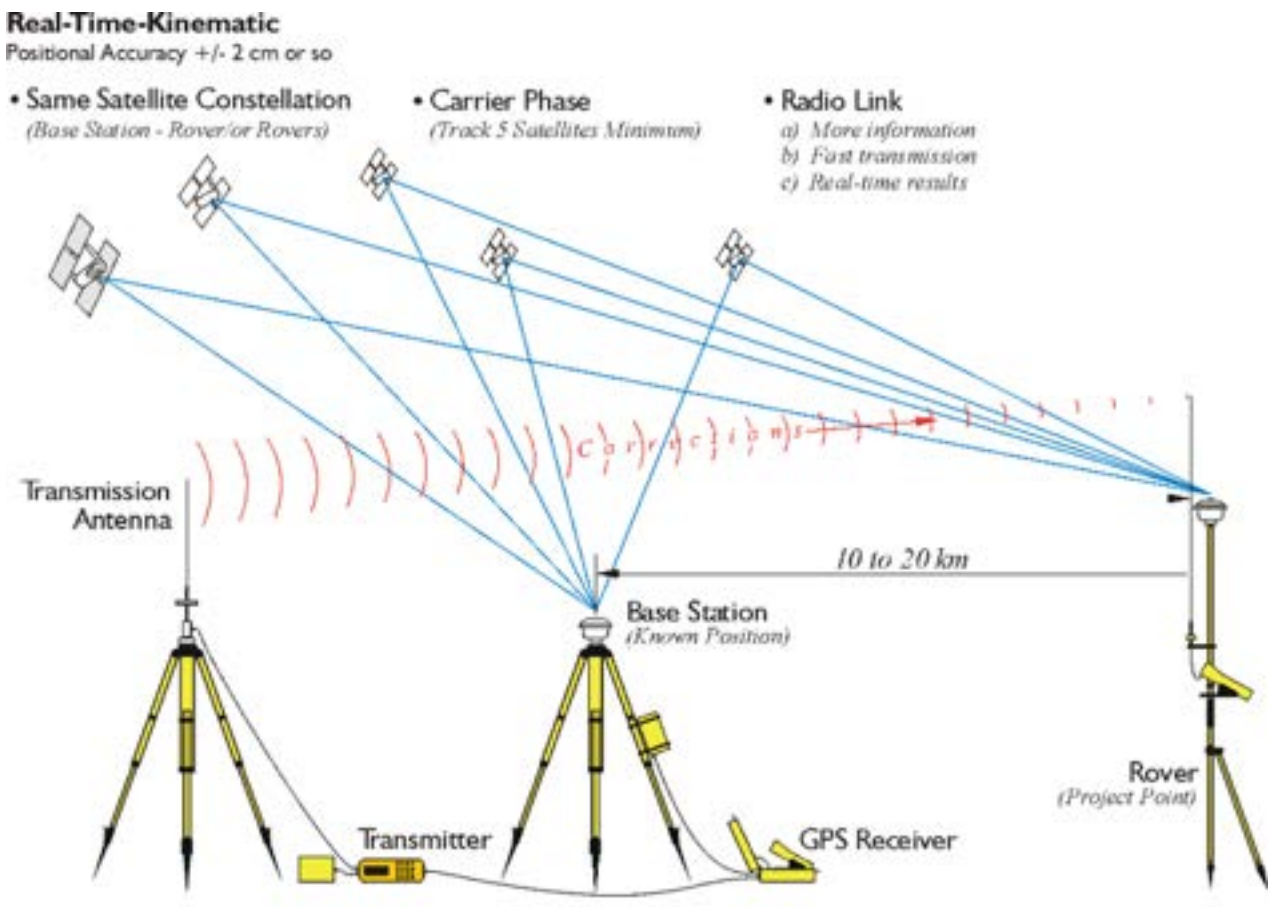
CONTENT

Classification of GPS Positioning

Classification of GPS Positioning

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Static and Kinematic GPS



Real-time kinematic

GPS work is sometimes divided into three categories; positioning, navigation, and timing (PNT). Most often, GPS surveying is concerned with the first of these, positioning. In general, there are two

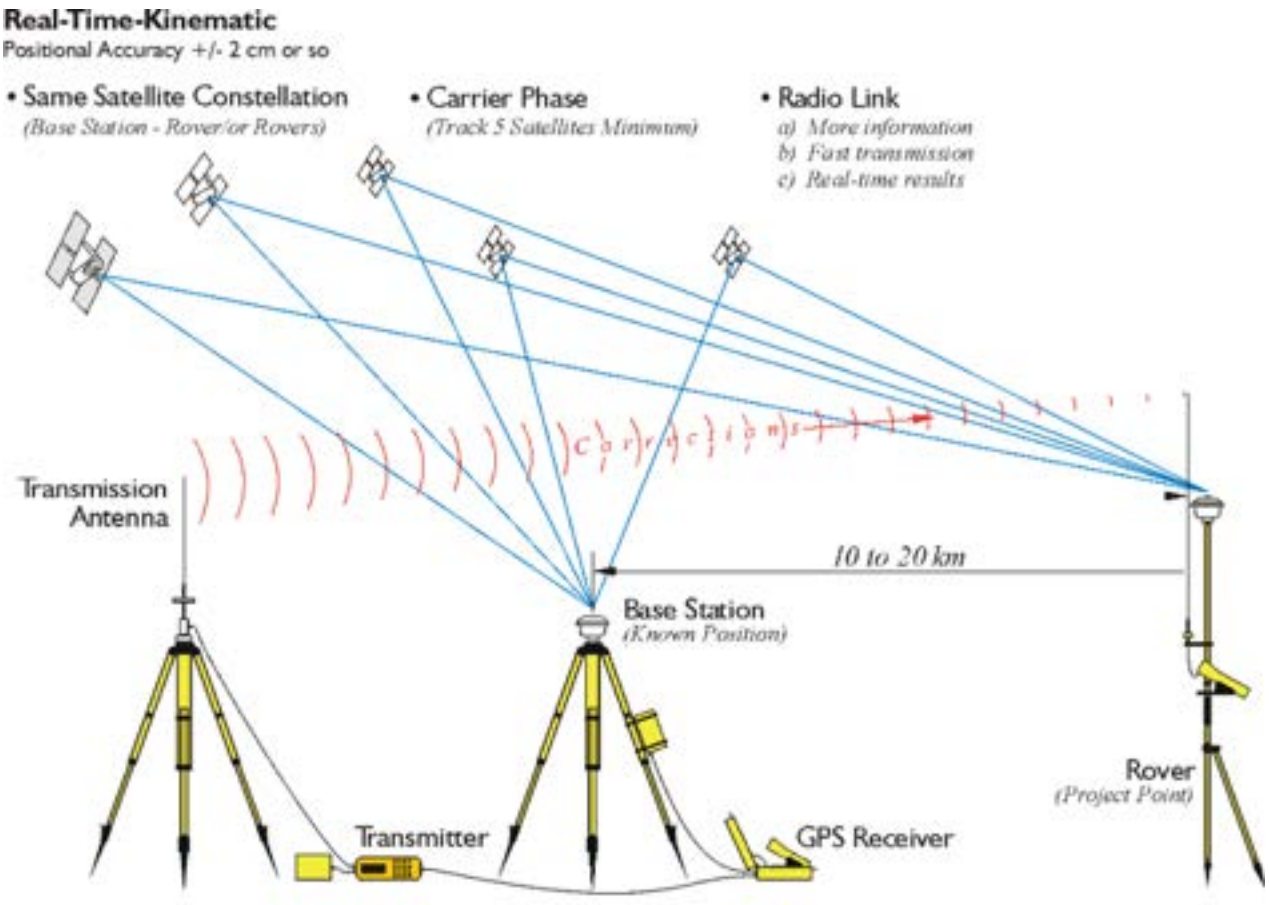
techniques used in surveying. They are kinematic and static. In static GPS surveying sessions, the receivers are motionless on the Earth during the observation. Because static work most often provides higher accuracy and more redundancy than kinematic work, it is usually done to establish control. The results of static GPS surveying are processed after the session is completed. In other words, the data is typically *post-processed*. The majority of GPS surveying control and geodetic work still relies on static applications.

In kinematic GPS surveying, the receivers are either in periodic or continuous motion. Kinematic GPS is done when real-time, or near real-time, results are needed. When the singular objective of kinematic work is positioning, the receivers move periodically using the start and stop methodology originated by Dr. Benjamin Remondi in the 1980s. When the receivers are in continuous motion, the objective may be acquisition of the location, attitude, and velocity of a moving platform (i.e., navigation), or positioning. The distinction between navigation and positioning is lessening.

Kinematic applications imply movement, one or more GPS receivers actually in motion during their observations. A moving GPS receiver on land, sea, or air is characteristic of kinematic GPS. Other characteristics of the application include results in real time and little redundancy. Hydrography, aerial mapping, gravimetric, and more and more land surveying projects are done using kinematic GPS.

Relative positioning is different than autonomous positioning. An independent receiver, an autonomous receiver, must rely on the information in the Navigation Message. In that sort of configuration, it has no way to improve on the corrections. In relative (aka differential) positioning, there are two receivers; one on a known point whose coordinates are well-established, and a second receiver on an unknown point. With this arrangement, the new unknown position can be established relative to the known point. Relative (differential) positioning creates solutions that are positionally superior to autonomous positioning, in large part because it is possible to improve on the corrections available in the Navigation Message.

📡 Differential GPS



Differential GPS

The term *differential* GPS, or *DGPS*, has come into common usage as well. Use of this acronym usually indicates a method of relative positioning where **usually** coded pseudorange measurements are used rather than carrier phase, though it is sometimes used in context of carrier phase solutions.

CONTENT

📚 The Navigation Solution



📍 Navigation Solution

Absolute (aka Autonomous, aka Single Point) Solution

Another type of point positioning is known as absolute positioning, single-point positioning, or the Navigation solution. It is characterized by a single receiver measuring its range to a minimum of four satellites simultaneously. An autonomous GPS positioning is not post-processed, and there is really no communication with a station that's on a known point. In other words, autonomous positioning is not differential (relative); the receiver is operating independently. As always, four satellites are required, as indicated here. The receiver is on an unknown position and must solve for its own receiver clock error. It needs to solve for x, y, z, and time. The Navigation solution is, in one sense, the fulfillment of the original idea behind GPS. It relies on a coded pseudorange measurement and can be used for virtually instantaneous positioning. In this method, the positions of the satellites are available from the data in their broadcast ephemerides. The satellite clock offset, and the ionospheric correction are also available from the Navigation messages of all four satellites. As you know, these corrections from that message are not ideal. Even if all the data in the navigation message contained no errors, and they surely do, four unknowns remain: the position of the receiver in three Cartesian coordinates, u_x , u_y , and u_z , and the receiver's clock error dT_u . Three pseudoranges provide enough data to solve for u_x , u_y , and u_z . And the fourth pseudorange provides the information for the solution of the receiver's clock offset. The ability to measure dT, the receiver's clock error, is one reason the moderate stability of quartz crystal clock technology is entirely adequate as a receiver oscillator. A unique solution is found here because the number of unknowns is not greater than the number of observations. The receiver tracks a minimum of 4 satellites simultaneously; therefore, these four equations can be solved simultaneously for every *epoch* of the observation. An epoch in GPS is a very short period of observation time (i.e. 1 second or perhaps less), and is generally just a small part of a longer measurement. However, theoretically, there is enough information in any single epoch to solve these equations. This is the reason the trajectory of a receiver in a moving vehicle can be determined by this method. With 4 satellites available, resolution of a receiver's position and velocity are both available through the simultaneous solution of these four equations.

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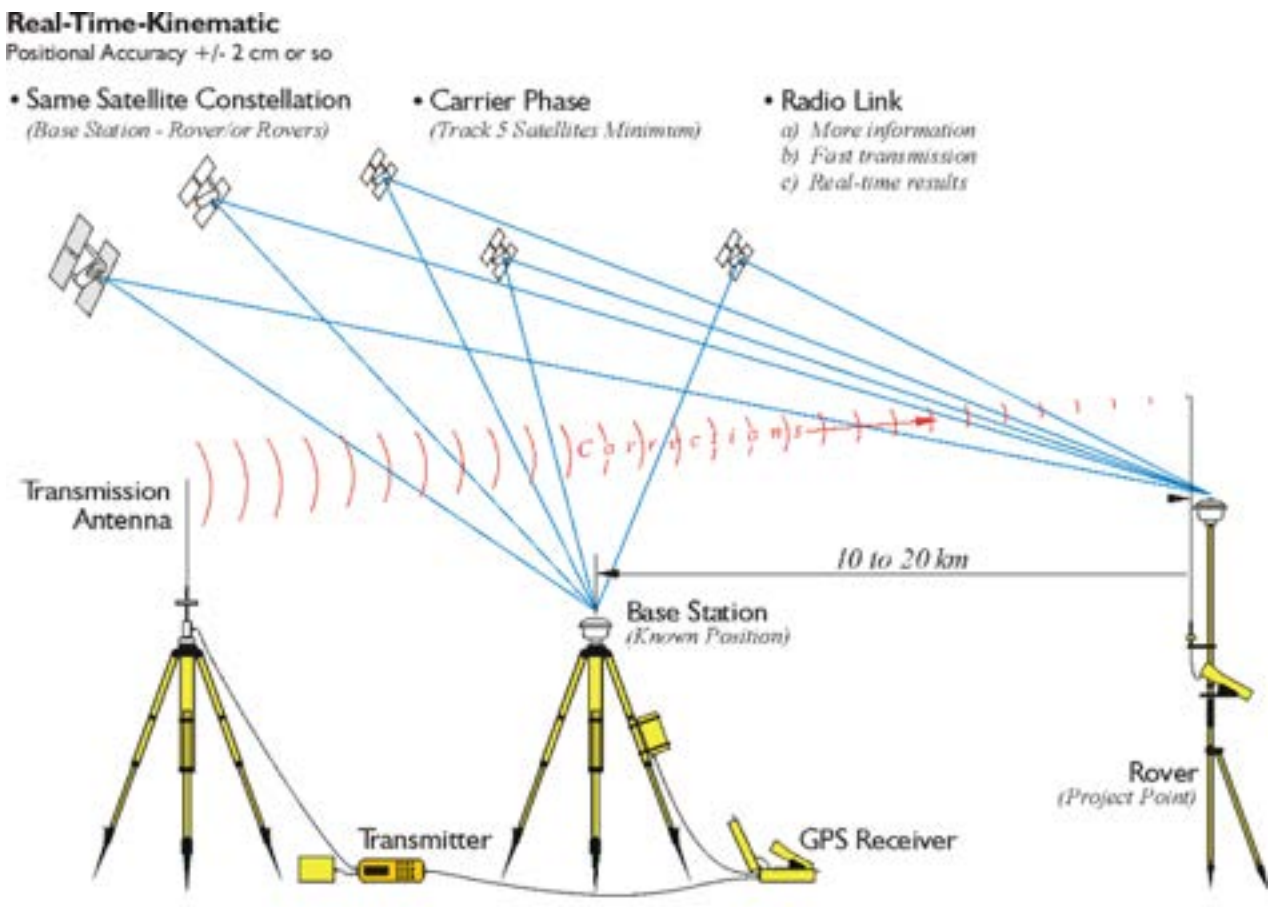
CONTENT

📚 Relative Positioning

Relative Positioning with RTK

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🔄 Real-Time Kinematic Positioning



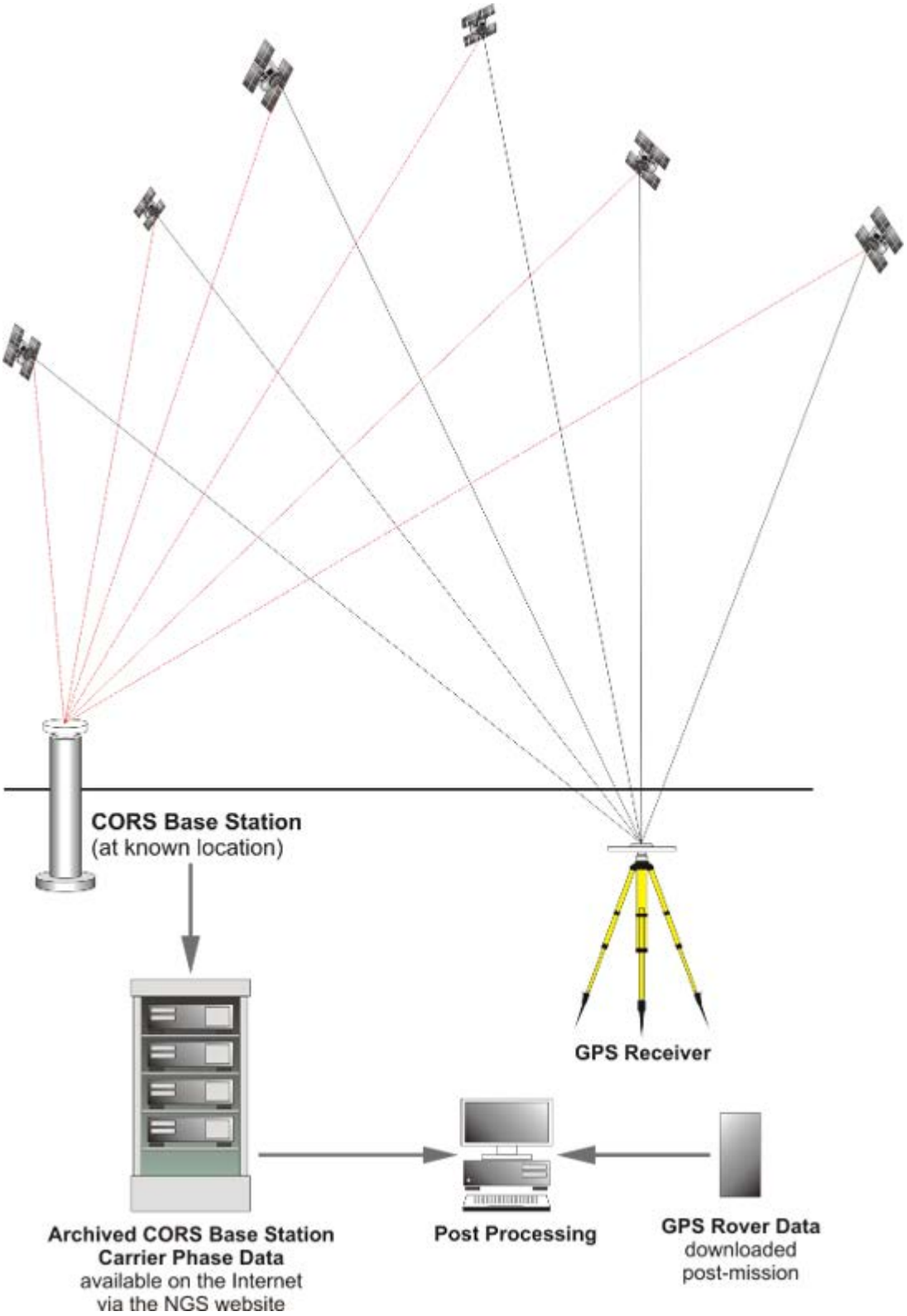
Relative (Differential) with RTK Positioning

One receiver is employed in single point positioning. A minimum of two receivers are involved in *relative* or *differential* positioning. The term *differential* GPS, or DGPS, sometimes indicates the

application of this technique with coded pseudorange measurements, whereas *relative* GPS **usually** indicates the application of this technique with carrier phase measurements. However, these definitions are by no means universal, and the use of the terms relative and differential GPS have become virtually interchangeable.

In relative positioning, one of the two receivers involved occupies a known position during the session. It is the base. The objective of the work is the determination of the position of the other, the rover, relative to the base. Both receivers observe the same constellation of satellites at the same time, and because, in typical applications, the vector between the base and the rover, known as a *baseline*, is so short compared with the 20,000-km altitude of the GPS satellites, there is extensive correlation between observations at the base and the rover. In other words, the two receivers record very similar errors (biases), and since the base's position is known, corrections can be generated there that can be used to improve the solution at the rover.

If the carrier phase observable is used in relative positioning, baseline measurement accuracies of ± (1 cm + 2 ppm) are achievable. It is possible for GPS measurements of baselines to be as accurate as 1 ppm or even 0.1 ppm. If realized, that would mean that the measurement of a nine-mile baseline would approach its actual length within ±0.05 ft. (1 ppm) or ±0.005 ft. (0.1 ppm).



Relative (Differential) with A Continuously Operating Reference Station and Static Positioning

The autonomous solution can be contrasted with relative positioning or differencing. In this image, you see a GPS receiver working at the same time as a continuously operating reference station that is on a known point. Both are collecting the same constellation of six satellites. And the CORS station (Continuously Operating Reference Station), downloads its observations to an archive. This is a base station archive that, in fact, can be made available on the Internet, as the National Geodetic Survey

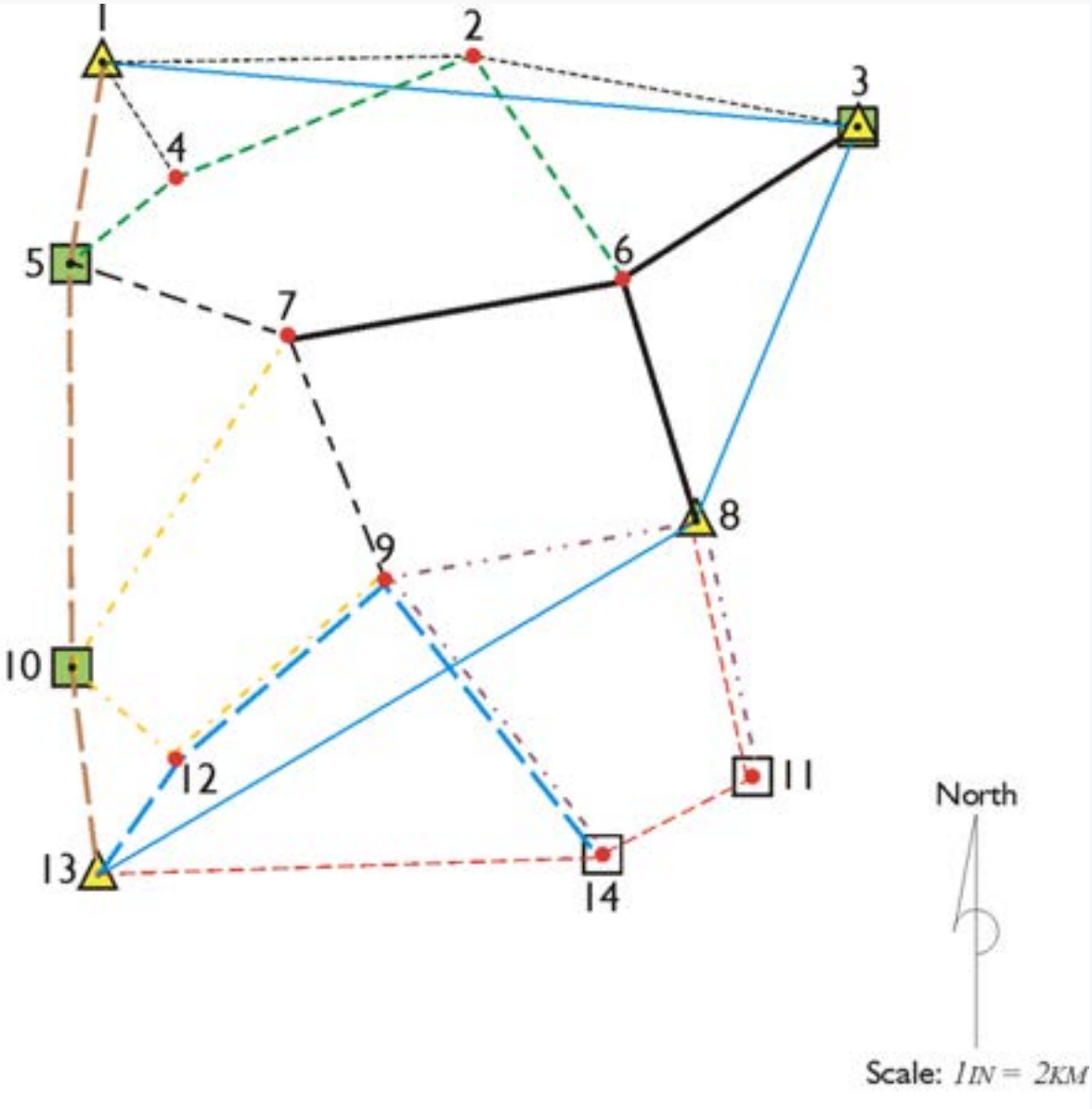
does. The GPS receiver can download its data at an unknown point to its memory. Post-mission, the two data sets can be brought together and post-processed on a computer. The corrections can be derived from the fact that the Continuously Operating Reference Station knows where it is. This allows the solution at the GPS receiver at the unknown point to be corrected to its actual position relative to the CORS. This is relative positioning. The vector between such a pair of receivers is known as a baseline. It's the simultaneity —in other words, the fact that they're observing the same constellation at the same time— that makes it possible to do accurate GPS in this way.

CONTENT

📚 Networks



Network and Multireceiver Positioning



A GPS Control Network

Network and *multireceiver positioning* are obvious extensions of relative positioning. Both the creation of a closed network of points by combining individually observed baselines and the operation of three or more receivers simultaneously have advantages. For example, the baselines have redundant measurements and similar, if not identical, range errors (biases). The processing methods in such an arrangement can nearly eliminate many of the biases introduced by imperfect clocks and the

atmosphere. These processing strategies are based on computing the differences between simultaneous GPS carrier phase observations.

📚 Differencing

GPS Differencing Techniques

GISC 4315

📊 Between-Receivers Single Difference

Between Receivers Single Difference

Please recall that the biases in both the pseudorange and the carrier phase equations discussed at the top obscure the true geometric ranges between the receivers to the satellites, which then contaminate the measurement of the baseline between the receivers. In other words, to reveal the actual vectors between two or more receivers used in relative positioning, those errors must be diminished to the degree that is possible. Fortunately, some of those embedded biases can be virtually eliminated by combining the simultaneous observables from the receivers in processes known as *differencing*. Even though the noise is increased by a factor of two with each differencing operation, it is typically used in commercial data processing software for both pseudorange and carrier phase measurements. There are three types of differencing, the *single difference*, *double difference*, and *triple difference*. Within the single difference category, there are the *between-receivers single difference* and the *between-satellites single difference*. Both require that all the receivers observed the same satellites at the same time. In GPS, the word *differencing* has come to represent several types of simultaneous baseline solutions of combined measurements. One of the foundations of differencing is the idea of the baseline as it is used in GPS. For example, a single difference, also known as a between-receivers difference, can refer to the difference in the simultaneous carrier phase measurements from one GPS satellite as measured by two different receivers. In the illustration, there are two receivers —q and r— observing the same satellite.

Between-Receivers Single Difference

A between-receivers single difference reduces the effect of biases, even though it doesn't eliminate them. Since the two receivers are both observing the same satellite at the same time, the difference between the satellite clock bias, *dt*, at the first receiver and *dt* at the second receiver, *Δdt*, is obviously zero. Also, since the baseline is typically short compared with the 20,000-km altitude of the GPS satellites, the atmospheric biases and the orbital errors, i.e., ephemeris errors, recorded by the two

receivers at each end are similar. This correlation obviously decreases as the length of the baseline increases. Generally speaking, this correlation allows centimeter level carrier phase positioning with baselines up to 10 km or so, and meter level positioning with baselines of a few hundreds of kilometers using pseudorange observations.

The between-receivers single difference provides better position estimates for the receivers by subtracting, *differencing*, each receiver's observation equation from the other. For example, if one of the receivers is a base standing at a control station whose position is known, it follows that the size of the positional error of the receiver there is knowable. Therefore, the positional error at the other end of the baseline can be estimated by finding the *difference* between the biases at the base and the biases at the rover. Corrections can then be generated which can reduce the three-dimensional positional error at the unknown point by reducing the level of the biases there. It is primarily this correlation and the subsequent ability to reduce the level of error that distinguishes differenced relative positioning from single point positioning.

🛰️ Between-Satellites Single Difference

Between Satellites Single Difference

The between-satellites single difference involves a single receiver observing two GPS satellites simultaneously and the code and/or phase measurement of one satellite are *differenced*, subtracted, from the other. The data available from the between-satellites difference allows the elimination of the receiver clock error because there is only one involved. And the atmospheric effects on the two satellite signals are again nearly identical as they come into the lone receiver, so the effects of the ionospheric and tropospheric delays are reduced. However, unlike the between-receivers single difference, the between-satellites single difference does not provide a better position estimate for the receiver involved. In fact, the resulting position of the receiver is not better than would be derived from single point positioning.

🔄 Double Difference

Double Difference

When the two types of single differences are combined, the result is known as a double difference. A double difference can be said to be a between-satellite single difference of a between-receiver single difference. The improved positions from the between-receiver single difference step are not further enhanced by the combination with the between-satellite single difference. Still, including the between-satellite single difference is useful because the combination virtually eliminates clock errors, both the satellite and receiver clock errors. The removal of the receiver clock bias in the double difference makes it possible to segregate the errors attributable to the receiver clock biases from those from other sources. This segregation improves the efficiency of the estimation of the integer cycle ambiguity in a carrier phase observation, N. In other words, the reduction of all the non-integer biases makes the computation of the final accurate positions more efficient.

The double difference, for all practical purposes, has eliminated the receiver clock errors and the satellite clock errors. This is used in most GPS post-processing and software. The integer ambiguity, N, still remains with the carrier phase observation.

⏱️ Triple Difference

Triple Difference

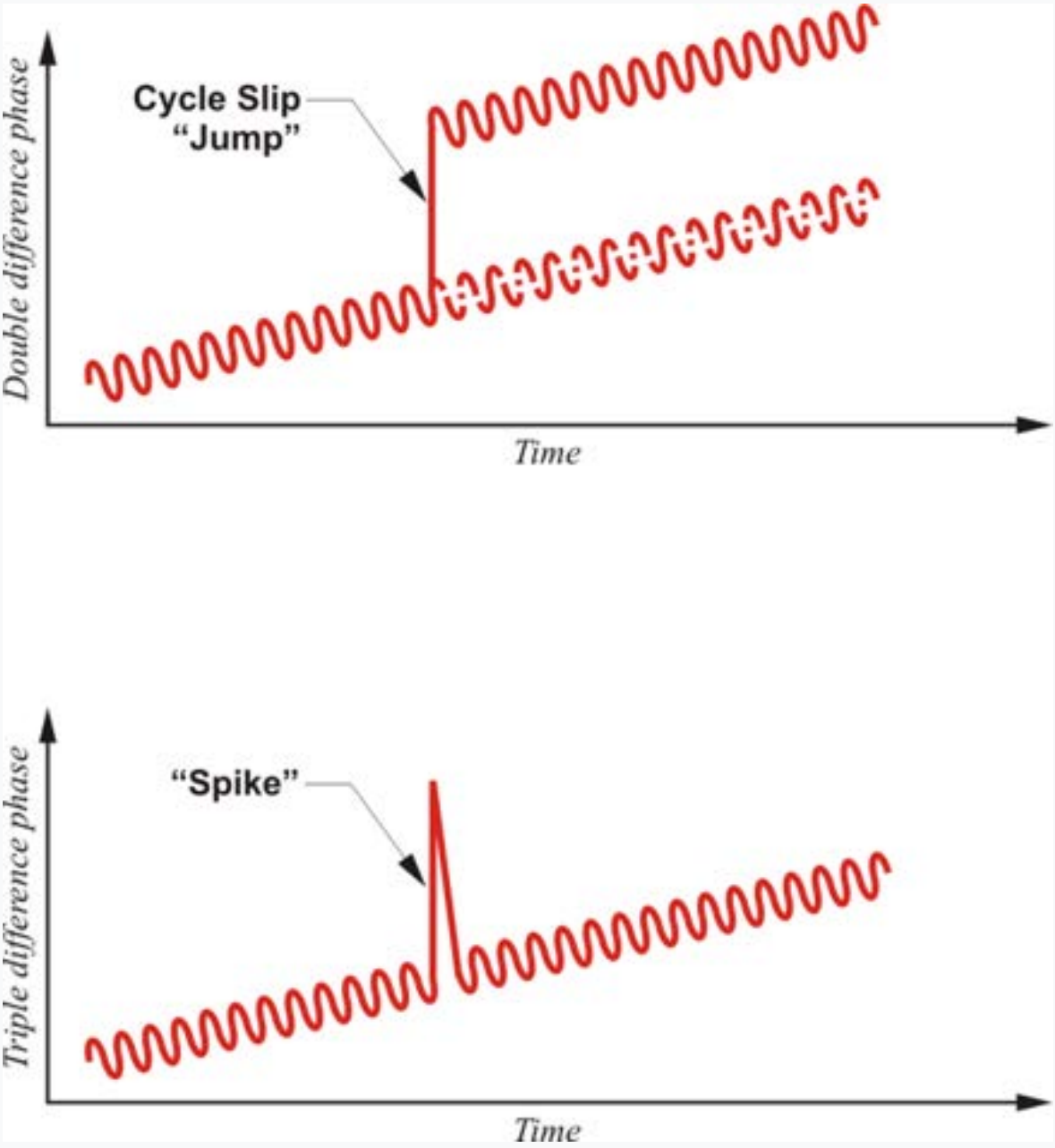
A triple difference is the difference of two double differences over two different epochs. The triple difference has other names. It is also known as the *receiver satellite-time triple difference* and the *between-epochs* difference. Triple differencing serves as a good pre-processing step, because it can be used to detect and repair cycle slips.

CONTENT

📚 Cycle Slip



📊 Understanding Cycle Slips



A Cycle Slip

Here's a graphic to give the idea of a cycle slip. As you see here, the cycle slip is indicated in a double difference, here, as a jump from one place to another. And in a triple difference, it's a spike.

A cycle slip is a discontinuity in a receiver's phase lock on a satellite's signal. A power loss, a very low signal-to-noise ratio, a failure of the receiver software, a malfunctioning satellite oscillator can cause a cycle slip. It can also be caused by severe ionospheric conditions. Most common, however, are obstructions such as buildings, trees, etc., that are so solid they prevent the satellite signal from being tracked by the receiver. Under such circumstances, when the satellite reappears, the tracking resumes.

Coded pseudorange measurements are not as affected by cycle slips as are carrier phase measurements. On the other hand, carrier phase positioning accuracy suffers if cycle slips are not detected and repaired. A cycle slip causes the critical component for successful carrier phase positioning, a resolved integer cycle ambiguity, N, to become instantly unknown again. In other words, lock is lost. When that happens, correct positioning requires that N be reestablished.

There are several methods of handling cycle slips. They are often controlled in post-processing rather than real-time.

🔧 Repairing Cycle Slips

Repairing Cycle Slips

In post-processing, the location and their size of cycle slips must be determined; then the data set can be repaired with the application of a fixed quantity to all the subsequent phase observations. One approach is to hold the initial positions of the stations occupied by the receivers as fixed, and edit the data manually. This has proven to work, but would does try patience. Another approach is to model the data on a satellite-dependent basis with continuous polynomials to find the breaks and then manually edit the data set a few cycles at a time. In fact, several methods are available to find the lost integer phase value, but they all involve testing quantities.

One of the most convenient of these methods is based on the triple difference. It can provide an automated cycle slip detection system that is not confused by clock drift and, once least-squares convergence has been achieved, it can provide initial station positions even using the unrepaired phase combinations. They may still contain cycle slips, but the data can nevertheless be used to process approximate baseline vectors. Then the residuals of these solutions are tested, sometimes through several iterations. Proceeding from its own station solutions, the triple difference can predict how many cycles will occur over a particular time interval. Therefore, by evaluating triple difference residuals over that particular interval, it is not only possible to determine which satellites have integer jumps, but also the number of cycles that have actually been lost. In a sound triple difference solution without cycle slips, the residuals are usually limited to fractions of a cycle. Only those containing cycle slips have residuals close to one cycle or larger. Once cycle slips are discovered, their correction can be systematic.

For example, suppose the residuals of one component double difference of a triple-difference solution revealed that the residual of satellite PRN 16 minus the residual of satellite PRN 17 was 8.96 cycles. Further, suppose that the residuals from the second component double difference showed that the

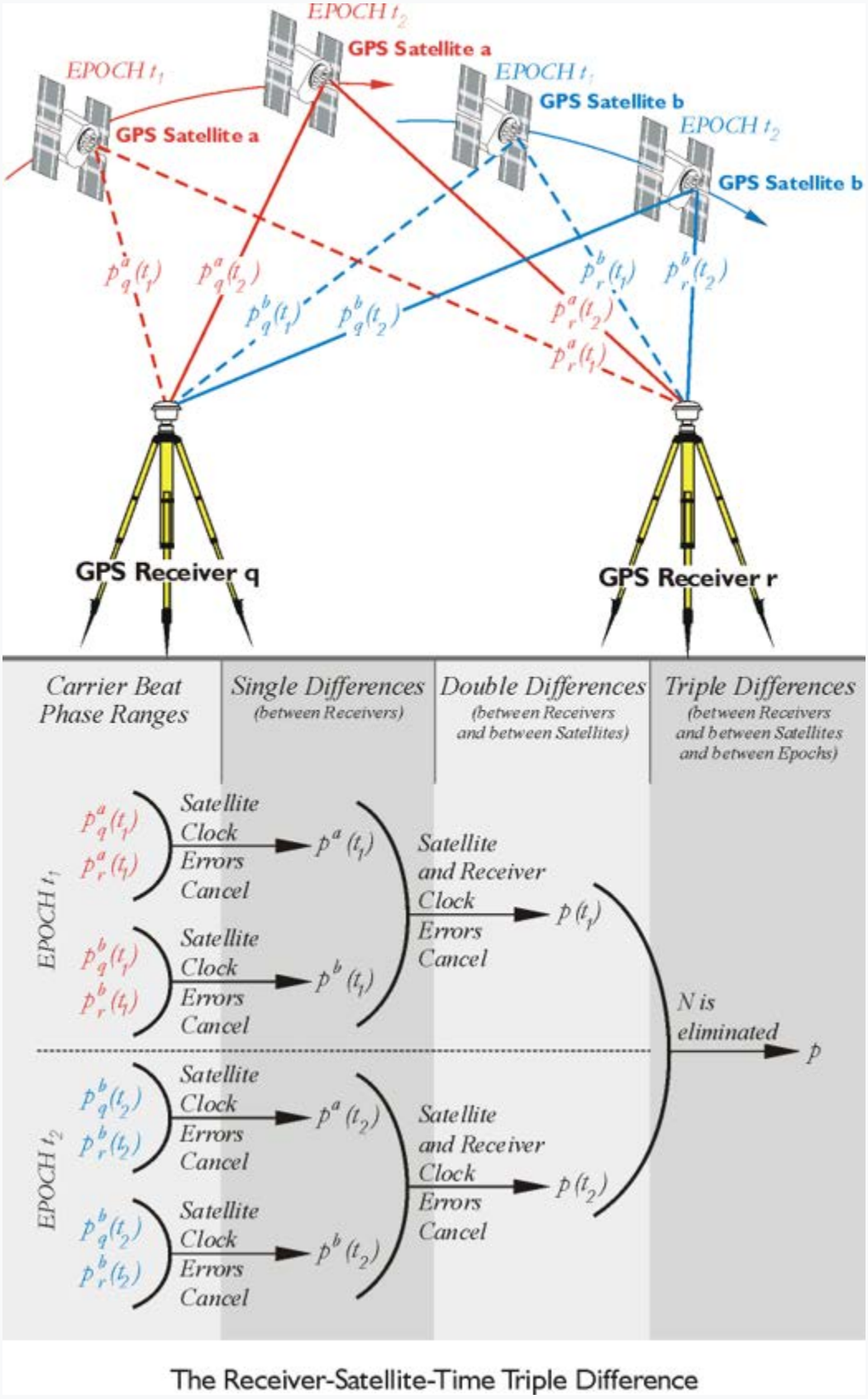
residual of satellite PRN 17 minus the residual of satellite PRN 20 was 14.04 cycles. Then, one might remove 9 cycles from PRN 16 and 14 cycles from PRN 20 for all the subsequent epochs of the observation. However, the process might result in a common integer error for PRNS 16, 17, and 20. Still, small jumps of a couple of cycles can be detected and fixed in the double-difference solutions. In other words, before attempting double difference solutions, the observations should be corrected for cycle slips identified from the triple difference solution. And even though small jumps undiscovered in the triple difference solution might remain in the data sets, the double difference residuals will reveal them at the epoch where they occurred. However, some conditions may prevent the resolution of cycle slips down to the one-cycle level. Inaccurate satellite ephemerides, noisy data, errors in the receiver's initial positions, or severe ionospheric effects all can limit the effectiveness of cycle-slip fixing. In difficult cases, a detailed inspection of the residuals might be the best way to locate the problem.

CONTENT

📚 Fixing Cycle Slips



🛠️ Triple Difference Processing



The Receiver-Satellite-Time Triple Difference

A Triple Difference

In carrier phase post-processing, the first solution is typically a simple pseudorange single point position at each end of the baseline. These code solutions provide the approximate position of each of the two receiver's antennas. Each position establishes a search area, a three-dimensional volume of uncertainty, the size of which is defined by the accuracy of the code solution. The correct position of the receiver is contained within it somewhere. The computational time required to find it depends on the size of the search area. Next, cycle slips are addressed because the subsequent instantaneous phase jumps will defeat the upcoming double differencing. Triple differencing is not as affected by cycle slips and is insensitive to integer ambiguities, so it typically precedes double differencing and is used to clean the data by detecting and repairing cycle slips. In fact, pre-processing using the triple difference to weed out and fix cycle slips is one of its primary appeals. As to detection, when a large residual appears in one of the triple difference's component double differences, it is likely caused by a cycle slip, so the satellite pairs can be sorted until the offending signal is singled-out and repaired. Triple differencing can also provide first positions for the receivers. While the receiver coordinates that result are usually more accurate than pseudorange solutions, their geometric strength is weak and not sufficiently accurate for determining short baselines. Nevertheless, they provide a starting point for the more accurate double-difference solutions that follow.

📚 Components of the Carrier Phase Observable



📡 Carrier Phase Observable Components

Components of the Carrier Phase Observable

Components of the carrier phase observable

From the moment a receiver locks onto a satellite to the end of the observation, the carrier phase observable can be divided into three parts. Two of them do not change during the session, and one of them does change. The *fractional initial phase* is established at the first instant of the lock-on. When the receiver starts tracking the satellite, it is highly unlikely to acquire the satellite's signal precisely at the beginning of a wavelength's phase cycle. It will grab on at some fractional part of a phase, and this fractional phase will remain unchanged for the duration of the observation. It is called the fractional initial phase or the phase measurement and is symbolized in the equation above by α α . The integer cycle ambiguity *N* represents the number of full phase cycles between the receiver and the satellite at the first instant of the receiver's lock-on. It can also be labeled the carrier phase ambiguity or the cycle count at lock-on. It does not change from the moment of the lock to the end of the observation, unless that lock is lost. But when there is a cycle slip, lock is lost, and by the time the receiver reacquires the signal, the normally constant integer ambiguity has changed. In that case, the receiver loses its place in its count of the integer number of cycles is lost. The value ß is the count of the observed phase accumulation coming in throughout the observation. Of course, the count grows from the moment of lock on until the end of the observation. In other words, ß is the receiver's record of the consecutive change in cycles between the receiver and the satellite as the satellite flies over. If the observation proceeds without cycle slips, the observed cycle count is the only one of three numbers that changes.

You may remember from the first Module that when a GPS receiver locks onto a signal from a satellite, it can acquire the fractional initial phase, α. It can interpret that fractional initial phase by

comparing what's coming in from the satellite with the replica frequency that it has on-board in its circuits. That comparison can take place pretty quickly, and it doesn't need to change through the observation of that satellite by that receiver. You may also remember that in an EDM measurement, since it is two-way, the EDM can send out frequencies of different lengths to resolve the cycle ambiguity. GPS doesn't have that advantage. It's a one-way system with set carrier wavelengths, so N is somewhat elusive. As the satellite moves along its orbit, the GPS receiver does observed cycle count. In other words, it can count the number of full wavelengths, ß, since its lock-on, and how much they have changed as the satellite moves from one place to another in its orbit. Armed with an initial pseudorange solution, the receiver can estimate the integer ambiguity (carrier phase ambiguity), N. The initial pseudorange observation gives the receiver an idea how many full cycles there are in that N value. From that starting place, it can use a double difference to nail down that carrier phased ambiguity. Perhaps you have heard the *fixed solution* terminology in post-processing software in GPS. It is often mentioned as opposed to a *float solution*. A fixed solution means that the integers are fixed. The N has been solved. A fixed solution is superior to a float solution.

Just to reiterate this— the first fractional initial phase occurs at the first instant of lock-on. The receiver starts tracking the incoming phase from the satellite. It can't yet know how to achieve the perfect synchronization. Lacking this knowledge, the receiver grabs on to the satellite's incoming signal at some fractional part. This fractional part does not change for the duration of the observation. The integer number of cycles (the carrier phase ambiguity) is the integer number of cycles of phase that are present at the moment of lock-on. The observed cycle count is the receiver's consecutive counting of the change in phase cycles between the receiver and the satellite as the satellite flies over. Of these three terms, ß is the only number that changes. That is, if the observation proceeds correctly. In other words, without any cycle slips. In the carrier phase observation, there are two values that do not change: the fractional initial phase and the integer cycle ambiguity, N. Only the cycle count changes, unless, of course, there's a cycle slip. When lock is lost, a cycle slip occurs. It can be a power loss, an obstruction, low signal to noise ratio (SNR), anything that breaks the signal from the satellite to the receiver. When there is a cycle slip, the receiver loses its place and its count of the number of cycles, the integer lock is also lost, and re-initialization is necessary.

INTRODUCTION & INSTRUCTIONS

📡 Overview Module 06: Receivers and Methods

Module 6: Receivers and Methods

GISC 4315

📖 Overview

GPS Receivers and Survey Techniques

The characteristics and capabilities of GPS receivers influence the techniques available to the user throughout the work, from the initial planning to processing. There are literally hundreds of different GPS receivers on the market. Aside from recreational receivers, all are generally capable of accuracies from sub-meter to sub-centimeter. They are capable of differential GPS, DGPS, real-time GPS, static GPS and other hybrid techniques. They usually are accompanied by post-processing software and network adjustment software. And many are equipped with capacity for extra batteries, external data collectors, external antennas, and tripod mounting hardware. Just as there are many types of GPS receivers, there are many ways to apply them in obtaining GPS positions. Each of these several very different techniques makes unique demands on the receivers used to support it.

This lesson is about those techniques and the fundamentals of GPS receivers.

🎯 Accuracy Range

- **Sub-meter** to **sub-centimeter**
- **Professional-grade** capabilities
- **Survey quality** precision
- **Multiple applications**

⚙️ Receiver Capabilities

- **Differential GPS** (DGPS)
- **Real-time GPS**
- **Static GPS**
- **Hybrid techniques**

🔧 Equipment Features

- **Post-processing software**
- **Network adjustment**
- **External accessories**
- **Mounting hardware**

🎯 **Lesson Focus**

This lesson explores receiver fundamentals and the diverse surveying techniques that leverage different GPS receiver capabilities for specific positioning applications.

- **Read Chapter 4 in *GPS and GNSS for Land Surveyors*.**

🎯 Learning Objectives

What You Will Learn

At the successful completion of this lesson, students should be able to:

📡 Receiver Components

- **recognize** the basic functions of the common features of GPS receivers, the antenna, the preamplifier, the RF section, the microprocessor, the CDU, the storage and the power
- **recognize** some of the important issues in choosing a GPS receiver
- **discuss** some of the trends in receiver development

📐 Survey Methods

- **explain** some GPS surveying methods
- **demonstrate** static
- **explain** differential GPS. DGPS
- **explain** kinematic

🚀 Advanced Techniques

- **describe** pseudokinematic
- **identify** rapid-static
- **define** on-the-fly
- **recognize** real-time-kinematic

🔧 Technical Applications

- **Receiver selection** criteria
- **Method optimization**
- **Technology trends**
- **Professional applications**

💡 **Learning Approach**

These objectives progress from understanding receiver hardware components to mastering the diverse surveying techniques that leverage different receiver capabilities.

🔍 Key Topic Categories

Understanding Receiver Technology and Methods

The learning objectives organize GPS receiver technology and surveying methods into logical categories that build from hardware understanding to advanced field techniques.

📡 Hardware

- Antenna systems
- RF processing
- Microprocessors
- Power systems

⚙️ Basic Methods

- Static surveying
- Differential GPS
- Kinematic methods
- Survey techniques

🚀 Advanced

- Pseudokinematic
- Rapid-static
- On-the-fly
- Real-time-kinematic

🎯 Selection

- Receiver choice
- Method matching
- Technology trends
- Application needs

🎯 Professional Relevance

Understanding receiver capabilities and matching appropriate surveying methods to project requirements is essential for efficient, accurate GPS surveying operations.

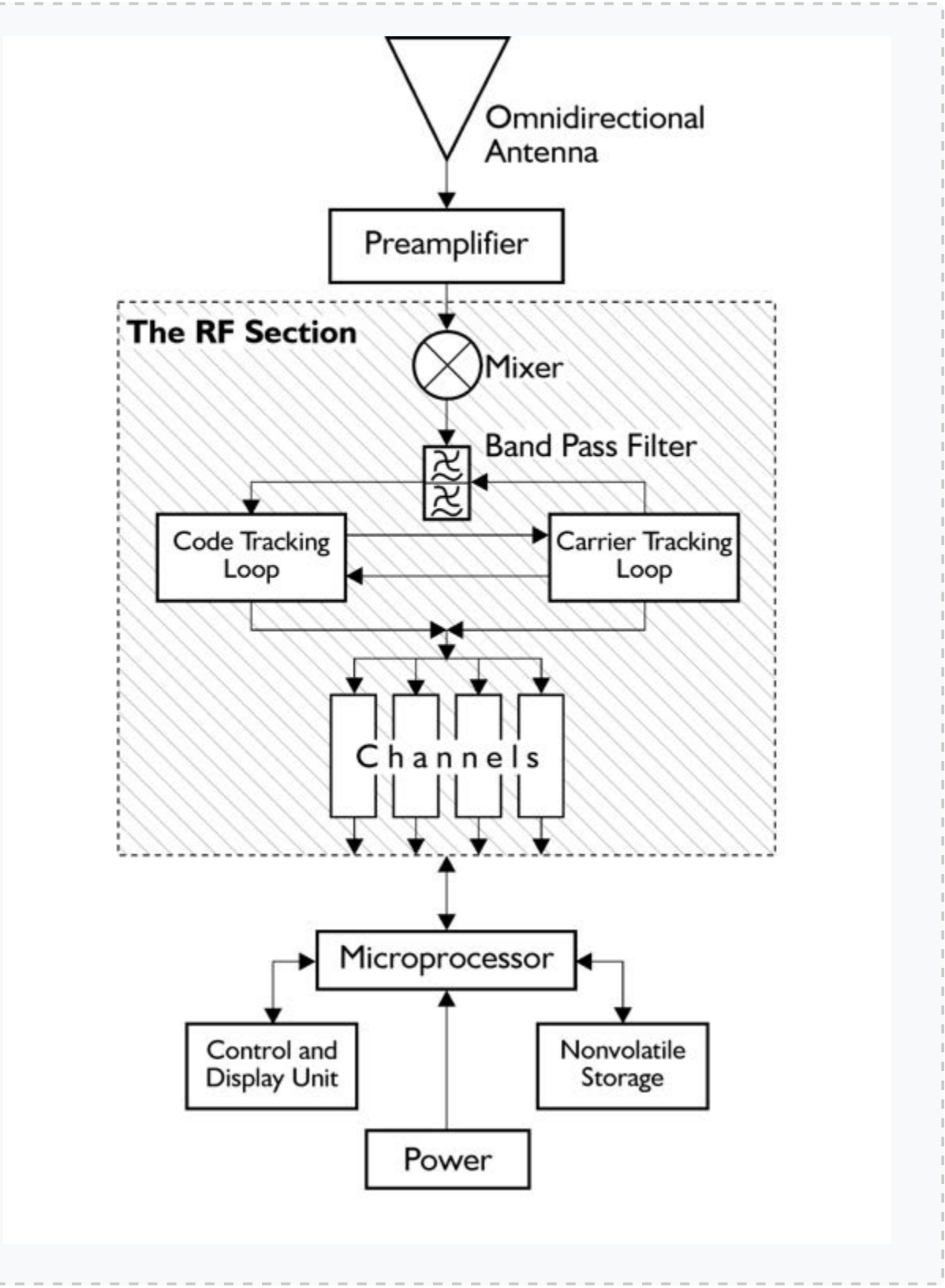
CONTENT

📚 Common Features of GPS Receivers



🔧 Receivers for GPS Surveying

Receivers for GPS Surveying



GPS Receiver Schematic

The receivers are the most important hardware in a GPS surveying operation. Their characteristics and capabilities influence the techniques available to the user throughout the work. There are many different GPS receivers on the market. Some of them are appropriate for surveying, and they share some fundamental elements. Though no level of accuracy is ever guaranteed, with proper procedures and data handling they are generally capable of accuracies from sub meter to centimeters. Most are also capable of performing differential GPS, real-time GPS, static GPS, etc., and are usually accompanied by processing and network adjustment software and so on.

GPS receivers come in a variety of shapes and sizes. Some have external batteries, data collectors. Some are tripod mounted. Some are hand-held and have all components built in, and some can be used in both ways, with externals and without. Nevertheless, most have similar characteristics. Here is a schematic drawing of a GPS receiver. It includes some of the common components.

A GPS receiver must collect and then convert signals from GPS satellites into measurements of position, velocity, and time. There is a challenge in that the GPS signal has low power. An orbiting GPS satellite broadcasts its signal across a cone of approximately 28° of arc. From the satellite's point of view, about 11,000 miles up, that cone covers a substantial portion of the whole planet. It is instructive to contrast this arrangement with a typical communication satellite that not only has much more power, but also broadcasts a very directional signal. Its signals are usually collected by a large dish antenna, but the typical GPS receiver has a small, relatively non-directional antenna. Stated another way, a GPS satellite spreads a low power signal over a large area rather than directing a high power signal at a very specific area. Fortunately, antennas used for GPS receivers do not have to be pointed directly at the signal source. The GPS signal also intentionally occupies a broader bandwidth than it must, to carry its information. This characteristic is used to prevent jamming and mitigate multipath but, most importantly, the GPS signal itself would be completely obscured by the variety of electromagnetic noise that surrounds us if it were not a spread spectrum coded signal. In fact, when a GPS signal reaches a receiver, its power is actually less than the receiver's noise level; fortunately, the receiver can still extract the signal and achieve unambiguous satellite tracking using the correlation techniques. To do this job, the elements of a GPS receiver function cooperatively and iteratively. That means that the data stream is repeatedly refined by the several components of the device working together as it makes its way through the receiver.

CONTENT

📚 The Antenna and The Right Hand Circular Polarized Signal

GPS Antenna Polarization

GISC 4315

📡 Right Hand Circular Polarized Signal

Right Hand Circular Polarized Signal

The antenna, *radio frequency* (RF) section, filtering and intermediate frequency elements are in the front of a GPS receiver. The antenna collects the satellite's signals and converts the incoming electromagnetic waves into electric currents sensible to the RF section of the receiver. Several antenna designs are possible in GPS, but the satellite's signal has such a low power density, especially after propagating through the atmosphere, that antenna efficiency is critical. Therefore, GPS antennas must have high sensitivity, also known as high gain. They can be designed to collect only the L1 frequency, L1 and L2, or all signals, including L5. In all cases, they must be ***Right Hand Circular Polarized, (RHCP)***, as are the GPS signals broadcast from the satellites.

Polarized waves oscillate in more than one direction. The electrical field vectors of the GPS Signal have a constant magnitude, but their direction rotates so that the electrical field vector of the wave describes a helix in the direction of propagation. Said another way, circularly polarized waves are those where the angle of the electric vector rotates around an imaginary line traveling in the direction of the propagation of the wave. The rotation may be either to the right or left. The GPS signal is a Right-Hand Circularly Polarized (RHCP) wave. You can illustrate it this way. With your right hand, give the *thumbs up* signal. Now, instead of pointing your thumb up, point it in the direction that the GPS signal is propagating. Your curling fingers show you the direction of the rotation of the field.

CONTENT

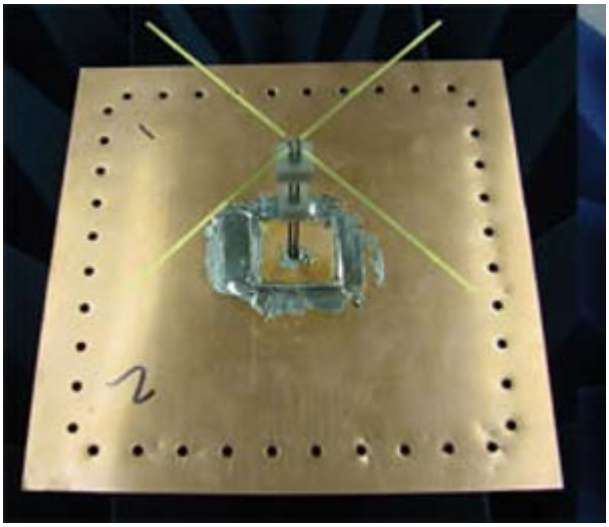
📚 The Antenna



📶 Antenna Types









Top left: Microstrip patch antenna, Top right: Quadrifilar antenna

Bottom left: Dipole antenna, Bottom right: Helix antenna

The illustration shows antenna types. They are not specifically GPS antennas.

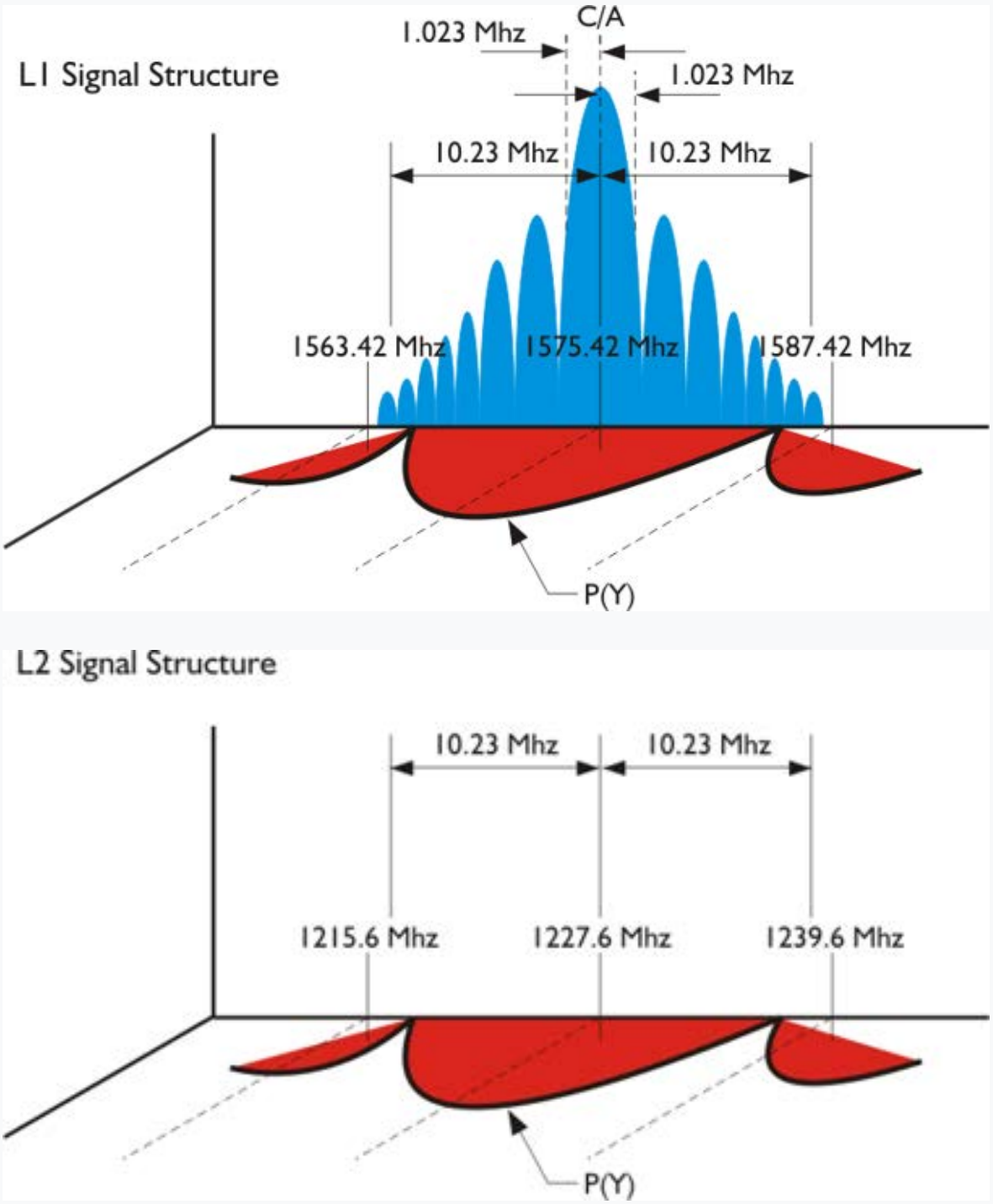
Most receivers have an antenna built in, but many can accommodate a separate tripod-mounted or range pole-mounted antenna as well. These separate antennas sometimes require connecting coaxial cables. The cables are an important detail. The longer the cable, the more of the GPS signal is lost traveling through it. They are usually in standard lengths to make sure that the impedance of the trip through the cable can be calibrated.

As mentioned earlier, the wavelengths of the GPS carriers are 19 cm (L1), 24 cm (L2) and 25 cm (L5), and antennas that are a quarter or half wavelength tend to be the most practical and efficient, so GPS antenna elements can be as small as 4 or 5 cm. Most of the receiver manufacturers use a *microstrip* antenna. These are also known as patch antennas. The microstrip may have a patch for each frequency, so it can receive one or all of the GPS carriers. Microstrip antennas are durable, compact, have a simple construction and a low profile. The next most commonly used antenna is known as a *dipole*. You may recall that this is the kind of antenna that was used with the Macrometer, the first commercial GPS receiver. A dipole antenna has a stable phase center and simple

construction, but needs a good ground plane. A ground plane also facilitates the use of a microstrip antenna where it not only ameliorates multipath, but also tends to increase the antenna's zenith gain--in other words, the gain of the antenna straight up. A *quadrifilar* antenna is a single frequency antenna that has two orthogonal bifilar helical loops on a common axis. Quadrifilar antennas perform better than a microstrip on crafts that pitch and roll, like boats and airplanes. They are also used in many recreational handheld GPS receivers. Such antennas have a good gain pattern, do not require a ground plane, but are not azimuthally symmetric. The least common design is the helix antenna. A *helix* is a dual frequency antenna. It has a good gain pattern, but a high profile.

Bandwidth

Bandwidth

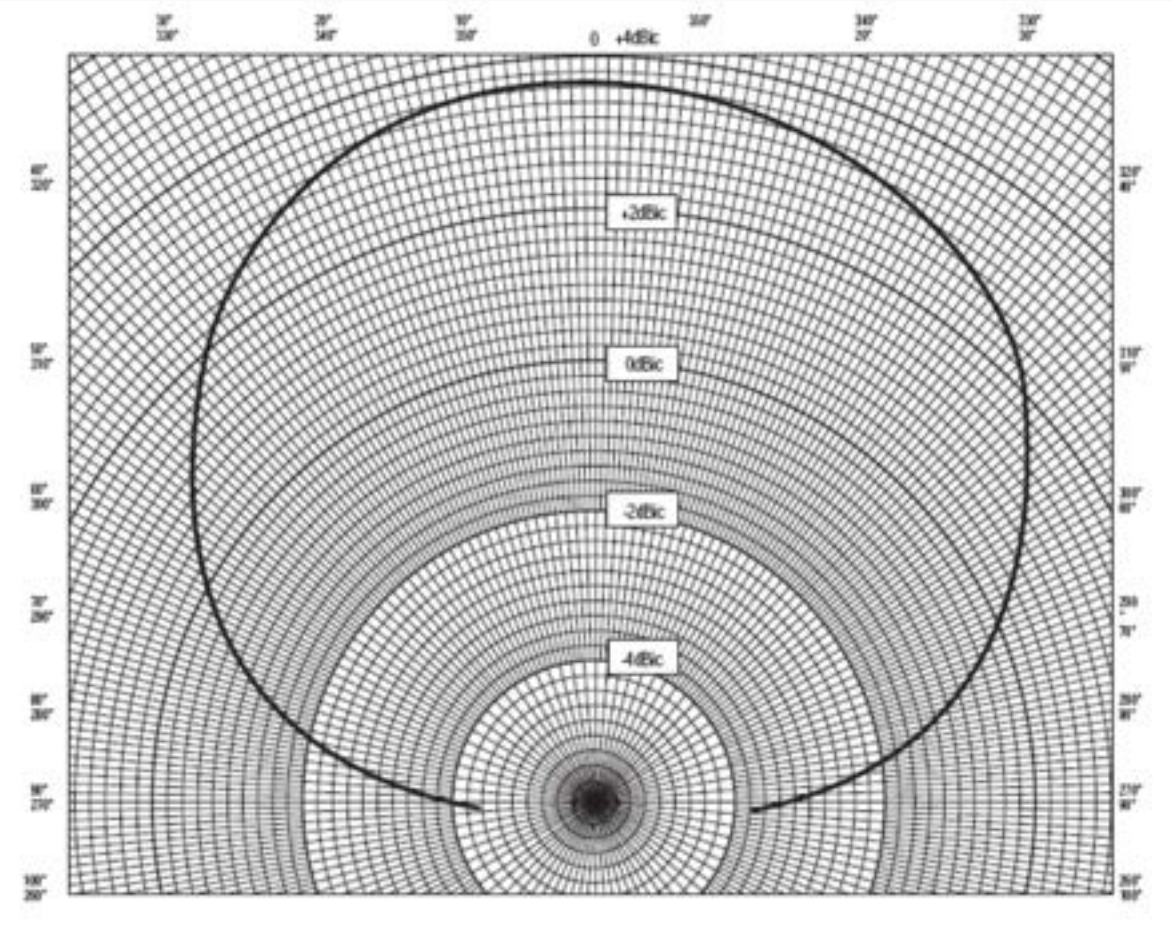


Power Spectral Density (PSD) Diagrams for L1 and L2

An antenna ought to have a bandwidth commensurate with its application. In general, the larger the bandwidth, the better the performance; however, there is downside. Increased bandwidth degrades the signal-to-noise ratio by including more interference. These PSD diagrams illustrate the C/A and P code signals power per bandwidth in Watts per Hertz as a function of frequency. GPS microstrip antennas usually operate in a range from about 2 to 20 MHz, which corresponds with the null-to-null bandwidth of the GPS signals. For example, the L2C signal, like the C/A, has the span of 2.046 megahertz. You can see it in the highest portion or the central lobe of the diagram. L5, like the P code, has a bandwidth of 20.46 megahertz, as you see in red. So an antenna on the front end of the receiver has to be able to accommodate that bandwidth of 20.46 megahertz if it is to track all of these signals. If the system's tracking C/A code or the L2C only, it could have a narrower bandwidth. It would need 2.046 MHz for the central lobe of the C/A code, or if it were designed to track the L1C signal, its bandwidth would need to be 4.092 MHz. A dual frequency microstrip antenna would likely operate in a bandwidth from 10 to 20 MHz.

🌐 Nearly Hemispheric Coverage

Nearly Hemispheric Coverage



GPS Antenna Pattern

Since a GPS antenna is designed to be omnidirectional, its gain pattern, that is the change in gain over a range of azimuths and elevations, ought to be nearly a full hemisphere, but not perfectly hemispheric. For example, most surveying applications filter the signals from very low elevations to reduce the effects of multipath and atmospheric delays. A portion of the GPS signal may come into the antenna from below the mask angle; therefore, the antenna's gain pattern is specifically designed to reject such signals. Second, the contours of equal phase around the antenna's electronic center, that is, the phase center, are not themselves perfectly spherical.

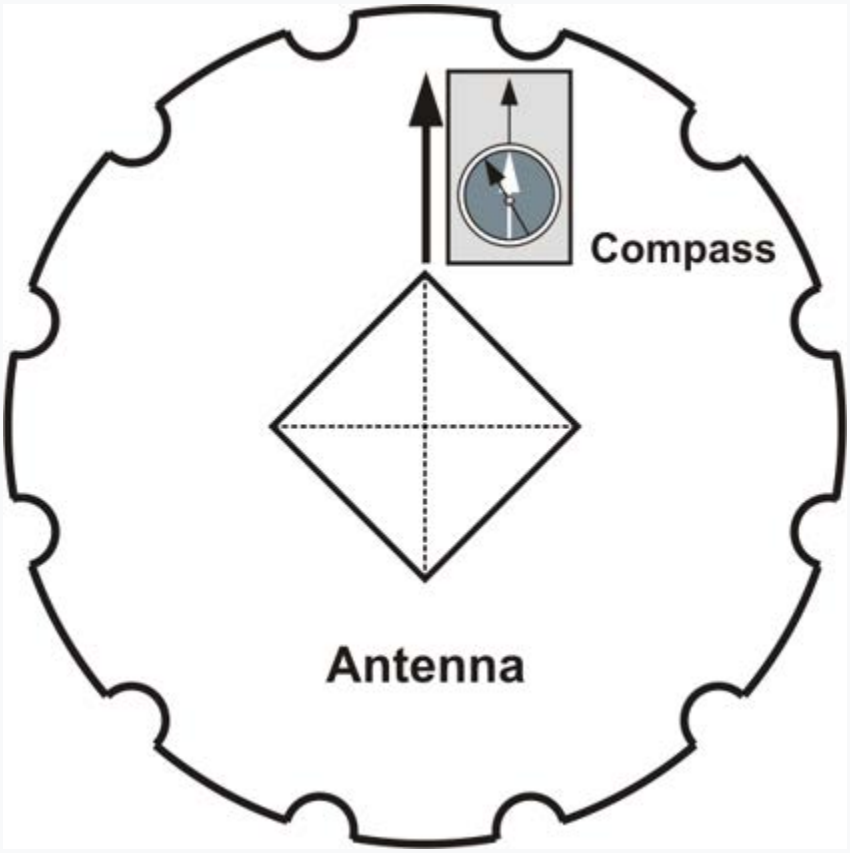
The gain, or gain pattern, describes the success of a GPS antenna in collecting more energy from above the mask angle, and less from below the mask angle. A gain of about 3 to 5 *decibels (dB)* is

typical for a GPS antenna. Just a brief description of a decibel-- we'll be seeing a little bit more of it. The decibel is a tenth of a bell, which was named for Alexander Graham Bell. It is a logarithmic dimensionless unit used to make a comparison. In this case, the gain of a real GPS antenna is compared to a theoretical lossless antenna that has perfectly equal capabilities in all directions. This imaginary perfection is known as an isotropic antenna. By the way a 3 decibel increase indicates a doubling of signal strength, and a 3 decibel decrease indicates a halving of signal strength so a typical omnidirectional GPS antenna with a gain of about 3 dB (decibels) has twice the capability of a perfect isotropic antenna in terms of receiving signal power in its preferred direction.

A decibel watt, dbW, indicates the actual power of a signal compared to a reference of one watt, The minimum power received from the C/A code on L1 is about -160dBW, -160 decibel Watts and the minimum power received from the P code on L2 is even less at -166 dBW. It is important that the GPS receiver antennas and pre-amplifiers be as efficient as possible, because the power received from the GPS satellites is low.

🧭 Antenna Orientation

Antenna Orientation



Antenna Oriented to North

In a perfect GPS antenna, the phase center of the gain pattern would be exactly coincident with its actual, physical, center. If such a thing were possible, the centering of the antenna over a point on the earth would ensure its electronic centering as well. But that absolute certainty remains elusive for several reasons.

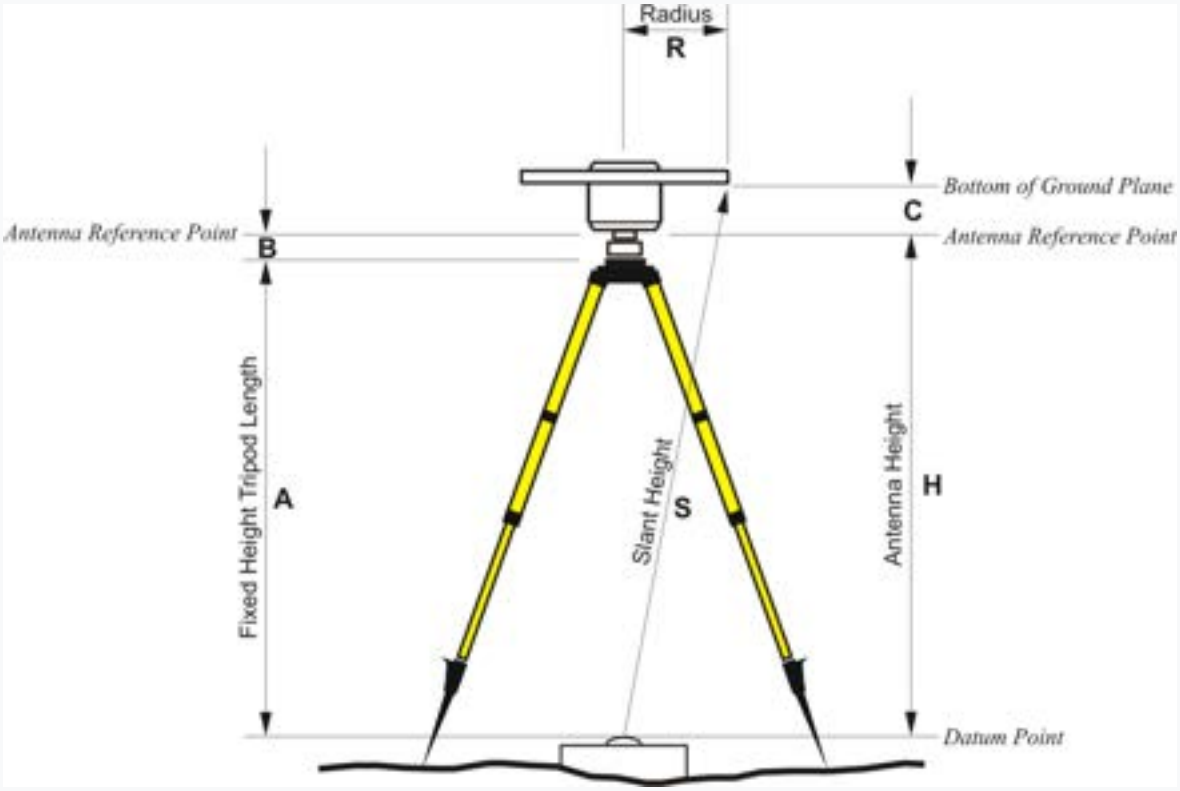
It is important to remember that the position at each end of a GPS baseline is the position of the phase center of the antenna at each end, not their physical centers, and the phase center is not an immovable point. The location of the phase center actually changes slightly with the satellite's signal. For example, it is different for the L2 than for L1 or L5. In addition, as the azimuth, intensity, and elevation of the received signal changes, so does the difference between the phase center and the physical center. Small azimuthal effects can also be brought on by the local environment around the antenna. But most phase center variation is attributable to changes in satellite elevation. In the end,

the physical center and the phase center of an antenna may be as much as a couple of centimeters from one another. On the other hand, with today's patch antennas, it can be as little as a few millimeters. It is fortunate that the shifts are systematic. To compensate for some of this offset error, most receiver manufacturers recommend users make sure their antennas are all oriented in the same direction when making simultaneous observations on a network of points. Several manufacturers provide reference marks on the antenna, so that each one may be rotated to the same azimuth, usually north, to maintain the same relative position between their physical and phase, electronic, centers. By orienting all the antennas in the same direction, the offset between the phase center and the physical center is in the same direction at each point. Therefore, the baselines come out exactly as if the physical and the phase center were coincident.

It's only when we're talking about control work, work that needs to be quite accurate, that all of the antennas need to be oriented to the same direction.

📏 Height of Instrument

Height of Instrument



Measuring the Antenna Height

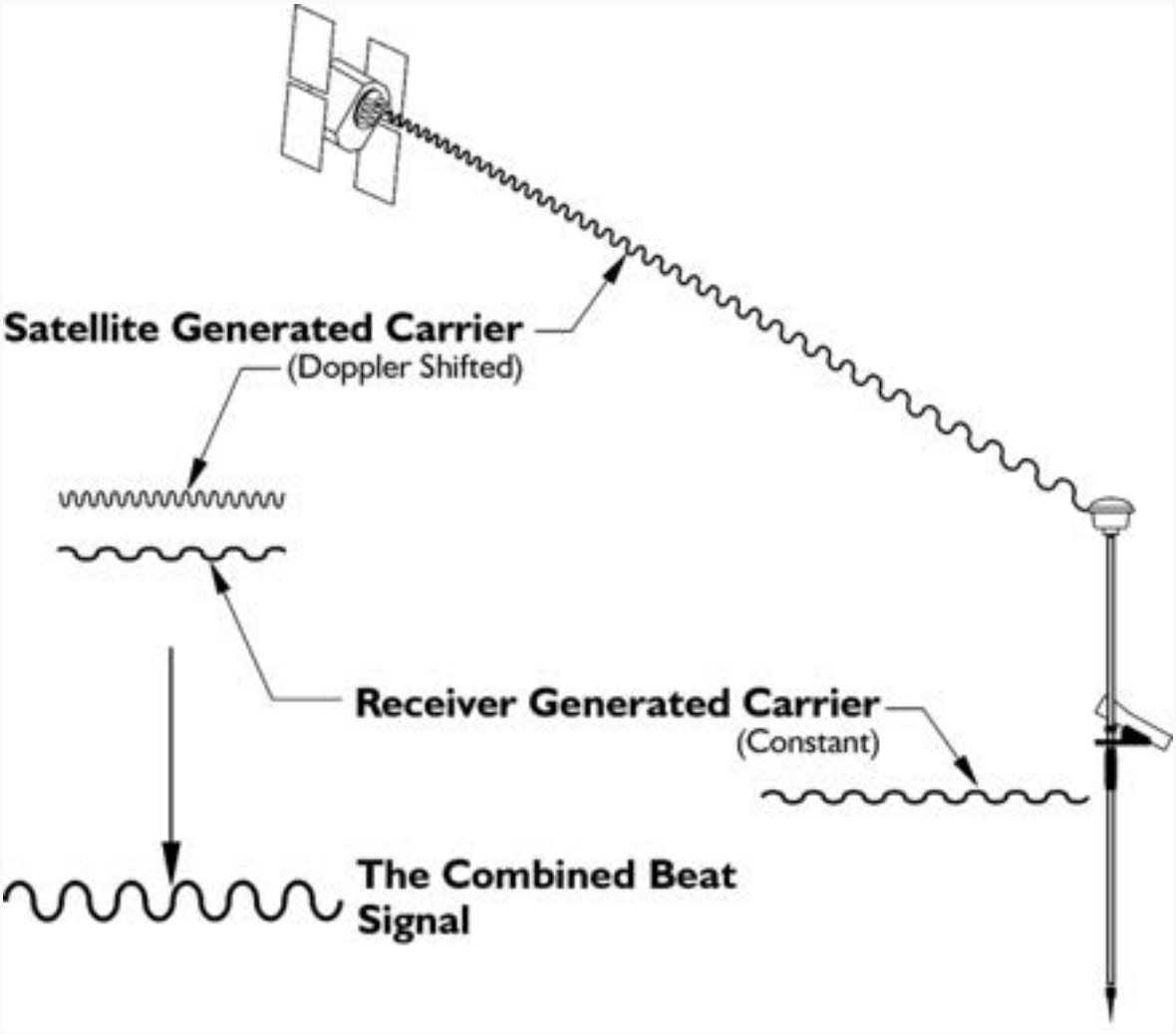
The antenna's configuration also affects another measurement critical to successful GPS work - the height of the instrument. An incorrect height of instrument is one of the most easily avoided errors. The measurement is normally made to some reference mark on the antenna. In this diagram, you see many ways of measuring that height. It can be measured at slant height or measured with a tape, usually to the antenna reference point. The ARP, or the antenna reference point, is frequently the bottom of the mount of the antenna. There's usually a correction to be added to actually bring that measurement up to the phase center of the antenna. This also becomes part of the necessary information when one is using continuously operating reference stations. It is possible to take the downloads from NGS managed Continuously Operating Reference Station (CORS) stations that are available on the Internet and post-process the observations taken with a roving GPS receiver. If you are doing so, it is also necessary to know the height of the antenna at the CORS. Of course, that is not an antenna that you would've set up, but that information is available along with the files from the base station.

CONTENT

📚 The RF Section



🔊 Signal Processing and Beat Frequency

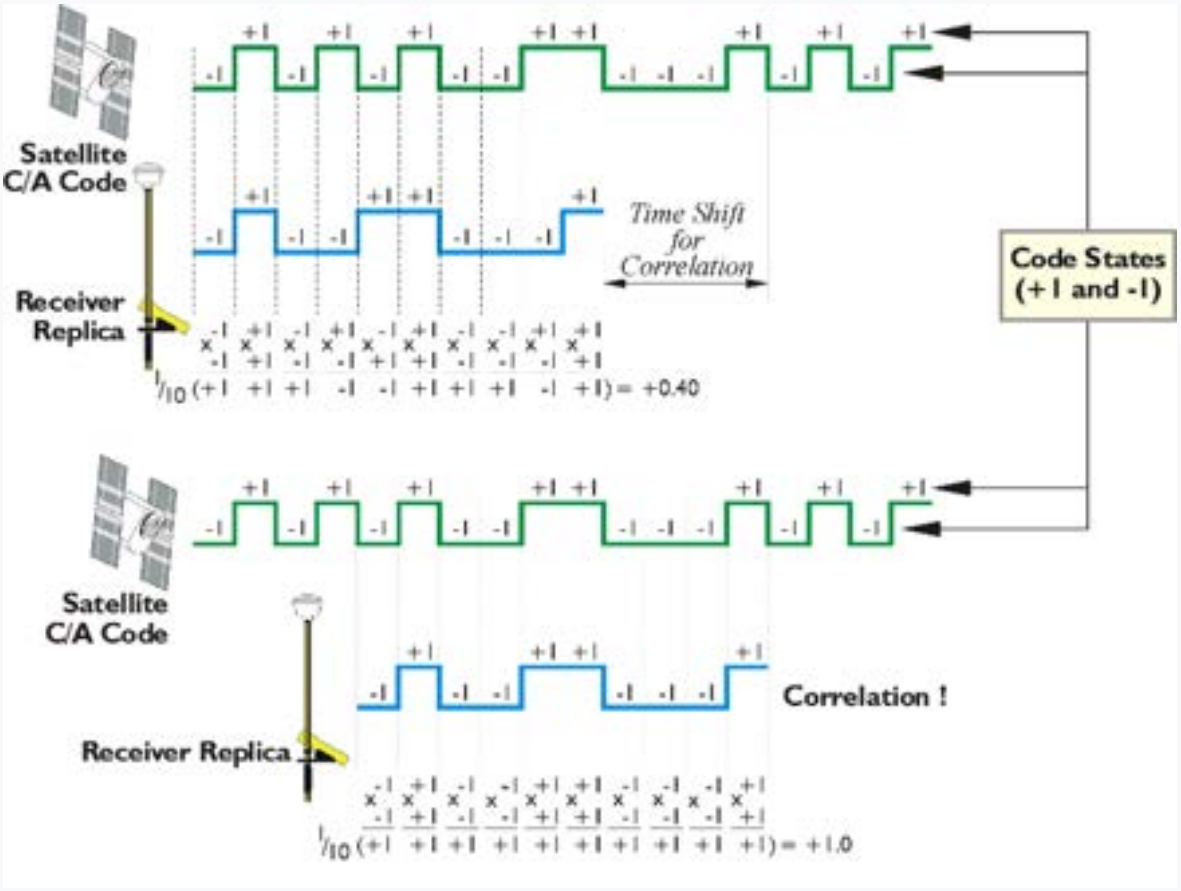


The Combined Beat Signal

The pre-amplifier is necessary, because the signal coming in from the GPS satellite is weak. The preamplifier increases the signal's power, but it is important that the gain in the signal coming out of the preamplifier is considerably higher than the noise. Noise is always part of the signal. Since signal processing is easier if the signals arriving from the antenna are in a common frequency band, the incoming frequency is combined with a signal at a harmonic frequency. This latter, pure sinusoidal signal is the reference signal generated by the receiver's oscillator. The two frequencies are multiplied together in a device known as a *mixer*. Two frequencies emerge: one of them is the sum of the two that went in, and the other is the difference between them. The sum and difference frequencies then go through a *bandpass filter*, an electronic filter that removes the unwanted high frequencies and

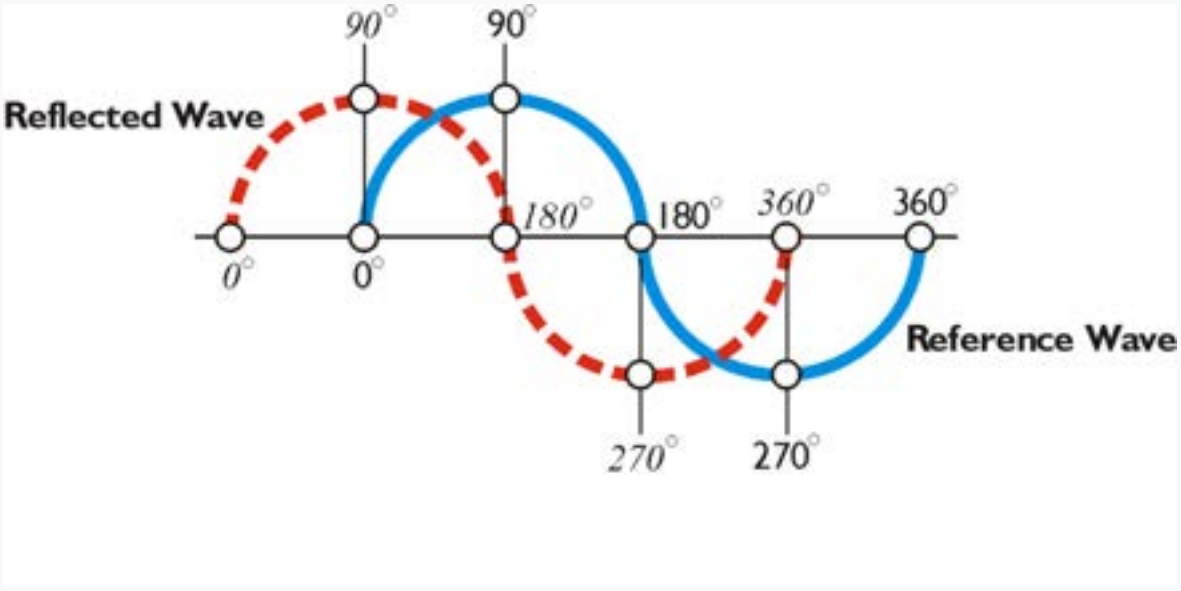
selects the lower of the two. It also eliminates some of the noise from the signal. For tracking the P(Y)-code, this filter will have a bandwidth of about 20 MHz, but it will be around 2 MHz if the C/A code is required. In any case, the signal that results is known as the *intermediate frequency (IF),* or beat frequency signal. This beat frequency is the difference between the Doppler-shifted carrier frequency that came from the satellite and the frequency generated by the receiver's own oscillator. In fact, to make sure that it embraces the full range of the Doppler effect on the signals coming in from the GPS satellites, the bandwidth of the IF itself can vary from 5 to 10 kHz Doppler. That spread is typically lessened after tracking is achieved.

🔄 Autocorrelation Process



Autocorrelation

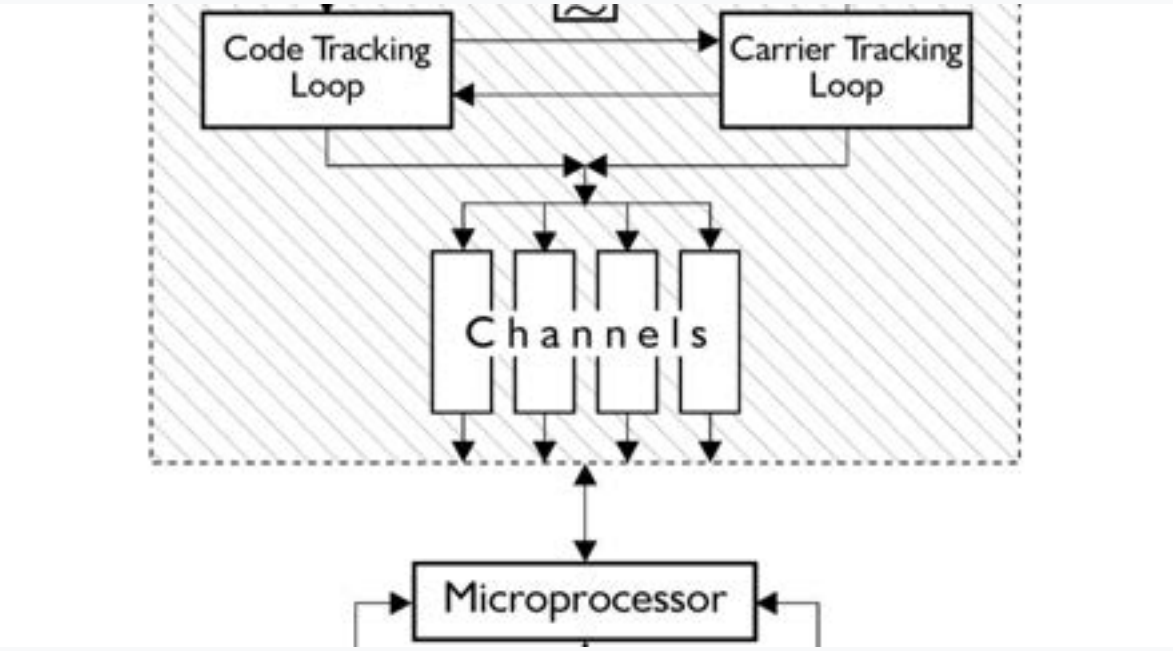
📐 Phase Difference



Phase Difference

As mentioned before, a replica of the C/A or P(Y) code is generated by the receiver's oscillator and that is correlated with the IF signal. It is at this point that the pseudorange is measured. Remember, the pseudorange is the time shift required to align the internally generated code with the IF signal, multiplied by the speed of light. The receiver also generates another replica, this time a replica of the carrier. That carrier is correlated with the IF signal and the shift in phase can be measured. The continuous phase observable, or observed cycle count, is obtained by counting the elapsed cycles since lock-on and by measuring the fractional part of the phase of the receiver generated carrier.

📡 Tracking Loops and Channels



Tracking Loops, Channels and the Microprocessor

We're not talking about just receiving one signal from one satellite but rather a minimum of four, and perhaps many more, from four to more satellites and the antenna itself does not sort the information it gathers. The signals from several satellites enter the receiver simultaneously. But in the *channels* of the RF section, the undifferentiated signals are identified and segregated from one another. There are usually several IF stages before the copies are sent into the separate channels, each of which extract the code and carrier information from a particular satellite. A channel in a continuous tracking GPS receiver is not unlike a channel in a television set. It is hardware, or a combination of hardware and software, designed to separate one signal from all the others. A receiver may have 6 channels, 12 channels, or hundreds of channels. At any given moment, one frequency from one satellite can have its own dedicated channel, and the channels operate in parallel. This approach allows the receiver to maintain accuracy when it is on a moving platform; it provides anti-jamming capability and shortens the time to first fix (TTFF). Each channel typically operates in one of two ways; working to acquire the signal or to track it. Once the signal is acquired, it is continuously tracked unless lock is lost. If that happens, the channel goes back to acquisition mode and the process is repeated.

While a parallel receiver has dedicated separate channels to receive the signals from each satellite that it needs for a solution, a *multiplexing (aka muxing)* receiver gathers some data from one satellite and then switches to another satellite and gathers more data and so on. Such a receiver can usually perform this switching quickly enough that it appears to be tracking all of the satellites simultaneously. A multiplexing receiver must still dedicate one frequency from one satellite to one channel at a time; it

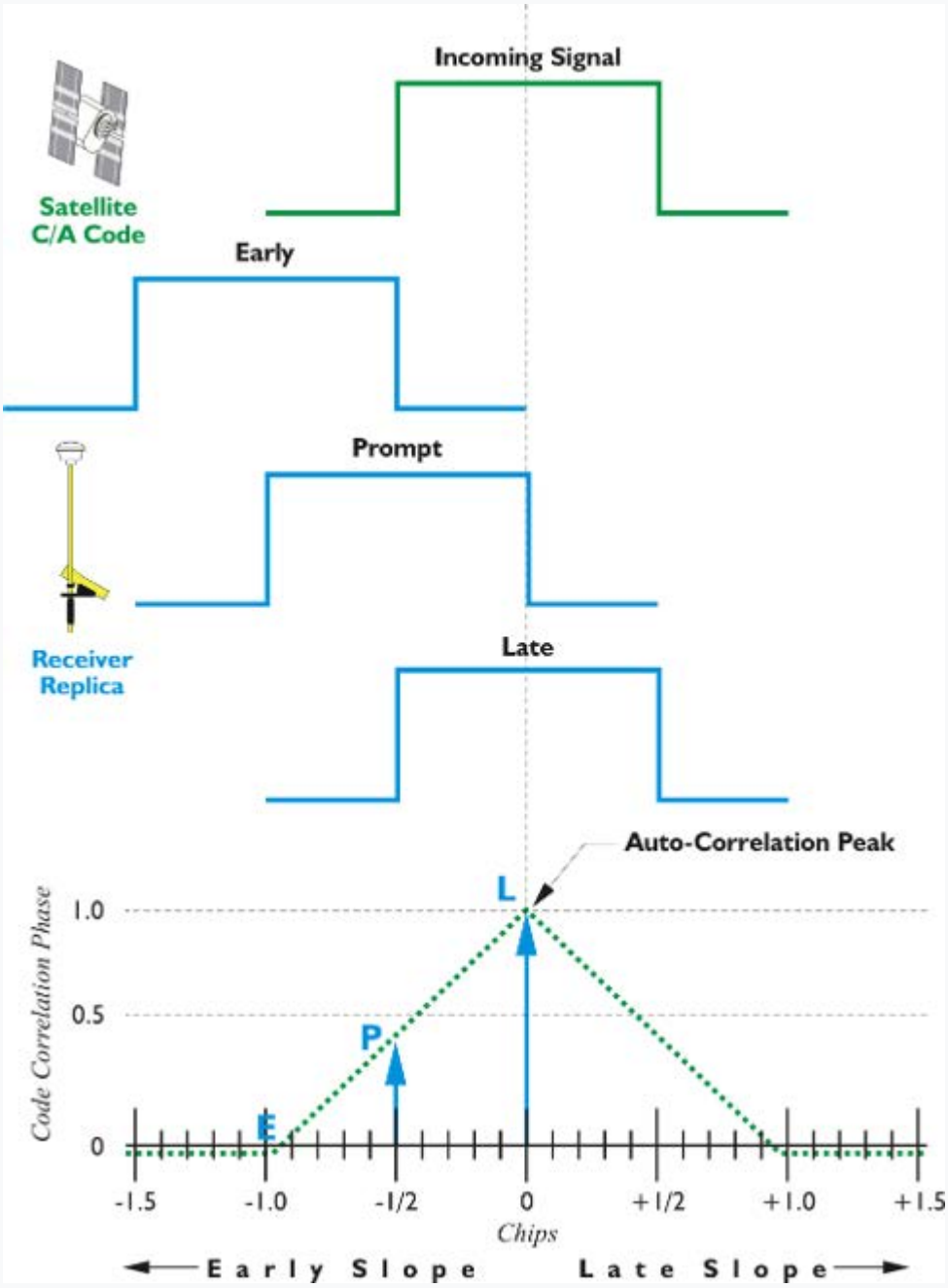
just makes that time very short. It typically switches at a rapid pace, i.e., 50 Hz. Even though multiplexing is generally less expensive, this strategy of switching channels is now little used. There are several reasons. While a parallel receiver does not necessarily offer more accurate results, parallel receivers with dedicated channels are faster; a parallel receiver has a more certain phase lock; and there is redundancy if a channel fails and they possess a superior signal-to-noise ratio (SNR). A multiplexing receiver also has a lower resistance to jamming and interference compared to continuous tracking receivers. Whether continuous or switching channels are used, a receiver must be able to discriminate between the incoming signals. They may be differentiated by their unique C/A codes on L1, their Doppler shifts, or some other method, but, in the end, each signal is assigned to its own channel.

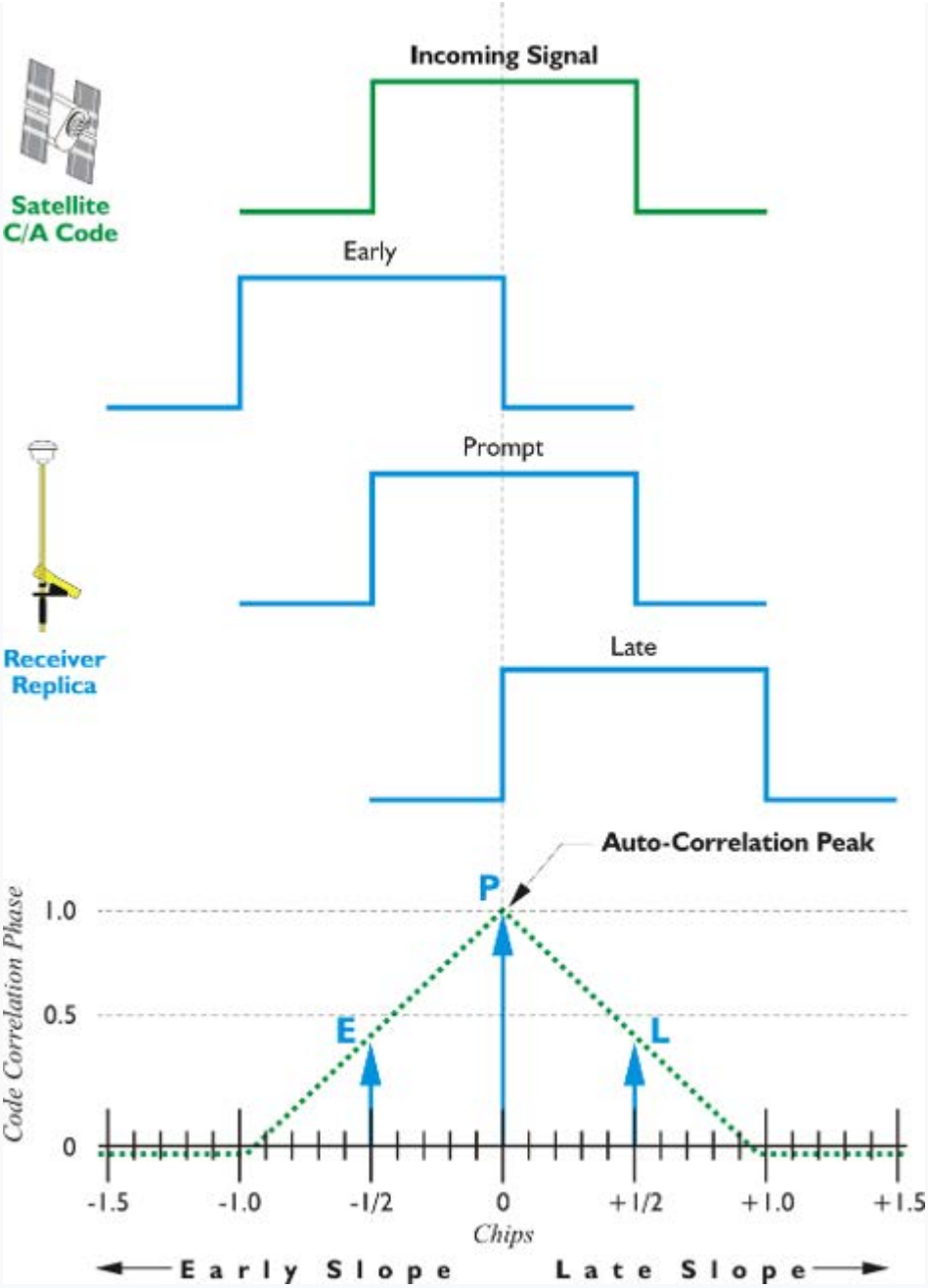
CONTENT

📚 Tracking Loops



📊 Correlation Peak


Incoming Signal
Satellite
C/A Code
Early
Prompt
Receiver
Replica
Late
Auto-Correlation Peak
L
P
E
1.0
0.5
0
Code Correlation Phase
-1.5
-1.0
-1/2
0
+1/2
+1.0
+1.5
Chips
Early Slope
Late Slope



Correlation Peak

There are *code tracking loops,* aka the *delay lock loops (DLL),* and *carrier tracking loops,* aka *the phase lock loops (PLL),* in the receiver. Typically, both the code and the carrier are being tracked in phase lock. The tracking loops connected to each of the receiver's channels also work cooperatively with each other. As mentioned, dual frequency receivers have dedicated channels and tracking loops for each frequency.

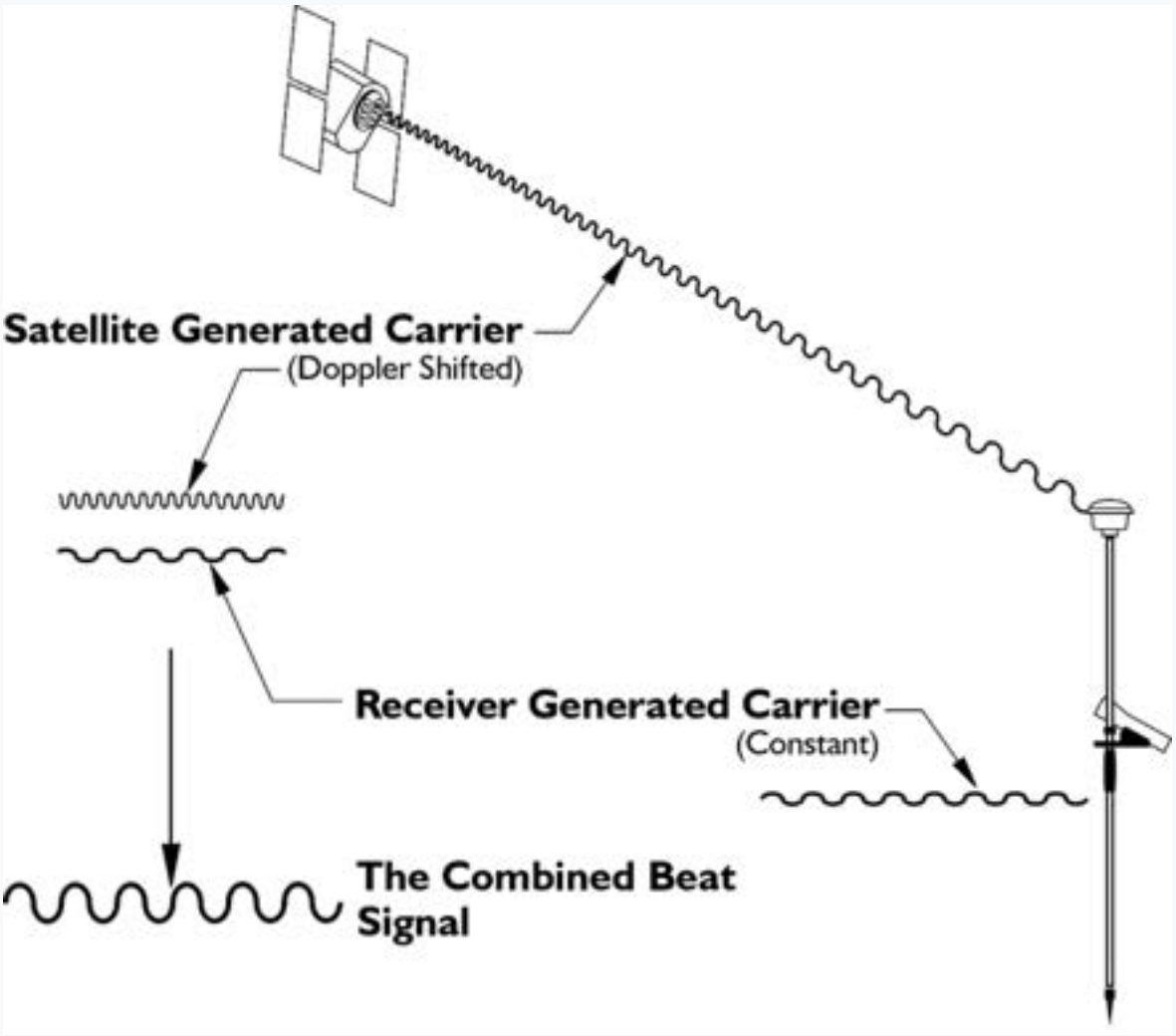
📡 Pseudoranging

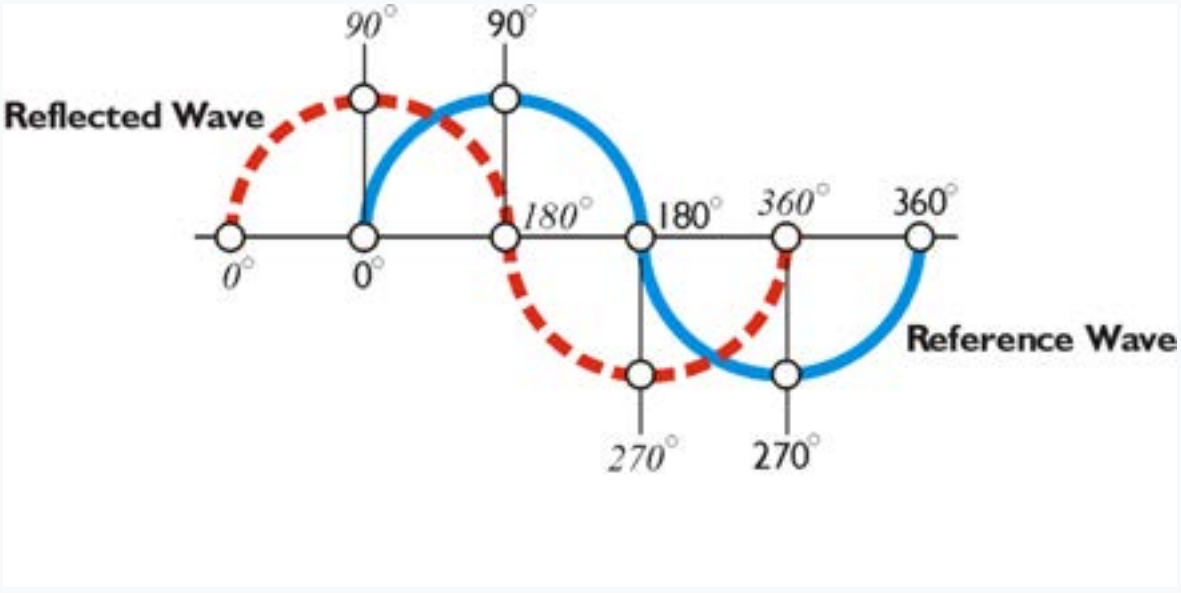
Pseudoranging

In most receivers, the first procedure in processing an incoming satellite signal is synchronization of the C/A code from the satellite's L1 broadcast, with a replica C/A code generated by the receiver itself, i.e., the *code-phase* measurement. When there is no initial match between the satellite's code and the receiver's replica, the receiver time shifts, or *slews*, the code it is generating until the optimum correlation is found. Then a code tracking loop, the delay lock loop, keeps them aligned. The time shift discovered in that process is a measure of the signal's travel time from the satellite to the phase center of the receiver's antenna. Multiplying this time delay by the speed of light gives a range. But it is called a pseudorange in recognition of the fact that it is contaminated by the errors and biases set out last week. The tracking loops that we see in the schematic diagram at the top are the code tracking loops.

Tracking loops are connected to each of the receiver's channels and also work cooperatively with each other. Multiple frequency receivers have dedicated channels and tracking loops for each frequency. Most receivers use the pseudorange range (the C/A code, on the L1) as the front door, so to speak, for the incoming satellite's synchronization. As we've discussed, the replica code is used to accomplish correlation. As you see in the diagram, the incoming signal is in the green at the top. The receiver replica is in blue. There is illustration of the replica as early, prompt, and late. When it is prompt, the replica and the incoming signal are correlated.

🔊 Combined Beat Signal and Phase Correlation


Satellite Generated Carrier
(Doppler Shifted)
Receiver Generated Carrier
(Constant)
The Combined Beat Signal


Reflected Wave
90°
90°
180°
180°
360°
360°
0°
0°
270°
270°
Reference Wave

Combined Beat Signal and Phase Correlation

📐 Carrier Phase Measurement

Carrier Phase Measurement

Once the receiver acquires the C/A code, it has access to the NAV message, or the newer navigation messages such as CNAV. It can read the ephemeris and the almanac information, use GPS time. But the code's pseudoranges, alone, are not adequate for the majority of applications. Therefore, the next step in signal processing for most receivers involves the carrier phase observable. As stated earlier, just as they produce a replica of the incoming code, receivers also produce a replica of the incoming carrier wave. And the foundation of carrier phase measurement is the combination of these two frequencies. Remember, the incoming signal from the satellite is subject to an ever-changing Doppler shift, while the replica within the receiver is nominally constant.

🔄 Carrier Tracking Loop

Carrier Tracking Loop

The process begins after the PRN code has done its job and the code tracking loop is locked. By mixing the satellite's signal with the replica carrier, this process eliminates all the phase modulations, strips the codes from the incoming carrier, and simultaneously creates two intermediate or beat-frequencies—one is the sum of the combined frequencies, and the other is the difference. The receiver selects the latter, the difference, with a bandpass filter. Then, this signal is sent on to the *carrier tracking loop* also known as the phase lock loop, PLL, where the voltage-controlled oscillator is continuously adjusted to follow the beat frequency exactly. This is basically how a GPS receiver locks on to that carrier and stays locked unless there is a loss of signal or a cycle slip.

CONTENT

📚 Doppler Shift

Doppler Shift in GPS

GISC 4315

📡 Typical Doppler Shift

Typical Doppler Shift

We've talked about the Doppler shift in several different contexts. One was the original transit system, NNSS system, that operated on the Doppler shift. As the satellite passes overhead, the range between the receiver and the satellite changes; that steady change is reflected in a smooth and continuous movement of the phase of the signal coming into the receiver. GPS uses the Doppler shift as an observable. It has broad applications in signal processing. It can be used to discriminate between the signals from various GPS satellites, to determine integer ambiguities in kinematic surveying, as a help in the detection of cycle slips, and as an additional independent observable for autonomous point positioning. But perhaps the most important application of Doppler data is the determination of the *range rate* between a receiver and a satellite. *Range rate* is a term used to mean the rate at which the range between a satellite and a receiver changes over a particular period of time.

📈 Typical Change in Doppler Shift

Typical Change in the Doppler Shift

This graphic shows the typical rate of change in the Doppler shift with regard to a stationary, static, GPS receiver. The signal received would have its maximum Doppler shift, +4 1/2 to 5 cycles per millisecond, when the satellite is at its maximum range, just as it is rising or setting. The Doppler shift

continuously changes throughout the overhead pass. Immediately after the satellite rises, relative to a particular receiver, its Doppler shift gets smaller and smaller, until the satellite reaches its closest approach, at zenith. At that moment its radial velocity with respect to the receiver is zero, the Doppler shift of the signal is zero as well. But as the satellite recedes, it grows again, negatively, until the Doppler shift once again reaches its maximum extent just as the satellite sets, -4 1/2 to 5 cycles per millisecond. It is very predictable. That predictability, the constant variation of the signal's Doppler shift, makes it a good observable. If the receiver's oscillator frequency is adjusted to match these variations exactly, as they're happening, it will duplicate the incoming signal's shift and phase. This strategy of making measurements using the carrier beat phase observable is a matter of counting the elapsed cycles and adding the fractional phase of the receiver's own oscillator. This is one way that the phase lock loop maintains its lock on the signal as the Doppler shift occurs with each of the satellites that it is tracking.

With respect to the receiver, the satellite is always in motion even if the receiver is *static*. But the receiver may be in motion in another sense, as it is in kinematic GPS. The ability to determine the instantaneous velocity of a moving vehicle has always been a primary application of GPS, and is based on the fact that the Doppler-shift frequency of a satellite's signal is nearly proportional to its range rate. The Doppler shift can be used to discriminate between the signals of the various satellites to help in the determination of the integer ambiguity. It can help in detection of a loss of lock due to a cycle slip.

🔄 Continuously Integrated Doppler

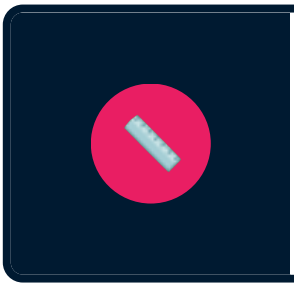
Continuously Integrated Doppler

The Doppler-shift and the carrier phase are measured by first combining the received frequencies with the nominally constant reference frequency created by the receiver's oscillator. The difference between the two is the often mentioned *beat frequency*, an intermediate frequency, and the number of beats over a given time interval is known as the Doppler count for that interval. Since the beats can be counted much more precisely than their continuously changing frequency can be measured, most GPS receivers just keep track of the accumulated cycles, the *Doppler count*. The sum of consecutive Doppler counts from an entire satellite pass is often stored, and the data can then be treated like a sequential series of biased range differences. *Continuously Integrated Doppler* is such a process. The rate of the change in the continuously integrated Doppler shift of the incoming signal is the same as that of the reconstructed carrier phase. Integration of the Doppler frequency offset results in an accurate measurement of the advance in carrier phase between epochs. Using double-differences in processing the carrier phase observables removes most of the error sources other than multipath and receiver noise.

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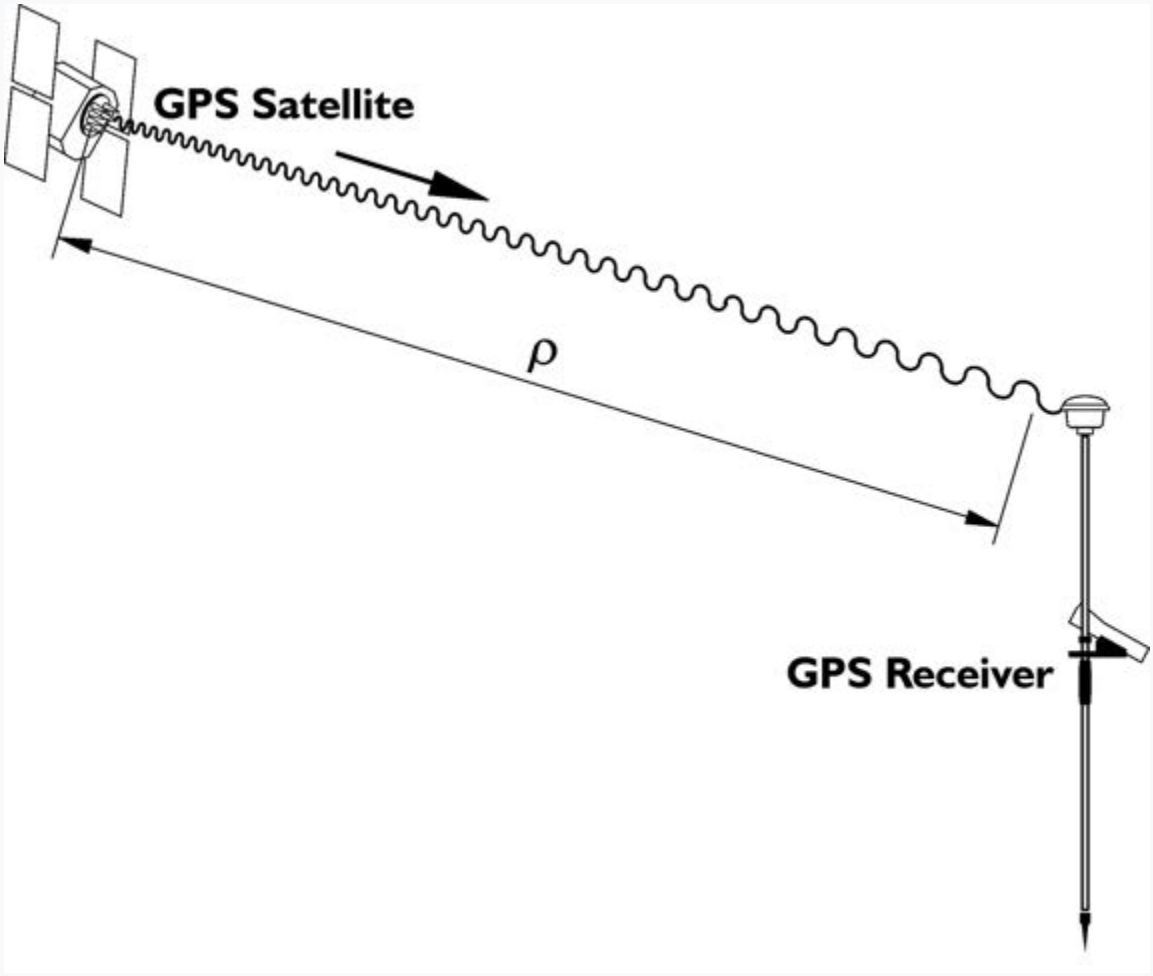
📚 The Integer Ambiguity



One-Way Ranging and Signal Processing

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📐 One-Way Ranging



One-Way Ranging

The solution of the integer ambiguity, the number of whole cycles on the path from satellite to receiver, would be more difficult if it was not preceded by pseudoranges, or code phase measurements in most receivers. This allows the centering of the subsequent double-difference solution. In other words, a pseudorange solution provides an initial estimate of the candidates for the integer ambiguity within a smaller range than would otherwise be the case, and, as more measurements become available, it can reduce them even further. After the code-phase measurements narrows the field, there are several methods used to solve the integer ambiguity. In the geometric method, the carrier phase data from multiple epochs are processed, and the constantly changing satellite geometry is used to find an estimate of the actual position of the receiver. This approach is also used to show the error in the estimate by calculating how its results hold up as the geometry of the constellation changes. This strategy requires a significant amount of satellite motion

to succeed, and, therefore, takes time to converge on a solution. It works pretty well, but requires satellite motion and takes time to converge. Another approach to solving the integer ambiguity is filtering. Independent measurements are averaged to find the estimated position with the lowest noise level. A third uses a search through the range of possible integer ambiguity combinations from which it calculates the one with the lowest residuals. These approaches can't assess the correctness of the particular answer, but they can provide the probability with certain conditions, that the answer is within given limits. Most GPS receivers use a combination of methods. Nearly all narrow the field by beginning with an initial position established by the code phase measurements. They then use one or more of the methods in combination to come up with the most probable value for the solution of the integer ambiguity, the N, the number of full wave cycles between the receiver and the satellite at lock on, the key to carrier phase observations.

🔄 Signal Squaring

Signal Squaring

There is a method that does not use the codes carried by the satellite's signal. It is called codeless tracking, or *signal squaring*. It was first used in the earliest civilian GPS receivers, supplanting proposals for a TRANSIT-like Doppler solution. It makes no use of pseudoranging and relies exclusively on the carrier phase observable. Like other methods, it also depends on the creation of an intermediate or beat frequency. But with signal squaring, the beat frequency is created by multiplying the incoming carrier by itself. The result has double the frequency and half the wavelength of the original. It is squared. There are some drawbacks to the method. For example, in the process of squaring the carrier, it is stripped of all its codes. The chips of the P(Y) code, the C/A code, and the Navigation message normally modulated onto the carrier by 180° phase shifts are eliminated entirely. As discussed earlier, the signals broadcast by the satellites have phase shifts called *code states* that change from +1 to –1 and vice versa, but squaring the carrier converts them all to exactly 1. The result is that the codes themselves are wiped out. Therefore, this method must acquire information such as almanac data and clock corrections from other sources. Other drawbacks of squaring the carrier include the deterioration of the signal-to-noise ratio, because when the carrier is squared, the background noise is squared, too. And cycle slips occur at twice the original carrier frequency. But signal squaring has its up-side as well. It reduces susceptibility to multipath. It has no dependence on PRN codes and is not hindered by the encryption of the P code. The technique works as well on L2 as it does on L1 or L5, and that facilitates ionospheric delay correction. Therefore, signal squaring can provide high accuracy over long baselines. So, there is a cursory look at some of the different techniques used to process the signal in the RF section. Now let's look at the microprocessor of the receiver.

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CONTENT

📚 The Microprocessor



GPS Microprocessor and Positioning Methods

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🖥️ GPS Microprocessor



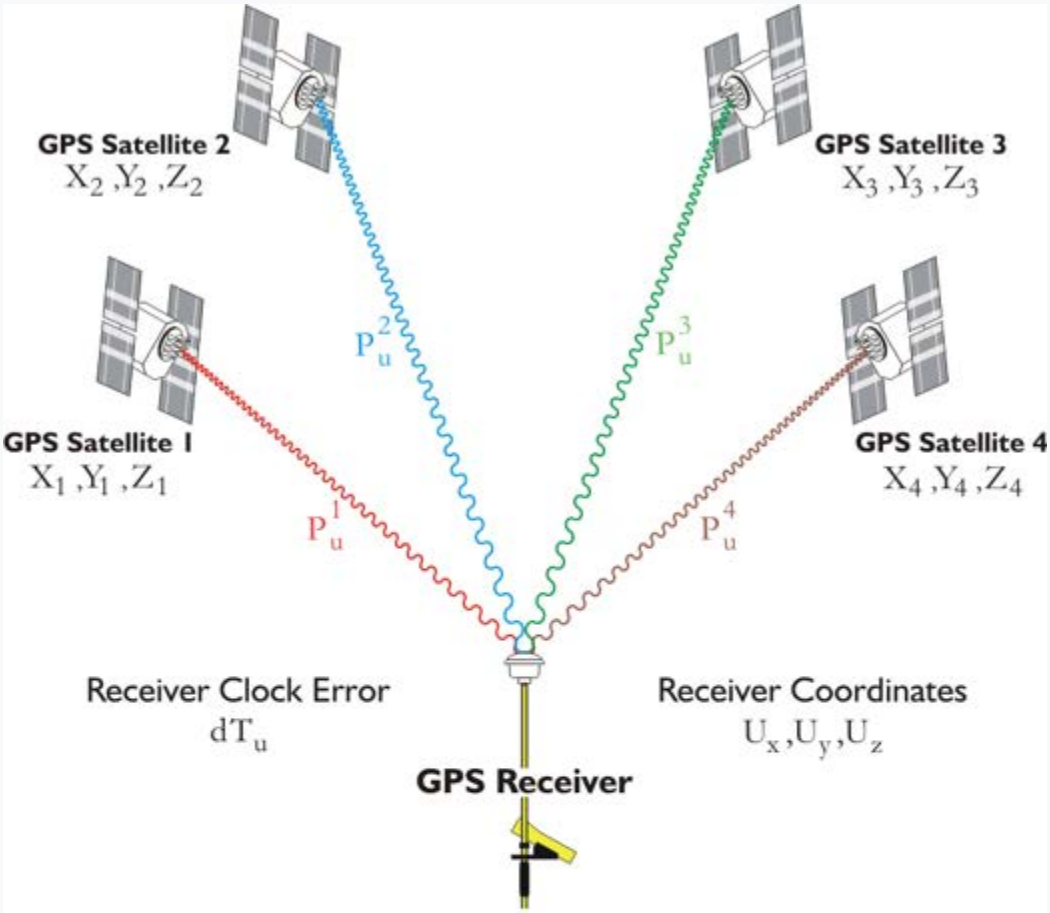
GPS Board

The microprocessor in a GPS receiver is the computer that manages data collection and is the home of the applications that mitigate multipath, noise, extract the ephemerides and other information from the Navigation message or newer navigation messages such as CNAV. It controls the entire receiver: the digital circuits, the tracking, and measurements. The receiver also has storage. However, more and more, the microprocessor is expected to produce the position in real time, instantaneously, or

near real-time by processing the ranging data, doing reference frame (datum) conversion, and sending the position to the control and display unit (CDU). There is a two-way street between the microprocessor and the CDU; each can receive information from or send information to the other.

📍 Differential Positioning

Differential Positioning



$$p_u^1 = \sqrt{(X_1 - U_x)^2 + (Y_1 - U_y)^2 + (Z_1 - U_z)^2} + c(dT_u)$$

$$p_u^2 = \sqrt{(X_2 - U_x)^2 + (Y_2 - U_y)^2 + (Z_2 - U_z)^2} + c(dT_u)$$

$$p_u^3 = \sqrt{(X_3 - U_x)^2 + (Y_3 - U_y)^2 + (Z_3 - U_z)^2} + c(dT_u)$$

$$p_u^4 = \sqrt{(X_4 - U_x)^2 + (Y_4 - U_y)^2 + (Z_4 - U_z)^2} + c(dT_u)$$

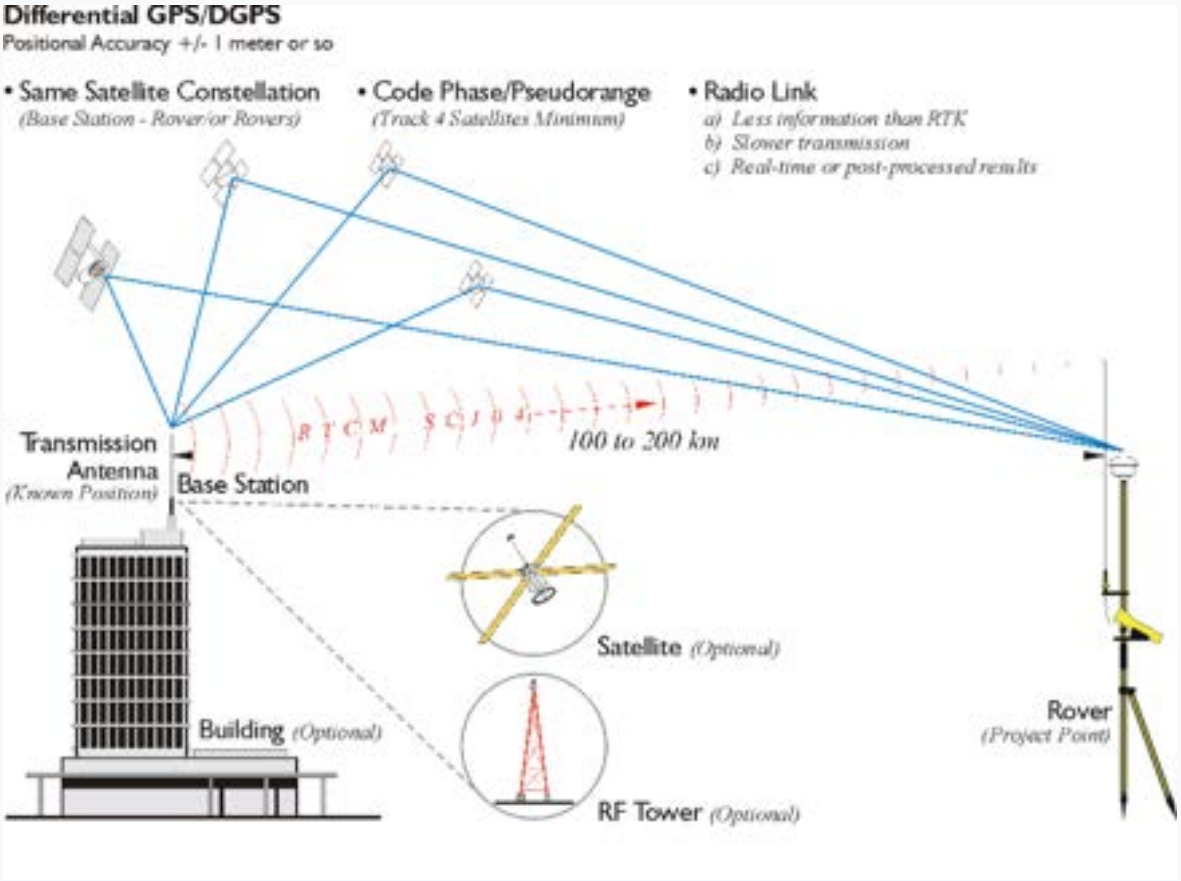
Autonomous GPS

Autonomous single-point positioning using unsmoothed code pseudoranges are based upon corrections provided in the navigation messages, which, as you know, have a certain amount of

weakness. Though they have improved since Selective Availability has been switched off, they are generally not as accurate as differentially corrected positions. Code-based pseudoranges using DGPS, differential GPS, can achieve good real-time, or post-processed results. DGPS is often used in collecting data for *Geographical Information Systems, GIS*.

DGPS

DGPS



Differential GPS (DGPS)

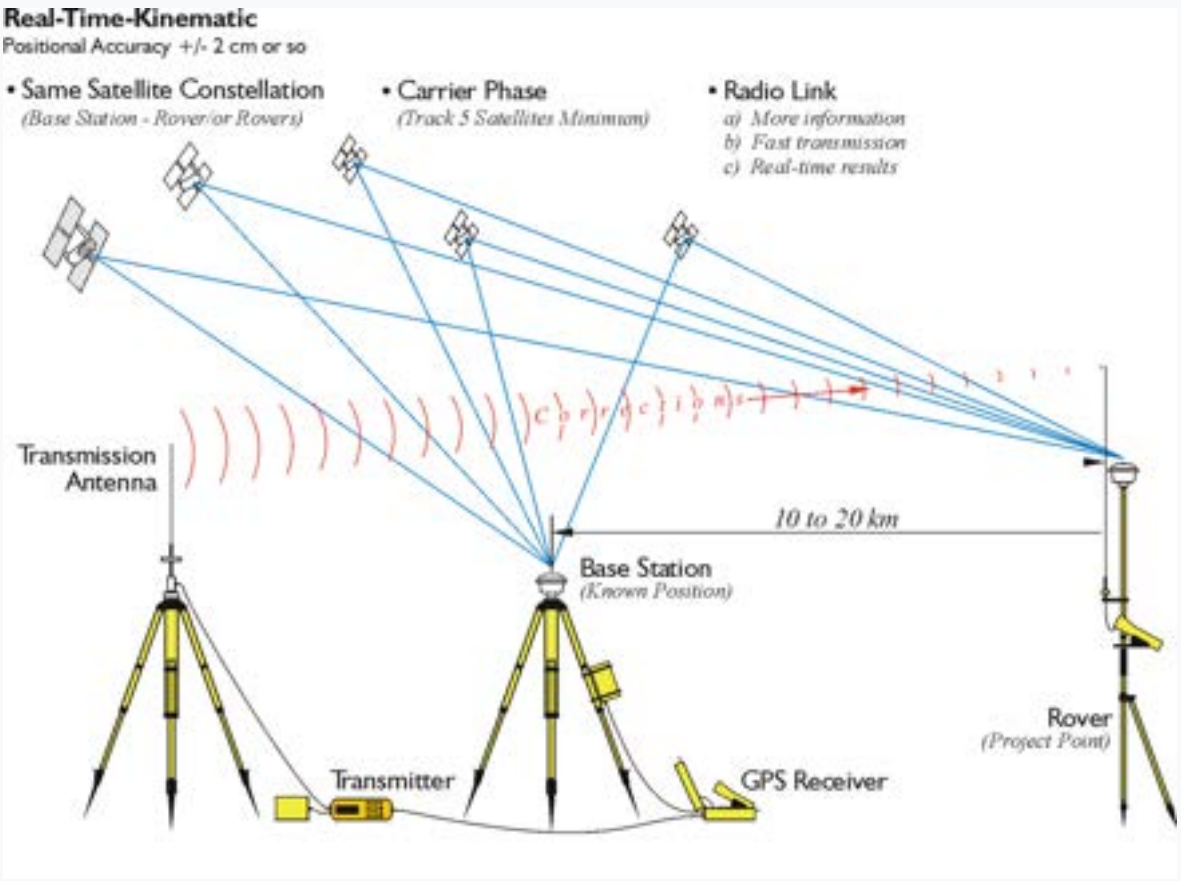
The type of differential positioning sometimes known as DGPS depends on code pseudorange observations, but requires at least two receivers. One receiver is placed on a control station, the base, and another on an unknown position, the rover. The base station antenna needn't be on a building, as illustrated. It could be on a tower. It could be on a non-GPS satellite. There are services

that allow you to subscribe to a correction signal. In those cases, there is a network of base stations collecting the signals from the constellation; the correction signal is compiled, beamed up to the non-GPS satellite, and that same message sent back down from the non-GPS satellite to subscribers. In all configurations, the base and the rover simultaneously receive the same signals from the same constellation of at least four GPS/GNSS satellites. That's important in that the way many of the errors in the observations are common to both the base and the rover receivers, the errors are correlated and tend to cancel each other to some degree. The data from such an arrangement can be post-processed; although, with a link between the base and the rover, a correction signal can be sent to the rover so that differential results can be had in real-time. Improvements in this technology have refined the technique's accuracy markedly, and meter- or even submeter results are possible. Still, the positions are not as reliable as those achieved with the carrier phase observable.

With GIS, corner search and mapping work excepted, much GPS work requires a higher standard of accuracy. Certainly, GPS control surveying often employs several static receivers that simultaneously collect and store carrier phase data from the same satellites for a period of time known as a *session*. After all the sessions for a day are completed, their data are usually downloaded in a general binary format to the hard disk of a PC for post-processing. However, carrier phase positioning can also be done in real-time.

🚀 Real-Time Kinematic

Real-Time Kinematic



Real-Time Kinematic (RTK)

Real-time kinematic surveying uses the carrier phase solution. Often, there is a radio link between the base and rover. While the baseline length with DGPS is often 100 to 200 kilometers or longer, the baseline length in RTK is more typically 10 to 20 kilometers, often less. However, the arrangement of receivers is similar: the constellation of satellites being tracked at the base station and also tracked at the rover, the base station is at a known point. There is a transmission antenna used in RTK. It transmits the correction signals to the rover in real time correction. While the arrangement is similar to DGPS, the solution is carrier phase as opposed to pseudorange. The minimum number of satellites with RTK is five. It is five, so that one can be lost and still assured that there will be a solution.

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CONTENT

📚 The Control and Display Unit

Control and Display Unit

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📱 Control and Display Unit (CDU)



Control and Display Unit (CDU)

A GPS receiver will often have a control and display unit. From handheld keyboards to soft keys around a screen to digital map displays and interfaces to other instrumentation, there are a variety of configurations. Nevertheless, they all have the same fundamental purpose, facilitation of the interaction between the operator and the receiver's microprocessor. A CDU typically displays status, position data, velocity, and time. It may also be used to select different surveying methods, waypoint navigation, and/or set parameters such as epoch interval, mask angle, and antenna height. The CDU can offer a combination of help menus, prompts, reference frame (datum) conversions, readouts of survey results, estimated positional error, and so forth. The information available from the CDU varies from receiver to receiver. But when four or more satellites are available, they can generally be expected to display the PRN numbers of the satellites being tracked, the receiver's position in three dimensions, and velocity information. Most of them also display the dilution of precision and GPS time or UTC.

CONTENT

📚 The Storage



💾 Internal Data Logging

Most GPS receivers today have internal data logging. The amount of storage required for a particular session depends on several things: the length of the session, the number of satellites above the horizon, the epoch interval, and so forth. For example, presuming the amount of data received from a single GPS satellite is ~100 bytes per epoch, a typical twelve channel dual-frequency receiver observing 6 satellites and using a 1-second epoch interval over the course of a 1-hour session would require ~2MB of storage capacity for that session. The miniaturization of storage continues. The cassettes, floppy disks, and drives used with the Macrometer are past. Extraordinary amounts of data can be stored in small, convenient devices or sent to the cloud.

📚 The Power



⚡ Power Requirements and Operation

It is fortunate that GPS receivers operate at low power. From 9 to 36 volts DC is generally required. This allows longer observations with fewer, and lighter, batteries than might be otherwise required. It also increases the longevity of the GPS receivers, themselves. About half of the available GPS carrier phase receivers have an internal power supply, and most will operate 5½ hours or longer on a fully charged 6-amp-hour battery. Most code-tracking receivers, those that do not also use the carrier phase observable, could operate for about 15 hours on the same size battery. Since most receivers in the field operate on battery power, batteries and their characteristics are fundamental to GPS/GNSS. A variety of batteries are used, and there are various configurations. For example, some units are powered by rechargeable batteries. Lithium, Nickel Cadmium, and Nickel Metal-Hydride may be the most common categories, but lead-acid car batteries still have an application as well. The obvious drawbacks to lead-acid batteries are size and weight. And there are a few others—the corrosive acid, the need to store them charged, and their low cycle life. Nevertheless, lead-acid batteries are especially hard to beat when high power is required. They are economical and long-lasting.

🔋 Battery Technologies and Characteristics

Nickel Cadmium batteries (NiCd) cost more than lead-acid batteries, but are small and operate well at low temperatures. Their capacity does decline as the temperature drops. Like lead-acid batteries, NiCd batteries are quite toxic. They self-discharge at the rate of about 10% per month, and even though they do require periodic full discharge, these batteries have an excellent cycle life. Nickel Metal-Hydride (NiMH) batteries self-discharge a bit more rapidly than NiCd batteries and have a less robust cycle life, but are not as toxic. Lithium–ion batteries overcome several of the limitations of the others. They have a relatively low self-discharge rate. They do not require periodic discharging and

do not have a memory issues as do NiCd batteries. They are light, have a good cycle life and low toxicity. On the other hand, the others tolerate overcharging, and the lithium-ion battery does not. It is best to not charge lithium-ion batteries at temperatures at or below freezing. These batteries require a protection circuit to limit current and voltage, but are widely used in powering electronic devices, including GPS/GNSS receivers.

CONTENT

📚 Some GPS Surveying Methods



📡 Static GPS Surveying Fundamentals

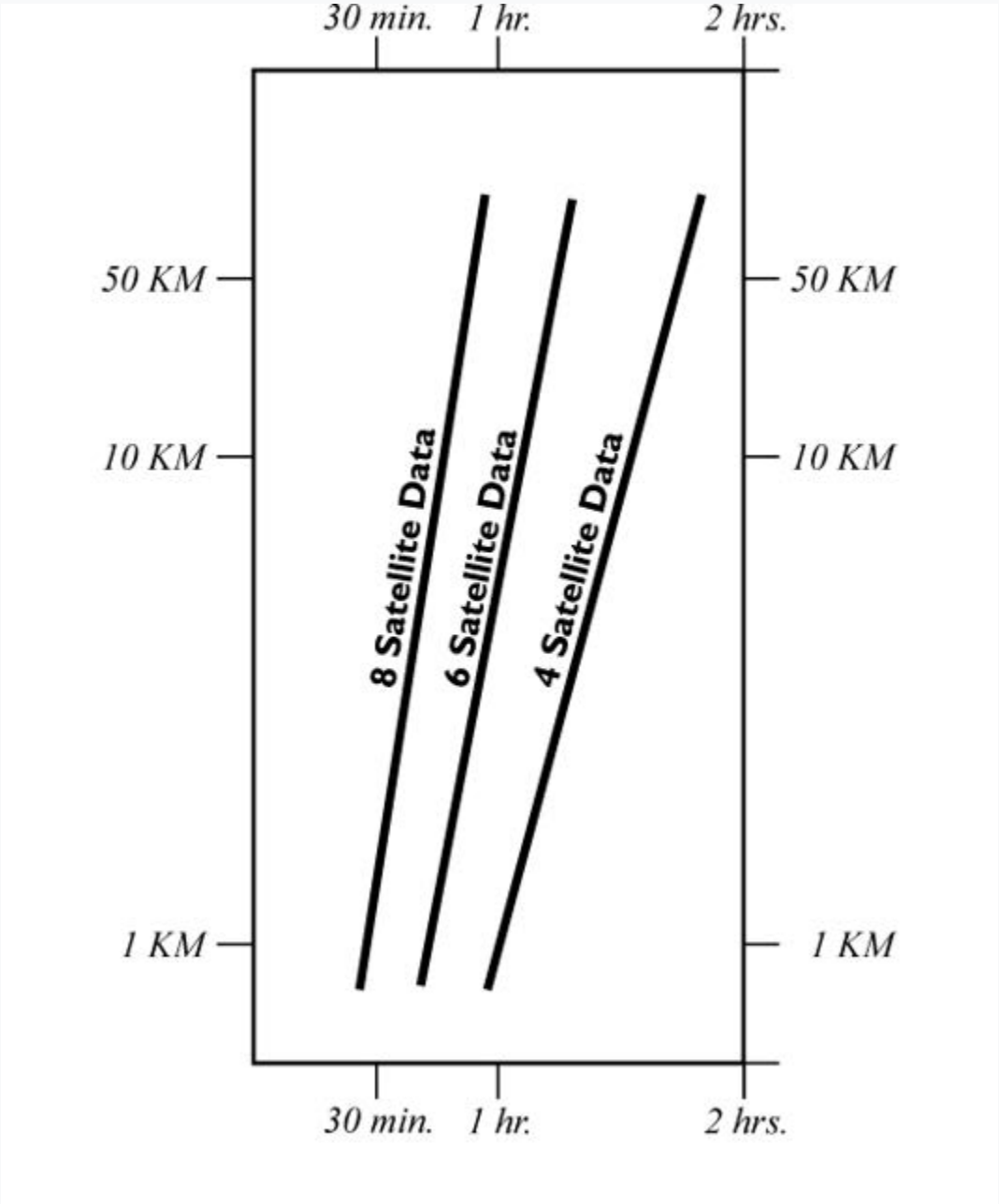


GPS Receiver in Senegal

Static GPS surveying was the first method of GPS surveying used in the field, and it continues to be the primary technique for GPS/GNSS control today. Relative static positioning involves several

stationary receivers simultaneously collecting data from at least four satellites during observation sessions that usually last from 30 minutes to 2 hours. A typical application of this method would be the determination of vectors, or baselines as they are called, between several static receivers to accuracies from 1 ppm to 0.1 ppm over tens of kilometers. There are few absolute requirements for relative static positioning. The requisites include: more than one receiver, four or more satellites, and a mostly unobstructed sky above the stations to be occupied. But, as in most of surveying, the rest of the elements of the system are dependent on several other considerations.

⏱️ Session Length Requirements



Static Sessions Lengths

Here is a diagram of approximate session lengths required for GPS static surveying. For example, with an eight satellite constellation and a 10 kilometer baseline, something between a half an hour and an hour would typically be required. A four satellite constellation might require a session between one and two hours for a baseline of the same length. The assessment of the productivity of a GPS

survey almost always hinges, in part at least, on the length of the observation sessions required to satisfy the survey specifications. The determination of the session's duration depends on several particulars, such as the length of the baseline and the relative position, that is the geometry, of the satellites among others. Generally speaking, the larger the constellation of satellites, the better the available geometry, the lower the positioning dilution of precision (PDOP), and the shorter the length of the session needed to achieve the required accuracy. For example, given six satellites and good geometry, baselines of 10 km or less might require a session of 45 minutes to 1 hour, whereas, under exactly the same conditions, baselines over 20 km might require a session of 2 hours or more. Alternatively, 45 minutes of six-satellite data may be worth an hour of four-satellite data, depending on the arrangement of the satellites in the sky.

🔗 Receiver Compatibility and Data Exchange

Relative static positioning, just as all the subsequent surveying methods discussed here, involves several receivers occupying many sites. Problems can be avoided as long as the receivers on a project are compatible. For example, it is helpful if they have the same number of channels and signal processing techniques. The subject comes up from time to time as to whether or not data from receivers from various manufacture can be used on the same project together at the ends of even the same baseline. The answer is, yes of course because fortunately, there is a file format, that allows one to use receivers of many types together. It is the Receiver Independent Exchange Format, RINEX , developed by the Astronomical Institute of the University of Berne in 1989. It allows different receivers and post-processing software to work together. Almost all GPS/GNSS processing software will output RINEX files.

CONTENT

📚 Receiver Categories



📡 Receiver Classification System



Receiver Categories

Receivers are generally categorized by their physical characteristics, the elements of the GPS signal they can use with advantage, and by the claims about their accuracy. There are receivers that use only the C/A code on the L1 frequency and receivers that cross-correlate with the P(Y). There are L1 carrier phase tracking receivers, dual-frequency and multi-frequency carrier phase tracking receivers, receivers that track all in view, and GPS/GNSS receivers. The more aspects of the GPS signal a receiver can employ, the greater its flexibility, but so, too, the greater its cost. It is important to understand receiver capabilities and limitations to ensure that the systematic capability of a receiver is matched to the required outcome of a project. As shown in the illustration, it is possible to divide receivers into three categories. They are: recreation, mapping, and surveying.

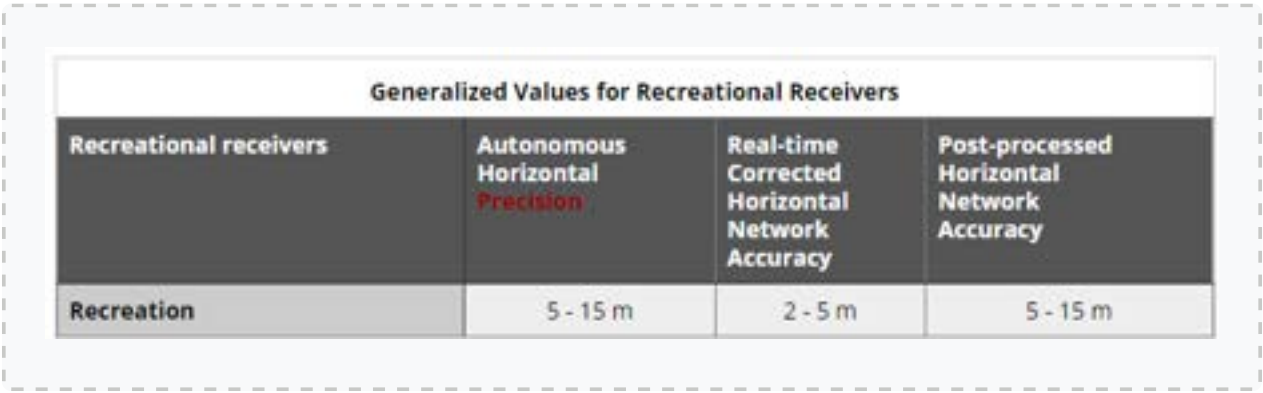
CONTENT

📚 Recreational Receivers

Recreation Receivers

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📊 Basic Characteristics and Capabilities



Generalized Values for Recreational Receivers

Recreational receivers	Autonomous Horizontal Precision	Real-time Corrected Horizontal Network Accuracy	Post-processed Horizontal Network Accuracy
Recreation	5 - 15 m	2 - 5 m	5 - 15 m

Generalized Values for Recreational Receivers

Recreation Receivers

These receivers are generally defined as L1 Code receivers which are typically not user configurable for settings such as mask angle, PDOP, the rate at which measurements are downloaded, the *logging rate*, also known as the epoch interval, and *signal to noise ratio SNR*. As you might expect, SNR is the ratio of the received signal power to the noise floor of a GPS observation. It is typical for the antenna, receiver, and CDU to be integrated into the device in these receivers. Generally speaking, receivers that track the C/A code provide only relatively low accuracy. Most are not capable of tracking the carrier phase observable. These receivers were typically developed with basic navigation in mind. Most are designed for autonomous (stand-alone) operation to navigate, record tracks, waypoints, and routes aided by the display of onboard maps. They are sometimes categorized by the number of waypoints they can store. *Waypoint* is a term that grew out of military usage. It means the coordinate of an intermediate position a person, vehicle, or airplane must pass to reach a desired destination. With such a receiver, a user may call up a distance and direction from his present location to the next waypoint.

📡 Augmentation Systems and Precision

A single receiver operating without augmentation produces positions which are not relative to any ground control, local or national. In that context, it is more appropriate to discuss the precision of the results than it is to discuss accuracy. Despite the limitations, some recreational receivers have capabilities that enhance their systematic precision and achieve a quantifiable accuracy using correction signals available from earth-orbiting satellites such as the *Wide Area Augmentation System, WAAS* correction. In other words, some have differential capability. The Wide Area Augmentation System is a *U.S. Federal Aviation Authority FAA* and the *U.S. Department of Transportation DOT* system that augments GPS accuracy, availability, and integrity. The system relies on a network of ground-based reference stations at known positions that observe the GPS constellation constantly. From their data, a correction message is calculated at two master stations. This message is uploaded to satellites on geostationary orbits. The satellites broadcast the message, and WAAS-enabled receivers can collect it and use the correction provided in it. For example, it provides a correction signal for precision approach aircraft navigation. Similar systems are Europe's *European Geostationary Navigation Overlay System, EGNOS*, and Japan's *Multifunction Transport Satellite, MTSAT*.

⚠️ Limitations and Performance Ranges

Recreational grade receivers typically do not have on-board feature data collection capabilities. They also do not usually have adequate on-board storage for recording the features (coordinates and attributes) required for a mapping project. Such capabilities are not needed for their designed applications. When a recreation receiver is used to obtain an autonomous, or stand alone, position, its precision may be within a range of 5-15m, as noted in the table above. Under less than optimal field conditions: tree cover and other obstructions, less than favorable GPS satellite geometry, etc., users can expect the precision of autonomous positions to lessen, sometimes substantially. The network accuracy had with real-time differential correction of 2-5m is also not always achievable due to the tendency of the Wide Area Augmentation Signal being difficult to acquire, particularly in the northern US and then only with a southern sky clear of obstruction.

📚 Local and Network Accuracy



📐 Accuracy Concepts and Definitions

Local and Network Accuracy

Local and Network Accuracy

The use of the phrase *network accuracy* is used to define its difference from local accuracy. *Network accuracy,* here, concerns the uncertainty of a position relative to a reference frame (datum). *Local accuracy* is not about a position relative to a reference frame (datum), but it represents the uncertainty of a position relative to other positions nearby. In other words, *local accuracy* would be useful in knowing the accuracy of a line between the two positions at each end. *Network accuracy* would not be about the accuracy of the positions at each end of the line relative to each other, but, rather, relative to the whole reference frame (datum). *Local Accuracy* is also known as *relative accuracy,* and *network accuracy* is also known as *absolute accuracy*.

🌍 Practical Applications and Control Networks

Within a well-defined geographical area, local accuracy may be the most immediate concern. However, those tasked with constructing a control network that embraces a wide geographical scope will most often need to know the position's relationship to the realization of the reference frame (datum) on which they are working. A point with good local accuracy may not have good network accuracy. Typically, network horizontal and vertical accuracies require that a point's accuracy be specified with respect to an appropriate national geodetic datum. In the United States, as a practical

matter, this most often means that the work is tied to at least one of the Continuously Operating Reference Stations (CORS) which represent the most accessible realization of the National Spatial Reference System (NSRS) in the nation.

Mapping Receivers

Mapping Receivers

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Mapping Receiver Capabilities and Applications

Mapping Receivers	Autonomous Horizontal Precision	Real-time Corrected Horizontal Network Accuracy	Post-processed Horizontal Network Accuracy
Mapping (L1 Code)	2 - 10m	0.5 - 5m	0.3 - 15m
Mapping (L1 Code & Carrier)	2 - 10m	0.5 - 3m	0.2 - 1m
Mapping (All GPS Code & Carrier)	2 - 10m	0.5 - 3m	0.02 - 0.9m
Mapping (GNSS)	2 - 10m	0.5 - 3m	0.02 - 0.9m

Generalized Values for Mapping Receivers

Mapping Receivers

These receivers are generally defined as those that allow the user to configure some settings such as PDOP, SNR, elevation mask and the logging rate. They most often have an integrated antenna and CDU in the receiver. They generally record pseudoranges and also can log data suitable for differential corrections either in real-time or for post-processing; many record carrier data. Mapping receivers are often capable of storing mapped features (coordinates and attributes) and usually have adequate capacity for mapping applications. This memory is required for differential GPS, DGPS receivers, even those that track only code. For many applications, a receiver must be capable of collecting the same information as is simultaneously collected at a base station, and storing it for post-processing. Receivers typically depend on proprietary post-processing software, which also includes utilities to enable GPS data to be transferred to a PC and exported in standard GIS file format(s) either over a cable or a wireless connection. Some mapping grade receivers for spatial data

collection are single frequency with code only or both code and carrier. Most mapping grade receivers of all tracking configurations are WAAS (or other *Satellite Based Augmentation Signal SBAS*) enabled, and thereby offer real-time results. Such differentially corrected mapping receivers may be capable of achieving a network accuracy of ~0.5 to 5 meters. Positions of sub-meter post-processed network accuracy can be achieved with mapping grade receivers. As noted, some mapping receivers offer tracking of the *Global Navigation Satellite System, GNSS*, constellations.

CONTENT

Surveying Receivers

Surveying Receivers

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Survey Grade Receiver Capabilities

Survey Receivers	Autonomous Horizontal Precision	Real-time Corrected Horizontal Network Accuracy	Post-processed Horizontal Network Accuracy
Survey	2 - 10 m	> 1 m	> 0.1 m

Generalized Values for Surveying Receivers

Surveying Receivers

Survey grade receivers are designed for the achievement of consistent network accuracy in the static or real-time mode. Positions determined by these receivers will generally provide the best accuracy of the categories listed. The components of these receivers can usually be configured in a variety of ways. The receivers are typically multi-frequency, code and carrier receivers. Many are GNSS receivers. They generally provide more options for setting the observational parameters than recreational or mapping receivers. Surveying receivers are typically capable of observing the civilian code and carrier phase of all frequencies, and are appropriate for collecting positions on long baselines. Survey grade receivers are capable of producing network accuracies of better than 1 m with real-time differential correction and better than 0.1 m with post-processing.

🔧 Practical Characteristics and Performance

Most share some practical characteristics: they have multiple independent channels that track the satellites continuously, and they begin acquiring satellites' signals from a few seconds to less than a minute from the moment they are switched on. Most acquire all the satellites above their mask angle in a very few minutes, with the time usually lessened by a warm start, and most provide some sort of alert to the user that data is being recorded, and so forth. About three-quarters of them can have their sessions preprogrammed in the office before going to their field sites. Nearly all allow the user to select the logging rate, also known as epoch interval and also known as sampling interval. While a 1 second interval is often used, faster rates of 0.1 second (10 Hz) and more increments of tenths of a second are often available. This feature allows the user to stipulate the short period of time between each of the microprocessor's downloads to storage. The faster the data-sampling rate, the larger the volume of data a receiver collects, and the larger the amount of storage it needs. A fast rate is helpful in cycle slip detection, and that improves the receiver's performance on baselines longer than 50 km, where the detection and repair of cycle slips can be particularly difficult.

INTRODUCTION & INSTRUCTIONS

📡 Overview Module 07: Static Positioning System Surveying

Module 07 Overview: Static GPS Surveying - GISC 4315



🧭 Module Overview

Static GPS/GNSS, where the receiver is stationary, is the original GPS/GNSS method. It is still the preferred approach for establishing the most accurate positions, the control. In many ways. For example, some processing and data checking should be performed on a daily basis during a GPS/GNSS project. Blunders from operators, noisy data, and unhealthy satellites can corrupt entire sessions. In any case, left undetected, such dissolution can jeopardize an entire project. These weaknesses in the data can sometimes be prevented before they happen by best practices such as optimization of the observation schedule. GPS/GNSS measurements are still composed of fundamentally biased ranges. Therefore, the goal remains to mitigate the biases and extract the true ranges. In this lesson, we start to get into the real details of how static GPS/GNSS work is done.

📚 Module Learning Objectives

At the successful completion of this lesson, students should be able to:

- discuss the difference between precision and accuracy
- discuss the basics of planning a static GPS/GNSS survey;
- recognize the role of NGS control;
- explain Continuously Operating Reference Stations;
- explain Static GPS/GNSS project design;
- demonstrate drawing GPS/GNSS baselines
- describe the difference between dependent and independent baselines;
- describe how to calculate the number of sessions necessary for a static survey;
- discuss some of the components of Static GPS/GNSS control such as equipment, station data sheets, visibility diagrams, monumentation and logistics

Read Chapter 6 in *GPS and GNSS for Land Surveyors*.

CONTENT

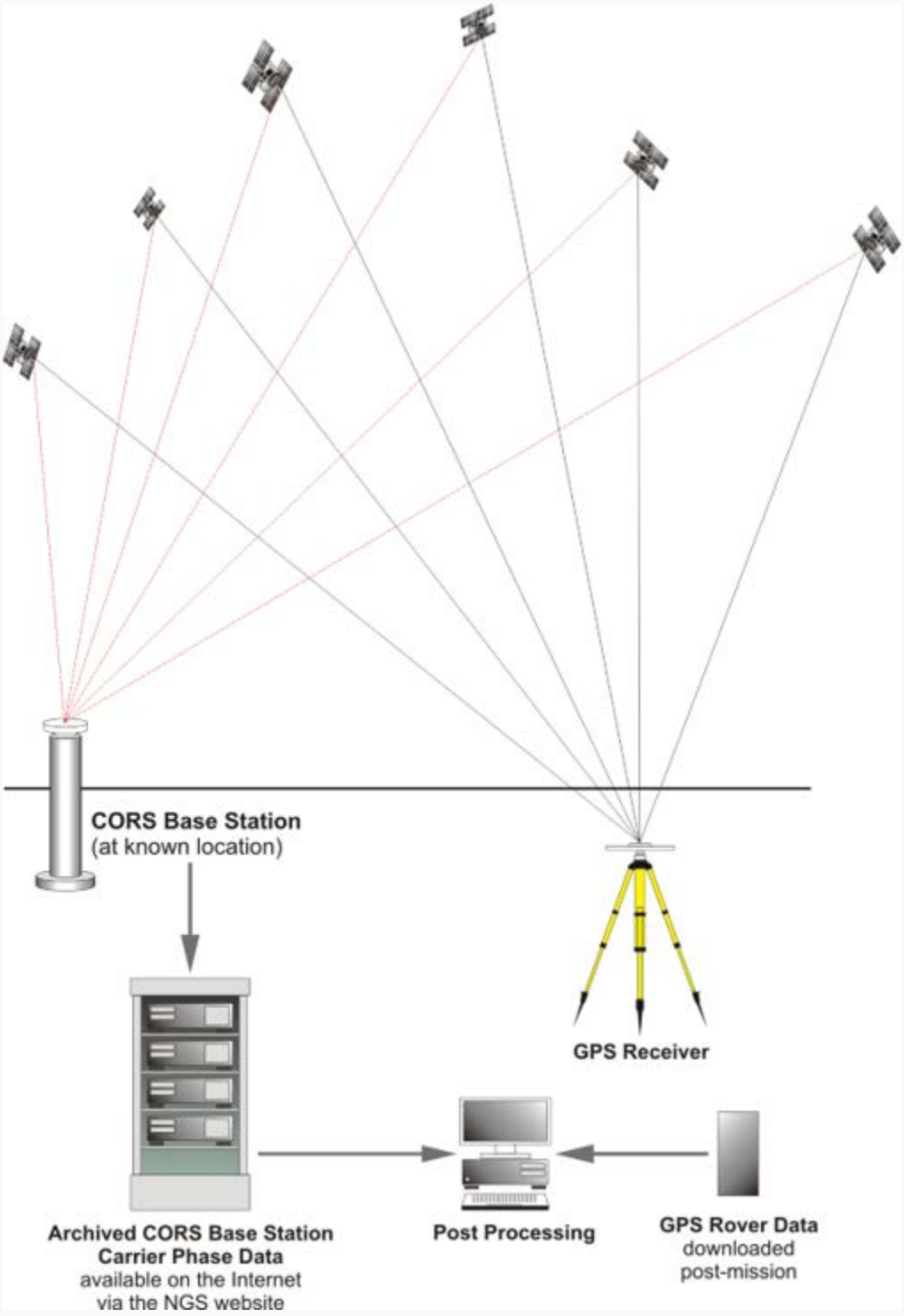
📚 Static GPS Control Surveying



Static GPS Surveying Overview

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📡 Static GPS/GNSS Fundamentals



Static GPS with Post-Processed Differential Correction

Static GPS/GNSS surveying has been used on control surveys from local to statewide to continental extent, and will probably continue to be the preferred technique in those categories. In static GPS/GNSS surveying, the receivers is motionless for a time, usually a relatively long occupation. If a static GPS/GNSS control survey is carefully planned, it usually progresses smoothly. The technology has virtually conquered two stumbling blocks that have defeated the plans of conventional surveyors for generations. Inclement weather does not disrupt GPS/GNSS observations, and a lack of intervisibility between stations is of no concern whatsoever, at least in postprocessed GPS/GNSS. Still, GPS/GNSS is far from so independent of conditions in the sky and on the ground that the process of designing a survey can now be reduced to points-per-day formulas, as some would like. Even with falling costs, the initial investment in GPS/GNSS remains large by most surveyors' standards. However, there is seldom anything more expensive in a GPS/GNSS project than a surprise.

CONTENT

📚 Planning a Static GPS/GNSS Control Survey: Accuracy and Precision

Precision and Accuracy in GPS Surveying

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🎯 Fundamental Concepts: Precision vs Accuracy

Precision and accuracy 1

A Few Words about Accuracy

When planning a GPS/GNSS survey, one of the most important parameters is the accuracy specification. A clear accuracy goal avoids ambiguity both during and after the work is done. First, it is important to remember that there is a difference between precision and accuracy. One aspect of precision can be visualized as the tightness of the clustering of measurements; the closer the grouping, the more precise the measurement. Accuracy, on the other hand, requires one more element. It has to have a *truth set.* For example, the *truth* in illustration for A, B and C is the center of the target - without that accuracy is indefinable. In other words, accuracy is not determined by measurement alone. There must also be a standard value or values involved is through the comparison of the measurements with such standard values that the outcome of the work can be found to be sufficiently near the ideal or *true* value, or not.

📡 GPS Precision and Accuracy Examples

Precision and accuracy 2

For example, on the left in the illustration it may seem at first that the average of the measurements in the GPS-A group are more accurate than the average of those in GPS-B because the GPS-A group is more precise. However, when the true position is introduced on the right it is revealed that the GPS-B group's average is the more accurate of the two, because accuracy and precision are not the same. When it comes to accuracy, there are other important details too. Local accuracy and network accuracy are not the same. As mentioned earlier, local accuracy, also known as relative accuracy, represents the uncertainty in the positions relative to the other adjacent points to which they are directly connected. Network accuracy, also known as absolute accuracy, requires that a position's accuracy be specified with respect to an appropriate *truth* set such as a national geodetic datum. Differentially corrected GPS/GNSS survey procedures which are tied to CORS stations, which represent the National Spatial Reference System of the United States, provide information from which network accuracy can be derived. However, autonomous GPS/GNSS positioning, that is a single receiver without augmentation is not operating relative to any control, local or national. In that context, it is more appropriate to discuss the precision of the results than it is to discuss accuracy.

📏 Horizontal and Vertical Accuracy Standards

Horizontal and vertical accuracy

It is typical for uncertainty in horizontal accuracies to be expressed in a number that is radial. The uncertainties in vertical accuracies are given similarly, but they are linear, not radial. In both cases, the limits are always plus or minus (±). In other words, the reporting standard in the horizontal component is the radius of a circle of uncertainty, such that the true location of the point falls within that circle at some level of reliability, i.e. 95-percent of the time. Also, the reporting standard in the vertical component is a linear uncertainty value, such that the true location of the point falls within ± of that linear uncertainty to some degree of reliability. GPS positioning it is reasonable to expect that the vertical accuracy will be about 1/3 that of horizontal accuracy. If the absolute horizontal accuracy of a

GPS position is ±1m then the estimate of the absolute vertical accuracy of the same GPS position would be ~±3m.

⭕ Circular Error Probable (CEP)

Circular Error Probable (CEP)

Here is a bit more on horizontal accuracy. The illustration shows a spread of positions around a center of the range. As the radius of the error circle grows larger, the certainty that the center of the range is the true position increases (it never reaches 100%).

It is not correct to say that every job suddenly requires the highest achievable accuracy, nor is it correct to say that every GPS/GNSS survey now demands an elaborate design. In some situations, a crew of two, or even one surveyor on-site may carry a GPS/GNSS survey from start to finish with no more planning than minute-to-minute decisions can provide, even though the basis and the content of those decisions may be quite different from those made in a conventional survey.

🚁 Survey Planning and Field Conditions







Helicopter

In areas that are not heavily treed and generally free of overhead obstructions, sufficient accuracy may be possible without a prior design of any significance. But while it is certainly unlikely that a survey of photocontrol or work on a cleared construction site would present overhead obstructions problems comparable with a static GPS/GNSS control survey in the Rocky Mountains, even such open work may demand preliminary attention. For example, just the location of appropriate vertical and horizontal control stations or obtaining permits for access across privately owned property or government installations can be critical to the success of the work. An initial visit to the site of the survey is not always possible. Today, online mapping browsers are making virtual site evaluation possible as well. Topography as it affects the line of sight between stations is of no concern on a static GPS/GNSS project, but its influence on transportation from station to station is a primary consideration. Perhaps some areas are only accessible by helicopter or other special vehicle. Initial inquiries can be made. Roads may be excellent in one area of the project and poor in another. The

general density of vegetation, buildings, or fences may open general questions of overhead obstruction or multipath. The pattern of land ownership relative to the location of project points may raise or lower the level of concern about obtaining permission to cross property. It is now relatively easy to do GPS/GNSS surveying, that is, when everything works the way it's supposed to. However, if there are going to be troubles like access to the points, overhead obstructions, work in trees, or helicopter transport, planning needs to be part of the process.

CONTENT

📚 NGS Control Data Sheets



📊 Understanding Existing Control

NGS Data Sheet

One of the things that is very useful is understanding the existing control that is currently available. Brass tablets and brass caps have been set around the United States by the National Geodetic Survey also known as passive marks. You can retrieve sheets that are similar to what you see in the illustration. These sheets have a great deal of excellent information concerning the coordinates, the quality, and many of the details about existing control points that can be used as checkpoints in any GPS/GNSS survey. Monuments that are the physical manifestation of the National Spatial Reference System (NSRS) and can be occupied with survey equipment are known as *passive* marks. They can provide reliable control when properly utilized. That utilization should be informed by an understanding of the datasheet that accompanies each station and is easily available online. There is a good deal of information about the passive survey monuments on each individual sheet. In addition to the latitude and longitude, the published data include the state plane coordinates in the appropriate zones. The first line of each data sheet includes the retrieval date. Then the station's category is indicated. There are several, and among them are Continuously Operating Reference Station, Federal Base Network Control Station, and Cooperative Base Network Control Station. This is followed by the station's designation, which is its name, and its Point Identification, PID. Either of these may be used to search for the station in the NGS database. The PID is also found all along the left side of each data sheet record and is always two upper case letters followed by four numbers. The state, county, country and U.S. Geological Survey (USGS)-7.5 minute quad name follows. Even though the station is located in the area covered by the quad sheet, it may not actually appear in the map. Under the heading, "Current Survey Control," you will find the latitude and longitude of the station in NAD83 which is fixed to the North American plate, currently in NAD83 (2011), and its height in NAVD88. The orthometric height in meters is listed as "ORTHO HEIGHT" and followed by the same

in feet. When the height is derived from GPS/GNSS observation, a geoid model must be used to determine the orthometric height. The model used is given. Adjustments to NAD27 and NGVD29 datums are a thing of the past. However, these old values may be shown under Superseded Survey Control. Horizontal values may be either *Scaled*, if the station is a benchmark or *Adjusted*, if the station is indeed a horizontal control point.

📏 Vertical Control Categories

When a date is shown in parentheses after NAD83 in the data sheet, it means that the position has been readjusted. There are 13 sources of vertical control values shown on NGS data sheets. Here are a few of the categories. There is *Adjusted*, which are given to 3 decimal places and are derived from least squares adjustment of precise leveling. Another category is *Posted*, which indicates that the station was adjusted after the general NAVD adjustment in 1991. When a station's elevation has been found by precise leveling but non-rigorous adjustment, it is called *Computed*. Stations' vertical values are given to 1 decimal place if they are from GPS/GNSS observation (Obs) or vertical angle measurements (Vert Ang). And they have no decimal places if they were scaled from topographic map, *Scaled*, or found by conversion from NGVD29 values using the program known as VERTCON.

🌍 ECEF Coordinates and Geodetic Information

When they are available, earth-centered earth-fixed (ECEF) coordinates are shown in X,Y and Z. These are right-handed system, 3-D Cartesian coordinates and are computed from the position and the ellipsoidal height. They are the same type of X, Y, and Z coordinates presented earlier. These values are followed by the quantity which, when added to an astronomic azimuth, yields a geodetic azimuth, is known as the Laplace correction. It is important to note that NGS uses a clockwise rotation regarding the Laplace correction. The ellipsoid height per the NAD83 ellipsoid is shown followed by the geoid height where the position is covered by NGS's GEOID program. Photographs of the station may also be available in some cases. When the data sheet is retrieved online, one can use the link provided to bring them up. Also, the geoidal model used is noted. Of course, the first retrieval date at the top of the sheet is valuable to know when the point was last recovered. There's much detail here. I leave it to you to read. But suffice it to say for the objective here, the NGS data sheet is helpful to learn what control is available in a particular area.

Coordinates

NGS data sheets also provide State Plane and Universal Transverse Mercator (UTM) coordinates, the latter only for horizontal control stations. State Plane Coordinates are given in both U.S. Survey Feet or International Feet, and UTM coordinates are given in meters. Azimuths to the primary azimuth

mark are clockwise from north and scale factors for conversion from ellipsoidal distances to grid distances. This information may be followed by distances to reference objects. Coordinates are not given for azimuth marks or reference objects on the data sheet.

🗿 Station Mark Information

The Station Mark

Along with mark setting information, the type of monument and the history of mark recovery, the NGS data sheets provide a valuable to-reach description. It begins with the general location of the station. Then starting at a well-known location, the route is described with right and left turns, directions, road names, and the distance traveled along each leg in kilometers. When the mark is reached, the monument is described, and horizontal and vertical ties are shown. Finally, there may be notes about obstructions to GPS/GNSS visibility, and so forth.

Significance of the Information

The value of the description of the monument's location and the route used to reach it is directly proportional to the date it was prepared and the remoteness of its location. The conditions around older stations often change dramatically when the area has become accessible to the public. If the age and location of a station increases the probability that it has been disturbed or destroyed, then reference monuments can be noted as alternatives worthy of on-site investigation. However, special care ought to be taken to ensure that the reference monuments are not confused with the station marks themselves.

🔄 Future of Passive Marks

The role of passive marks is undergoing a change as a new North American reference frame takes shape (North American Terrestrial Reference Frame: NATRF). NGS is no longer setting nor maintaining passive marks.

CONTENT

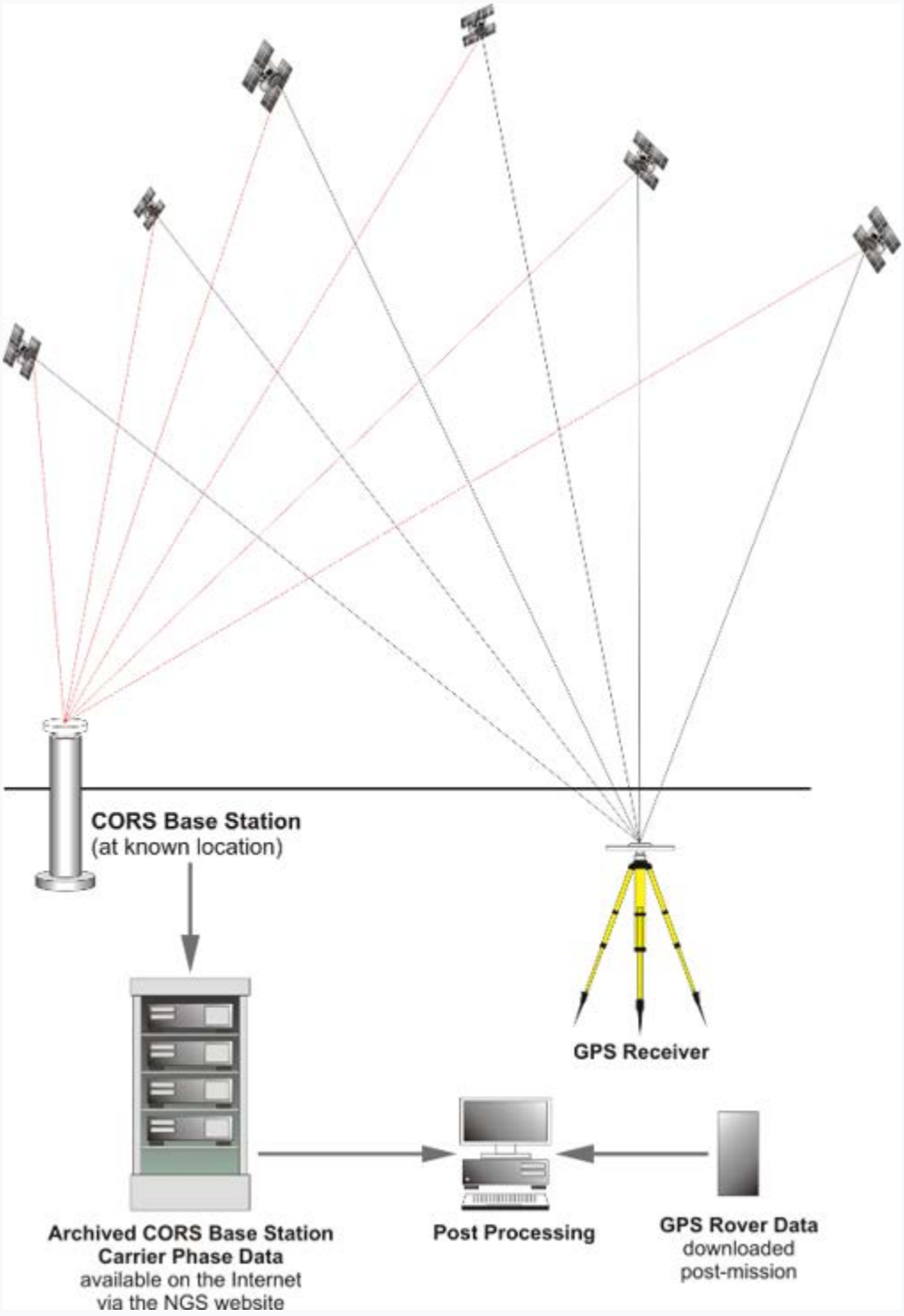
📚 Control from Continuously Operating Networks



Continuously Operating Reference Stations

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📊 CORS Networks and Active Stations



Static Surveying

The requirement to occupy physical geodetic monuments in the field can be obviated by downloading the tracking data available online from appropriate continuously operating reference stations (CORS) where their density is sufficient. These stations, also known as Active Stations, comprise fiducial networks that support a variety of GPS/GNSS applications. While they are frequently administered by governmental organizations, some are managed by public-private organizations and some are commercial ventures. The most straightforward benefit of CORS is the user's ability to do relative positioning without operating his own base station, by depending on that role being fulfilled by the network's reference stations. While CORS can be configured to support differential GPS (DGPS) and real-time kinematic (RTK) applications, as in Real-Time Networks, most networks constantly collect GPS/GNSS tracking data from known positions and archive the observations for subsequent download by users from the Internet.

These days it is quite possible to not have to occupy an existing brass tablet, brass cap, such as indicated on this NGS data sheet. The reason for this is because of the continuously operating reference stations that have been established by the National Geodetic Survey are very useful. These active stations are a network and the most straightforward benefit is that a user can do relative positioning without operating his own base station and by depending on the CORS station, which as you see in this illustration downloads its reference information from tracking the GPS/GNSS network to an archived base station that is available on the Internet. You can access this and then combine it with the GPS/GNSS data that you have collected and post process for a differential, position a relative position, without having to occupy an existing control point on the ground.

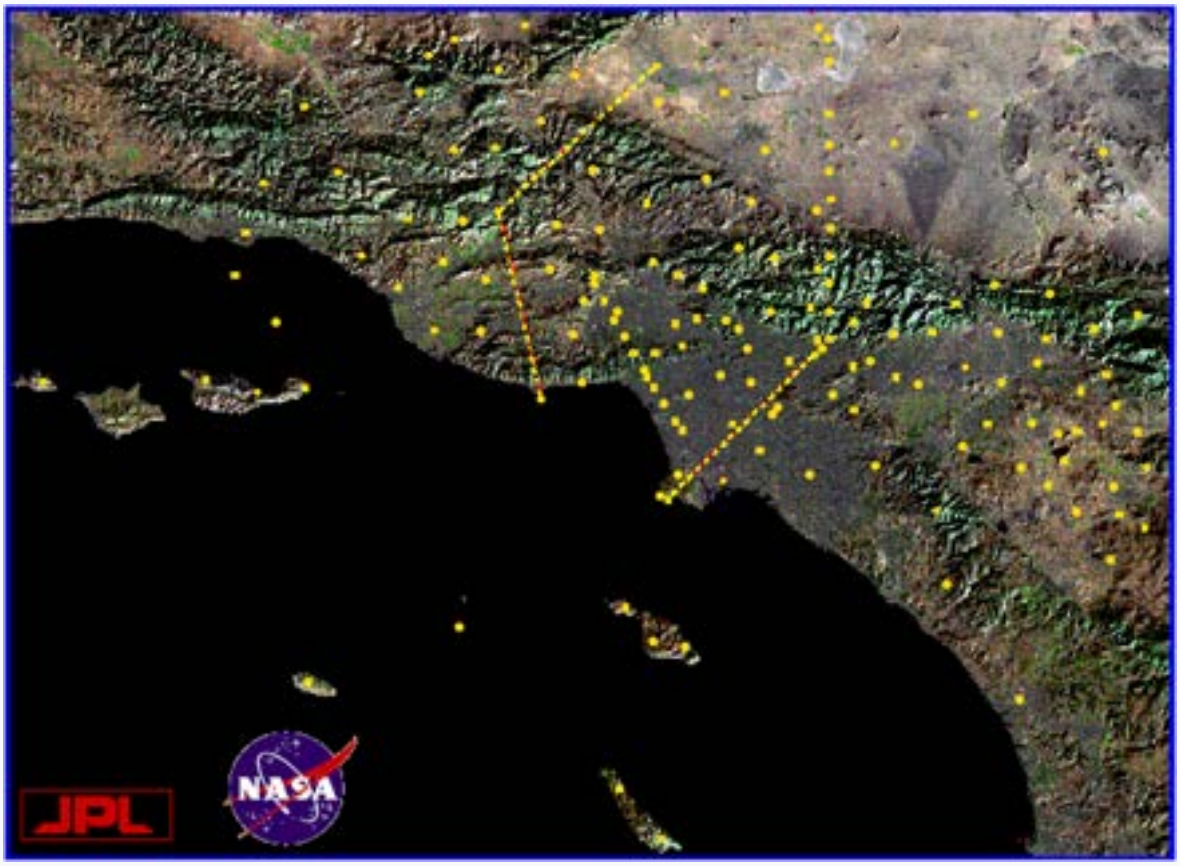
🌍 International CORS Networks



IGS Network

In many instances the original impetus of a network of CORS was geodynamic monitoring as illustrated by the GEONET established by the Geographical Survey Institute, GSI, in Japan after the Kobe earthquake. Networks that support the monitoring of the International Terrestrial Reference System, ITRS have been created around the world by the International GNSS Service, IGS, which is a service of the International Association of Geodesy and the Federation of Astronomical and Geophysical Data Analysis Services originally established in 1993. The United States is by no means the only place in which such networks exist.

🏛️ Regional Networks and Partnerships



Southern California Integrated GPS Network

Here is an illustration of the Southern California Integrated GPS/GNSS Network, SCIGN is a network run by a government-university partnership. More and more of these networks are appearing across the United States, sometimes on a statewide basis, county basis, and other regional designations. These continuously operating stations occupy a known position and then track the full constellation of GPS/GNSS satellites continuously. Then this data is made available, sometimes with a subscription cost, but in the case of NGS for free, and it is available then to do differential positioning of static GPS/GNSS sessions.